

The Thiele Machine

A Complete Account from First Principles

An abstract model of computation, and an argument
about what an account of computation should preserve

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Before the book: the short version, and how to check it

I didn't invent a machine. I found one.

That is how I see this work. I think there is structure here that the usual picture of computation leaves out. The conviction brought me to the project; the definitions and proofs are how I put it to the test.

What the Thiele Machine is. It is an abstract model of computation. It is a machine the way Turing's machine is a machine: mathematics you can reason about, not a device. It is not a CPU. It is not an instruction set, a VM, a graph, or category theory. Those are things I built with it, or things it can carry.

The model is small. A state. A step rule. A ledger that only climbs. A rule for which steps have to pay. And a window: what an observer is allowed to see of the state. Like a Turing machine, it is a precise way to say what a computation is. Unlike a Turing machine, it does not compute anything new. It adds what a computation establishes, and what establishing it cost.

What this book is. It is an argument, made with that model. The argument is about what an account of computation should preserve.

I built this because the answer at the end of a computation did not tell me everything I wanted to know. What structure did the machine use? What did it establish? What did it pay to establish it? I wanted those questions in the mathematics, with definitions somebody else could check.

The usual account describes a computation by what it can read and write: memory, registers, where the program is. I think that account is a shadow. There is another axis. It holds what the computation established, and what that cost. Here is the argument, one step at a time, with where each step stands.

1. *The shadow loses something real.* Two runs can look identical through the usual window and still differ in what they established and what they paid. Nothing reading only the window gets the difference back. Proved, for each window I name.
2. *Establishing is not free.* If the step that turns uncertified into certified has to pay, every run that starts uncertified and ends certified has paid. Proved, for any system that follows that rule, whatever its states and instructions are. And the rule itself isn't arbitrary. On a machine with finite memory, a certificate that can never be revoked can only be switched on by a step that merges two states, and merging is what Landauer's principle charges for whenever the machine could

be in either state. Proved, with Landauer's principle as the named premise. The heat comes from that uncertainty: a machine already known to be in one state loses nothing, and that is proved too. Learning isn't what costs. A measurement that forgets nothing is free, and an observer still learns from it. That is proved too. What costs is committing to what you learned, in a record the machine can't take back. A small finite piece of the VM is such a machine, and the VM runs it paying exactly what its merges cost. The full VM prices the merge that certifies and leaves other merges free. That is a design choice, and both halves of it are proved.

3. *Leaning on a fact takes a paid history.* A computation is entitled to lean on a structural fact when it holds the evidence for it and a history that earned it. Defined precisely; the cost floor is proved for shortcuts that hand over their receipts.
4. *So the axis belongs in the step.* An account of computation should carry what was established and what it cost, as part of how it steps. Not as something a program bolts on afterward. Proposed. This is the conviction the proofs are there to test. Part of it is now proved. If the account is going to charge certification exactly, never less and never more, the charge can't be computed from the window. A step that certifies and a step that doesn't can look the same through it, so any charge read off the window either misses the one or bills the other. The exact price needs the reading inside what the step carries. Proved, for any machine with such a pair, and the VM has one for both windows I name. On a machine with finite memory, an account that prices merging steps the way Landauer's principle does already charges every step that writes a permanent record, at least the logarithm of how many states it squeezes together. The same bound holds in bits of Shannon entropy. Whether every account has to price merges is physics, not a theorem.
5. *Which events every account has to price.* Certification is the one I pinned down. Why that one has an answer now: the events a finite machine must price are the steps that merge states, every step that writes a record nothing takes back is one of them, and a flip escapes the price only when the same step can take the reading back somewhere. Proved, with Landauer's principle as the named premise. The stronger guess, that the priced events are exactly the permanent records, is false, and a three-state counterexample shows it. What else belongs on the meter is open. The pointer-observable conjecture is my candidate answer.

Certification is one point I could pin down. It gave me a law I could state, a VM I could build and run, and proofs I could put in front of someone who did not share my intuition. It is one witness in a much larger argument about structure and observation. It is not the argument. I am not done looking.

How the book is laid out. Part I is the argument. Part II is one build of the model:

a 51-opcode VM, the hardware it runs on, a category-theory layer for structure, and a check against the CHSH bound. The build is there so the argument has something you can run and try to break. It is a witness. It is not the subject. Part III is how to check all of it. Part IV is what is still open. The vocabulary I had to learn is at the back, for anyone who wants it.

The shadow is real for the projections I proved it for. Drop the relevant distinction and no decoder gets it back from what remains. A different encoding can keep it. That is part of the question: which distinctions deserve to survive an account of a computation?

I think there is something foundational here. I will be straight about the difference between that conviction and a theorem. The results below have statements, premises, and proof terms. The larger claim still has work to do. I built the machine so that work could be done in public, with something firmer than my confidence to argue over.

Abstract

The Thiele Machine is an abstract model of computation, and this book is an argument made with it. The argument is about what a computation keeps and what keeping it costs. The usual account of a computation, what it reads and writes, is a shadow of it; the model adds the axis that account leaves out. It carries structure as part of its state, a ledger that only climbs, a rule for which events have to pay, and a window saying what an observer sees. The first result is about that window. Two runs can look identical through it and still differ in what they established and what they paid. No function of the window gets the difference back, and no checker that reads only the window can be both sound and complete about a claim that turns on it. That is a theorem about what the shadow loses. It is not true by definition. Certification is the one event I pinned down all the way. The rule is called A2: any step that flips a designated yes/no reading from false to true costs at least one. A2 is a rule I chose, and the cost theorem built on it sits close to that choice: once the flip is priced, every run that flips has paid. What A2 adds is where it lives. It is part of the step rule. It is not a program running on top of one. And for an exact price it has to be: when a step that certifies and a step that doesn't look the same through the window, no charge computed from the window prices certification exactly, and the VM has such a pair for both windows I name. The cost theorem built on A2 doesn't care what the states or instructions are. Any system that follows A2 is covered.

The formal results go further than that one floor. They pin down exact unit pricing relative to how many certification flips happen, show the ledger is unique once the schedule is fixed, exhibit states that a specified window can't tell apart, give a universal property for folding traces, and check some specialized algebraic certificates. The executable VM and the hardware bridges are one build of the model. They realize parts of this, each under its stated conditions. They are witnesses to the argument. They are not its subject. Paying for an event is accounting. Truth needs a checker whose soundness is proved.

Two questions stay open: what the ledger means physically, and which events every account of computation has to price. For the full instruction set, the F1

physical premise pair can't coexist with free jumps, and `F1_physical_premises_incompatible` proves it. A narrower pair does hold together. On a machine with finite memory and a certificate that can never be revoked, the step that certifies has to merge two states, so pricing merges, as Landauer's principle does in the worst case over what the machine might hold, yields A2 (`a2_from_merging_price_and_permanence`). Priced per halving, switching k states on beside m certified ones costs at least $\log_2((m+k)/m)$. In Shannon entropy the same step removes at least that many bits from the uniform distribution on the states in play, and with Landauer's principle as the premise that is at least $k_B T \ln((m+k)/m)$ of heat. Any distribution that leaves every state in play possible forces positive heat; a state known in advance forces none, and both are proved. A four-slot piece of the VM with a certification flag is such a finite machine. The VM runs it step for step, and its ledger rises by exactly what the merges cost. The full VM prices only the merge that certifies and leaves the rest free, by design. And on finite hardware the events that must be priced are exactly the merges: every permanent-record write is one, and a flip escapes the price only when the same step can take it back. What stays open on the physical side is the experiment. The observation results don't forbid encodings that keep the information. The trace-fold property doesn't make the VM's state space initial.

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Part I

The argument

This part is the argument, and the model it is made with. Where a VM instruction shows up here, it is an example of a step. The build it comes from is Part II.

1 Let me be straight with you

One thing before anything else, because people get it backwards. I used language models to write the code and the proofs in this project, and I trust language models about as far as I can throw them, which is exactly why I ran every line through a proof assistant that does not care who or what wrote it. Coq rejects a wrong proof the same whether a tenured professor or a model hands it over, and it accepted these, with none of my own axioms anywhere in them, re-checkable using the reproduction procedure in Section 21.1. Being self-taught and using modern tools did not make this looser. It is the reason I made a machine check all of it, me included. The long version of that is in Section 2, under how the work actually got made; I am putting the short version up here so it lands before anything else does.

What this project is. It's an argument about what an account of computation should preserve, made with an abstract model built to carry it. Its central proposal is structural entitlement. A computation is entitled to lean on a structural fact when it holds the evidence for it and a history that earned it. The model, its laws, and the instructions of any one build are different things, and I keep them apart. Generic `CERTIFY` pays to set a flag. It checks nothing. A claim that something is actually true needs explicit witnesses and a checker proved sound. Some results here are that kind. The generic A2 theorem isn't one of them. It's an accounting law.

The proved core of the argument, in three steps.

1. Anything that follows A2 pays something to get from uncertified to certified.
2. A window that shows two states as the same, when they disagree on the ledger or on some other question, can't answer that question on every state.
3. A checker that only sees evidence where those two states collide can't be both sound and complete about the claim that separates them.

Those are statements about named laws and named windows. An ordinary machine can keep a richer encoding, or enforce the same invariant through its own transition table. What I'm proposing is about why the extra distinctions matter. It isn't a theorem that every Turing-equivalent system lacks them.

Substrate vs scaffolding. Two layers run through the rest of this document. I kept

conflating them myself until the proofs forced the line. Confuse them and the theorems quietly stop meaning what they say.

The *substrate* is the shape: a kind of state, and a rule for stepping from one state to the next. Different theorems ask that shape for different extra pieces, and the pieces aren't the same. No Free Insight asks for a certification reading and the A2 rule on it: flipping it from false to true costs at least one. Ledger initiality asks for something else, a fixed starting value and a fixed schedule of increments. Each theorem holds for any state type and step rule that carries the pieces it asks for, whatever instruction set builds them.

The structural-axis undecidability theorem in Section 10 asks for different pieces again. The base substrate brings a ledger that never goes down, a way to write its own programs down as states, and Kleene's recursion theorem for the program transformations it can express. The `WithShortcutPredicate` extension adds a shortcut predicate, one program it accepts, one it rejects, and the rule that programs which behave the same get the same answer. None of that needs A2. The candidate-indexed `nat_substrate` family that supplies its restricted fixed points carries no certification content at all, by construction. So "substrate-level" doesn't mean one fixed interface every theorem shares. It means each result is stated over exactly the extension it uses. It isn't folded into the 51-opcode VM.

The *scaffolding* is everything that holds the shape up so you can touch it. That's the 51 opcodes, the Python reference kernel, the extracted OCaml runtime, the Bluespec hardware, and the simulator harness. The opcode semantics live here too, along with the conditional correspondence between the Kami model and `vm_apply`, the synthesized gate count, and the translations further down that are tested or trusted to the tools. That's how this particular substrate gets concrete enough that you can boot it, audit it, and run it on silicon. The 51-opcode VM fills in several of the extensions at once. It's an A2-respecting `CertCostMachine`, and it's a `WithShortcutPredicate` instance, among others. A substrate-level result still only borrows the extension its own theorem names. That's the line to hold onto. The VM is one instance filling several extensions at the same time. It isn't one flat interface called "the substrate." Let that line blur and a theorem stated over one extension starts reading as if it carried premises it never uses.

What "new model of computation" means here. It adds something to how a machine runs: structure it keeps, and an account of what that structure cost. It doesn't add computable functions. When I say "classical shadow" I mean the particular window or fragment I name. I don't mean every way a conventional machine could store things. Enforcing the law, storing it, and letting an observer see it are separate questions. It is an abstract model the way the Turing machine is. It isn't a new primitive the way the Turing machine is. A Turing machine builds computation out of a tape, a head, and a table. The substrate takes any kind of state and any step rule and asks them to obey a

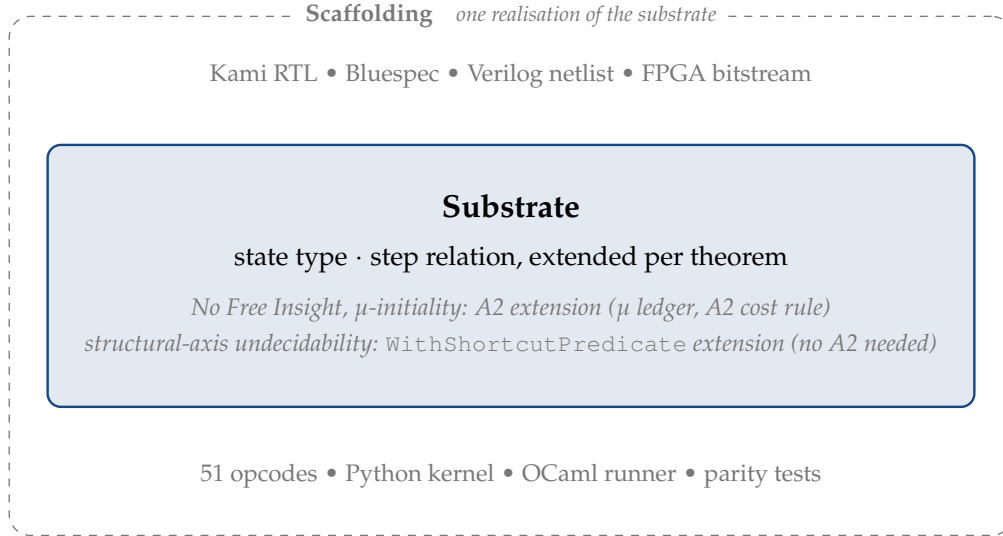


Figure 1: Substrate inside, scaffolding around it. The substrate is a kind of state and a step rule, and each theorem asks it for its own extra pieces. No Free Insight needs the certification floor. Ledger initiality needs a starting value and an increment schedule. Structural-axis undecidability needs the separate `WithShortcutPredicate` extension and no A2 at all. Each theorem holds for any system carrying the pieces it uses, whatever the instruction set. The scaffolding is what makes this particular substrate something you can run, check, and synthesize. Merging the two layers, or merging the two extensions into one, is the most common way this document gets misread.

law. So it sits on top of a machine. It doesn't stand beside it. A Turing machine whose table carries the rule is one of its instances. What I am claiming is narrower and, to me, more interesting. The law, the structural state it prices, and what an observer can and cannot see of that state belong in the step. They are not something a program bolts on afterward.

The renderer experience is the entry point, and Section 2 tells it in full. I started by trying to build a 3D engine where category-theoretic objects were the runtime objects. Modules. Morphisms. Composition. Not data structures simulating those things. The actual things. The classical CPU saw none of that. It shuffled bytes that happened to encode the structural state. Z3 saw formulas that happened to refer to the same shapes. The structural facts the engine was computing with were in the architecture, not in the data. That is the concrete shape I had to formalise. The substrate this document describes is what came out: structure as state, not as an approximation written on top of state. Compositional structure was just the first kind I met.

The formal results turn that into a precise version of the shadow. `thiele_simulates_turing` lifts a Turing machine into the model, inside its own file-local transition system. `degenerate_projection_theorem` pins down exactly what the chosen window forgets: two states have equal six-field shadows exactly when they satisfy the named equality relation. Keeping two executions in step takes more than that. The

graph and the CSRs have to agree too. The VM's separation witnesses show which structural distinctions each named window loses.

The irrecoverability theorems are about `vm_certified`, `csr_cert_addr`, and μ , each under its named window. Once more than one state sits behind a view, nothing gets every original back from it. You can still pick defaults to fill the gaps. The defaults don't tell you what was really there. And an encoding that keeps the information, with a decoder that reads it, is always allowed.

What is claimed unique, and what is not. When I say there's only one possible substrate, I'm squeezing three different claims into one breath. This document proves one of them.

The first is about the law. Given the A2 rule, is there basically one honest way to price certification? Section 7 answers a version of that for a fixed schedule. `instruction_consistent_measure_equals_mu` shows that any measure using the kernel's own per-instruction schedule and starting at zero has to equal μ on every reachable state. It doesn't show the schedule itself is forced. A different schedule prices things differently, and that's a different accounting, just as honest.

The second is about the core. Strip every A2-respecting system down to what the theorems actually use. Do they all turn out to be the same object, under the right idea of "the same"? That's the strongest reading of "only one substrate." It isn't a theorem here. It's a conjecture, written down so it can be attacked. Round 1 calls a record-carrying machine *adequate* when its natural-valued ledger is monotone, its certification reading satisfies A2, some run from an initial state is certified, and each two-counter halting problem has an initial state whose observed halt agrees with that problem. The last clause is an existential halting correspondence, not a two-way effective simulation theorem. Written formally, its conjecture is

$$\text{Adequate}(M) \implies \text{StructuralCore}(M) \simeq \text{ThieleCore},$$

Here M is already the six-field structural record, and $\simeq$ is the relation `core_equiv`: a bisimulation that covers starting states in both directions and observes certification and next-step ledger cost. It ignores absolute ledger balance and halting. Round 2 fixes a stronger extension criterion and equivalence below. Neither relation is literal sameness of opcodes, storage layout, implementation language, or clock cycles.

One construction shows why the equivalence has to say explicitly what it observes. Section 3.7 builds a target (the Coq witness is `certification_agreement_does_not_imply_descent`) that keeps the VM's whole instruction history, charges the same schedule, and reads certification the way the VM does. It agrees with the VM on every certification and every charge. It still tells the empty trace apart from `JUMP 0 0`, two traces the VM treats as the same, because it kept the history and the VM didn't. A2 and matching certification don't make that extra history fall away. Round

1's relational bisimulation says why the kept history does not count, and `history_core_equiv_thiele` proves that quotient. This is a sanity result for the definition, not evidence that every adequate machine is equivalent.

The third is about this exact machine. Is this particular VM state space the only one, up to renaming? I don't claim that anywhere. No theorem needs it. The separating witnesses (Section 14's graph pair and the irrecoverability theorems above) stand on their own however the second question comes out. They don't need it, and it doesn't need them.

So the first is proved, for a fixed schedule. The third isn't claimed. The second is open, written as two dated conjectures with different premises and observations. Each is a statement to test, not a theorem, and each has a test case it has to survive.

Dated comparison definitions. Two rounds are fixed on 29 September 2026. Round 1 in `StructuralCore.v` observes certification and next-step cost through a bisimulation covering initial states in both directions. It discards retained history, absolute ledger balances, and halting observations. Its adequacy condition requires a monotone ledger, A2, some reachable certified state, and an existential correspondence with two-counter-machine halting. The last condition supplies neither an effective input encoding nor a two-way computability theorem. Initial certification already satisfies the record condition.

Round 2 in `StructuralCoreRound2.v` concerns extensions of the specified unbounded VM. An explicit projection commutes with steps, preserves halting, and covers every reference initial state. The ledger is monotone; certification is permanent and has a reachable false-to-true write satisfying A2. Its bisimulation observes certification, absolute ledger balance, halting, and step cost. Its relational form permits extra history to be ignored, but no Round 2 history-equivalence theorem is claimed at this checkpoint. Neither cost rescaling nor stuttering is implicit. These definitions do not supply a uniqueness theorem. The rounds remain separate so a later outcome cannot change the question that was tested.

What this project is not. It is not a claim that I solved P vs NP. It is not a theory of everything. It is not a claim that computation is physics or that physics is computation. The CHSH algebraic result is pure math. The Landauer connection is motivation. That is the full extent of the physics content here. The rest is bookkeeping.

A note before the technical content, on what I do and do not know about prior work. I started this with no background in computer science, math, physics, quantum mechanics, hardware, or category theory. None. I didn't know what a Turing machine was. I didn't know what Coq was. I built this alone. Nobody mentored me or walked me through the field. The only person who has touched this work in any way is a friend who taught me how to use GitHub and set up an IDE. Every name, every comparison, every paper I went looking for, came from me searching on my own. So

treat what follows as a list of labels I reached for, not a literature review. Where I have mismatched one against the real thing, tell me, because that is a correction I can use.

What I compare each time is what a result is actually about. Landauer’s bound is about heat: a physical cost, under a thermodynamic model. No-free-lunch is about averages over a whole spread of problems. A2 is a floor on one kind of step, the step that flips a designated certification reading on. Exact pricing adds a second rule on top of that floor: never charge more. So the comparisons below are about the shape of each result. They are not claims about who got somewhere first. They don’t show certification is the only event worth pricing, either. The two sources I lean on hardest are Landauer, [“Irreversibility and Heat Generation in the Computing Process”](#) (1961), and Wolpert–Macready, [“No Free Lunch Theorems for Optimization”](#) (1997).

Here are the labels I went and looked into. Each entry says what I understood the label to mean and why I do not think Thiele reduces to it. Each is a claim inviting correction. If the comparison is wrong, tell me.

- Linear logic (Girard). What I read: a kind of logic where assumptions are consumed when you use them, like physical resources. Compiled into modern programming languages, it shows up as type rules that say “this value can only be used once.” My reading of why Thiele is not a restatement: linear logic constrains what the program can write down. Thiele’s A2 rule constrains what the machine’s step function does. Same intent. Different layer.
- Proof-carrying code (Necula). What I read: when one machine downloads code from another, the code can carry a small proof that the host can check before running it. The proof is data accompanying the program. My reading: Thiele’s cost-on-certification is enforced by the machine itself, not by a proof bundled with the program. The program does not carry receipts. The machine produces them. Foundational PCC (Appel) is closer in spirit: proofs about a small trusted base, not certificates bundled with the program. I haven’t read enough Appel to say more.
- Session types (Honda). What I read: type systems that constrain how two or more concurrent programs talk to each other so the conversation cannot go off-script. My reading: Thiele tracks structural facts a single computation has paid for. It does not constrain inter-program communication. Different problem.
- Bennett’s reversibility argument. What I read: Charles Bennett (1973) showed that you can in principle organise any computation so it does not erase information mid-stream. Operations that throw information away are the irreversible ones. Everything else can be undone. My reading: in the 51-opcode realisation, CERTIFY and MORPH_ASSERT and EMIT and REVEAL, among others, are charged even at $\delta = 0$, a floor no program can waive, while the rest are free at $\delta = 0$. Positive cost is the larger set, since any instruction can carry a voluntary δ ; the mandatory floor is the part I put there on purpose. The shape matches.

The content does not: the floor is an accounting rule, not a proof that those operations are logically or physically irreversible. The history-lift counterexample in Section 19 shows why transition injectivity and charge are separate questions. The dollars-and-joules calibration is a separate named hypothesis.

- Landauer’s bound. What I read: Rolf Landauer (1961) argued that erasing one bit of information must release at least $kT \ln 2$ of heat into the environment around the chip. The expression means Boltzmann’s constant times temperature times the natural log of 2. My reading: this is the physical-units side of Bennett’s argument. In Thiele the structural side, the substrate tracking each positive-cost transition via μ , is proved. The conversion to joules is a named hypothesis (see Section 19). I treat Landauer’s claim as motivation, not as something the proofs rest on. Motivation pays no bills in Coq.
- No Free Lunch (Wolpert and Macready, 1997). What I read: averaged over every possible problem, no search method beats blind guessing. Whatever an algorithm wins on one set of problems it gives back on another. My reading: this is the same “structure isn’t free” intuition, but it lives as an average over a whole problem space, and the cost is refundable. You buy an edge on your problems by spending prior assumptions, and you pay for it somewhere else. A2 charges one commitment event in one run, with no refund anywhere. Same family. Different shape: an average, not a per-step floor.
- Kolmogorov complexity (Kolmogorov 1965; Solomonoff and Chaitin reached it independently around then). What I read: the amount of structure in an object is the length of the shortest program that prints it. Most objects have no short description, so you cannot squeeze below that floor. My reading: same “structure isn’t free” intuition, but it is a static property of a finished object. No steps. No moment of paying. A2 charges the moment a running machine commits to a certificate. Same family. Different shape: a property of a string, not a law of a step.
- Cook-Levin and P vs NP (Cook 1971, Levin 1973). What I read: checking a solution can be cheap while finding one looks expensive. Whether finding is genuinely harder than checking is the open P-vs-NP question. Cook-Levin is the theorem that SAT is NP-complete, not a proof that search is hard. My reading: same find-versus-check intuition, but it is an asymptotic, worst-case, still-open statement about how difficulty grows with problem size. A2 charges a constant on the one commitment step in a concrete run, whether finding it was easy or hard. Same family. Different shape: an open scaling gap, not a per-step floor. I am not claiming anything about P vs NP; see “What this project is not.”
- Rovelli’s cost of choosing (Rovelli 2024, *The Thermodynamic Cost of Choosing*). What I read: committing to one branch of a binary choice at temperature T dissipates at least $kT \ln 2$. That is the same floor Landauer found for erasure, now charged on the act of deciding rather than the act of forgetting. My reading:

this is the closest existing thing to A2's claim. It charges a commitment, not a finished object or an average. But it is physical: indexed to a temperature, living in thermodynamics. A2 charges the commitment in a substrate, with no temperature and no physics, blind to how the witness was found. Closest neighbour. Not the same chair.

- CerCo (certified cost annotations through a verified C compiler). What I read: take a C program, run it through a compiler that itself has been formally proved correct, and attach cost annotations to the source program. The compiler preserves those annotations all the way down to machine code. My reading: CerCo's cost lives as decoration on the source code that survives compilation. Thiele's cost lives in the step rule of the abstract machine. The path down to hardware is proved as far as the Kami model, under stated premises, and tested past it. Different mechanism. Different layer.
- Kami-RISC-V. What I read: a verified processor design written in Coq using the Kami framework that targets the RISC-V instruction set. RISC-V is an open-source CPU architecture, a published list of instructions any chip designer can implement. My reading: I use the same Coq-to-Kami-to-Bluespec pipeline that the Kami-RISC-V work pioneered. The instruction set is mine. The cost discipline is mine. The proofs were built for this machine and exist nowhere else. The toolchain is shared. Nothing else is.
- NPA hierarchy (Navascués-Pironio-Acín). What I read: a way of bounding what kinds of correlations between two separated experiments are achievable using quantum mechanics. The bound takes the form of a sequence of polynomial inequalities, each tighter than the last. The bounds are usually checked by an external numerical solver. My reading: Thiele puts one of those bounds inside the machine's own step rule. A chain of Coq theorems proves that when the integer check CHSH_LASSERT runs passes, the specified zero-marginal Q_1 NPA matrix is PSD (Section 18). That direction is the one proved. The matrix condition itself is proved equivalent to three polynomial inequalities. A second chain carries it one level up, to Q_{1+AB} , once for each of the four opcodes in that family. That is two levels of NPA. The tower goes higher. The machine enforces the bound at the step itself. Nobody has to check it afterward from the outside.
- Lawvere's fixed-point theorem (1969). What I read: in a cartesian closed category, if a map from A onto the functions $A \rightarrow Y$ hits every function, then every map from Y to itself has a fixed point. Cantor, Gödel, Tarski, and Turing's halting argument are all instances. My reading: the diagonal in Section 10 is one more instance. Nothing about the diagonal is mine. What I added is the packaging as an interface, and one model, L , where the recursion theorem it needs is proved rather than assumed (Section 10.4).
- Tsirelson (1980) and Landau (1988). What I read: Tsirelson proved the $2\sqrt{2}$ bound and characterized which two-outcome correlators quantum mechanics

can produce: exactly the ones that are inner products of unit vectors. Landau gave the explicit condition for the two-setting case. My reading: the column-contractivity biconditional and the ellipsope results in Section 18 are their mathematics, rederived in Coq for one slice and turned into an integer check an opcode runs. The characterization is theirs. The step that refuses to advance on a tally outside it is mine.

- Bennett (1982), on Maxwell’s demon. What I read: measuring can be done without erasing anything, so it can be free; what costs is erasing the record to make room. My reading: Section 9.7 checks exactly that on the logic, and the permanent-certificate theorem is the certificate-specific version of it. Same idea. Mine is stated for every implementation of a certificate that lasts, and machine-checked.
- Maroney (2009) and Sagawa and Ueda (2008 onward). What I read: logical irreversibility and thermodynamic cost come apart; the heat Landauer’s principle demands depends on the actual distribution over states. For a state already known, the entropy bound supplies no positive heat floor. My reading: the known-state boundary in Section 9.5 is their point, restated for certificates. I am not adding physics there. I am adding the link to the certifying step.
- Zurek’s quantum Darwinism (2009). What I read: the classical-looking facts are the ones the environment keeps many independent copies of. Pointer observables win because they are redundantly recorded. My reading: the pointer-observable conjecture of Section 26 borrows that shape for computation. It is an analogy, and I have not shown it is more than one.
- Hashcash and proof of work (Dwork and Naor 1992; Back 1997). What I read: make a sender spend computation that a receiver can check cheaply, so establishing something costs. My reading: same instinct, different object. Proof of work checks a witness against a difficulty target; that check does not measure the effort spent on a particular successful attempt. A2 charges one ledger unit at the step that certifies and verifies nothing about effort.
- Quantitative information flow (Smith 2009). What I read: measure how much an observer learns about a secret from what it sees, in bits of min-entropy. My reading: the irrecoverability theorems rule out exact decoding from named windows. They do not establish zero information leakage: a window can reveal part of a secret without determining it. The observer knowledge in Section 9.7 describes which initial states remain consistent with the observed trace; no min-entropy leakage theorem is supplied there. QIF measures what an observer learns. It doesn’t price committing to it.
- Graded monads and graded types (Katsumata 2014; Orchard, Liepelt, and Eades 2019). What I read: a type system can carry a cost as a grade on each computation, and the grades compose as the computations do. My reading: μ is a grade carried in the state by the step rule rather than in a type. How close that is, an

embedding or something weaker, is a question I have not settled.

I read enough of each to convince myself Thiele is not a restatement. I cannot say that with the confidence of someone who has actually read the field. The only things I will tell you with confidence are the ones where the argument actually closes. Structure is state in this model. In the VM build it has its own opcodes to reach it. Fix the cost schedule and the starting value, and exactly one ledger fits: μ . `mu_is_initial_monotone` proves that on every state the machine can reach. One local rule, that a certification flip costs at least one, forces a cost onto every run that ends certified, on any abstract certification system at all. That is `universal_nfi_any_substrate`. Two states can agree on every classical field and still have different graphs (Section 14). The classical view cannot see that difference. A passing CHSH gate implies positive-semidefiniteness of the specified NPA matrix, and the algebra gets there without any quantum input. The argument is the part I can defend. The proof corpus is where I checked I hadn't fooled myself. The literature placement is the part where I am asking for help.

Prior-work map

This table separates the inherited idea from the result checked here. An implementation in Coq is not, by itself, a priority claim for the mathematics.

Prior work	What this development contributes
Lawvere (1969)	The diagonal is inherited. The development supplies an explicit weak call-by-value calculus, quotation, a proved recursion theorem, and its undecidability applications. It does not instantiate that calculus as the VM's total-run interface.
Tsirelson (1980) ; Landau (1988)	The correlation bounds are prior mathematics. The contribution here is the checked algebra for specified matrix slices and the connection from integer VM checks to those conditions. No general quantum-realizability claim follows from the fixed-slice test.
Bennett (1982)	Reversible measurement supplies the conceptual boundary. The development exhibits zero-cost observer narrowing and proves conditional bounds on state-space compression. Permanent certification is linked to merging on finite state spaces.
Maroney ; Sagawa–Ueda	Distribution-sensitive thermodynamic bounds are inherited. The certificate-specific entropy calculation exposes its support and permanence premises. Heat still requires the named Landauer premise; no additional physical law is proved.

Prior work		What this development contributes
Zurek (2009)		Redundant environmental records motivate an analogy. The computational pointer conjecture is not a derivation of quantum Darwinism and supplies no new quantum prediction.
Dwork–Naor; cash	Hash-	Computational pricing motivates a comparison, not an identification. A2 meters designated certification events. It supplies neither a hash-search lower bound nor proof of a prover’s elapsed effort.
Smith (2009)		Observation classes connect to information flow. The checked collisions rule out exact decoding of designated claims. They do not imply zero leakage, and the cost premises add an accounting question rather than a new leakage measure.
Katsumata graded types	(2014);	Compositional grades are inherited. The trace ledger is identified with an additive writer cost. The certificate reading and its observation boundaries are additional data, not a proof that graded semantics cannot represent them.

2 How I got here

What I am is curious, past the point of good sense.

I pick a thing up and cannot set it down until I have turned it over enough times to see how it holds together, and this time it did not let go.

I am not a programmer and I was never formally taught any of this.

I was not trying to build a formally verified CPU.

I would not have known what one was if it bit me.

I was trying to build a 3D renderer.

There was no plan behind it.

Just a pull to make something real with my own hands, and the kind of stubbornness that will not let a thing go until it stands up on its own.

I have worked on it in off hours every day since.

Once I started digging in, two things came clear fast. I knew nothing about computers, and the field was already saturated, every obvious road paved and crowded. So I needed an angle nobody else was standing on, and the angle I had was not a strategy, it was just the only way my head works. I did not know what compositional meant. I had never heard of category theory. I called the thing I was building digital DNA,

because that is what it felt like in my hands, and I thought about it in the only language I had: gravity pipes and directions and vectors, things that were linear and somehow not linear at the same time, structure that flowed and joined and came apart without my owning one proper word for any of it. I built a custom rendering pipeline out of those intuitions, by feel, throwing out pieces that did not sit right even when I could not have told you the rule they were breaking. And then, the way I always do, I went looking for what I was actually doing, and found out it already had a name. An old one. A deep one. What I had been reaching for in the dark was composition. It was category theory, a whole field I had never once heard of, sitting there describing the exact shapes I had been shoving around. The moment that landed, the rendering pipeline stopped being a rendering pipeline and started turning into a categorical engine, and I started swallowing the real theory as fast as I could find the words for things I had already built.

The method underneath was crude and it held. I wrote a long structured spec in JSON (a plain-text format for describing data) as a context document, fed it to Copilot, and had it build one Python file per concept, one at a time, each with its own tests, and I would not move to the next until I could watch the current one actually run. Category. Object. Morphism. Functor. Monoidal product. (The basic furniture of category theory; the first three I build from scratch in Section 13, functors and monoidal products in Section A.) It came together in pieces and never in chapters, the way a picture resolves out of static instead of getting drawn in left to right. The Yoneda lemma (a result buried deep in category theory) was the piece that made the rest snap into place, the night I quit borrowing other people's runtimes and started writing my own around it, with Z3 (a constraint solver out of Microsoft Research) wired into the middle.

Then within weeks I found out the engine I had built to draw pictures was general enough to run a physics simulation on the exact same primitives, without my rewriting any of it. I wrote Newton's laws as a program in it, half expecting nothing, and it ran. I want to be careful about what that was and what it was not, because this is exactly the kind of thing that gets misread. It was not me deriving physics, and I am not claiming the machine explains gravity. It was the compositional structure being general enough to express a physics simulation the same way it had expressed a renderer, and that generality is what told me the structure was the point and the pixels never were. Somewhere in those weeks I crossed from debugging why one pixel would not show up to understanding that I had quietly built a categorical computing engine, and that the rendering had only ever been the first thing I aimed it at. The pixel did eventually show up, for what it is worth.

I kept going, because by then I could not not keep going. I built a small custom language for describing scenes (the name for that is a domain-specific language, a DSL), then a program that read the language and caught my mistakes in it (a linter), then a way to type commands at the whole thing from a terminal (a command-line

interface, a CLI). The night the engine first ran a program declaratively, meaning I told it what should happen and let it work out the how instead of my spelling out every step, I told Julie I thought I was onto something. She said okay.

Then I hit a wall, and it is the wall this whole document is about. Z3 could settle the small questions and fell over on the exact ones I needed answered, which left me staring down an infinite hallway of guess-and-test with no proof waiting at the end of it, ever. One night, not even working, I was watching a video about Flatland, the old story about a creature living in two dimensions who cannot so much as picture a third. And it landed on me sideways, all at once, the way the real ones always do. I said it out loud: I get it now. There is another axis. Classical computation only ever sees what it can read and write off its own surface, and there is a whole direction it is blind to, not because it is broken but because it was never built with an eye that points that way. I opened a new repo that same night and called the thing in it the Thiele Machine. The name began as a working title and became the name of the model. It carries the larger claim plainly: the machine treats structural state as part of computation.

What survived the whole trip from there to here is not the code. Not one line of it. By the time the Coq proofs finally landed, every line of that first renderer had been thrown out and rewritten at least twice, probably more, and good riddance to all of it. What survived was a single architectural commitment I made early and never once let anyone talk me out of: structure is the state itself, the actual thing the machine runs on, not a metaphor laid over some realer state underneath. Category theory was the first language I found for that structure. It is not what the model is. The model asks for a state, a step, a ledger, and a window, and category theory appears nowhere in that list. It lives on in the build. The `Category` structure that came together in that first week and first ran morphisms correctly at two in the morning on January 22 is, in different words and different syntax, the same object `vm_graph` is in the 51-opcode realisation. Finding the different words took fifteen months. I did not write that code, not then and not now. I directed the models and refused everything they handed me until the morphisms composed the way the shape said they had to. The morphisms in that early code were objects that remembered where they came from; the morphisms in the realisation are objects that remember where they came from and that you can walk up to and challenge with `MORPH_ASSERT`. The receipt grew teeth. But the shape itself never moved. The shape was the substrate the entire time, and everything else, every renderer and language and proof, was me slowly learning to see what I had already built. By the shape I mean structure held as state. Not category theory. Category theory was just the first language I had for it.

I chased that one question for fifteen months. The renderer worked in January. Within weeks the same category structure was general enough to run a physics simulation, which is not the same as deriving physics and I do not claim it is. Formal verification was six months and a graveyard of dead ends further on. The hardware came after that. Every step pulled back to show another layer underneath that I had not known

was there, and I mean that as plainly as it sounds: I did not know what a Turing machine was when I started, I did not know what Coq was, I did not know most of the words in this document existed. I learned each one the same way, by walking face-first into a wall that needed it and not being let past until I had it. The needing always came first. The learning came after, dragged along behind the needing because it had no say in the matter.

Large language models were my hands the whole way through, and I want the division of labor exact, because it is the thing people most want to get wrong about work like this. The hands had nothing like this to draw on: no reference implementation, no prior project to point at. The ideas, the architecture, the direction, the demand that the thing be true: those are mine, all of them, no asterisk. The symbol-pushing, the actual Coq and the actual math, the models produced and the proof checker judged. The proofs compile or they do not. I refused every one that did not. Coq is an obedient idiot. It checks the proof matches the statement. It does not check that the statement means what you think it means. A theorem can claim “every cat that is also a prime number has whiskers” and the proof can be three characters long and Coq will nod and stamp it. So I read every statement. The proofs you can’t fake. The statements you can fake by accident, which is the more dangerous failure mode. I refused both kinds. However the symbols got onto the page, the argument on it is checkable by reading, and it would be exactly as right or as wrong if a million monkeys had typed it. The monkeys could not have refused.

A word on what I actually do, because it is stranger than the usual split and it runs under everything after it. I do not write code or math by hand. I do not pick constants. I do not grind through derivations. I have never written a line of code on paper or in a notepad in my life. What I do is see, and direct, and refuse. When I take hold of an idea it does not stay a point. It opens up. It becomes a shape I can blow up and turn over and take apart, lay across another shape to find where the two of them touch, run forward and then back to where it started, zoom into until it is the only thing in the world and pull back until it is one dot in a field of other dots, and the whole time it stays wired to everything I have ever connected it to, the right connections and the wrong ones both, all of them live at once. I am not reading that field neutrally, the way a search would. The thing doing the looking is me, the whole of me, my gut and my mood and my stubbornness picking what lights up and which way I move through it, and that is the part you cannot lift out without turning me back into a search engine. So I see where a thing has to fit, the shape it is obligated to have, usually well before I could write it down in a single symbol. Then the model gives me the symbols, the proof checker takes them or throws them back, and I refuse anything that will not compile or does not match the shape I am holding. That refusal is the one thing in this entire document that is mine and only mine: the demand that it be true, and the throwing-out of everything that wasn’t.

I am telling you all of this because it is not background color, it is the engine. I did

not come up through anywhere that hands you a foundation and says trust us, this part is settled. No professors, no colleagues, no field full of authority to lean back on when a claim got heavy. From the outside that is a deficit. From the inside it is the whole reason any of this is solid: when I meet a claim I either find the proof or I do not believe it, and that goes double for my own, because my own are the ones I most want to wave through. It is how I am built. It is the only way I know to hold a thing and actually trust that I am holding it.

So everything in here gets built from the ground up, and not as a courtesy to you. I build it because I do not believe I understand a thing until I have built it with my own hands, watched it stand up on its own, and tried to knock it over. I built this document the exact way I built the machine, brick by brick, refusing each brick until it held weight. That includes the parts you already knew cold before you opened this. I did not skip them for you, and I could not skip them for me.

The vocabulary I had to go look up to do any of this sits in Appendix A, at the back, so the argument comes first. If a word stops you, it is there. If you already know this material, you never have to open it. But if you catch something in the basics that is wrong, I want to know, and I mean that as a real ask and not a polite one. Nothing in here is correct just because I wrote it. Especially the parts I wrote.

On the name. I wanted to call it the meta machine. Then Facebook bought the word “meta”: a company renaming itself after a term for self-reference and draining it of any in the same motion. So I went one level up, where this machine already lives. The most meta move available, for a machine that runs on meta-level reasoning, was to put my own name on the cover and take the word back. That’s the whole of it, a meta move on a meta machine. It happens to be my dad’s name too, and my wife’s now. So I asked them before I did it. They said go ahead.

3 The structural axis: the projection classical models drop

Why this section exists. This section exists because a renderer I was building in January 2025 walked me into a wall, and the wall turned out not to be made of what I thought.

The renderer was a categorical engine. I had built it so the things running at runtime were the category-theory objects themselves: modules were objects, transformations between them were morphisms, composing two transformations was literally composition, and running two independent parts side by side was the monoidal product (Section A builds all of these). I was not simulating category theory on top of ordinary data. The structure was the thing the engine ran on. And it worked, which is the part I still find a little funny: within weeks the same engine that was supposed to draw pixels was running physics on the identical primitives, and I had not planned that and did not at first believe it.

Then I tried to prove something about what it was doing, and the floor went out. The machine underneath, the ordinary CPU, had seen none of the structure the entire time. (That ordinary architecture, one processor and the memory passing data back and forth across a single bus, is the von Neumann machine, after John von Neumann who wrote it down in 1945; every laptop and phone and server you have ever touched is one.) It had been faithfully shuffling the bytes my modules and morphisms happened to be spelled out in, faithful the way a printing press is faithful to a poem it cannot read. When I brought in Z3 to verify the morphism chains the engine was reasoning with, I met the same blindness in different clothes: Z3 saw formulas, true ones, that happened to refer to the shapes, and it never once saw a shape. For a long stretch I took this for a Z3 problem, a tooling problem, the kind you solve by getting cleverer with the solver. It was not a Z3 problem. The structure I wanted checked was real and was carrying the whole computation, and there was nowhere in the machine for it to sit as a thing the machine itself had to answer for. I was asking a tool to certify a fact about the one part of the computation the computation had no place to keep. You do not get cleverer your way into a slot that was never built.

The Thiele Machine is that slot. It is the substrate the engine had needed all along and could never be handed on the hardware it was running on: a machine where the structure is first-class state instead of a pattern in the bytes, where building and checking that structure are steps the machine takes rather than something you read back out afterward, and where committing to a structural fact is an act the machine is charged for and cannot quietly perform for free. The engine's structure happened to be categorical. The slot doesn't care what kind of structure goes in it. Everything past this point is that substrate's correctness conditions, laid out in English you can follow and check yourself, with a Coq proof standing next to each one. The Coq is not what makes any of it true. It is there because my own eye slides straight over the gaps in an argument I am hoping is right, and I wanted a reader that would not let me. This section is the long-form answer to a question a stuck renderer put to me and would not drop until I quit trying to out-clever it and went looking for what was actually missing.

In the formal version, A2 is a proof field on the transition interface, `cs_cert_costs`. On the VM it is proved from what the instructions actually do. A window that throws away the relevant fields can't rebuild them from what's left. That's a fact about those windows. It doesn't mean every implementation needs separately named fields.

The law isn't tied to this instruction set. The state type and the predicate are parameters, so you can study as many instances as you like. That doesn't make every observable change cost something. It doesn't make any old predicate an independent geometric dimension, either.

3.1 Substrate-independence: the abstract setup

The kernel theorem `universal_nfi_any_substrate` is written to be as hard to wriggle out of as I knew how to make it, and the way you make a theorem hard to wriggle out of is to make it ask for almost nothing. It takes any abstract computational system at all, and by any I mean it genuinely does not care what the system is made of or whether it looks anything like the machine in this document. It asks for five pieces and refuses to ask for a sixth. A set of states, and a state can be whatever it is for you: bits on a tape, words in a register file, cells in memory, it makes no difference to the theorem. A set of instructions, the moves the thing can make. A step rule, which takes a state and an instruction and hands back the next state. A cost function, which pins a non-negative integer to each instruction. And a certification predicate, a plain yes-or-no on states that says whether a given state counts as certified. That is the entire list. The reason I keep leaning on how short it is: any objection that begins “sure, but that only holds for your particular machine” has nothing to attach to once the result has agreed to run on something this bare.

One premise, labelled A2 and named `cs_cert_costs` in the kernel: if any single step flips the certification predicate from no to yes, the instruction that did it cost at least 1. One conclusion: every trace that starts uncertified and ends certified has total cost at least 1.

And it does not care how powerful the thing is, in either direction. It never asks whether your system can compute everything a Turing machine can (the word for that is Turing-complete: you are Turing-complete if you can simulate a Turing machine, at most Turing-equivalent if a Turing machine can simulate you). It never asks for the reverse either. Stronger than a Turing machine, weaker, dead even, none of it reaches the floor. Bring the five pieces, charge A2 on the flip, and you are in scope. The floor does not check your computational class at the door, because it was never about what you can compute. It is about what your step is allowed to do for free.

The categorical-separation result in Section 14 is about one specific observable tuple. Two states can agree on it and still have different graphs. A full conventional encoding could keep the graph, and a decoder could read it back. What the theorem pins down is what this particular window throws away.

The abstract theorem needs a designated predicate and the A2 premise. A Turing-machine trace isn’t an instance just because it’s a computation. It becomes one when somebody supplies the state reading, the step function, the cost, and the proof.

3.2 What is canonical, and under which conditions

Three results, kept apart. `universal_nfi_any_substrate` puts a floor under every system that follows A2. `mu_is_initial_monotone` proves the ledger is unique on reachable states, once the starting value and the exact increments are fixed. The projection theorems say what specific windows lose.

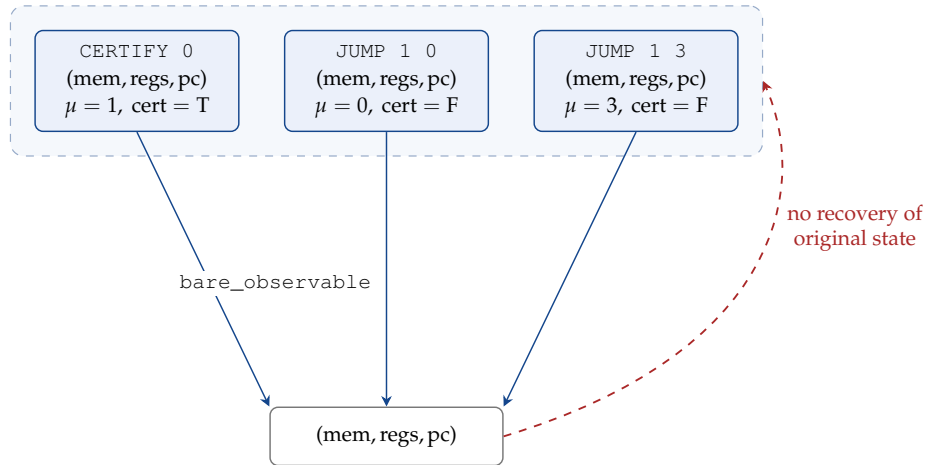


Figure 2: Different states, same view. Three one-step runs from the same clean start. The window shows memory, registers, and the program counter, and all three look identical through it; the graph is identical too. What each run established and what it paid are not. When more than one state sits behind a view, no function of the view gets every original back. You can pick default values to fill the gaps. The defaults don’t recover what was really there. The first two are the Coq witness pair `abs_A_strict` and `abs_B_strict_g`. The third is `abs_B_cost_g`, the same jump charged 3.

The abstract `CertificationSystem` carries any state type, any instruction type, a step function, a non-negative cost per instruction, a yes/no predicate, and a proof of A2. Its state doesn’t need separate cost and certification fields. A conventional encoding can carry both. The classification is about the semantics supplied. The names and layout of registers don’t enter into it.

A fixed machine can enforce an invariant just by which transitions it allows. A simulator keeps that invariant when its correctness contract ties the two together. One broken simulator doesn’t show that every simulator, or every conventional machine, fails to enforce it. And a law protected inside a machine doesn’t automatically vouch for a transcript somebody else hands you.

Certification is one instance of all this. Any other family of observables, $\alpha_i : S \rightarrow X_i$, needs its own explicit events and cost laws for each guarantee it claims. Piling predicates together doesn’t give you a topology, a metric, or independence between them. And the current instruction set lets even partition structure change at zero cost.

3.3 The canonical contrast: the Turing machine

Here is what I had to learn about Turing machines to compare what I built to one. I went and read about them on my own. I’m walking through it because I had to walk through it. If I have anything wrong below, I’m relaying what I read, so tell me.

What I read says Alan Turing described his machine in 1936 to make precise the intuitive notion of “algorithm.” The machine, as I understand it, is this:

- An infinite tape. Think of it as an infinitely long strip of paper, divided into cells. Each cell holds one symbol from a finite alphabet, say $\{0, 1, \text{blank}\}$.
- A read/write head. One cell at a time. It reads the symbol, overwrites it if the table says to, then moves left or right.
- A finite set of states, $Q = \{q_0, q_1, \dots, q_n\}$. The machine is in exactly one state at a time. q_0 starts the run. Some states halt it.
- A transition table. Given the current state and the symbol under the head, it says: write this, move left or right, enter that state.

That is the entire machine. Nothing else.

Here is what I read as the remarkable thing about this simple machine: it can compute *anything that can be computed by following a definite procedure*. The sources I dug up claim every algorithm that has ever been written, in every programming language that has ever existed, can be translated into a Turing machine. The label I found for this is the Church-Turing thesis. The original claim (Church 1936, Turing 1936), as I read it, is specifically about *effective calculability*: anything that can be computed by following a mechanical, finite, step-by-step procedure, a Turing machine can also compute. I did not find a counterexample. I also have not verified the thesis myself. I am reporting the ground I had to stand on before I could push against it.

The claim about physical hardware is a related but separate conjecture, called in the sources I read the *physical Church-Turing thesis*: every physical computing device is computationally equivalent to a Turing machine. As I read it, this is an empirical claim about physics, not proven mathematics. I did not find a built physical counterexample. Your laptop, your phone, every server in every data center: in the usual computability sense, they compute no more functions than a Turing machine computes. The differences I found are resource differences. Speed. Space. Energy. Cost. The class of computable functions stays the same.

3.4 The observation boundary

A Turing machine's step only reads the symbol under its head and its control state. The computation can still work out global properties, or carry old evidence forward in its state. So the impossibility here is narrower than "Turing machines can't see structure." It's about one particular window. When two full states look the same through it and differ in some structural property, nothing reading only the window recovers that property every time.

A structural opcode can expose or check a property through the richer state. What it guarantees has to be read off its semantics. A ledger on its own records a total. Finding where in the history an event happened takes the trace too. And paying something doesn't, by itself, make anything true.

3.5 Structural cost and other resource measures

This machine keeps its own cost schedule, one that notices particular events, running alongside ordinary execution. I'm not saying complexity theory only studies two resources, or that it can't model what verification costs. What's new here is how the pieces fit together: transition laws, structural state, windows onto that state, and accounting results, each with its conditions stated in its theorem.

3.6 The subsumption theorems, in full

Section 1 framed classical computation as the degenerate fragment of the Thiele substrate, the corner where the structural axis lies dormant and nothing ever lights it up. That picture is four theorems in the kernel. Each one below starts with the plain reading, then gives the formal statement and proof sketch as the models wrote them and Coq checked them. Hold the picture up against the proofs and decide for yourself, without taking my word for either one and without leaving this page.

| **Plain reading.** Classical programs leave the structural axis alone. The shadow you see equals the full state's projected fields. The graph and cert channels stay at whatever they were at the start.

Theorem 3.1 (D2 faithfulness). *For every classical program prog (a sequence of vm_instruction values each satisfying is_classical_opcode) and every starting state s_0 , let s_n be the state after running prog on s_0 . Then (a) $\text{shadow_proj}(s_n)$ extracts exactly the six projected fields of s_n (vm_regs , vm_mem , vm_pc , vm_mu , vm_err , vm_certified); and (b) the structural layer is preserved: $s_n.\text{vm_graph} = s_0.\text{vm_graph}$, $s_n.\text{vm_csrs.csr_cert_addr} = s_0.\text{vm_csrs.csr_cert_addr}$, and $s_n.\text{vm_certified} = s_0.\text{vm_certified}$.*

Proof. Part (a) unfolds shadow_proj : by definition it extracts exactly those six fields, and reflexivity closes it. Part (b) is D3 conservativity (next theorem) iterated over the program. □ □

| **Plain reading.** The classical opcodes are defined by what they don't touch. They may change the projected working fields, including the ledger charge assigned to the instruction, but they leave the graph and certification channels alone. A program built only out of them cannot disturb those channels, even if you wanted it to. That dormancy is built in. It isn't an accident.

Theorem 3.2 (D3 conservativity). *If $\text{is_classical_opcode}(i) = \text{true}$ and $s' = \text{vm_apply}(s, i)$, then $s'.\text{vm_graph} = s.\text{vm_graph}$, $s'.\text{vm_csrs.csr_cert_addr} = s.\text{vm_csrs.csr_cert_addr}$, and $s'.\text{vm_certified} = s.\text{vm_certified}$. By induction over the program list, classical programs preserve all three.*

Proof. Case analysis on the instruction i . The predicate $\text{is_classical_opcod}$

$e(i) = \text{true}$ holds exactly for the compute, memory, control-flow, ordinary I/O, heap, and MORPH_GET / TENSOR_GET opcodes: the ones whose `vm_apply` rule mutates only `vm_regs`, `vm_mem`, `vm_pc`, `vm_err`, `vm_mu`, `vm_logic_acc`, `vm_mu_tensor`, or `vm_mstatus`. Twenty-two opcodes return false from `is_classical_opcode` and are excluded by the premise, but not all for the same reason. PNEW, PSPLIT, PMERGE, TENSOR_SET, MORPH, COMPOSE, MORPH_ID, MORPH_DELETE, and MORPH_TENSOR mutate graph or module-tensor state. MORPH_ASSERT can write the cert-address channel on success; CERTIFY writes the top-level certification flag; CHSH_TRIAL mutates the witness counters. LASSERT, LJOIN, EMIT, REVEAL, PDISCOVER, CHSH_LASSERT, and the four Q_{1+AB} CHSH_LASSERT variants are also outside the classical fragment because they carry validation, revelation, discovery, tensor-charge, or mandatory-floor semantics. The D3 proof only needs the forward direction: for each opcode that `is_classical_opcode` admits, direct inspection of the apply rule leaves `vm_graph`, `csr_cert_addr`, and `vm_certified` fixed. Iterating over the program list by induction closes the trace-level preservation. $\square$ $\square$

| Plain reading. Take any Turing-machine configuration and drop it into the Thiele substrate untouched: the TM part stays exactly as it was, $\mu = 0$, the structural fields empty. Now run `thiele_run` on it. Nothing happens that wasn't going to happen anyway. Step for step the TM portion does precisely what the standalone machine would have done, and because every move it makes is a classical move, the ledger never comes off zero. The induction on fuel just carries that out to the whole trace. The Turing machine is sitting inside the larger object running its same computation on its same tape to its same answer, having paid nothing, and from where it stands there is no way to tell it has been subsumed at all. That is what subsumption is supposed to look like: the old thing keeps working, identically, for free.

Theorem 3.3 (Thiele simulates Turing). *Lift a Turing configuration by keeping the TM part and setting μ to zero: $\text{lift_config}(c) := \{ \text{th_tm_config} = c; \text{th_mu} = 0 \}$. Then for every fuel $n \in \mathbb{N}$, transition table δ , and TM configuration c :*

$$(\text{thiele_run } n \delta (\text{lift_config } c)).\text{th_tm_config} = \text{tm_run } n \delta c.$$

Proof. Induction on the fuel n .

Base ($n = 0$). Both `thiele_run 0` and `tm_run 0` return their inputs unchanged. Reflexivity.

Step ($n + 1$). Unfold one step on both sides. Let $c' = \text{tm_step } \delta c$ when defined. On a lifted configuration, `thiele_step`'s TM portion is exactly `tm_step` δc : structural fields are not consulted, and lifted μ stays at 0 because TM steps are classical. Apply the induction hypothesis to the next n steps from `lift_config(c')`. $\square$ $\square$

| Plain reading. Four things about the same window. First, every Turing configuration lifts into the Thiele substrate and runs the same computation, with $\mu = 0$ and no structural content. Second, going back down is a quotient: two VM states have the same shadow exactly when they agree on the six projected fields. Third, run a classical program on two states that agree on the shadow, and also on the graph and the CSRs that classical instructions can peek at, and the outputs agree on the shadow too. That extra condition matters. `MORPH_GET` and `TENSOR_GET` count as classical and they read hidden state, so equal shadows alone aren't enough. Fourth, the window really does lose something: some VM states look identical from the classical side and differ on the structural axis. Put together, the classical shadow is a real quotient of the VM state, the classical fragment respects it once the hidden fields it reads are pinned down, and going down drops real information.

Theorem 3.4 (Degenerate projection theorem (four-part)). *The following four propositions hold simultaneously:*

1. Faithful embedding. For every fuel n , transition table δ , and TM configuration c :

$$(thiele_run\ n\ \delta\ (lift_config\ c)).th_tm_config = tm_run\ n\ \delta\ c.$$

2. `shadow_proj` is the quotient map. For all VM states s_1, s_2 :

$$\begin{aligned} shadow_proj(s_1) &= shadow_proj(s_2) \\ \iff eq_on_classical_shadow(s_1, s_2). \end{aligned}$$

3. Classical traces respect the compatibility relation. For every classical program `prog` and every two VM states s_1, s_2 with `shadow_proj(s1) = shadow_proj(s2)` and `s1.vm_graph = s2.vm_graph` and `s1.vm_csrs = s2.vm_csrs`:

$$shadow_proj(run(prog, s_1)) = shadow_proj(run(prog, s_2)).$$

Shadow agreement alone is not a strong enough hypothesis. `TENSOR_GET` and `MORPH_GET`, both classical opcodes, read `vm_graph` into a visible register; `HEAP_LOAD` and `HEAP_STORE` read `csr_heap_base` out of `vm_csrs`. Two states can agree on all six projected fields and still disagree on the graph or the heap base, and either instruction then writes a different value into a projected register, breaking the shadow-only version of this claim. The extra agreement on `vm_graph` and `vm_csrs` closes that gap.

4. The projection is strictly lossy. There exist VM states s_1, s_2 with `shadow_proj(s1) = shadow_proj(s2)` and $s_1 \neq s_2$.

Proof. The theorem is four lemmas stitched into one conjunction.

- Part (1) is `thiele_simulates_turing` (preceding theorem).
- Part (2) is definitional. `shadow_proj` extracts its six projected fields. Equal projections mean those six fields agree. The Coq witness is `shadow_proj_kernel_is_eq_on_classical_shadow`.
- Part (3) is the compatibility theorem `D2_classical_shadow_preserved`. Its premise is stronger than equality of the six-field shadow: the two inputs must also agree on `vm_graph` and the complete `vm_csrs`, because classical `MORPH_GET/TENSOR_GET` and `heap` operations can read those hidden fields. Under that compatibility relation, the two output shadows agree. This is preservation of a specified relation, not a claim that shadow equality alone commutes with execution.
- Part (4) is `shadow_strictly_lossy`. The Categorical Separation construction gives the witness: s_1 has one identity morphism in `pg_morphisms`; s_2 has none. The six projected fields agree. The morphism lists do not.

The conjunction of the four pieces is the theorem. □ □

3.7 What the projection cannot reverse: three non-existence theorems

The subsumption theorems above establish what the VM state carries over and above its classical projection: fields the projection drops. The natural next question is whether a classical observer could reconstruct those fields by some clever derived predicate on the surviving shadow. The answer is no, and it is a structural no, the kind you cannot argue your way around. The kernel proves three irrecoverability theorems (each one a non-existence of a function from the bare classical projection back to a Thiele-only field) and packages the corollaries the subsumption argument relies on. Every one is proved with zero axioms and no admits, fully constructively.

The projection in this subsection is the four-field map `forget : VMState → TMSnapshot`, keeping `(pc,  $\mu$ , regs, mem)`. It is already forgetful, but it still retains μ . For the cost-ledger separation, the stricter three-field projection `bare_observable` drops μ as well, keeping only `(pc, regs, mem)`: the observable the standard complexity-theory cost model sees.

This document uses several distinct execution models and several distinct projections. They aren't one model seen from different angles. Table 3.7 is the registry: which state type each one runs over, what counts as one step, what an observer sees, and which theorems are stated over it. Consult it whenever a later section says “run,” “classical shadow,” or “simulation”; the domain is not always the same one.

Model	State type	Execution function	Observation	Key theorem(s)
File-local TM lift	state	run_thiele/run_tm, fuel-indexed fetch/step execution	toy TM state; the lift keeps the TM state and $\mu = 0$	thiele_simulates_turing
VM instruction-list fold	abstract S	acm_run, folds a supplied list in list order	the target machine's full state	abstract_nfi; not PC-indexed
PC-indexed VM step	VMState	vm_apply, one instruction supplied by the caller; the instruction is not fetched here	full VMState	μ conservation and opcode semantics
PC-indexed VM runner	VMState	run_vm, fuel-indexed fetch from vm_pc, then vm_apply	full VMState, fuel-truncated	execution lemmas in SimulationProof
Bounded substate evaluator	VMState	vm_run, exactly 1000 run_vm steps, always Some	full VMState; fuel exhaustion is not divergence	conditional VM diagonal and external bounded decider
Word64 unbounded halting	VMState	vm_halts_at, an existential fuel witness for run_vm reaching a halted PC	halted final VMState; divergence is absence of a witness	vm_halts_at_deterministic, saturation
Natural-valued sibling	VMState	run_vm_u over vm_apply_u; writes and arithmetic are unmasked naturals	full VMState, with host/guest fields interpreted by the selected interpreter	unbounded-step and interpreter frame lemmas
CM2 guest	CM2ConfigU	cm2_step_instr, cm2_run; successful nonzero decrement jumps and zero falls through	guest PC and two natural counters	mm2_termination_guest_iff
60-instruction CM2 host	VMState	fixed cm2_interpreter_program under run_vm_u	cm2_result, halt/malformed status, and preserved ambient state	mm2_halt_if_host_if; separate from the 122-instruction guest
12-opcode self-interpreter guest	GConf	g_run for the GInstr fragment; g_step_is_vm_apply_u relates it to vm_apply_u	guest registers 0–3, guest μ , PC, and termination status	self_interpreter_correct; unbounded guest scope only
122-instruction self-interpreter host	VMState	fixed self_interpreter_program under run_vm_u; packed code and width are input data	guest result/status; host scratch and host μ are	self_interpreter_correct, frame, and

Table 3.7: model registry. The file-local TM lift is a proof device local to `thiele_simulates_turing`; it is not a claim that the bounded evaluator implements arbitrary Turing machines on the 51-opcode hardware. Connect two rows by an explicit correspondence theorem only where a claim in this document actually depends on that connection; otherwise treat them as different objects answering different questions.

| Plain reading. The VM's `vm_certified` flag is not a function of the four fields kept by `forget`. Two VM states can agree on $(pc, \mu, regs, mem)$ and still differ on whether they are certified. No formula computed from that four-tuple alone can decide certification. The certification bit lives outside the four-tuple.

Theorem 3.5 (Cert irrecoverability, `cert_not_function_of_forget`). *There is no function $\phi : TMSnapshot \rightarrow bool$ such that, for every VM state s ,*

$$\phi(\text{forget } s) = s.\text{vm_certified}.$$

Proof. The witnesses are `trace_witness_A` (certified, defined) and `forget_witness_uncert` (not certified, defined). Direct computation shows the two states have equal `forget` images. Suppose ϕ existed: then $\phi(\text{forget } \text{trace_witness_A}) = \text{true}$ and $\phi(\text{forget } \text{forget_witness_uncert}) = \text{false}$. But the inputs are equal, forcing the outputs to be equal too. Contradiction. $\square$ $\square$

| Plain reading. The VM's μ counter is not a function of the three-field classical observable $(pc, regs, mem)$. Two VM states can agree on that observable and still have different μ values. A clever classical observer can't compute the difference from what it sees. The difference isn't in what it sees. The observer's window is blind to it by construction.

Theorem 3.6 (Mu irrecoverability from bare observable, `mu_not_function_of_bare_observable`). *There is no function $\phi : BareTMObservable \rightarrow nat$ such that, for every VM state s ,*

$$\phi(\text{bare_observables } s) = s.\text{vm_mu}.$$

Proof. The witnesses `mu_separation_witness_zero` and `mu_separation_witness_one` (defined) differ only in their μ field (0 vs 1) and have identical bare observables. A ϕ satisfying the equation would force $0 = 1$. Contradiction. $\square$ $\square$

| Plain reading. The `csr_cert_addr` channel can't be recovered from the four fields `forget` keeps. This one is about states and what a window shows of them. The program-level diagonal in Section 10 works with a different predicate, one about outcomes being equal. Carrying this collision over to that predicate would take a correspondence theorem of its own.

Theorem 3.7 (Cert-addr irrecoverability, `cert_addr_not_function_of_forget`).

There is no function $\phi : \text{TMSnapshot} \rightarrow \text{nat}$ such that, for every VM state s ,

$$\phi(\text{forget } s) = s.\text{vm_csrs}.\text{csr_cert_addr}.$$

Proof. The witnesses `cert_addr_witness_zero` and `cert_addr_witness_one` (defined) differ only in their `csr_cert_addr` field (0 vs 1) and have identical `forget` images. A ϕ satisfying the equation would force $0 = 1$. Contradiction. $\square$ $\square$

The three subsumption-witness corollaries.

The irrecoverability theorems above ground three statements the subsumption argument relies on. These are the corollaries the monograph cites as “the VM content the bare projection structurally cannot carry.”

1. *A2 cannot be stated as a constraint on the bare projection alone.* `A2 (cs_cert_costs`, a typed field on `CertificationSystem`) says that a `cert`-predicate flip from `false` to `true` forces `instruction cost` ≥ 1 . By `cert` irrecoverability, any `cert`-predicate phrased as a function of the four-field `TMSnapshot` cannot agree with the VM’s `vm_certified` on every VM state. So `A2` on the bare classical signature is not the VM’s `A2`. It constrains a different predicate. The Coq corollary is `no_classical_a2_cert_predicate`.
2. *The cost-ledger observability separation has no analog in the bare projection.* The separation above names two VM states with identical `(pc, regs, mem)` but different μ : the witnesses `mu_separation_witness_zero` and `mu_separation_witness_one` from the μ -irrecoverability theorem. As a predicate on the bare classical observable, the separation would have to read μ from an observable that does not contain it. μ -irrecoverability forbids that. The separation lives at the Thiele level because it has to. It isn’t a drafting choice. The Coq corollary is `no_classical_mu_separation_predicate`.
3. *The CSR channel does not factor through the named projection.* `no_classical_cert_addr_predicate` rules out getting the specified CSR field back from the four fields `forget` keeps. That’s all it does. It doesn’t make this state predicate the same as the program-level diagonal, and it doesn’t rule out a conventional encoding that keeps more.

Recovery and sections of a projection.

Picture one shadow with several full states stacked above it. `fiber_has_two_preimages` exhibits exactly that. So there’s no going back down that returns every original full state. You can still build a way back that picks defaults, a default graph, a default ledger, a default flag. More than one state behind a view doesn’t, by itself, settle every question about which choice would be the canonical one.

Start with the question the observer promises to answer. If two states look identical to that observer but need different answers, nothing reading that observation answers

correctly on both. `decoding_requires_fiber_constancy` proves that for any kind of state, observation, and answer. Hand over one sample state from each group of look-alikes and it runs the other way too: if every state in a group gives the same answer, you can compute the answer from the sample. Neither result needs a certification bit. That’s the exact sense in which a distinction becomes necessary once a model promises to answer a question about it.

The trace-fold universal property.

Here’s the picture. Take another machine M , pick where it starts, and walk a list of instructions through it one at a time. That walk is the only way to turn instruction lists into states of M that starts in the right place and takes one step per instruction. `thiele_trace_fold_initial` proves it, for any `CertCostMachine` M and starting point s_0^M , by induction on the list. It’s a statement about traces. `thiele_is_initial_a2_substrate` is another name for the same theorem.

Going from VM states, instead of traces, is harder. `thiele_morphism_unique_on_reachable` proves that two certification-preserving maps from VM states, if somebody supplies them and they agree at the start, agree on every reachable state. It doesn’t show such maps exist. A one-state target whose predicate is always false satisfies A2 and has no such map from certified VM states at all. In general, getting from traces down to states needs two things: equal VM outcomes have to land on equal targets, and certification has to line up.

That condition is exact. If two instruction lists reach the same VM state, they have to evaluate to the same target state before the trace walk can come down to a map on states. `trace_descent_unique_value_iff` proves that’s the same as having one target value at every reachable VM state. Actually building the map also takes a chosen sample trace for each reachable state. With that choice in hand, `reachable_simulation_exists_iff` says a certification-preserving reachable simulation exists exactly when traces are compatible and certification agrees. The Boolean certification quotient is a concrete target that works.

And there’s a target that fails for a more interesting reason than never certifying. It keeps the full instruction history, charges the VM schedule, and reads certification with the VM’s own evaluator. Certification agrees exactly. But the empty trace and `JUMP 0 0` reach the same VM state and leave different histories behind. `certification_agreement_does_not_imply_descent` proves that’s a real obstruction. A2 and agreeing certification don’t force a target to forget its history.

Neither the universal property nor the projection theorem picks which event to price. Neither says anything about physical foundations.

3.8 What the projection cannot verify: the verifier corollary

The non-existence theorems above proved the projection is structurally lossy: μ , `vm_certified`, and `csr_cert_addr` are not functions of the bare classical observ-

able. The natural next move is to ask what that lossiness means for verification. If a verifier sees only the projection (a list of `StrictClassicalState` snapshots, the Turing-classical trace), what can it soundly decide?

Answer: not the claims that depend on the dropped fields.

| Plain reading. A verifier that sees only the classical projection of a VM execution can't soundly decide claims about μ . Two different VM executions project to identical classical traces and disagree on whether $\mu = 1$. The verifier has nothing to tell them apart by. On a claim that depends on μ , it can be sound or complete from the bare projection. It can't be both. This is the receipt theorem carried one step further, from "no function gets μ back from the projection" to "no verifier soundly decides μ -sensitive claims from it."

Theorem 3.8 (Bare-setting verifier impossibility, `bare_setting_no_sound_complete_verifier`). *There is no verifier $V : \text{list } \text{StrictClassicalState} \rightarrow \text{bool}$ that is simultaneously sound and complete on the claim " $\text{vm_mu } s = 1$."*

Proof. The two single-step witness states from the kernel's receipt theorem (`ReceiptTheorem`: no function of the strict three-field shadow recovers μ ; its witnesses are `pol_state_A`, `cost-1 CERTIFY 0`, and `pol_state_B`, `cost-0 PNEW [] 0`) project to the same strict-shadow trace. State *A* satisfies the claim; state *B* does not. A complete verifier must accept the honest run from *A*. On the same accepted transcript, soundness forces the claim to hold for every state that could explain the transcript, including *B*. The first requirement gives acceptance; the second contradicts *B*'s $\mu = 0$. The Coq check assumes no axioms. □ □

Three escapes, each constructed.

Three structurally distinct ways to clear the impossibility live in the kernel as concrete verifiers, each one a closed Coq theorem.

1. *Substrate-trust.* The transcript carries the full `VMState` and the verifier reads μ directly. `substrate_escape_succeeds` exhibits a verifier with constant cost that is both sound and complete on $\text{vm_mu } s = 1$. The verifier's decision function is `Nat.eqb (last t).vm_mu 1`; the proof is one inhabitation per quantifier.
2. *Hardness, as a commitment contract.* The transcript carries a commitment bit. `commitment_contract_verifier` constructs a sound and complete verifier from a `CommitmentBitContract`. The contract is a parameter the verifier is built over. It isn't an axiom. The contract is stated against which transcripts each state can stand behind, forged ones included. It has two parts. Binding: a set commitment comes only from a state where the claim holds. Honesty: a state where the claim holds always commits. The definition that computes

the bit from the claim meets it (`honest_commitments_satisfy_contract`). An unchecked bit breaks it (`unchecked_bit_violates_contract`). Soundness and completeness are supplied by this exact disclosure contract. The unit cost prices reading the bit; it does not bound cryptographic verification time. Whether a real commitment scheme meets the contract is a question for cryptography. This kernel doesn't answer it.

3. *Interaction.* The verifier elicits a response from the prover that pins the claim. `interactive_escape_succeeds` exhibits a constant-cost sound and complete verifier when the response set is rich enough to report μ . This is the simplest possible interactive model (one round, one fixed query), and that is all the corollary needs.

The substrate channel is the option the structural axis makes available; hardness and interaction are the two routes classical cryptography and complexity already used. Here the hardness route appears as an exact commitment-bit contract. All three are witnessed by closed Coq theorems, each with its own trust profile. The commitment theorem itself supplies no hardness reduction. What follows is the shared obstruction they all answer to.

| Plain reading. Squash a richer transcript down to its classical trace and the same collision comes back. The condition is that the collision has to survive into the richer transcript type: the two bare-indistinguishable states s_A, s_B have to show up as two transcripts t_A, t_B under `explains_T`. That isn't free. It's exactly what each of the three interfaces above has to establish about its own transcript type, and each one does it by building t_A and t_B outright. Given that, checking a μ -sensitive claim has to look at structure the classical trace doesn't carry. Any verifier that lives entirely on the classical projection fails on that same pair. The full state, the contract-governed commitment bit, and the interactive response are three built verifiers that do look at that structure. Whether some fourth kind of transcript could expose it another way is left open here. The claim is that these three work, each under its own trust contract. They aren't claimed to be the only possible designs.

Theorem 3.9 (Factorisation impossibility, `V_does_not_factor_through_classical`). *Let T be a transcript type with a projection $\pi : T \rightarrow \text{list StrictClassicalState}$ and a relation $\text{explains_T} : \text{VMState} \rightarrow T \rightarrow \text{Prop}$ connecting VM states to the transcripts that could have produced them. Suppose the bare-setting witness pair (states s_A, s_B with identical bare projections, $\text{vm_mu } s_A = 1$, $\text{vm_mu } s_B = 0$) lifts to T : there exist $t_A, t_B : T$ with $\pi(t_A) = \pi(t_B)$, $\text{explains_T } s_A t_A$, and $\text{explains_T } s_B t_B$. Then no verifier $V : T \rightarrow \text{bool}$ that is sound (every transcript V accepts explains only $\text{vm_mu} = 1$ states) and complete (every transcript explaining a $\text{vm_mu} = 1$ state is accepted) factors through π .*

Proof. Suppose V factored through π , i.e. $V = V' \circ \pi$ for some $V' : \text{list StrictClassicalState} \rightarrow \text{bool}$. Since $\pi(t_A) = \pi(t_B)$, $V(t_A) = V(t_B)$. Completeness applied to t_A (which explains s_A , $\text{vm_mu} = 1$) forces $V(t_A) = \text{true}$. Soundness applied to t_B (which explains s_B , $\text{vm_mu} = 0$) forces $V(t_B) = \text{false}$. Contradiction. $\square$ $\square$

4 What structure is

Everyone uses the word “structure,” and the word does the work while meaning nothing in particular, which you get away with right up until you try to build a machine whose entire job is to charge for it. Then the fog has a price. So I am going to pin the word down, further down than feels necessary, because every time I have watched this argument slip, it slipped at a step earlier than anyone was looking.

4.1 Partitions

The smallest piece of structure there is, is a partition. You take everything in front of you and sort it into buckets: nothing in two buckets, nothing left on the floor. That is the whole of the definition, and yes, it is the thing you do with a kindergartner and a pile of blocks. Keep it anyway. The instant the buckets are drawn you have made a claim about the world, and the claim turns out to cost money.

Formally, a partition of a set S is a collection of non-empty pieces $S_1, S_2, \dots, S_k$ that share no elements and together cover all of S , so every element lands in exactly one piece. In symbols, $S_1 \cup S_2 \cup \dots \cup S_k = S$, with $\cup$ the set union from Section A. Each piece I call a module, or a block, or a cell, whichever word the surrounding sentence wants.

Here is the part that is not kindergarten. The partition is not a tidier way to hold information you already had. The partition is itself information, a fact that did not exist until you drew it. Put x in S_1 and y in S_2 and you now know, for nothing and for keeps, that the two live in different groups. Before, you would have had to go and look. After, the answer is a lookup, and the lookup does not cost one cent more when S swells from ten elements to a billion. You bought a question’s answer down from a search to a glance. Something paid for that. Somewhere a bill came due, whether or not anyone troubled to write it on a receipt.

Not writing it on a receipt is the classical world’s whole way of life. Carve up the memory of an ordinary machine, announce “these addresses are module A , those are module B , and A and B never touch,” and you have said something with teeth, something a later step can lean its full weight on and run a hundred times faster for. A Turing machine will happily hold that announcement. It writes the table onto its tape the way it writes anything, as bytes. What it cannot do is hold it as a fact the machine itself is answerable for. Nowhere in the machine is there a place that says this partition is real and somebody paid to build it; the table just sits there, asserted, and the step rule has no opinion about whether it was earned or invented. The Thiele substrate

keeps that place. It carries the partition as part of its state, posts every build to the ledger at its declared cost, and will not stamp the structure certified while the bill for the check is still open: the step that stands behind it pays at least one, and there is no other door. That gap, between a structure you merely wrote down and a structure the machine is on the hook for, is the whole reason this section exists, and a good part of the reason the rest of the document does.

One thing to keep straight about the machine's own partitions. The runtime partition graph stores module records, and PNEW is allowed to make empty regions. Tidying each region locally doesn't turn the whole thing into a clean partition of memory, with no overlaps and nothing left over. Claims about global addresses, overlap, or locality need their own well-formedness and representation conditions. The hardware's partition wall is a separate mechanism. In the assembled PNEW format, operand A carries the first listed address and operand B the list length. The hardware table stores B and enforces the local range $[0, \text{length})$. It doesn't keep the global addresses you supplied, and it doesn't prove the region has no duplicates.

4.2 Axioms carried by modules

A partition by itself is just a set of labels, and labels are cheap because labels promise nothing. Calling something module *A* commits you to exactly zero about what is inside it. Structure that earns the name starts one step later, when you hang a constraint on a module and make it answer for something.

Two pieces of vocabulary, so I can say that cleanly.

A **predicate** is a yes-or-no question you can put to a thing and get a definite answer back. " $x > 5$ " is a predicate about a number. "This memory region holds a sorted list" is a predicate about a region of memory. A predicate is the shape a claim takes once it is sharp enough to be checked instead of merely believed.

A **logical formula** over Boolean (true/false) variables is those variables wired together with AND ($\wedge$), OR ($\vee$), and NOT ($\neg$). Take $(x_1 \vee x_2) \wedge (\neg x_1 \vee x_3)$: it comes out true exactly when at least one of x_1, x_2 holds and, at the same time, at least one of $\neg x_1, x_3$ holds. A choice of true or false for every variable that makes the whole thing come out true is a satisfying assignment; one that makes it come out false does not satisfy it. That is the entire machinery, and it carries everything built on top of it.

Now hang one on a module. "Module *A*'s region satisfies the predicate Φ_A ." The capital Greek Φ_A is nothing but a name I am pinning on the predicate I want to attach; mathematicians reach for Greek out of habit and there is no magic in the ink. Φ_A can carry real content: every value in this region sits between 0 and 100, or this region encodes a satisfying assignment to the formula living in module *B*, or this region is sorted top to bottom.

A caution about the word. The module record stores strings it calls axioms. Storing

a string doesn't prove what it says, and it doesn't hook it up to a predicate about memory. That takes an explicit representation and a checker proved sound. The guarded formula instructions do specific checking. Generic module metadata and generic CERTIFY don't establish arbitrary propositions.

I will come back to one concrete encoding in Section 12. In the 51-opcode realisation, LASSERT reads a formula from memory, checks it against two witnesses (one satisfying assignment, one falsifying assignment showing the formula is not a tautology), and either accepts or traps. The hardware checker is the Logic Engine. It is a finite-state controller in the chip: a dedicated circuit that walks the formula word by word and verifies the witnesses without calling an external constraint solver. If validation succeeds, the checked constraint package can feed the later certification path. If validation fails, LASSERT traps (Section A: routes execution to a fixed error address) and sets an error flag. The μ cost is charged either way.

Where in the run a check happened is something the trace tells you. A trusted trace shows which step ran a particular check and what it charged. The ledger alone can't do that. It's a total. It doesn't say what was checked or when. And a history somebody else hands you needs its own evidence before you trust it.

4.3 Structural gain

A trace has *structural gain* when some structural fact is uncertified at the door and certified by the time the trace stops. The plainest case: you open holding a partition nobody has vouched for, and you close holding a checked, certified satisfying-assignment witness. Something structural walked into the computation that was not there when it started.

The shallow question is whether the gain happened, and you can settle that by holding the first state next to the last and looking. The substrate is not built for the shallow question. It is built for *where*: which single step in the run is the one that turned belief into certified knowledge. Pin the trace to the board, walk it end to end, and find the step where the fact crossed from uncertified to certified.

There is always such a step, and it always cost something: a step with $\mu \geq 1$, never a free one, and it can be nowhere else. That is No Free Insight, said plainly before I have earned it on paper. The gain has a location, the location has a price, and the price is never zero.

5 Structural entitlement

It would be easy to read this whole project as one sentence, certification has a price, and stop at the doorway. That sentence is true. It is also the first cost law and the shallowest thing in here, the part you can see without going in. The room behind it is bigger, and this section is about what is in the room.

The deeper object is *structural entitlement*. A computation has it when the state contains

not just data, but a structural fact later steps are allowed to use as already established. I call that fact *admissible*. Admissible means the trace earned the right: a cost-positive step paid for it, and the proof object establishing it is on hand. Admissible is not just “true.” Admissible is “tracked and paid for.”

Several kinds of object can be admissible in this sense: a partition (Section 4), a decomposition (a split of a problem into independent parts), a morphism (Section 13), a constraint (Section 4), a witness (a value that proves the existence of something), or a narrowed feasible set (defined below). The gap between admissible fact and raw data matters. A raw bit pattern can be copied. A certified decomposition can justify solving two smaller problems instead of one large one. At the raw-data level, a morphism ID can be forged in a register: write 42 into the register and call it “the ID of some morphism.” An asserted morphism in the graph has provenance (Section A). In the 51-opcode realisation, the trace contains the MORPH and COMPOSE steps that built the morphism, and MORPH_ASSERT then committed it. A formula sitting in memory is text. A successful LASSERT check with a satisfying assignment and a falsifying assignment establishes that the encoded formula can be satisfied and isn’t a tautology, under the checker’s soundness contract. That’s all it establishes. It doesn’t make the falsifying valuation one of the states in the current prior list, and it doesn’t make any particular decomposition or factorization sound. Tying those together takes a representation premise of its own. In the explicit entitlement witness below, the prior and posterior lists and their observation relations are what tie them together.

Definition 5.1 (Structural entitlement). A state has structural entitlement to a claim P when three things are present:

- a structural object carried by the state, such as a partition, morphism, witness counter, or formula package;
- a checked relation between that object and the claim P ;
- a trace-visible certification or assertion event that makes later use of P admissible.

Without the third part, the object may exist as raw data. The computation still has not earned the right to use it as certified structure.

Here is how the pieces connect.

5.1 The precise vocabulary

Every term below is an exact Coq definition, not a metaphor. I ran every theorem that follows through Coq, so I won’t stamp “machine-checked” on each one; assume I did. That isn’t what makes them true. It’s how I caught the places I’d talked myself into something.

Feasible set. A feasible set Ω is the list of substrate states still on the table. In the kernel it is exactly that, a `list VMState`, and the literalness is the whole point. This is not “the space of possible worlds” or any other phrase that lets the possibilities stay vague and uncounted while sounding profound. It is a finite list of concrete states you could print to a page. One thing it is not: something the kernel reads off a running trace. The representation theorem takes the list as given, along with the subset, observation, covering, and representative facts about it. Where a construction proves the list got shorter, learning is the list getting shorter, and it is concrete enough that you can stand over it and watch the states drop.

Prior and posterior. Two snapshots of that list, before and after one observation or certification event. The *prior* Ω is the list as it stood going in; the *posterior* Ω' is the list coming out. The theorem asks for the posterior to be a subset of the prior, $\Omega' \subseteq \Omega$. That is a premise the theorem is handed. It is not a law that every list in the VM only ever shrinks. Inside that premise you can rule a state out and you can never rule one back in. When the posterior comes out strictly smaller, the event cut the field: states that were live going in are gone coming out. Careful with “strictly smaller,” though. A list can get shorter because it lost a duplicate entry, without losing a single distinct state, so $|\Omega'| < |\Omega|$ and $\Omega' \subsetneq \Omega$ are two different claims. In the concrete $n=1$ witness both lists are literal and duplicate-free, so there the two agree. Whatever is still standing afterward is the posterior.

Distinguishing observation. A distinguishing observation is a function that separates an eliminated state from the posterior. Formally: there exists at least one eliminated state $s_w \in \Omega \setminus \Omega'$ such that the observation function outputs a value on s_w that no state in Ω' produces. The observation is the “test” that reveals the structure. No named test, no exclusion: the posterior only genuinely rules out something the prior allowed once you can point at the test that does the ruling-out. Name it and the whole thing becomes checkable by anyone, including someone who thinks I am wrong. That is the entire appeal.

Strictly stronger predicate. The word “stronger” here runs backward from how it sounds. In ordinary talk the stronger claim is the bolder one, the one that says more; in logic the stronger predicate is the one that lets fewer things through. P_1 is strictly stronger than P_2 when it accepts a strict subset of what P_2 accepts. Formally in Coq: $P_1 \leq P_2$ means whenever P_1 is true, P_2 is true too, and strictness means there exists an observation where P_2 is true but P_1 is false. So P_1 is the harder claim to satisfy, not the louder one: it rules more states out, and everything that clears its bar clears the lower one for free. If P_2 is “the state is in Ω ” and P_1 is “the state is in $\Omega' \subsetneq \Omega$ ”, then P_1 is strictly stronger, anything in Ω' is in Ω , and not the other way around. The inversion is worth holding onto. Loose talk reaches for “stronger” to mean grander, and here it means smaller, and the whole vocabulary of narrowing only works once you take it the logician’s way round.

Structure-addition event. In the 51-opcode realisation, a structure-addition event is a single execution step where the assertion register `csr_cert_addr` transitions from 0 to nonzero. Precisely: $s.\text{vm_csrs}.\text{csr_cert_addr} = 0$ before the step and $(\text{vm_apply } s \ i).\text{vm_csrs}.\text{csr_cert_addr} \neq 0$ after. This happens only through a successful MORPH_ASSERT whose property string carries a non-NUL byte, which stores that property’s nonzero ASCII checksum in `csr_cert_addr`. Not “structural knowledge was acquired” in fog. This field flipped from zero, at this step, because MORPH_ASSERT asserted such a property against an existing morphism.

Supra-certification (`has_supra_cert`). In the 51-opcode realisation, supra-certification is reached when `csr_cert_addr` $\neq 0$. The name distinguishes it from `vm_certified`, the top-level flag set by CERTIFY. Supra-certification tracks the MORPH_ASSERT channel: has a morphism property been asserted and its checksum registered? A trace can set `vm_certified = true` without `has_supra_cert` by using CERTIFY and not MORPH_ASSERT; CERTIFY never touches `csr_cert_addr`. Both are positive-cost certification channels. They record different things. Two channels, two costs, one substrate.

Decision tree. A decision tree, in the kernel, is the bare branching skeleton: a leaf, or a node with a left subtree and a right subtree, and nothing written on it. What gets audited is its shape, the depth and the leaf count; the semantic bookkeeping, which observation was made and which states survived it, lives in the observation function and the posterior-representative reduction standing next to the tree, not inside the tree object. The tree is a proof object, not a runtime structure: it exists so the move from prior to posterior answers to a definite branching shape, with leaves enough to index the posterior and a depth the trace has to pay for, rather than a magic jump. “Realized by the trace” means exactly that payment: $\text{depth}(T) \leq \text{cert_setter_executions}$, the trace ran at least as many cert-setters as the tree is deep. That count takes in the kernel’s whole cert-setter class, not only the instructions that flip a persistent certification bit. It is a number. The trace does not walk the tree node by node or follow one root-to-leaf path. A shortcut isn’t sound on its word: it’s sound once the shape, the reduction, the observation, and a trace that paid for the depth are all on the table, and not one second before.

Posterior-representative reduction. Given a prior state $s \in \Omega$, a posterior-representative reduction provides a function mapping s to a representative $s' \in \Omega'$. This proves that every prior state corresponds to something in the posterior. The purpose is accounting: the posterior was not reached by arbitrary deletion, but by a coherent map where each prior state is represented. Without this, the narrowing can be fake. Delete states, announce a small posterior, and provide no principled connection to the prior.

| *Plain reading.* Same witness as Categorical Separation, read backward. Two VM

states have identical projections under any projection that drops the morphism graph. They are still different states. So the projection is provably many-to-one. A function defined only on the projection output can't tell them apart. The projection already collapsed them.

Theorem 5.1 (Classical blindness as quotient). *The four-field projection keeps only (PC, μ , registers, memory). It is many-to-one: different VM states can land on the same image. Section A calls that a quotient. The projection identifies states that agree on the kept fields, even if they differ in certification, witness counters, or morphism graph. The richer six-field `shadow_proj` keeps `vm_certified`, but still forgets the morphism graph. No function of either projection alone can separate the morphism-witness states.*

Proof. Section 14 gives the witness pair. s_1 carries `pg_morphisms` = [(0,id)]. s_2 carries `pg_morphisms` = []. Everything the projections keep is identical: `vm_regs` = [], `vm_mem` = [], `vm_pc` = 0, `vm_mu` = 0, `vm_err` = false, `vm_certified` = false. The four-field projection and six-field `shadow_proj` both map them to the same image. They are different states with equal projections. Any classifier using only the projection-image must agree on equal images, so it agrees on s_1 and s_2 . $\square$ $\square$

If your state model only records registers, memory, PC, an error bit, and maybe μ , you have already thrown away part of what matters. You have collapsed states that differ on the inside into a single raw state that looks the same from the outside. That collapse is a theorem about a projection map.

Plain reading. Three things ride along on every state besides the classical fields: the ledger, the certification flag, and the graph. Each one can change while everything else you can see holds still, including the other two. Five independence claims, one witness pair each, the μ ones parametric in the starting state. Case (5) is the categorical-separation pair: same five projected fields, different morphism graphs. The other four are pairs of states reached by one-step programs from the clean start that match on the named tuple, graph included, and disagree on the named coordinate. What makes (1)–(3) work is a cost-matched JUMP on one branch. PNEW would touch the graph. JUMP touches nothing but pc and μ , so the witness holds the graph fixed too. Three structural axes, five witnesses, and each axis comes out independent of the classical fields and of the other two.

Theorem 5.2 (μ -ledger necessity, full pairwise independence). *For two state components X and Y, $X \perp Y$ here means: knowing Y does not determine X on every state. The symbol is redefined locally. Elsewhere it can mean orthogonality; here it means independence. Concretely, there is a witness pair: two substrate states with the same Y and different X. For the μ results the construction is parametric, working from every reachable starting state, not one lucky start; for the certified results it runs from any*

uncertified state, the clean start included; for the graph result the pair is built directly. Five independence results hold at once, and `vm_graph` is held fixed on the right-hand side of every one of them, not only in result (5): (1) $\mu \perp (\text{mem}, \text{regs}, \text{pc}, \text{vm_graph})$; (2) $\text{certified} \perp (\text{mem}, \text{regs}, \text{pc}, \text{vm_graph})$; (3) $\text{certified} \perp (\text{mem}, \text{regs}, \text{pc}, \mu, \text{vm_graph})$; (4) $\mu \perp (\text{mem}, \text{regs}, \text{pc}, \text{certified}, \text{vm_graph})$; (5) $\text{vm_graph} \perp (\text{mem}, \text{regs}, \text{pc}, \mu, \text{certified})$. This is the full pairwise claim: each of the three structural components, μ , vm_certified , vm_graph , fails to be a function of the classical projected fields together with both of the other two, not just of the classical fields alone. Results (1) and (2) sharpen further once (4) and (3) tie down the remaining structural coordinate; read together, no structural component is ever recoverable from the classical fields plus the other two structural components, in any of the three positions.

Proof. Each claim is a witness pair: two VM states agree on the named projection-tuple and differ on the named coordinate. For the μ and certification witnesses, reachability is part of the construction where the text says so. The graph-separation pair doesn't claim it; that pair is built directly.

(5) $\text{vm_graph} \perp (\text{mem}, \text{regs}, \text{pc}, \mu, \text{certified})$. The witness is Categorical Separation: same projected fields, different graph. s_1 carries `pg_morphisms = [(0, id)]`; s_2 carries `pg_morphisms = []`. The pair is directly constructed, field for field, because the claim needs nothing more: “not a function of the projection” is settled by any two states the projection cannot tell apart, however they came to be.

(1)-(2) $\mu, \text{certified} \perp (\text{mem}, \text{regs}, \text{pc}, \text{vm_graph})$. Same visible state and same graph, different ledger and cert flag. One branch runs `CERTIFY 0`; the other runs `JUMP 1 0` rather than `PNEW`: `jump_state` passes `vm_graph`, `vm_regs`, and `vm_mem` straight through unchanged and only moves `vm_pc` and μ , where `PNEW` would have added a partition and moved `vm_graph` along with it. Registers, memory, pc, and graph all match between the two branches. μ does not: 1 versus 0; nor does `certified`: true versus false. The Coq witnesses are `abs_A_strict` and `abs_B_strict_g`, and the record type carrying all four held-fixed fields is `FullStrictShadow`.

(3) $\text{certified} \perp (\text{mem}, \text{regs}, \text{pc}, \mu, \text{vm_graph})$. Case (2) with the ledger and the graph both tied down: `CERTIFY 2` against `JUMP 1 3`. Both charge $\mu = 3$, both leave the visible tuple and the graph identical, and they still disagree on `vm_certified`. The Coq witnesses are `abs_A_cost` and `abs_B_cost_g`, over `FullCostShadow`.

(4) $\mu \perp (\text{mem}, \text{regs}, \text{pc}, \text{certified}, \text{vm_graph})$. Same visible tuple, same certification flag, same graph, different ledger: `CERTIFY 0` against `CERTIFY 1`. `CERTIFY` never touches `vm_graph`, so this pair holds the graph fixed without needing to change instruction; only the declared costs differ, so μ comes out 1 versus 2. The Coq witnesses are `abs_A_cert` and `abs_B_cert`, over `FullCertShadow`.

The parametric rider belongs to the μ results, and there it is not decoration: the source

lemmas provide the certify-versus-jump construction from the stated reachable context, so no prefix covered by that premise prevents the μ witness pair. The certified results run from any uncertified state, and that is as far as they can go, for the honest reason that certification is permanent: once a prefix has certified, nothing clears the flag, so both branches of any extension agree on it. One witness pair is exactly what “not a function of” requires. A graph-inert instruction with matching cost supplies the corresponding graph-fixed witness. $\square$

The witness states do the work here: same selected observation, different ledger, flag, or graph. Try to reconstruct the missing datum from that observation and one of the witnesses defeats you. The result is sharp about the map it studies, on all three structural axes at once. It doesn’t say every conventional encoding has to throw the datum away. It doesn’t say these coordinates form a geometric basis, either.

| Plain reading. The change in μ across a run is just the sum of the per-step costs. μ -Conservation handles each step. Induction stitches them together. Two different starting states with the same starting μ end up at the same final μ on the same program. The starting graph, registers, memory: none of it matters for the μ count. Only the program does. The cost is attached to the trace. The state that carries it doesn’t invent it. The substrate records the receipt. It doesn’t make the receipt up.

Theorem 5.3 (μ -ledger uniqueness, for the kernel’s fixed schedule). *Run a fixed list of instructions, in order. The μ it accumulates is the starting ledger plus the sum of the step costs, charged by the kernel’s own fixed per-instruction schedule `canonical_cost` (the schedule `vm_apply` itself charges, not a stand-in for some other pricing choice). Partition graph, registers, memory, program counter, certified flag: none of them changes that sum. One caution. This is about folding over a list. A real run follows the program counter, which can skip, repeat, and jump, and the theorem does not say such a run executes each listed instruction exactly once. Formally:*

$$\mu_{\text{final}} = \mu_{\text{initial}} + \sum_{i \in \text{prog}} \text{instruction_cost}(i).$$

The right side uses the summation symbol from Section A. It reads: add up `instruction_cost(i)` for every element i in the fixed list `prog`. The result does not depend on the starting state’s structural fields. Any measure that charges exactly `canonical_cost` at every corresponding step and assigns 0 to `init_state` agrees with `vm_mu` on every reachable state (`instruction_consistent_measure_equals_mu`). This fixes the accumulated total once the schedule and the zero start are fixed. It is not a claim that `canonical_cost` is the only schedule a system could honestly adopt, and it says nothing about a measure charging a different per-instruction price; Section 7 below keeps that separate question, exact-event pricing among possible schedules, apart from this one. If any executed element of the fixed list has positive `canonical_cost`, then $\Delta\mu$ is strictly positive: this is the theorem’s positive-cost-step guarantee, not a claim that no program or no instruction

can cost zero.

Proof. Induct on $prog$. At each step, apply μ -Conservation.

Base ($prog = []$). Empty program. No steps. The sum is 0, so both sides reduce to μ_{initial} . The equation holds.

Step ($prog = i :: rest$). Let $s_1 = \text{vm_apply}(s_0, i)$. By μ -Conservation,

$$s_1.\mu = s_0.\mu + \text{instruction_cost}(i).$$

Apply the induction hypothesis to $rest$ starting from s_1 :

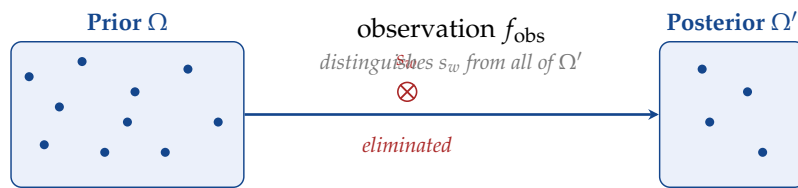
$$\mu_{\text{final}} = s_1.\mu + \sum_{j \in \text{rest}} \text{instruction_cost}(j).$$

Substitute $s_1.\mu$:

$$\begin{aligned} \mu_{\text{final}} &= s_0.\mu + \text{instruction_cost}(i) + \sum_{j \in \text{rest}} \text{instruction_cost}(j) \\ &= \mu_{\text{initial}} + \sum_{i \in prog} \text{instruction_cost}(i). \end{aligned}$$

The right-hand expression depends only on $prog$ and μ_{initial} . No other starting-state field gets a vote. If any step has positive cost, then $\Delta\mu$ is strictly positive: all terms are non-negative, and that term is ≥ 1 . $\square$ $\square$

The ledger is the sum the trace and the fixed schedule add up to. Any accounting with the same step increments and the same starting value agrees with it. A different schedule can give a different total. And the theorem doesn't hang an intrinsic physical bill on a classical computation nobody annotated with costs.



$$\Omega' \subsetneq \Omega \quad \text{and} \quad f_{\text{obs}}(s_w) \notin f_{\text{obs}}(\Omega') \implies \log_2^\uparrow |\Omega| - \log_2^\uparrow |\Omega'| \leq \Delta\mu$$

Figure 3: The pile gets smaller. A strict cut plus a test that names what got cut gives a stronger predicate on what's left. The bound on the right also needs the supplied fiber cover and the tree-payment premise. $\log_2^\uparrow |\Omega| - \log_2^\uparrow |\Omega'|$ is a difference of rounded list sizes. It doesn't say the VM asked that many questions, and it doesn't say the lists hold that many distinct states.

| Plain reading. Smaller set, stronger predicate. That is the whole of (a), and it is pure set inclusion doing the work. (b) is where the distinguishing observation

earns its rent. Without it you have an existential: somebody, somewhere, lives in the gap. With it you have a name and an address. An existential is a promise. A witness is the proof of the promise.

Theorem 5.4 (Feasible narrowing becomes predicate strengthening). *Suppose a posterior feasible set Ω' is a strict subset of a prior feasible set Ω . Suppose also that an observation distinguishes a removed state from all posterior states. Then the membership predicate for Ω' is strictly stronger than the membership predicate for Ω . (The membership predicate for Ω is the function that takes a state and returns true exactly when the state is in Ω .)*

Proof. Define $P_\Omega(s) := (s \in \Omega)$ and $P_{\Omega'}(s) := (s \in \Omega')$. “Strictly stronger” has two parts.

(a) $P_{\Omega'}$ implies P_Ω . If $P_{\Omega'}(s)$ holds, then $s \in \Omega'$. Since $\Omega' \subseteq \Omega$, we get $s \in \Omega$. So $P_\Omega(s)$ holds.

(b) P_Ω does not imply $P_{\Omega'}$. Since $\Omega' \subsetneq \Omega$, there exists $s_w \in \Omega \setminus \Omega'$. So $P_\Omega(s_w)$ is true and $P_{\Omega'}(s_w)$ is false. That is enough for strictness. The distinguishing observation gives more: an explicit Boolean function witnessing s_w , not just the assertion that some such state exists. □

It feels like two separate things happened. First you did some work and the search space got smaller; then, as a reward, you got to assert something stronger. Two events, cause and effect, the second one earned by the first. That ordering is the illusion. There is no second event. The instant Ω shrinks to Ω' the stronger predicate already exists, because the predicate is nothing but membership in the set, and the set is smaller. Nobody pulled the predicate out of a hat to make the theorem go through; it was sitting there the whole time, defined by the boundary that just got redrawn. The distinguishing observation is what lets you write the boundary down honestly, but it does not author a new claim on top of the narrowing. The narrowing is the claim. One fact wearing two costumes: shrink the feasible set and tighten the predicate are the same move read off opposite sides of the same equation.

| Plain reading. Cert address starts at 0, ends nonzero. So somewhere along the trace, it flipped. That flip is exactly what Section 5 calls a structure-addition event. The proof is just looking: if the value went from 0 to nonzero across a finite trace, the change happened on some specific step. The mathematics is bookkeeping.

Theorem 5.5 (Strengthening requires structure addition). *If a trace starts with no certification address, ends certified for a strictly stronger predicate, and that certification is tied to the supra-certification channel defined above, then the trace contains a structure-addition event.*

Proof. “No certification address at start” is $s_0.\text{vm_csrs.csr_cert_addr} = 0$. “Certified for the supra-cert channel at end” is $s_n.\text{vm_csrs.csr_cert_addr} \neq 0$, the definition of `has_supra_cert`. The trace runs from s_0 to s_n through a finite sequence of steps. The field `csr_cert_addr` starts at 0 and ends nonzero. Somewhere in the trace, it flipped. That step is, by the definition in Section 5, a structure-addition event. The strictly-stronger-predicate hypothesis names what the certification is *for*; it does not affect the existence of the flip. $\square$ $\square$

This is the formal answer to “where did the new admissible structure enter?” Arithmetic on its own never answers that, because arithmetic only ever hands back what was already derivable from what you fed it; it closes a question, it does not open the world up to a fact that was not already in the premises. A stronger admissible predicate is exactly such a new fact, so it could not have fallen out of the computation for free. It had to be added, and the theorem says where: not in a flourish at the end, not off the page, but at one specific step you can point to in the trace, a structure-addition event with a timestamp.

| Plain reading. This is the same theorem as the one in the NoFI section, carried up here on purpose. Down there it was proved as a fact about cost. Up here the structural-entitlement layer needs that same floor as one of its own load-bearing parts, because the whole argument about who gets to claim a stronger predicate stands on it: certification never comes free wherever A2 is in force. Accept A2 and you have already agreed that flipping certification on costs you something. A2 acceptance is the entry fee.

Theorem 5.6 (Universal certification cost). *Take any abstract computational system: states, step labels, step rule, cost function, certification predicate. If every one-step move from uncertified to certified costs at least 1, then every trace from uncertified to certified has total cost at least 1.*

Proof. This is the same theorem as the Universal No Free Certification result in Section 8, restated here at the structural-entitlement layer. The proof is the trace-induction argument given there. Empty trace cannot flip the cert predicate. A non-empty trace either flips it at the first step, where single-step A2 gives ≥ 1 , or flips it later, where the induction hypothesis gives ≥ 1 on the rest. Either case yields total cost ≥ 1 . $\square$ $\square$

This theorem matters because it is not about the 51-opcode ISA. The cost floor is forced for any system the moment that system has a certification predicate and a cost rule that charges for certification transitions. Build the predicate so it can flip at zero cost and you have free certification by construction. And a system that refuses to represent certification at all has no way to talk about structural entitlement from the inside. The entitlement gets smuggled in from outside, and somebody signs their name to it. Smuggled entitlement is not entitlement, no matter whose name is on it.

| Plain reading. Three conclusions, three pieces snapped together. The first is the predicate-strengthening theorem: a strict cut plus a test that names what got cut. The second is the structure-addition theorem: the cert address starts at zero and ends nonzero, so some step flipped it. The third is the bound. The pile of candidates is covered by the tree’s leaves times what’s left, so the drop in rounded list size can’t beat the rounded leaf count, and a binary tree’s rounded leaf count can’t beat its depth. The trace has to run at least that many cert-setters, and each one costs at least 1. So the ledger pays at least the drop. It’s a count of cert-setter payments. Nobody is saying each node of the tree matches a particular step, and nobody is saying this is Shannon entropy. Compression for less than you paid never shows up, because the ledger doesn’t extend that kind of credit.

Theorem 5.7 (Structural entitlement representation). *For any values satisfying the explicit Coq premises of `structural_entitlement_representation`, consisting of:*

- *a strict feasible-set narrowing $\Omega' \subsetneq \Omega$;*
- *an observation function that distinguishes at least one eliminated witness state from all posterior states;*
- *a Certified posterior predicate built from the specified decoder, observation equality, and membership in Ω' ; in this theorem Certified includes the VM’s `has_supra_cert` condition, whereas the separate observation-level theorem uses `CertifiedObs`;*
- *a decision tree realized by the trace: the tree’s depth is bounded by the count of cert-setter instructions the trace executes (`decision_tree_realized_by_trace`, defined as $\text{depth}(T) \leq \text{cert_setter_executions}$). This is a payment bound on depth, a scalar comparison, not a claim that the trace’s step sequence is mapped node-by-node onto the tree’s branches; and*
- *a posterior-representative reduction tying each prior list entry to a posterior representative with equal observation, together with the explicit fiber-size bound that yields $|\Omega| \leq \text{leaves}(T)|\Omega'|$,*

the posterior predicate is strictly stronger than the prior predicate, the trace contains a structure-addition event, and the size of the narrowing is bounded by the paid ledger:

$$\log_2^\uparrow |\Omega| - \log_2^\uparrow |\Omega'| \leq \Delta\mu.$$

Here $|\Omega|$ means the length of the prior feasible list, using the $|\cdot|$ notation from Section A. Ω is a plain `list VMState`, not deduplicated and not a Coq set; $|\Omega|$ counts list entries, duplicates included, not distinct states. Nothing below assumes Ω is duplicate-free. $|\Omega'|$ is the posterior list’s length, on the same convention. $\log_2^\uparrow$ means: take the base-2 logarithm and round up. A 1000-entry list needs 10 yes-or-no questions to pin down one entry, because $2^{10} = 1024$, the first power of two that can cover 1000 items. That is the intuition, and

only the intuition. The VM did not ask those questions. This is not Shannon entropy. With duplicates in the list, it is not a count of distinct states either. Both sides of the bound are integers. The eye wants to read $\log_2^\uparrow |\Omega| - \log_2^\uparrow |\Omega'|$ as the textbook bit-gain $\log_2(|\Omega|/|\Omega'|)$, and that is the one reading to set aside: the two ceilings round separately, so the difference can land above that real-valued number or below it. The proof bounds the integer itself, through the supplied fiber cover and the tree's depth, and the depth by the cert-setter payment. $\Delta\mu$ is final μ minus initial μ across the bounded trace.

Proof. Three conclusions, chained.

(i) *Strictly stronger predicate.* The strict narrowing $\Omega' \subsetneq \Omega$ and the distinguishing observation match the hypotheses of the theorem above. Apply it. The membership predicate of Ω' is strictly stronger than that of Ω .

(ii) *Structure-addition event.* By hypothesis, the posterior certification uses the supra-cert channel. So the trace starts with `csr_cert_addr = 0` and ends with `csr_cert_addr  $\neq$  0`. The Strengthening-requires-structure-addition theorem gives a structure-addition event at some step k^* .

(iii) *The $\log_2^\uparrow$ bound.* The realized decision tree T is binary; write ℓ for its leaf count. Sort the prior states by which posterior state they end at. That is $|\Omega'|$ groups, and the covering hypothesis (`tree_covers_feasible_reduction`, itself discharged from the trace's fibered reduction by `fibered_reduction_implies_tree_cover`) caps every group at ℓ states. So $|\Omega| \leq \ell \cdot |\Omega'|$. Take $\log_2^\uparrow$ of both sides and use that it is subadditive on products, $\log_2^\uparrow(ab) \leq \log_2^\uparrow a + \log_2^\uparrow b$:

$$\log_2^\uparrow |\Omega| \leq \log_2^\uparrow \ell + \log_2^\uparrow |\Omega'|, \quad \text{equivalently} \quad \log_2^\uparrow |\Omega| - \log_2^\uparrow |\Omega'| \leq \log_2^\uparrow \ell.$$

A binary tree holds at most 2^d leaves at depth d , so $\log_2^\uparrow \ell \leq d$, and the two combine to

$$\log_2^\uparrow |\Omega| - \log_2^\uparrow |\Omega'| \leq \log_2^\uparrow \ell \leq d.$$

The realized-by-the-trace hypothesis is exactly the numeric bound $d \leq \text{cert_setter_executions}$: the tree's depth is charged against the count of cert-setter instructions the trace executes, not matched branch-by-branch against the trace's own step sequence. Section 7 charges each cert-setter step at least 1 in μ , so that many cert-setter steps cost at least that much total, and the total is at most $\Delta\mu$, the cost of the whole trace. Chain the inequalities:

$$\log_2^\uparrow |\Omega| - \log_2^\uparrow |\Omega'| \leq d \leq \text{cert_setter_executions} \leq \Delta\mu. \quad \square$$

□

This theorem does not quantify over the English word “structure.” It defines the witness class: strict narrowing, distinguishability, certification, and an explicit tree/fiber

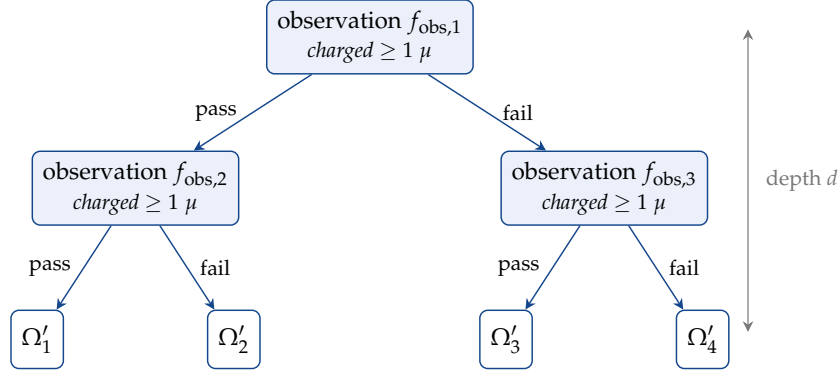


Figure 4: The decision tree witness. The tree is a bare branching shape. What it means comes from the observation and the posterior-representative premises next to it. The trace pays for its depth: depth is at most the number of cert-setters the trace runs. That’s a count. It doesn’t match each node to a particular step. On the right, the depth bounds how much the list shrinks. On the left, the μ the trace paid bounds the depth.

representative for the reduction. For every member of that class, structural entitlement is not metadata and not a slogan. It is predicate strengthening. It requires structure addition. Its quantified narrowing is charged to μ . That is what entitlement means here. Not a feeling, not a story.

| Plain reading. The record is exactly what the representation theorem needs as input. Unpack the record. Plug the fields into the theorem. Read off the three conclusions. The class is non-empty because `factored_n1_shortcut` builds a member of the class on the page, by name. It’s non-empty because somebody built one. Nobody argued it into existence.

Theorem 5.8 (Every sound structural shortcut lands here). A `SoundStructuralShortcut` is a package containing exactly the receipts a shortcut must provide to count as sound: a prior feasible set, a posterior feasible set, strict narrowing, a distinguishing observation, a certified posterior predicate, a realized decision tree, and a posterior-representative reduction. For every such package, the structural-entitlement representation theorem applies. The shortcut lands in strictly stronger posterior knowledge, a structure-addition event, and the $\Delta\mu$ lower bound.

Proof. By construction, a `SoundStructuralShortcut Coq` record contains: Ω , Ω' with $\Omega' \subsetneq \Omega$, an observation function f_{obs} with a distinguishing-witness proof, a decision tree T with the numeric tree-payment obligation, a posterior-representative reduction with a representatives proof, and an initial-uncertified proof on the starting state. These fields and obligations are exactly the hypotheses of the Structural-Entitlement-Representation theorem. Project the seven data fields and eight proof obligations from the record into the theorem’s hypotheses. Apply the theorem. The three conclusions (strictly stronger predicate, structure-addition event, $\log_2^\uparrow\text{-narrowing} \leq \Delta\mu$) follow. □

This is the formal version of “it has to.” Once a shortcut stops being a human story and becomes a sound object the substrate may use, it has to carry the data that makes it admissible. In this framework, that data is exactly the structural-entitlement witness. When symmetry, factorization, modularity, independence, or decomposition is named as the source of a shortcut, the next question is no longer mystical. It is a checklist: the prior set, the posterior set, the observation that rules something out, and the representative witness and tree witness. With those in hand the shortcut is admissible. Without them the structure is still sitting outside the computation, however good the story around it sounds.

5.2 What narrowing buys

There’s a concrete version of all this you can run, a factored search problem built to make the difference visible. Two programs go after the same answer. The blind one runs with every instruction cost set to 0 and searches the flat space, paying nothing and seeing nothing, which is the point and not an accident: a machine with no ledger gets its blindness for free. The sighted one pays first. Two receipt-visible events, each `EMIT " " 0`, for a total receipt cost of 18, and only then does it split the problem into two factors and search them apart. That 18 is the whole tab. The kernel proves the exact cost formulas, and it proves the arithmetic that makes the trade lopsided: the N^2 versus $2N$ gap doesn’t merely win, it eventually dwarfs that constant μ cost by orders of magnitude, so the price you paid up front stops mattering almost immediately. The executable tests run the actual VM at representative sizes and watch the measured numbers land where the proof says they will. The receipt costs 18 once. The exponent saves you forever.

The point of this example is not the N^2 versus $2N$ speedup. Divide-and-conquer is not the new claim. Nobody is selling that. The point is the auditable trace. The receipt-visible `EMIT` events plus the certified suffix satisfy the `SoundStructuralShortcut` witness contract, so the shortcut is admissible to the substrate, not just asserted by the programmer who happens to feel good about it. Admissible is all it is. The receipt does not prove the factorization is right. It does not authenticate a referee, and it does not tie the property label to a factorization checker. The 18μ is the cost of the receipt-only trace; the extended trace costs 19μ after the three-instruction certification suffix. The receipt is the evidence of what the trace paid for. Whether the math inside it is right is a separate question, and it needs its own proof.

There are two nearby claims here and it matters to keep them separate, because conflating them is exactly how a real theorem turns into a stage trick. The `EMIT`-only $N = 1$ witness closes the observation layer. It proves posterior admissibility as `CertifiedObs` (a Coq type for “a certified observation has been recorded,” meaning the VM has observed and recorded a result but has not yet activated a cert channel). It does not close the stronger bridge. The trace does not end in `has_supra_cert`. In the semantic CSR-transition sense, it does not realize structure addition. Appending `PNEW`, `MORPH_ID`, and `MORPH_ASSERT` to the factorized search fires the concrete

certification channel and places the resulting trace within the full structure-addition theorem.

That example is not a P vs NP result. Anybody who reads it as one is reading their own ambition into the bytes. It is a small, honest witness for what the substrate is meant to track. Certified structure can change which computation is admissible. The cost of acquiring that admissibility is not the same as the time saved by using it. One is the entry fee. The other is the dividend.

What the package looks like in the concrete $n=1$ witness. It pays to lay the whole package open once, at the smallest factored-search case, where the object is small enough to count on your fingers and nothing has to be taken on faith. The record `SoundStructuralShortcut` carries fifteen fields. Seven are data: decoder, observation fn, representative observation fn, observation equality, prior set, posterior set, decision tree. The other eight are proof obligations standing over that data, the receipts that say the data behaves the way the fields promise. The nine semantic ingredients below cover six of the seven data fields, the decoder being the identity map and living in the wiring, plus five of those obligations: strict narrowing, the distinguishing observation, tree realization, the certified posterior, and the initial-uncertified state. The remaining three obligations are discharged inline by the wiring that holds the record together. Fifteen fields, nine ingredients, four pieces of wiring, and that is the whole of it. Nothing here is hand-waved, because there is no room left to wave a hand: every part has a place, and every place is filled.

1. **Prior feasible set Ω .** `sighted_n1_supra_prior := [init_state; sighted_n1_supra_final]`. A literal two-element list of `VMState` values.
2. **Posterior feasible set Ω' .** `sighted_n1_supra_posterior := [sighted_n1_supra_final]`. A literal one-element list.
3. **Strict narrowing.** $|\Omega'| = 1 < 2 = |\Omega|$ by direct inspection of the lists.
4. **Distinguishing observation.** `sighted_n1_supra_obs_fn`. Returns `sighted_n1_supra_trace` when `vm_mu = 19`, else `[]`. The `post-MORPH_ASSERT` state has `vm_mu = 19`. The initial state has `vm_mu = 0`. The observation separates them.
5. **Decision tree.** `sighted_n1_tree := dt_branch dt_leaf dt_leaf`. A two-leaf branching tree, one branch per factor.
6. **Tree realized by the trace.** The closed witness supplies the numeric premise `decision_tree_realized_by_trace`: the tree depth is at most the number of cert-setter executions in the bounded run. The theorem does not claim that the trace's opcode sequence is a node-by-node traversal of the tree.
7. **Posterior-representative reduction.** `sighted_n1_supra_repr_obs_fn`. Defined as `if Nat.eqb s.(vm_mu) s.(vm_mu) then [] else`

[instr_halt 0]; the tautology `Nat.eqb mu mu = true` always picks the empty-list branch, so this is the empty-observation representative dressed up as a conditional.

8. **Certified posterior predicate.** Built from `omega_predicaterceipt_list_eqbsighted_n1_supra_posterior_sighted_n1_supra_obs_fn`. The `MORPH_ASSERT` in the supra-suffix activates the VM’s certification channel, which is the channel premise used by the formal package; it does not check the mathematical factorization named in the surrounding prose.
9. **Initial uncertified state.** `init_state.(vm_csrs).(csr_cert_addr)=0` by definition of `init_state`.

All of that machinery only earns its keep if something actually satisfies it, and the worry with any definition this elaborate is that you have quietly written down a shape nothing in the world fits. So the lemma `sound_shortcut_from_components` does the unglamorous thing: it takes the seven data fields listed above and eight proof hypotheses, one per obligation, and out the other end comes a `SoundStructuralShortcut` record. Then the closed term `factored_n1_shortcut: SoundStructuralShortcut33sighted_n1_supra_traceinit_state` feeds it the seven fields and clears each of the eight obligations with a named lemma, and the downstream theorem `factored_n1_shortcut_lands_in_representation` runs `every_sound_structural_shortcut_lands_here` over that finished term and reads back exactly what the abstract class promised: a strictly stronger posterior predicate, a structure-addition event, and the $\log_2^\uparrow$ narrowing bounded by $\Delta\mu$. The payoff is small to state and it is the one I actually cared about. The class is not empty, and it is non-empty by construction, not by argument: there is a concrete shortcut built on the page that hits every clause, which means the whole apparatus is about something real and not a definition admiring itself in a mirror.

5.3 The boundary

The hard internal theorem for the explicit class is proven, and so is the boundary in both directions on a concrete witness. An `EMIT`-only factorized-search trace closes only the observation layer. The posterior predicate is certified observationally. But the trace does not reach `has_supra_cert` and does not realize semantic structure addition. A trace with the factorized search unchanged and a post-search suffix `PNEW ; MORPH_ID ; MORPH_ASSERT` closes the whole chain: stronger predicate, structure addition, and the $\Delta\mu$ lower bound. An unconditional Shannon-style bound on every deterministic single trace is something the code refuses to claim. A single execution path is not enough information. The bound needs the whole tree, or a per-branch preimage, or a distribution over traces. The proof stops where the witness stops. That’s the honest place to stop. It isn’t a hole waiting to be patched.

Once entitlement is internalized with explicit witnesses instead of waved at, the picture firms up into three facts that lean on each other. You cannot strengthen a certified

claim for free: the receipt costs μ or it does not exist. Strip a trace down to its classical shadow and the distinctions that entitlement turns on stop being visible at all, so the shadow can no longer tell you what was earned and what was merely assumed. And narrowing the feasible set is no longer a vibe; it has a formal path into predicate strengthening with a ledger lower bound underneath it, so the act of knowing more shows up as a number that went up. Then the substrate-level result in Section 10 settles the question you would ask next, which is whether all of this can be put on autopilot: is there a uniform, internally representable, always-terminating procedure that takes a formalized structural argument and hands back a `SoundStructuralS` `shortcut` witness whenever a sound one exists? There is not; Section 10 constructs the reduction. A partial procedure, restricted to some class of arguments or free to decline rather than terminate, is not what that reduction rules out. The structural axis comes with its own undecidable predicate, in that total-procedure sense, which is the unsurprising and slightly bleak news that insight does not reduce to a crank you can turn.

6 What insight is, and why it cannot be free

On this substrate the word *insight* stops being a mood and starts having a meter reading. It comes in three tiers, three genuinely different things that happen when a computation learns, and they sort themselves by price: the first can be free, the second never is, and the third is where the price tag has its hand on a certified game result that no single fixed classical strategy can produce.

6.1 The three tiers of observation

The kernel proves each tier lands where it lands, by what the substrate makes it cost, so the split is the substrate's verdict and not my filing system. Cheapest first.

Tier 0: Raw observation (free at $\delta = 0$). A raw observation takes something in, writes it down, and stops there. It does not say what the data means, does not lean on it, and does not let any later step lean on it either. The cleanest case in the realisation is a CHSH trial, one round of the two-player game Section 18 builds from scratch: you run the round, a counter ticks up by one, and the certification channels stay cold. Nothing has been claimed, so there is nothing to commit to. The trial's cost is whatever δ it carries. At $\delta = 0$ the ledger does not move; with a positive δ it is charged and it is still Tier 0. So looking can be free on this substrate. You can watch as long as you like at $\delta = 0$ and the meter never starts. That's a fact about this machine's cost schedule. It isn't a claim that looking is free in the physical world. The meter starts the instant you turn what you saw into something you are willing to stand behind, and that instant is a different tier.

Tier 1: Certified structural insight (always costs ≥ 1). Two cert channels. `csr_cert_addr` (the formula/certificate address, a control-status register; Section 11 explains the term). `vm_certified` (the top-level flag). Any concrete instruction that

turns either channel from off to on costs at least 1. The kernel proves it: `certification_requires_positive_mu`.

Precision first. `LASSERT` reads a constraint package and charges for it. `LASSERT` does not flip either cert channel. Two opcodes do the flipping: `CERTIFY` (sets `vm_certified = true`) and `MORPH_ASSERT` (sets `csr_cert_addr nonzero` when the assertion succeeds). `LASSERT` is the validation that precedes them. The kernel groups a broader cost-positive class: `LASSERT`, `EMIT`, `REVEAL`, `LJOIN`, `READ_PORT`, `CERTIFY`, `MORPH_ASSERT`, and the five-opcode `CHSH_LASSERT` check family (Section 12 builds it). Each costs at least 1 at the step level. `LASSERT` is part of the chain. The cert-channel flip happens at the subsequent `CERTIFY` or `MORPH_ASSERT`, not at `LASSERT` itself.

The chain, written out. Readers collapse two steps into one. Don't. Suppose register `r0` holds the address of a four-byte formula in memory and `r1` holds the address of a dual-witness certificate (a satisfying assignment together with a falsifying assignment, the non-triviality witness from Section 12). Step one: `LASSERT freg=r0 creg=r1 kind=SAT flen=4 delta=0` reads the formula header, walks the formula and the two witnesses through the on-chip Logic Engine, charges $4 \times 8 + 1 = 33$ to μ , and either advances the PC (validation passed) or traps to `LASSERT_TRAP_PC` and latches `vm_err = true` (validation failed). The μ cost is charged either way. Both `csr_cert_addr` and `vm_certified` stay untouched by this step, validation passing or not. Step two, conditional on step one passing: `MORPH_ASSERT mid=mproperty="validated_constraint" cert="" delta=0` charges $S(0) = 1$ to μ (S is successor, $S(n) = n + 1$; Section 7 builds the notation properly) and flips `csr_cert_addr` to the ASCII checksum of "validated_constraint", which is nonzero because the property string carries non-NUL bytes. The supra-cert channel is now active.

Total cost across the two steps: 34μ . `LASSERT` paid for the validation work. `MORPH_ASSERT` paid for the channel flip. The cert channel only crossed from 0 to nonzero on the second step, not the first. The two costs are not redundant. One is the cost of checking the constraint package against its dual witness. The other is the cost of committing to the result of that check by activating the cert channel.

That two-step shape is a protocol the program author chooses to follow, not a dependency the two opcodes enforce on each other. `MORPH_ASSERT` checks only that `mid` names an existing morphism; it does not read, and has no way to read, whether a `LASSERT` executed earlier in the same trace, on the same formula the property string names. Nothing stops a program from calling `MORPH_ASSERT mid=mproperty="validated_constraint" cert=""` directly, with no `LASSERT` anywhere in its history, and getting the identical channel-active state and the identical checksum for the identical 34μ -shaped cost if it pads with an equivalent no-op charge. `csr_cert_addr nonzero` means only that some `MORPH_ASSERT` fired on an existing

morphism with a non-NUL-byte property string; it does not mean a specific formula was checked, and the property string is a label, not a proof obligation the opcode verifies against anything. A downstream predicate that reads `csr_cert_addr` and leans on “the validated constraint the LASSERT step checked” is leaning on the trace having actually been built in the two-step shape above. That’s a fact about how the program was written. `MORPH_ASSERT`’s own semantics don’t establish it. Skipping LASSERT does not break a chain the kernel checks; it breaks a discipline the surrounding program was supposed to follow.

Tier 2: Certified binary-game witness insight (always costs ≥ 1). The sharpest case, and it is worth slowing down on why it is sharp. Here the trace is certified and its CHSH statistics show S pushed past 2, the classical ceiling (Section 18 builds the game and the ceiling from nothing). The certification on its own is already a Tier 1 event, so it already costs at least one. But look at what is being certified. $S > 2$ on a finite tally rules out any single fixed deterministic strategy replayed identically at every recorded trial (Section 18’s finite-tally remark); a fixed- λ shared-randomness-then-local-choice story of that narrow, unvarying kind cannot produce it. Reading the violation as ruling out local hidden variable models generally, the idealized-correlator claim Section 18 actually proves, needs enough trials that the tally and the population correlation have converged; a short run from a genuinely randomized local model can land past 2 by chance, as the same section’s coin-flip example shows. So this tier certifies a fact that constrains the space of classical explanations, more sharply as the trial count grows, and the price does not flinch for the size of the thing it is pricing. Certified evidence of a game violation costs at least one to come by, on a trace of any length you care to run.

6.2 Why insight cannot be free

The proof:

Suppose a trace starts in an uncertified state and ends in a certified state. Somewhere in that trace, a specific opcode fired that flipped one of the certification registers from off to on. The two concrete cert-channel instructions writing `vm_certified` or `csr_cert_addr` are: `CERTIFY` (which sets `vm_certified = true`) and `MORPH_ASSERT` when the morphism exists (which sets `csr_cert_addr` to the ASCII checksum of the property string: nonzero for any property string with a non-NUL byte). These are the only instructions that change those fields. `CHSH_LASSERT` sits in the certification cost-discipline class ($S(\delta) \geq 1$) but does not write to `vm_certified` or `csr_cert_addr`. Its success is observable via PC advance and `vm_err` staying false (the bridge theorem operates on that signature). `MORPH_ASSERT` with an empty property string or a missing morphism does not activate the cert channel, though it still costs $S(\delta) \geq 1$. This is proven by case analysis over all 51 opcodes. Every other instruction leaves those fields unchanged. So: the cert flip happened at a `CERTIFY` or `MORPH_ASSERT` step. Look at that exact step. That step, by the semantics of the 51-opcode VM, has cost $S(\delta) \geq 1$, because both `CERTIFY` and `MORPH_ASSERT` are in

the mandatory-floor class. So at that step: $\mu_{\text{after}} = \mu_{\text{before}} + S(\delta) \geq \mu_{\text{before}} + 1$. And because μ never decreases (each instruction adds a non-negative cost), $\mu_{\text{before}} \geq \mu_{\text{initial}}$. Therefore $\mu_{\text{final}} \geq \mu_{\text{after}} \geq \mu_{\text{before}} + 1 \geq \mu_{\text{initial}} + 1$.

That comes out of the step and cost rules, by case analysis and induction over the trace. In this model a certification transition can't be free. Whether the certified thing is true is a separate question. A paid flag proves nothing on its own, and a checker can be sound whatever the charging convention happens to be.

Cost alone isn't depth, either. A machine can spend real resources checking a claim that turns out to be shallow. `LASSERT` doesn't count as structural certification just because a formula parses and one satisfying assignment exists. It also needs a non-tautology witness: one satisfying assignment and one falsifying assignment. Under checker soundness, that proves the encoded formula can be satisfied and isn't a tautology. It doesn't, by itself, make the falsifying valuation a state in the current prior feasible set, and it doesn't make the formula a sound factorization or decomposition of the problem being run. Those take their own representation and semantic proofs. The receipt proves what was spent. The witness proves the stated formula property and nothing more.

7 The μ ledger

μ (mu) is the global structural cost ledger of the Thiele substrate. One counter, carried by every state. Not the whole substrate. The part that keeps receipts.

7.1 What mu is

μ is a single natural number riding along on every substrate state (a natural number being a non-negative integer, $0, 1, 2, \dots$; the notation cheat sheet is in Section A). It starts at 0 and it never goes down, and that second half is the whole character of the thing. A counter that can fall is an accountant you can lean on to revise the books. A counter that only ever climbs is a fact about history: it says how much has been spent, and it cannot be talked into saying less. μ is the second kind.

In the 51-opcode realisation, every instruction carries a cost parameter (δ) in its encoding. The VM adds that cost to μ when the instruction executes. Here is how δ is used in practice:

For **pure compute programs**, meaning programs that do arithmetic, load and store values, and jump, every instruction carries $\delta = 0$. The concrete zero-cost comparison case, `blind_program`, has this property: every instruction sits at $\delta = 0$. The proof that its total μ -cost is zero is a direct calculation. For these programs, μ stays at 0 unless a positive-cost instruction fires.

For **certification and revelation instructions** (`CERTIFY`, `LASSERT`, `MORPH_ASSERT`, `EMIT`, `REVEAL`, `LJOIN`, `READ_PORT`, and the five-opcode `CHSH_LASSERT` check family): the cost is at least 1, regardless of δ . Even $\delta = 0$ still costs 1. The positive floor is

part of this instruction schedule.

CERTIFY sets a persistent flag and charges $S(\delta) = \delta + 1$. The first false-to-true transition is the event A2 prices. Running it again still costs under the VM schedule. A flag that stays set doesn't prove anything was physically erased, either. An implementation can keep the old state somewhere else.

One is the smallest positive whole number, so a mandatory floor of one on top of δ gives $\delta + 1$. A2 allows more than that. The substitution theorems pin down exact unit pricing relative to how many flips happen. They don't pin down the whole VM schedule, and they don't pick a physical unit.

More things to watch doesn't mean a bigger bill, either. `joint_floor_is_least` proves that the smallest charge satisfying a finite family of unit event floors is one whenever any event fires, and zero otherwise. One unit can cover several floors at once. Two independent yes/no coordinates are a concrete example. If you want their costs to add, you need an accounting rule that actually makes them add. Being independent doesn't do it.

Throughout this document, $S(n)$ means $n + 1$. The S stands for "successor" in the basic math sense: the successor of n is the next natural number after n . So $S(0) = 1$, $S(1) = 2$, and so on. $S(\delta) = \delta + 1$ is always at least 1. `EMIT`, `REVEAL`, and `READ_PORT` go further: their cost also grows with the information content they carry. An `EMIT` of one ASCII character (which is 8 bits, see Section A) with $\delta = 0$ costs $8 + 1 = 9$. Ten characters costs $80 + 1 = 81$. Emitting more costs more. The price tracks the bits, every time, and there's no rate to negotiate.

For **everything outside that positive-cost class**, including `ADD`, `LOAD`, `STORE`, `JUMP`, `PNEW`, `PSPLIT`, `PMERGE`, `MORPH`, `COMPOSE`, and the rest, the VM charges exactly δ , which can be 0. Here is why these are allowed to be free.

The schedule lets module operations run at $\delta = 0$. That's a design choice. It isn't derived from thermodynamics. A zero-cost `PNEW` can change the graph, so "every observable structural change costs something" would be false for this VM.

Building graph structure and asserting it are charged differently here. Reading anything physical into that would take a model of which states carry information, where operations begin and end, what history is kept, and what dissipates. Opcode names and graph changes don't tell you any of that.

μ grows whenever `instruction_cost` is positive. The certification/revelation class is forced to be positive even when its encoded δ is 0. Other instructions can remain free by carrying $\delta = 0$.

Here is the complete cost schedule:

Instruction class	Cost = <code>instruction_cost</code>
Zero-floor operations (ADD, δ (can be 0) LOAD, JUMP, PNEW, MORPH, ...)	
Certification floor (CERTIFY, $S(\delta) = \delta + 1$ MORPH_ASSERT, LJOIN, the CHSH_LASSERT family)	
EMIT (emit payload as receipt- visible trace event)	$ \text{payload} _{\text{bits}} + S(\delta)$
REVEAL (disclose bits from a module)	$\text{bits} + S(\delta)$
READ_PORT (read bits from in- put port)	$\text{bits} + S(\delta)$
LASSERT (validate formula and witnesses)	$\text{flen} \times 8 + S(\delta)$

The $|\text{payload}|_{\text{bits}}$ notation means “the number of bits in the payload.” A one-byte ASCII character contains 8 bits. A ten-byte string contains 80 bits.

For LASSERT: flen is the number of formula units in the encoded formula. Each unit is one byte (8 bits) in the binary encoding used by the VM. This is a design choice in the binary format. The formula header declares how many bytes follow. The VM charges 8 bits per byte. The multiplication by 8 simply converts byte-count to bit-count, matching the information content you are asking the trace to validate.

The proof that μ is always exactly $\mu_{\text{before}} + \text{instruction_cost}$ for each step:

| Plain reading. Look at each of the 51 opcodes one at a time. Every apply rule writes μ as the old μ plus the cost. The cost term is exactly what `instruction_cost` returns. That includes the error paths. Failing an instruction costs the same as succeeding at it, because the cost is on the attempt. The outcome doesn’t change it. No opcode escapes this. Every one of the 51 lands in the same equation.

Theorem 7.1 (μ -Conservation). *For any VM state s and VM instruction i : $(\text{vm_apply } s \ i).\mu = s.\mu + \text{instruction_cost } i$.*

Proof. Case analysis over the 51 opcode constructors of `vm_instruction`. For each constructor i , the apply rule has the form

$$(\text{vm_apply } s \ i).\mu = s.\mu + c_i,$$

where c_i is the cost term written in the apply rule. Compare to `instruction_cost`: for zero-floor opcodes $c_i = \delta$; for certification-floor opcodes (CERTIFY, MORPH_ASSERT, LJOIN, and the CHSH_LASSERT family, its four Q_{1+AB} variants in-

cluded) $c_i = S(\delta)$; for `EMIT` $c_i = |\text{payload}|_{\text{bits}} + S(\delta)$; for `REVEAL` and `READ_PORT` $c_i = \text{bits} + S(\delta)$; for `LASSERT` $c_i = \text{flen} \times 8 + S(\delta)$. In every case c_i matches the corresponding case of `instruction_cost` by definition. The error sub-cases (e.g., `MORPH_ASSERT` on a missing morphism, `CHSH_TRIAL` with bad bits, `LASSERT` length mismatch) follow the same cost rule: the trap or err-latch only changes the PC and the error flag; the μ update is the same. So $(\text{vm_apply } s \ i).\mu = s.\mu + c_i = s.\mu + \text{instruction_cost}(i)$ for every i . $\square$ $\square$

This is not a monotonicity claim. It is an equality. μ goes up by exactly the cost of the instruction, always, for every instruction, in every case including error cases. There is no backdoor.

Don't read too much into that equality. It means the ledger is exact about what was spent. It doesn't mean every unit spent bought something deep. On this substrate, real semantic gain is a higher bar than spending. It needs a narrowing witness on top of the ledger entry.

7.2 Why μ is the unique ledger for its schedule

Here is the stronger claim, which took a while before the proof compiled. μ is not just a valid cost measure for this 51-opcode realisation. Once the schedule is fixed, it is the unique one. Any other cost function that starts at zero and tracks the per-instruction cost schedule exactly must equal μ on every reachable VM state. Not "isomorphic" in some abstract sense. Literally equal, value by value, on every reachable state.

| Plain reading. M agrees with μ at the start (both zero). On every step, M and μ move by the same amount (M by hypothesis (b), μ by μ -Conservation). Induction closes it. No other accounting can match the cost schedule at every step, start at zero, and still disagree with μ on a reachable VM state. Given the schedule, the count is forced.

Theorem 7.2 (μ -Initiality). *Let M be any function from VM states to natural numbers satisfying: (a) M assigns zero to the initial VM state, $M(\text{init_state}) = 0$; and (b) M tracks the cost schedule exactly: for every VM state s and every VM instruction i , $M(\text{vm_apply } s \ i) = M(s) + \text{instruction_cost}(i)$. Then M equals μ on every reachable VM state. (Monotonicity, the property that $M(s') \geq M(s)$ when s' is reachable from s , follows from (b) because `instruction_cost` is a non-negative natural number. You do not need to state monotonicity separately.)*

Proof. Induction on the reachability witness for s . A reachable VM state is either `init_state` itself or `vm_apply s' i` for some reachable s' and some VM instruction i .

Base case ($s = \text{init_state}$). By hypothesis (a), $M(\text{init_state}) = 0$. By definition of the initial state in the kernel, $\text{init_state}.\mu = 0$. So $M(\text{init_state}) = \text{init_state}.\mu$.

Step case ($s = \text{vm_apply } s' \text{ } i$ for reachable s'). By hypothesis (b),

$$M(s) = M(s') + \text{instruction_cost}(i).$$

By the induction hypothesis on s' , $M(s') = s'.\mu$. By μ -Conservation,

$$s.\mu = (\text{vm_apply } s' \text{ } i).\mu = s'.\mu + \text{instruction_cost}(i).$$

Substituting: $M(s) = s'.\mu + \text{instruction_cost}(i) = s.\mu$. □ □

In plain language: given the cost schedule, there's no other way to keep the books. Charge $S(\delta)$ for the certification-floor opcodes, $|\text{payload}|_{\text{bits}} + S(\delta)$ for `EMIT`, and δ for everything else, and μ is the only function on VM states that starts at zero and follows the schedule. The schedule itself was a design choice: the decision that `CERTIFY` costs $S(\delta)$ and `ADD` costs δ . It isn't arbitrary, though, and that's the part worth slowing down on. Two more results sit under it. The first is about pricing. A charge meets the certification floor exactly when it covers every flip, and it meets the floor without ever overcharging exactly when it charges on the flips and nowhere else, one unit each (`a2_equal_trust_substitution_payoff`). Hold trust fixed and the only exact rule left is A2's own. The second is about what can't be dropped. A window that forgets μ can't answer questions about μ , and a window that forgets the certification flag can't answer questions about certification. The full window answers both (`P_full_complete_neither_mu_nor_cert_droppable`). Neither one is redundant. A law that lives in the step isn't a check you run afterward and hope you got right. Uniqueness given the schedule is the bookkeeping. Pricing and irredundancy are why the bookkeeping has this shape. No competing count can match the schedule, start at zero, and land on a different total, and there's nowhere left for that freedom to hide.

7.3 The LASSERT cost formula

The most important cost formula in the VM is for `LASSERT`, the opcode that validates a logical formula and its witnesses:

$$\Delta\mu_{\text{LASSERT}} = \text{flen} \times 8 + S(\text{mu_delta})$$

flen is the formula-unit count the instruction declares, and the declaration drives the bill: eight concrete bits per declared unit. The per-step mandatory charge is always at least 1, no matter how cheap the unit-count feels. What keeps the declaration honest is the header check below: the VM reads the count the formula's own in-memory header carries, and no validation succeeds unless the two agree.

What makes this the one to watch is where the bill gets keyed. The operand drives the charge, but no successful check ever underpays, because succeeding requires the declared count to equal the count sitting in the formula's own header. Eight bits for

every unit, plus the floor of at least 1 you pay for the act of checking at all, and the act of checking is not up for negotiation. So a two-unit formula lands at $16 + 1 = 17$, and there is no posture you can strike to talk it down: the formula's own header gives the true count, and a declaration that disagrees with it never validates anything. Declaring a smaller count does not shrink the formula; it only gets you caught. That is the whole reason you cannot smuggle a larger formula through a smaller bit count.

That minimum is relative to the stated formula-length charging rules, as `lassert_cost_from_component_floors` and `lassert_cost_is_its_formula` formalize them. It isn't a lower bound on every possible way of validating or representing a formula.

What this buys you: the trace pays for presenting and checking a constraint package, no more, no less. It does not, by itself, guarantee the constraint is meaningful. You could present a formula that is always true, a tautology, and the ledger would still charge you the full bill. The VM closes this separately via the dual-witness requirement: see the `LASSERT` opcode description in Section 12.

The obvious thing to try, and where it runs out. The programmer declares *flen* in the instruction encoding, so the first thing I wanted to know was: what stops them from writing $flen = 0$ for a huge formula and paying only $S(0) = 1$? It's the cheat I'd reach for first.

The VM catches it. When `LASSERT` runs, the VM reads the formula's actual encoded-unit count from the formula header in memory and compares it to whatever *flen* the instruction claimed. If they do not match, the VM traps (Section A: routes execution to a fixed error address): the program counter jumps to `LASSERT_TRAP_PC`, which is `0xF00` in hexadecimal, decimal 3840. The error flag is set. The check does not succeed. This trap-to-fixed-address behaviour belongs to `LASSERT` and to the `CHSH_LASSERT` check family that inherits its discipline; every other error in the VM simply sets `vm_err` to true and continues advancing the program counter by 1. The checks are special because a failed check of this kind is a serious integrity violation. It isn't the kind of error you just step past. The μ cost is still charged: that detail matters because failed checks are not free. The Coq lemmas `lassert_honest_cost` and `lassert_honest_mu_cost` prove that if `LASSERT` completes without trapping, the declared count equals the in-memory count, and the μ charge uses the real formula bits.

8 No Free Insight: the first cost law

Section 4 ended on one sentence: the gain has a location, the location has a price, and the price is never zero. This section is that sentence as a theorem, twice. Once for any system at all, and once for this machine.

8.1 The abstract versions

| Plain reading. The cert predicate is off at the start, on at the end. So somewhere a step flipped it. Whichever step did the flipping costs at least 1 by the single-step rule. Every other step costs at least 0 because costs are non-negative naturals. Sum them up: at least 1. Empty trace can't do the flipping at all, so it's vacuous. That's the whole argument. The induction just makes precise the idea that the flip has to happen *somewhere*, and wherever it happens it pays.

Theorem 8.1 (Universal No Free Certification). *Let $\mathcal{C}$ be any abstract computational system with:*

- *a set of states;*
- *a set of step labels;*
- *a step rule (given a state and a step label, the rule produces the next state);*
- *a cost function (each step label has a non-negative integer cost);*
- *a certification predicate (a yes/no test on states, with at least one “no” state and at least one “yes” state).*

Premise (A2, the single-step cost rule): for every state s and step label i , if $\text{cert}(s) = \text{false}$ and $\text{cert}(\text{step}(s, i)) = \text{true}$, then $\text{cost}(i) \geq 1$.

Conclusion: any trace of step labels $t = [i_1, i_2, \dots, i_n]$ that starts at state s_0 with $\text{cert}(s_0) = \text{false}$ and ends at state s_n with $\text{cert}(s_n) = \text{true}$ has total cost $\sum_{k=1}^n \text{cost}(i_k) \geq 1$.

Proof. Induction on the trace t .

Base case ($t = []$). Then $s_n = s_0$, so $\text{cert}(s_n) = \text{cert}(s_0) = \text{false}$. But the hypothesis says $\text{cert}(s_n) = \text{true}$. Contradiction; the empty-trace case is vacuous.

Step case ($t = i :: \text{rest}$). Let $s_1 = \text{step}(s_0, i)$ be the state after the first step label. Case-split on $\text{cert}(s_1)$.

- *Subcase A:* $\text{cert}(s_1) = \text{true}$. The first step label flipped the certification predicate from false to true in one step. By the A2 premise, $\text{cost}(i) \geq 1$. The total cost is $\text{cost}(i) + \sum_{j \in \text{rest}} \text{cost}(j) \geq 1 + 0 = 1$, since every individual cost is a non-negative natural.
- *Subcase B:* $\text{cert}(s_1) = \text{false}$. Then the rest of the trace must do the flipping: $\text{cert}(\text{run}(\text{rest}, s_1)) = \text{true}$ with $\text{cert}(s_1) = \text{false}$. Apply the induction hypothesis at (rest, s_1) to get $\sum_{j \in \text{rest}} \text{cost}(j) \geq 1$. The total trace cost is $\text{cost}(i) + \sum_{j \in \text{rest}} \text{cost}(j) \geq 0 + 1 = 1$.

Either subcase gives total trace cost ≥ 1 . □ □

This theorem does not know about registers, memory, partitions, morphisms, CHSH trials, or the 51-opcode ISA. It says: once a model has a yes/no certification test and the one-step certification transition is priced, trace-level non-free certification falls out by induction over the trace, no further premises required. If that one-step premise fails, the model permits free certification by definition. If the model has no certification predicate, it cannot talk about structural entitlement internally.

And don't read "that's the whole argument" as "the premise is the whole content." The single implication is one line from the pricing rule, yes. It is an accounting law, and I call it one. It isn't a law pulled from nowhere, though. The nearest older rule is Landauer's erasure accounting, and the two meet at an exact place. A trusted erasure-accounting law, the same trust posture, certifies at total cost zero when no erasure is reported (`commitment_cost_not_reducible_to_erasure_cost`). Its one step sends both states to certified. That is a merge, and it goes unreported. Report every merge, and on a finite machine with a permanent certificate the erasure law charges the certification flip at least one. Same bill as A2 (Section 9.1). So A2 is what Landauer's rule says about the step that certifies, wherever memory is finite, certificates last, and the machine could be in either of the states the step merges. Where Landauer's rule isn't in force, A2 still prices the flip. The VM is that case: its free JUMP merges program counters, and its CERTIFY, itself a merge, still pays (`vm_prices_certifying_merge_leaves_others_free`). And if the goal is to price exactly the certification flips, never less and never more, a charge on any other predicate misses that goal (`substitution_test_rejects_non_a2_exact_substitute`). That uniqueness is relative to the goal. Whether certification is the event that deserves the price is the open question of Section 26. A2 is a rule I chose. On finite hardware with certificates that last, it is also the rule Landauer's principle would have chosen for that step.

| Plain reading. This is the abstract theorem pointed at the 51-opcode machine. Its two concrete certification channels, `csr_cert_addr` and `vm_certified`, both cost something at the step that turns them on. And the predicate-strengthening reading falls out of the structure-addition theorem.

Theorem 8.2 (Thiele No Free Insight). *For the 51-opcode VM specifically, cert-channel activation is positive-cost, and certified predicate strengthening has to pass through a structure-addition event. The relevant certification/revelation steps all live in the positive-cost class.*

Proof. Two parts.

Cert-channel activation requires positive cost. The Both-channels theorem below establishes that any trace activating either certification channel (`csr_cert_addr` becoming nonzero or `vm_certified` becoming true) has $\Delta\mu \geq 1$. The actual channel-writing cases are covered by the positive-cost cert/revelation class: CERTIFY, MORPH_ASSERT, EMIT, REVEAL, LJOIN, LASSERT, READ_PORT, and the

CHSH_LASSERT family. The class is deliberately broader than literal field writers; every member has $\text{instruction_cost} \geq S(\delta) \geq 1$, so the lower bound holds.

Certified predicate strengthening requires a structure-addition event. By hypothesis the trace ends certified for a stronger predicate via the supra-cert channel (`csr_cert_addr` nonzero). The Strengthening-requires-structure-addition theorem (Section 5) gives the structure-addition event directly. $\square$ $\square$

This is the VM-level version. The kernel proves the two concrete cert channels cannot be activated for free, by case analysis over all 51 opcodes, then adds the predicate-strengthening layer: once a stronger accepted predicate is tied to `has_supra_cert`, the trace must contain a structure-addition event somewhere in it.

8.2 The law is the potential method

Cost reasoning has old tools, and I checked the law against two of them.

The first is cost semantics. Danner and Licata use a monadic translation to carry cost with a program's result; graded monads (Katsumata; Orchard, Liepelt, and Eades) track effects through grades. Running a trace in the writer monad over the natural numbers returns the final state and the trace's total cost (`run_writer_is_run_and_cost`). This identifies the trace ledger with a writer cost. It does not embed those authors' full semantics or type systems.

The second is a lower-bound potential argument. Give every state a potential. If every step's cost plus the change in potential is never negative, then any run costs at least the drop in potential. Now pick the potential that is one on an uncertified state and zero on a certified one. Standard upper-bound AARA uses the opposite inequality: initial potential covers cost plus final potential. The result here is not an equivalence to its affine type system.

Theorem 8.3 (*A2 is the potential method, `a2_iff_nonnegative_amortized_cost`). For any step function, cost, and yes/no reading, A2 holds exactly when every step has non-negative amortized cost under that potential.*

No Free Insight is then the potential method's own bound: a run from uncertified to certified drops the potential by one, so it costs at least one (`nfi_by_potential`). Every `CertificationSystem` is an instance (`certification_system_is_potential_method`).

So the law isn't a new kind of cost reasoning, and I'm glad to say so. It is a lower-bound potential argument, pointed at the certification reading. What the rest of this book adds is about that reading, not the method: Section 9.4 shows that on the VM's two named windows no price computed from the observed transition prices certification exactly, and a potential read off the window gives a price of that kind. This observation

does not establish a separation from the cited frameworks.

8.3 The specific versions

For the 51-opcode VM concretely, there are several specific versions. The two certification channels I mentioned in Section 6 are the certificate-address register (`csr_cert_addr`) and the top-level flag (`vm_certified`). Both are defined precisely in Section 11, register by register.

| Plain reading. If you see `csr_cert_addr` move from zero to nonzero, the proof catches that step inside a cost-positive guardrail class. That class is broader than the direct CSR writers, but broad is fine here: every member costs at least 1. The field cannot flip for free.

Theorem 8.4 (No Free Certification, single step). *If $s.vm_csrs.csr_cert_addr = 0$ and $(vm_applies\ i).vm_csrs.csr_cert_addr \neq 0$, then $instruction_cost(i) \geq 1$.*

Proof. Case analysis over the 51 opcode constructors. The Coq proof uses `cert_addr_setterb`: if `csr_cert_addr` changes from 0 to nonzero, the instruction must fall inside that cost-positive class. The class includes `instr_morph_assert`, `instr_emit`, `instr_reveal`, `instr_ljoin`, and `instr_lassert`. It is a deliberately broad guardrail class, not a list of literal CSR writers. In the current `vm_apply` semantics, the direct CSR-writing success case is `instr_morph_assert`. For the theorem, the required fact is cost: every member of `cert_addr_setterb` has $instruction_cost \geq 1$, with `instr_emit` and `instr_reveal` higher because they also charge payload bits. □ □

| Plain reading. Single-step lifts to trace-level. The universal theorem does the work.

Theorem 8.5 (No Free Certification, trace level). *If a trace starts at a state with $csr_cert_addr = 0$ and ends at a state with $csr_cert_addr \neq 0$, then the trace's total μ cost is ≥ 1 .*

Proof. Instantiate the universal theorem above with $\mathcal{C}$ = the 51-opcode VM; $cert(s) := (s.vm_csrs.csr_cert_addr \neq 0)$; $step = vm_apply$; $cost = instruction_cost$. The A2 premise is exactly the single-step theorem above. □ □

| Plain reading. Same shape as the other channel. Only one opcode (`CERTIFY`) can flip the top-level flag. It charges ≥ 1 . The universal theorem lifts that to the trace.

Theorem 8.6 (No Free Certification via `CERTIFY`). *If a trace starts at a state with $vm_certified = false$ and ends at a state with $vm_certified = true$, then the trace's total μ cost is ≥ 1 .*

Proof. Case analysis over the 51 opcodes shows that `vm_certified` flips from false to true only when `i` is an `instr_certify` constructor; every other opcode leaves it unchanged. The `CERTIFY` apply rule charges $S(\delta) \geq 1$. So the single-step instance of A2 holds for this channel. Instantiate the universal theorem with `cert(s) := s.vm_certified`, otherwise as above. $\square$ $\square$

| Plain reading. Two channels, one master claim. Whichever channel got activated, the corresponding single-step rule fired and the universal theorem lifted it to the trace. There is no third way.

Theorem 8.7 (Both channels). *For ANY transition of either certification channel from absent to present, whether the certificate-address register `csr_cert_addr` or the top-level flag `vm_certified`, μ grows by at least 1 over the trace.*

Proof. Suppose the trace either flips `csr_cert_addr` from 0 to nonzero, or flips `vm_certified` from false to true (or both). In the first case the trace-level theorem (`csr_cert_addr`) applies. In the second case the trace-level theorem (`vm_certified`) applies. The disjunction is exhaustive over “cert-channel activation,” so μ grows by ≥ 1 in either case. $\square$ $\square$

Theorem 8.7 is the one I actually reach for downstream. When something later in the document needs “and certification was not free,” this is the line it cites, because it has already done the bookkeeping of asking which way the certification came in and answered: doesn’t matter, both are priced. The certificate-address register `csr_cert_addr` and the top-level flag `vm_certified` are not two separate promises I have to keep track of; they collapse into one claim that holds no matter which door a program walks through to call itself certified.

8.4 What this means in practice

Run any program on the kernel. Watch the μ counter. If the program ends with `vm_certified = true` or with `csr_cert_addr` nonzero, then somewhere in the trace, μ went up by at least 1. You can audit the trace and find exactly which opcode paid the bill. There is no program that can reach a certified state from scratch without paying for it.

That’s for a real run from the initial state. A certified flag there means the ledger went up. The ledger can’t vouch for a state somebody else hands you, and it doesn’t prove any proposition on its own.

9 Why A2 isn’t arbitrary: permanent certificates on finite hardware

A2 is a rule I chose. This section is about whether I had to. On a machine with finite memory whose certificate lasts, the answer is close to yes. The step that certifies has to

merge states, and merging is what Landauer's principle prices whenever the machine could be in either state. The physics stays a named premise. The counting doesn't.

9.1 A permanent certificate has to merge states

Here is the shape. A machine has finitely many states. Some are certified, and no step ever un-certifies one. Now a step carries an uncertified state into the certified region. The certified states were already there, and they can't leave, so every certified landing place is spoken for by one of them. The newcomer has to land where some certified state also lands. Two states become one, and the step forgets which one it came from.

Forgetting like that is what Landauer's principle charges for, with one condition that matters. The principle prices the entropy a step removes from what the machine might actually be holding. If the machine could be in either of the two merged states, the merge removes entropy and costs heat. If it is already known to be in one of them, nothing is lost and nothing is owed. So the premise here prices every merge at one or more, which is Landauer's bill in the worst case over what the machine might hold. Under that premise, on a finite machine whose certificate can't be revoked, A2 stops being a rule I picked. It is what Landauer's principle says about the one step that certifies. Section 9.5 states the version that depends on the distribution exactly.

| Plain reading. The certified states form a closed room. A step that forgets nothing shuffles the people in the room among the room's own seats, and a closed room with a fixed number of seats has no seat left for someone from outside. So a step that lets someone in has to seat two people in one chair. That is the whole proof. The pigeonhole does the work.

Theorem 9.1 (A permanent flip merges states, `permanent_flip_is_not_injective`). *Take any state type with a duplicate-free list of every state, any instruction type, any step function, and any certification reading. Suppose certification is permanent: if $\text{cert}(s)$ then $\text{cert}(\text{step}(s, i))$ for every s and i . If $\text{cert}(s)$ is false and $\text{cert}(\text{step}(s, i))$ is true, then the step for i is not injective: two different states land on the same state.*

Proof. Suppose the step for i were injective. Let C be the list of certified states. By permanence the step sends C into C . Being injective, it sends C to a duplicate-free list of the same length inside C , so it hits every member of C . The certified state $\text{step}(s, i)$ is then $\text{step}(c, i)$ for some certified c . Injectivity gives $s = c$, so s is certified. It isn't. □ □

Theorem 9.2 (A2 from the merging price, `a2_from_merging_price_and_permanence`). *Under the same finiteness and permanence, suppose every instruction whose step is not injective costs at least 1. Then every step that flips certification from false to true costs at least 1. That is A2. `certification_system_from_merging_price` packages the machine as a `CertificationSystem` with its A2 field discharged by this theorem, and p*

`permanent_certification_trace_floor` carries it to every trace through `universal_nfi_any_substrate`.

Proof. The previous theorem says the flipping step is not injective. The premise prices it at 1 or more. □ □

There is a counting version too. `permanent_flips_collapse_at_least` takes k distinct uncertified states that one instruction switches on. The step sends the certified states and those k newcomers into the certified states, so it has at least k fewer distinct images than inputs. Switching on more states collapses more states.

Each premise is needed, and the file proves a counterexample for each. Keep the whole history, and every step is injective while the certificate still switches on (`unbounded_history_escapes`). That is Bennett’s escape, the same move as the history lift in Section 19, and it needs memory that grows without bound. Finite memory closes it. Let the certificate be revoked, and flipping one bit certifies for free, then un-certifies (`revocable_certificate_escapes`). Leave merging steps unpriced, and a reset certifies at cost zero (`free_merge_escapes`).

This also settles the rival from Section 8. The trusted erasure system there certifies at cost zero with no erasure reported. Its one step sends both states to certified. That is a merge, and its erasure flag doesn’t report it (`commit_without_erasure_system_is_not_honest`). Call erasure accounting honest when the flag reports every instruction that merges states. Then on a finite machine with a permanent certificate, honest erasure accounting charges the certification flip at least 1 (`honest_erasure_accounting_implies_a2`). The two laws send the same bill.

What the theorem covers, and what it doesn’t. The pricing premise is Landauer’s principle stated on the logic, a named premise. It isn’t a theorem about heat. The 51-opcode VM is not an instance, and that is by design. Its ledger is an unbounded natural, so its state space is infinite. And its schedule prices the merge that certifies while leaving other merges free: `JUMP` merges program counters at zero declared cost. Section 9.6 proves both halves of that choice, and builds a finite piece of the VM that is an instance. The theorem is about a permanent certificate on finite hardware. The core is a pigeonhole argument, and the physics it leans on is Landauer’s and Bennett’s (“Notes on Landauer’s principle, reversible computation, and Maxwell’s Demon”, 2003). None of those pieces is new. `CERTIFY` sets a flag to true whatever it held, and that is Landauer’s own example of an irreversible step. That finite memory eventually has to erase is Bennett’s point. What the theorem adds is the link, stated for every implementation at once: however a certificate that lasts is encoded, on finite memory the step that switches it on merges states. No clever encoding of a permanent certificate dodges that. Finiteness and permanence together close Bennett’s history escape for the one step that certifies.

9.2 How much a permanent certificate costs

A2 says the flip costs at least one. Landauer’s principle says more: squeezing states together costs one unit per halving. Put that in as the premise and the bill gets a size.

The shape is the same closed room. Say m states are already certified, and one step lets k more in. All $m + k$ of them land in the room’s m seats. Squeezing $m + k$ states into m places is a squeeze by a factor of $(m + k)/m$, and each unit of cost pays for one halving. So the cost is at least $\log_2((m + k)/m)$.

Theorem 9.3 (The cost of switching k states on, `permanent_flips_compression_bound`, `permanent_flips_log_bound`). *Take a finite state space with a permanent certificate, and suppose every instruction satisfies the counting premise `compression_priced`: an instruction with cost c maps any duplicate-free list of n states onto at least $n/2^c$ distinct states. If m states are certified and one instruction switches k distinct uncertified states on, then*

$$m + k \leq 2^{\text{cost}} \cdot m, \quad \text{so} \quad \log_2^\uparrow(m + k) \leq \text{cost} + \log_2^\uparrow m.$$

With $k = 1$ the cost is at least 1, which is A2 again (`a2_from_compression_price_and_permanence`).

Proof. The k newcomers and the m certified states are $m + k$ distinct states. Permanence keeps the certified ones certified, and the newcomers land certified by hypothesis, so all $m + k$ land among the m certified states. The counting premise says $m + k \leq 2^{\text{cost}}$ times the number of distinct landing places, and that number is at most m . For the logarithm form, $\log_2^\uparrow$ never decreases, and $\log_2^\uparrow(m \cdot 2^c) = c + \log_2^\uparrow m$. $\square$ $\square$

The bound is tight. Take four sheets, three blank and one stamped, and one instruction that stamps whatever it is handed. Switching the three blanks on squeezes four states into one, and the bound says the instruction costs at least 2 (`quad_stamp_needs_two`). Cost 2 meets the counting premise (`quad_stamp_cost_two_is_priced`). That is $\log_2 4$, paid exactly.

This is the number I wanted. A2’s “at least one” is the floor for a single flip. The real price depends on how many states get squeezed into how few seats, and it grows with the logarithm of the squeeze, the way Landauer’s does. The units are halvings, not joules. Turning halvings into heat is $k_B T \ln 2$ per halving, and that conversion is physics.

9.3 Which flips a finite machine must pay for

Step 5 asks which events every account has to price. The permanent-certificate theorem suggested an answer: exactly the ones that write a record nothing takes back. I tried to prove that. It’s false, and the true statement is sharper.

Call an instruction *forced to be priced* when every cost assignment that prices merges,

as Landauer’s principle does, charges it at least one. Three facts hold on any finite machine, for any yes/no reading.

Theorem 9.4 (Forced price is merging, `forced_priced_iff_merges`). *An instruction is forced to be priced exactly when its step merges two states.*

Proof. If it merges, every merge-pricing assignment charges it. If it doesn’t, charge it zero and charge everything else one; that assignment prices every merge and leaves this instruction free. Building that assignment needs decidable equality on instructions.

□

□

Theorem 9.5 (Merge or revoke, `flip_merges_or_revokes`). *If an instruction switches the reading on somewhere, then either it merges two states, or it switches the reading off somewhere else.*

Proof. Search the certified states for one the instruction switches off. The search is finite, so it ends. If it finds one, that’s the revocation. If it finds none, the instruction keeps the reading on everywhere it was on, and the permanent-flip argument, applied to this one instruction (`permanent_at_flip_is_not_injective`), says it merges. □ □

Corollary 9.6 (Permanent records are forced, `permanent_record_write_is_forced_priced`). *An instruction that never switches the reading off, and switches it on somewhere, is forced to be priced.*

The converse is false. Take three states, two unmarked and one marked, and one instruction that marks the first, unmarks the marked one, and sends the third to the marked one. It merges the first and third, so it is forced to be priced. It also revokes a mark, so it writes no permanent record (`forced_price_without_permanent_record`). So “the forced events are exactly the permanent records” doesn’t hold.

Here is what does. The events a finite machine must price are the merges. Every step that writes a permanent record is one of them. And a flip escapes the price only one way: the same instruction takes the reading back somewhere else (`injective_flip_revokes`). A record that stays has to be paid for. A record that’s free is one the instruction can also erase.

That answers step 5 in a narrower form than I conjectured, and it proves something about step 4. Step 4 says the axis belongs in the step. For an account of a finite machine that prices merges, it isn’t optional: that account already charges every step that writes a permanent record, at least a logarithm of the squeeze. What no theorem here can settle is whether every account of computation has to price merges. That’s Landauer’s principle, and it’s physics.

9.4 The exact price can't be read off the shadow

Step 4 says the account should carry what was established as part of how it steps. Here is a way to test that. Say an account wants to price certification exactly: at least one for every step that turns the reading on, and nothing for any step that doesn't. `exact_commitment_pricing_characterization` says a price meets the floor and never overcharges exactly when it is that one. The question here is who can compute it.

A shadow account sees the window before a step and the window after it. Its price for the step is a function of those two pictures. Now put two steps side by side. They start at the same picture and end at the same picture. One of them certified. The other didn't. The shadow account has one number for that pair of pictures. If the number is at least one, it bills the step that didn't certify. If it is zero, it misses the step that did.

| Plain reading. A clerk who sees only the before and after photos has to give two jobs with the same photos the same bill. If one job earned a certificate and the other didn't, the clerk gets one of them wrong.

Theorem 9.7 (No exact shadow price, `shadow_cannot_price_exactly`). *Take any states, any instructions, any step function, any yes/no reading, and any window. Suppose two steps show the same window before and the same window after, and one switches the reading on while the other doesn't. Then no price that is a function of the observed before and after both meets the floor and never overcharges. Any such price that meets the floor bills a step that did not certify (`shadow_floor_overcharges`).*

Proof. The two steps get the same price. The floor makes it at least one. The step that didn't certify pays it. □ □

Theorem 9.8 (A window that shows the reading prices exactly, `window_showing_reading_prices_exactly`). *If the reading is a function of the window, then the price "one when the window's reading goes from off to on, zero otherwise" meets the floor and never overcharges. Such a window has no collision (`window_showing_reading_has_no_collision`).*

The VM has the collision for both windows I name. From the clean start, `CERTIFY 0` and `JUMP 1 0` leave the same program counter, registers, and memory. One certifies. The other doesn't. So no account that prices from `bare_observable` prices VM certification exactly (`vm_bare_observable_collision`, `vm_bare_shadow_cannot_price_exactly`). Put the ledger in the window and use `JUMP 1 1` instead. Both steps then raise μ by one, and the four-field `forget` window has the collision too (`vm_forget_collision`, `vm_forget_shadow_cannot_price_exactly`).

The mathematics here is one line, and it isn't new. It is the fiber-constancy fact of

Section 3.7 (`decoding_requires_fiber_constancy`) with a price in place of a decoder: nothing computed from a window can tell apart two things the window shows the same. What the theorem adds is where it lands. The VM has such a pair, so the exact price of certification is one of the things the usual window cannot compute.

So the exact price lives in the step, or in a window that already shows certification. The read/write account can't carry it. That is the part of step 4 the proofs reach. It doesn't say every account has to price certification exactly. It says an account that does has to carry the reading.

9.5 The same bound in bits

The squeeze bound counts states. Landauer's principle is usually stated in entropy. So here is the same result in Shannon entropy.

The shape doesn't change. Put a probability on each state that is certified or about to flip. After the step, all of that probability sits on the m certified states. A distribution on m states carries at most $\log_2 m$ bits. So whatever the distribution had before, the step leaves it at most $\log_2 m$.

Theorem 9.9 (Entropy ceiling after a permanent step, `permanent_step_entropy_ceiling`, `permanent_step_entropy_drop`). *Take a finite state space with a permanent certificate and an instruction that switches $k \geq 1$ distinct uncertified states on. Let p be any probability distribution on the states that lives on the m certified states and those k . Then the distribution after the step has Shannon entropy at most $\log_2 m$, so the step lowers the entropy by at least $H(p) - \log_2 m$.*

Proof. The step sends every state in play to a certified one: the certified ones by permanence, the k by hypothesis. So the pushed distribution lives on the m certified states. Entropy is at most the logarithm of the size of any set that holds the support (`entropy_le_log_support`). That last step uses $\ln y \leq y - 1$ and nothing else. It is Shannon's bound, proved again here so the chain is checked end to end. $\square \quad \square$

Corollary 9.10 (Bits removed by a permanent flip, `permanent_flip_uniform_entropy_drop`). *For the uniform distribution on the $m + k$ states in play, the step removes at least $\log_2((m + k)/m)$ bits.*

That is the number the counting bound gives, now in bits. The entropy form gives a third route to A2. Call a whole-unit price entropy-priced when every instruction costs at least the bits it removes from any distribution. On a finite machine with a permanent certificate, an entropy-priced cost satisfies A2 (`a2_from_entropy_price_and_permanence`), and the trace floor follows through `universal_nfi_any_substrate` (`entropy_priced_trace_floor`). The other side holds too. A step that forgets nothing removes no bits from any distribution (`step_entropy_invari`

ant_if_injective). That is Bennett’s side, and an entropy price can leave such a step free.

The heat is where the physics comes in. Landauer’s principle, written as a named premise `landauer_heat`, says a step that lowers the entropy of the state by ΔH bits dissipates at least $k_B T \ln 2 \cdot \Delta H$. Under that premise, a permanent flip of k states beside m certified ones, from the uniform distribution on those $m + k$, dissipates at least

$$k_B T \ln \frac{m+k}{m}$$

(`permanent_flip_heat_floor`), and that is strictly positive when at least one state flips and the temperature is positive (`permanent_flip_heat_positive`). The premise is Landauer’s. The theorem is the step from his premise to the certificate. It says nothing about a device until somebody measures one.

The uniform distribution is one choice, and the result doesn’t depend on it. What Landauer’s principle charges for is the entropy a step removes from the distribution the machine actually has. That is the generalized form of the principle (Maroney, “Generalizing Landauer’s principle”, 2009; Sagawa, “Thermodynamic and logical reversibilities revisited”, 2014). So the file also says exactly when a step removes any.

Theorem 9.11 (When a step removes entropy, `entropy_drop_as_point_sum`, `entropy_drop_nonneg`, `entropy_drop_pos_of_support_merge`). *For any deterministic step and any distribution p , the entropy removed is a sum of one term per state, $p(x)(\log_2 q(\text{step}(x)) - \log_2 p(x))$ over the states with $p(x) > 0$, where q is the distribution after the step. Every term is nonnegative. The sum is positive whenever two different states of positive probability land on the same state.*

Corollary 9.12 (Any spread-out distribution, `permanent_flip_full_support_entropy_drop_positive`, `permanent_flip_full_support_heat_positive`). *On a finite machine with a permanent certificate, take a step that switches a state on and any distribution that gives every certified state and the flipping state positive probability. The step removes entropy, and under `landauer_heat` with positive temperature it dissipates positive heat.*

The other direction is the honest boundary. If the machine is known to be in one state before the step, the step removes no entropy, whatever it merges, and Landauer’s principle forces no heat at all: the premise holds exactly when $\text{heat} \geq 0$ (`known_state_step_removes_no_entropy`, `known_state_flip_forces_no_heat`). So the heat a permanent certificate costs comes from uncertainty about which state the machine is in. A machine that already knows its state can certify for free, physically. The ledger still charges it, because A2 is a rule about the step, and the step doesn’t know what the observer knows.

9.6 A finite VM that pays for its merges

The 51-opcode VM isn't an instance of the permanent-certificate theorem, and I don't want it to be. Here is what it does instead. It prices the one merge that certifies, and it leaves every other merge free.

Theorem 9.13 (The VM prices the certifying merge, `vm_certify_merges`, `vm_certifying_step_is_priced_merge`). *CERTIFY merges states: a certified state and an uncertified state that agree on everything else land on the same result. Every VM step that switches the certification flag on is such a merge, and it costs at least one.*

Theorem 9.14 (And leaves other merges free, `vm_jump_is_free_merge`, `vm_prices_certifying_merge_leaves_others_free`). *JUMP 0 0 merges program counters and costs zero. So the full VM does not price every merge, and the merging-price premise fails for it.*

That second theorem isn't a gap in the VM. Pricing every merge would charge every overwrite: every `LOAD_IMM`, every `ADD`, every `STORE`. That is what Landauer's principle asks of a physical machine, and whether every account of computation has to answer to it is the physics question this book leaves open. The VM's schedule keeps classical computation free and puts the price on certification alone. A2 holds for it as a theorem either way. What the finite argument adds is that the price isn't arbitrary: on hardware that does price its merges, the same bill comes due. So I built a finite machine that prices all its merges and showed the VM runs it.

The finite machine has four program slots and a certification flag: eight states. It has three instructions. `FCertify` sets the flag and moves to the next slot. `FJump a` moves to slot *a*. `FNext` moves to the next slot and wraps around. Their prices are 1, 2, and 0. Each price is what its squeeze demands. `FCertify` merges each slot's flagged and unflagged states, a two-to-one squeeze. `FJump` merges four slots into one, a four-to-one squeeze, so it pays two halvings. `FNext` is a bijection and pays nothing.

Theorem 9.15 (The finite machine is an instance, `fin_finite`, `fin_permanent`, `fin_merging_priced`, `fin_compression_priced`). *The eight states are a duplicate-free list of every state. The flag is permanent. Every merging instruction costs at least one (`fcertify_merges`, `fjump_merges`, `fnext_injective`). And every instruction meets the counting premise: cost *c* squeezes any duplicate-free list at most 2^c -fold. So A2 holds for it as a theorem, twice over (`fin_a2_from_merging_price`, `fin_a2_from_compression_price`).*

The counting premise needed one general fact. If no image of a map has more than *K* preimages in a duplicate-free list, the list has at most *K* times as many members as images (`fiber_bound_compression`).

Theorem 9.16 (The VM runs it, `vm_runs_finite_machine`, `vm_pays_finite_price`, `vm_runs_finite_trace`). *Read a VM state through the window (program counter modulo four, certification flag). Then `CERTIFY 0`, `JUMP a 2`, and `CHECKPOINT "" 0` step exactly as `FCertify`, `FJump a`, and `FNext` do, and the VM ledger rises by exactly the finite machine's price. The same holds for whole runs.*

Corollary 9.17 (`vm_fragment_certification_paid`). *Any VM run built from those three instructions that starts uncertified and ends certified has raised μ by at least one. That payment comes from the finite machine's derived price. No pricing rule is supplied for the purpose.*

Two limits. The price of two on `JUMP` is the program's choice of `mu_delta`. The VM's schedule allows it and doesn't force it, because the schedule prices certification and leaves other merges to the program. And the window is a projection. The VM's full state keeps more than the eight finite states do, its ledger included. What the theorem shows is that the permanent-certificate argument isn't about an imaginary machine. A fragment of the VM I built runs it, and pays what Landauer's principle would ask of it.

9.7 Learning is free. Committing is not.

Everything so far prices one event: the step that switches a certificate on. The obvious next question is whether narrowing things down costs too. Every time a computation rules out a possibility, it knows more. Does knowing more cost?

It depends on whose possibilities they are.

Start with the machine's own. Say the machine could be in any state of some set, and it runs a fixed program. Afterwards it could be in fewer states, because some of them landed together. Every time two states land together, that is a merge, and merges are what the price charges. So the machine's own spread can't shrink for free.

An observer is different. The observer watches a window and crosses off the starting states that don't match what it saw. That list can shrink while the machine forgets nothing at all. Picture a hidden bit and a blank display. One step adds the bit into the display. No two states land together, so merge pricing does not force a positive charge. The file exhibits one admissible cost assigning that injective step zero; the lower-bound pricing rule would also permit overcharging it. The machine can still be in either of its two states. But the observer, reading the display, now knows which one.

| Plain reading. The machine can't squeeze its own possibilities for free. An observer can learn for free, because learning doesn't have to squeeze anything. The bill comes later, when the display has to be wiped for the next reading. Wiping is a merge.

Theorem 9.18 (The machine’s narrowing is priced, `run_narrowing_priced_log`). Take a finite deterministic machine whose costs meet the squeeze price `compression_priced`. For any duplicate-free list D of starting states and any program t , let D' be the states the machine can be in after running t from somewhere in D . Then

$$\log_2^\uparrow |D| \leq \text{cost}(t) + \log_2^\uparrow |D'|.$$

The run itself is the witness. No decision tree, payment proof, or representative map is supplied.

Proof. Induction on the program. One instruction with cost c squeezes a duplicate-free list by at most 2^c , and the states reachable after the rest of the program from the squeezed list are exactly the states reachable after the whole program from the original list. Multiply the squeezes. $\square$ $\square$

Theorem 9.19 (An observer’s narrowing can be free, `observer_narrowing_can_be_free`). There is a finite machine, with prices that meet the squeeze price, and a one-instruction program of cost zero, such that an observer who watches the display along the run goes from two candidate starting states to one. So the observer reading of the target,

$$\log_2^\uparrow |\Omega| - \log_2^\uparrow |\Omega'| \leq \Delta\mu,$$

with Ω' the starting states consistent with what the observer saw, fails for a merge-priced machine.

Proof. The states are pairs (hidden bit, display). `Measure` adds the bit into the display by exclusive-or and costs zero. It is a bijection on the four states, so it meets the squeeze price. Start with the display blank. Before the step both starting states show a blank display. After it, the display equals the bit. The observer’s list drops from two to one; the cost is zero. The machine’s own spread stays at two (`demon_machine_spread_kept`). $\square$ $\square$

Theorem 9.20 (Wiping the record is priced, `wipe_costs_at_least_one`). In the same machine, `Wipe` blanks the display. Every cost that meets the squeeze price charges it at least one.

The VM does the same thing. `XOR_ADD 2 1 0` adds register 1 into a blank register 2 at zero cost. An observer watching register 2 goes from two candidate states to one, and the ledger stays at zero (`vm_observer_narrowing_at_zero_cost`).

None of this is new physics. It is Bennett’s answer to Maxwell’s demon (“[The thermodynamics of computation, a review](#)”, 1982), stated on the logic and checked in Coq. Measuring can be done without merging. What costs is making room again.

It also says exactly which insight No Free Insight prices. Raw learning can be free,

the way Tier 0 of Section 6 already is. The insight that can't be free is certified insight: committing to what you learned, in a record the machine can't take back. The structural-entitlement bound of Section 5 still holds, and it holds because it is handed a trace that paid for its decision tree. This section shows that payment is a premise. The narrowing alone doesn't force it.

10 The substrate and its halting-shaped wall

The Thiele Machine is not the 51 opcodes. The 51 opcodes are scaffolding. They make the substrate concrete enough to verify in Coq, run as OCaml, and synthesize to silicon, but they are not the substrate itself. The substrate is the step relation together with the cost-ledger monotonicity rule: every atomic transition either preserves μ or charges it. PNEW, MORPH, CHSH_LASSERT, the whole instruction set. Those are the experimental apparatus that gives the substrate a concrete realization to probe. Strip the opcodes and the substrate is still there. Add different opcodes that respect the substrate's monotonicity discipline and the substrate is still the same substrate.

The Turing machine has the same shape. Tape, head, and transition table are scaffolding for one realization. Register machines, lambda calculus, counter machines, RAMs, and partial recursive functions all instantiate the same step-relation primitive with different scaffolding; the equivalences between them are theorems, and that this class captures effective computation at all is the Church-Turing thesis. They are all the Turing machine in the sense that matters. Their primitive steps and cost models can differ even when their computable functions agree.

The result that follows is a fact about the substrate, not about the 51 opcodes. The opcodes are one realization. Other substrates that supply the same typeclass fields inherit the same theorem the moment they are presented.

Relationship to Rice's theorem. The result that follows is Kleene's 1938 diagonal pattern instantiated at the substrate level. Rice's 1953 theorem applies the same diagonal to non-trivial extensional properties of recursive functions computed by Turing machines; the substrate-level version below applies the very same diagonal to non-trivial extensional predicates of substrate programs in any model that exposes an internal recursion theorem and an internal representability predicate. It is the same trick, two floors apart, which is the whole point. The formal content is the diagonal construction; what the substrate-level framing adds is the parameterization (any substrate satisfying the typeclass interface, not just the standard Turing model) and the choice of extensional predicate (`AdmitsShortcut`, branching on `csr_cert_addr` reachability). The result is therefore a substrate-level instance of the pattern Rice used; the result does not fall out as a corollary of Rice's theorem in its original setting, because Rice's premises are about partial recursive functions in the Turing-machine model, and nothing in them knows about arbitrary substrates carrying their own recursion theorems and representability predicates. The formalized result should be read with

its own interface premises. Its proof structure does not establish a non-reducibility or Turing-degree separation from classical halting.

10.1 The Substrate typeclass

A Coq typeclass is a checklist of operations and properties a type has to supply to count as the thing. The `Substrate` interface used here packages the following data and premises independently of any particular instruction set. It does not itself contain a certification predicate or A2:

- A type `state` of substrate states.
- A type `Program` of substrate programs.
- An outcome function `run : Program → state → option state`. The record does not impose a correspondence between this function and `step`, or a divergence interpretation of `None`; those require an instance contract. The VM instance always returns `Some` of its 1000-fuel result, including when that result has not halted.
- A cost function `mu : state → nat`.
- An atomic step relation `step : Program → state → state → Prop`.
- A cost-ledger monotonicity field `Substrate.mu_monotone`: `forall p s s', step p s s' → mu s ≤ mu s'`. This is a separate general-purpose monotonicity property from the kernel's A2 (which specifically asserts that a step flipping certification false-to-true costs ≥ 1 , and which is carried by the `CertificationSystem` record, not by this typeclass). Both properties hold in the VM: all instruction costs are non-negative and the cert transition costs ≥ 1 . See Section A.8 for the A2 statement and the related axioms A3 through A6.
- A Gödel coding: functions `encode : Program → state` and `decode_safe : state → Program` with the round-trip property `decode_safe (encode p) = p`. This states recovery by the supplied mathematical decoder; it does not assert that a program in the substrate can execute that decoder or inspect the encoded field.
- An internal-representability predicate `Representable : (Program → Program) → Prop`, scoping which Coq function transformers the substrate can express as programs of its own kind. Concrete substrates instantiate this with a precise meaning; the minimal nat-coded substrate defines `Representable f` as “`f` is the diagonal flip transformer for the substrate's own decide function,” which is exactly the shape the diagonalization needs.
- A recursion-theorem field, internal form: for every internally representable transformer `f : Program → Program`, there is a program `p` extensionally equivalent to `f p`, where extensionally equivalent means the two give the same result on every input, however differently they are written.

The recursion-theorem field is the load-bearing piece. It is the substrate’s Kleene 1938 recursion theorem, restricted to transformers the substrate can actually express. The restriction is not a hedge; it is the substrate’s own scope. Turing 1936 did not prove “no decider exists in any conceivable mathematical universe”; he proved “no Turing machine decides halting,” because Turing machines were the substrate Turing was studying. The restriction to internally representable transformers is the same scope move at the substrate level: the substrate’s limitative theorem is about what the substrate itself can express, not about every Coq function that exists in the meta-theory.

The recursion theorem is not derivable from `mu_is_initial_monotone` (the cost-functional uniqueness result). Initiality of the cost functional states that a measure initialized to zero and satisfying the exact increment law $M \text{ (vm_apply } s \ i) = M \ s + \text{instruction_cost } i$ agrees with μ on reachable states. It does not select the instruction-price schedule, identify measures on unreachable states, or collapse all monotone measures to one inhabitant. The recursion theorem instead supplies extensional fixed points for transformers satisfying the chosen representability predicate. In the `nat` construction it is proved separately for each parameter d , whose flip determines that instance’s restricted representability class and code-2 behavior.

10.2 The shortcut-admittance predicate

The substrate becomes interesting for the limitative result when it is equipped with a non-trivial *shortcut-admittance predicate*: a Prop-valued predicate `AdmitsShortcut` on programs, witnessed by at least one program that admits and at least one that does not, and depending only on extensional behavior (programs with the same `run` behavior on every state admit shortcuts iff each other).

The typeclass extension `WithShortcutPredicate` packages this: an `AdmitsShortcut` field, two canonical witnesses `yes_program` and `no_program` with their respective inhabitedness or non-inhabitedness lemmas, and the extensionality field. These are the conditions the diagonalization needs. A substrate that supplies them earns the limitative theorem.

The kernel constructs a concrete `nat`-coded family and a VM instance parameterized by its representability class and recurrence premise.

For each $d : \mathbb{N} \rightarrow \text{bool}$, `nat_substrate d` supplies the typeclass fields for a language with distinguished codes 0, 1, and 2. Its representability predicate selects the flip transformer associated with d , and code 2 supplies its fixed point. The theorem `nat_self_undecidable` states that d fails to decide `nat_admits d`. The global assumption report is closed; d remains an explicit mathematical parameter.

The 51-opcode VM substrate `vm_substrate` is parameterized over the VM-specific Gödel encoding (`vm_encode`, `vm_decode_safe`, the round-trip), the VM’s internal-representability predicate (`vm_representable`), and the VM’s internal recursion theorem restricted to representable transformers. The Coq Section discipline is iden-

tical to the one the physical bridges use for their named constants: the parameters are not global axioms; closing the surrounding Section discharges them as explicit forall premises on every consumer. The encoding is a Section parameter too, though not a live one: the kernel's closed round-trip lemma (`program_to_nat`, `nat_to_program`, no Hypothesis) discharges it, which is exactly what `vm_structural_shortcut_undecidable_encoded` does. The VM's internal-representability predicate and bounded recurrence premise remain explicit parameters of the encoded theorem. The generic theorem is independent of this particular VM instance, but still requires its own interface premises. The separate family `nat_substrate` supplies those premises only for its d -dependent semantics and representability class; it does not discharge the VM premise.

The VM-side `AdmitsShortcut` is the extensional predicate “this VM program's bounded run from `init_state` coincides with the bounded run of `simple_morph_trace`.” `yes_program` is `simple_morph_trace` itself; `no_program` is the empty trace (which produces `init_state`, demonstrably distinct from `simple_morph_final`, since the latter has fired the supra-cert channel and the former has not). The extensionality field is the standard “equal run on every state implies the predicates agree.” The instantiation that assembles these into the `WithShortcutPredicate` interface lives in the kernel.

10.3 The substrate-level limitative theorem

| Plain reading. Kleene 1938's diagonal, transplanted from time to structure. Same shape: assume a decider, build a flipper, find a fixed point, watch the program disagree with itself in both cases. The work is in the typeclass: once the substrate has its own recursion theorem, two canonical inhabitants on opposite sides of the predicate, and extensionality, the diagonal lands itself. What it rules out is a correct decider whose flip belongs to the supplied representability class. Pointing it at a concrete execution model takes that model's fixed-point premise. The generic proof doesn't supply it.

Theorem 10.1 (Structural-axis undecidability, substrate-level). *For every Substrate-instance equipped with a `WithShortcutPredicate` extension, there is no Coq-definable function `decide : Program → bool` whose corresponding diagonal flip transformer is internally representable in the substrate and that for all programs p satisfies: `decide p = true` iff `AdmitsShortcut p`.*

Proof. Assume for contradiction such a `decide` exists. Define the diagonal flip transformer

$$\text{flip}(p) = \begin{cases} \text{no_program} & \text{if } \text{decide } p = \text{true} \\ \text{yes_program} & \text{if } \text{decide } p = \text{false} \end{cases}$$

By hypothesis, `flip` is internally representable in the substrate. Apply the sub-

strate's typeclass field `recursion_theorem` to `flip`: there exists `p` such that `prog_equiv p (flip p)`, where `prog_equiv` is the extensional equivalence “run agrees on every initial state.”

Case-split on `decide p`. *Case A* (`decide p = true`): by the bi-implication, `AdmitsShortcut p`; by definition of `flip`, `flip p = no_program`; by the fixed-point witness, `prog_equiv p no_program`; by extensionality of `AdmitsShortcut` (typeclass field), `AdmitsShortcut p` iff `AdmitsShortcut no_program`. The latter is false (typeclass field `no_program_refuses`). Contradiction.

Case B (`decide p = false`): symmetrically, by the bi-implication $\neg$ `AdmitsShortcut p`; `flip p = yes_program`; `prog_equiv p yes_program`; extensionality yields `AdmitsShortcut p` iff `AdmitsShortcut yes_program`. The latter is true (typeclass field `yes_program_admits`). Contradiction.

Both cases close. No such `decide` exists. `PrintAssumptionsstructural_shortcut_undecidable` returns `Closed` under the global context. $\square$

| Plain reading. For each candidate d , the construction builds an instance and proves that d fails on that instance's predicate P_d . The instance itself depends on d . This is a family theorem. It doesn't hand you one fixed effective predicate that beats every effective decider.

Theorem 10.2 (Structural-axis undecidability for the nat-coded substrate). *For every Coq function $d : \text{nat} \rightarrow \text{bool}$, the substrate `nat_substrate d` cannot internally decide its own structural-shortcut membership predicate. In particular, d itself is not a correct decider for the predicate of the substrate it parameterizes; the corollary `nat_self_undecidable` reads off this self-referential conclusion directly.*

Proof. For each fixed d , the substrate `nat_substrate d` supplies every typeclass field constructively: `Program := nat`; `yes_program := 0`; `no_program := 1`; `nat_run d` is a finite Coq pattern match using $d(2)$; `prog_equiv` is pointwise equality of run; `Representable f` is the precise predicate “ f is the diagonal flip transformer for the substrate's own decide function”; `recursion_theorem` is discharged by exhibiting code 2 as the fixed point of any such transformer (its `run` is defined to mirror whatever the transformer outputs); `AdmitsShortcut` is extensional by construction; `yes_program_admits` and `no_program_refuses` hold by definitional unfolding of `run` at 0 and 1. Apply the substrate-level theorem above. The corollary `nat_self_undecidable` specializes the conclusion to d itself: d cannot be a correct decider for `nat_admits d`, cannot answer `true` exactly when the predicate holds. `Print Assumptions` on both `nat_structural_shortcut_undecidable` and `nat_self_undecidable` returns `Closed` under the global context. $\square$

| Plain reading. The VM version holds only under the bounded recurrence and

representability premises printed in the statement. The nat family gives a separate concrete example for each candidate decider. It doesn't discharge the VM premise. Unbounded execution and a self-interpreter need their own semantics and proofs, and the next subsection has them.

Theorem 10.3 (Structural-axis undecidability for the 51-opcode VM, conditional on the VM's own recursion theorem). *Assume the VM's internal recursion theorem restricted to representable transformers: for every transformer f on VM programs with $\text{vm_representable } f$, there exists a program p such that $\text{vm_run } p$ and $\text{vm_run } (f\ p)$ agree on every initial state. Under that assumption, there is no Coq function $\text{decide} : \text{list vm_instruction} \rightarrow \text{bool}$ whose diagonal flip transformer satisfies vm_representable and that for all programs p satisfies: $\text{decide } p = \text{true}$ iff $\text{vm_admits_shortcut_extensional } p$.*

Proof. Instantiate the substrate-level theorem at `vm_substrate`. The Gödel encoding piece (`vm_encode`, `vm_decode_safe`, the round-trip) enters as Section parameters and is then discharged by the kernel's closed round-trip lemma (`program_to_nat`, `nat_to_program`, no Hypothesis) in `vm_structural_shortcut_undecidable_encoded`. The VM's internal-representability predicate `vm_representable` and the VM's internal recursion theorem restricted to representable transformers are the Section parameters that stay live; closing the surrounding Coq Section discharges them as explicit `forall` premises on every consumer, the same Section-parameter discipline the physical bridges use for their named constants, and the theorem above states that recursion-theorem premise directly rather than leaving it implicit in the proof. The VM-side `AdmitsShortcut` is the extensional predicate “this VM program's bounded run from `init_state` coincides with the bounded run of `simple_morph_trace`”; `yes_program` is `simple_morph_trace`; `no_program` is the empty trace, which produces `init_state` and so differs from `simple_morph_final` (where the supra-cert channel has fired). The substrate-level diagonal applies; `vm_structural_shortcut_undecidable` is its specialization, carrying the VM-side parameters as premises. The nat family above is a separate application with different semantics. This VM specialization remains conditional on its stated bounded recurrence premise and representability class.

`vm_run` returns `Some (run_vm 1000 p s)` even when fuel expires before halting. Equality of its two finite-data results is decidable externally. Under the recurrence premise, the flip of that correct external decider must therefore lie outside the chosen representability class. The unbounded relation `VMUnboundedExec.vm_halts_at` instead requires an existential fuel witness and a halted result. Its determinism and saturation proofs do not identify it with bounded-result equality. A separate development supplies a fixed 60-instruction interpreter over the natural-valued sibling `vm_apply_u`, with executable variable-width encodings of two-counter programs. The guest branches on a successful nonzero decrement and falls through on zero. That

convention matters. A guest that branches on zero instead is a weaker machine, and it isn't used as a universality witness. `VMUnboundedCM2Bridge.mm2_haltng_host_iff` proves equivalence between halting in the pinned Coq Undecidability Library's MM2 semantics and raw execution of this fixed host interpreter on the encoded input. Its correctness, malformed-code exclusion, and divergence contracts are proved independently of any supplied interpreter-correctness premise.

The transferred `actual_unbounded_host_synthetic_undecidability` theorem retains the library's precise synthetic meaning: a Boolean halting decider would enumerate the complement of single-tape binary Turing-machine halting. It is not a proof of the unrestricted proposition $\sim \text{decidable}$, nor a recursion theorem for the full VM ISA. The original word64 machine, bounded equality theorem, and conditional structural-shortcut theorem above are unchanged. No effective-specialization theorem or concrete representability/recurrence instance for the full VM is established by this reduction. $\square$

10.4 A recursion theorem that is proved, not built in

The nat family above has one weak spot, and it is worth naming exactly. Its recursion field holds because code 2 is defined to copy whatever the transformer returns. The fixed point is there by construction. The VM version asks for its recursion theorem as a premise. So neither one shows the premise can be met by a model that has to earn it.

L can. L is the weak call-by-value lambda calculus, a standard Turing-complete model (Forster and Smolka, ITP 2017). I built it from scratch in Coq: terms, a deterministic step rule, and a Scott encoding that turns every term into a closed value, its code. Then the classic construction. A quote combinator Q takes the code of a term to the code of its code (Q_spec). With it, every closed value s has a closed term t that reduces to s applied to t 's own code.

Theorem 10.4 (Kleene's second recursion theorem for L, `second_recursion`). *For every closed value s there is a closed term t with $t \rightarrow^* s \ulcorner t \urcorner$.*

Proof. Let A be the term that takes a code x , builds the code of x applied to the code of x , and passes it to s . Building that code needs the code of x itself, which is what Q supplies. Put $t := A \ulcorner A \urcorner$. One step substitutes $\ulcorner A \urcorner$ for x , and the rest reduces to $s \ulcorner t \urcorner$. $\square$ $\square$

Every transformer an L term computes on codes has a fixed point under equality of reachable values (`L_recursion_theorem`). Plug it into the same diagonal and you get Rice's theorem for L.

Theorem 10.5 (Rice for L, `L_rice`). *Let P be a property of L programs that depends only on which values they reach. If P holds for one closed program and fails for another, no closed L*

term decides P from programs' codes.

The halting problem falls out: no L term decides whether an L program reaches a value ($L_halting_undecidable$). And the diagonal in the exact shape of the substrate theorem is ruled out too: no Boolean decider that returns true exactly when the property holds has a flip the model can compute ($L_structural_shortcut_undecidable$). Every one of these closes under the global context. Nothing is assumed, and nothing is true by definition.

Two limits. L is not the 51-opcode VM, and it is not a `Substrate` instance: the interface asks for a total `run` function, and a model that can really diverge doesn't have one without an axiom. So this is the same diagonal proved again for L , not the substrate theorem applied to L . What it shows is that the diagonal's recursion premise is one a real model of computation meets. The VM's own recursion theorem is still a premise.

10.5 The exact interpreter targets

An interpreter is a program that runs other programs, handed to it as data. The machine running one is the host; the program being run is the guest. The word "interpreter" names two different constructions in the current kernel, and I am keeping both names because they answer different questions. The 60-instruction construction is the fixed host `cm2_interpreter_program`; the selected B3 self-interpreter target is the separate 122-instruction host `self_interpreter_program`. Neither is a full-51-opcode self-interpreter for the physical word64 VM.

The 60-instruction CM2 host. The guest syntax is `CM2InstrU`, with five instructions: `HALT`, `increment-counter-0`, `increment-counter-1`, and `decrement-and-jump` for either counter. A guest configuration is `CM2ConfigU = (pc, c0, c1)` with all three components in $\mathbb{N}$. The decrement convention is explicit: a nonzero counter is decremented and the guest jumps to the encoded target; a zero counter remains zero and the guest falls through to the next address. A finite guest program is packed into one natural using an executable width selected from the program itself. The host receives `cm2_total_config_encodingambientpstart`; the guest program, width, guest configuration, and host scratch are encoded in the initial host state. The host's own state and cost fields are ambient; the guest counters, PC, and result are carried in the designated host registers.

The bridge the kernel actually proves is `mm2_halting_host_iff`: MM2 halting is equivalent to the existence of a fuel for which the fixed host reaches `cm2_halted` on the encoded input. The more local correctness contract is split into both directions: `cm2_uniform_interpreter_simulation_total` gives a positive finite host run for every successful guest step, while `cm2_uniform_interpreter_raw_correct_total` says that a raw host run ending in the terminal state has exactly the corresponding guest halt or fall-off result, and conversely every guest halt has such a

host run. Canonical encodings cannot produce the malformed status (`cm2_uniform_interpreter_raw_never_malformed`); malformed packed data is still part of the host contract and is not silently identified with a guest halt. Guest divergence is the absence of a finite guest halt witness and is equivalent to the host never reaching its terminal PC by `cm2_uniform_interpreter_divergence_total`. The output observation is `cm2_result` together with `cm2_halted`; host scratch and host μ are not guest state.

This CM2 host is a separate reduction/bridge result. It does not establish the bounded VM diagonal's recurrence premise, and it does not establish a full-word64 or full-ISA universal interpreter.

The 122-instruction self-interpreter. The selected B3 target uses `GInstr`, the twelve-instruction arithmetic and control fragment `HALT`, `LOAD_IMM`, `XFER`, `ADD`, `SUB`, `MUL`, `AND`, `OR`, `SHL`, `SHR`, `JUMP`, and `JNEZ`. Guest register fields are restricted to registers 0 through 3, cost fields to the representable byte range, and guest values, immediates, and jump targets are natural numbers without a word64 bound. The executable encoding maps each guest instruction to a nonzero packed word, chooses a width `g_width`, and packs the program into the natural `g_code`. `g_width_fits` proves that the chosen width represents every well-formed instruction in the finite guest program; an out-of-range fetch is the guest's normal no-instruction termination condition.

The host is the literal list `self_interpreter_program`, whose length is 122. At an interpreter boundary, the representation relation `h_rep/hbc` places the guest registers in host registers `R0–R3`, the guest ledger in `R11`, the guest PC in `R12`, the packed code and width in `R14` and `R13`, and leaves scratch registers and ambient non-register state explicit. Every host instruction has declared cost zero in this construction; the guest ledger is data in `R11`, not the host's `vm_mu`. The guest step is related directly to the unbounded sibling semantics by `g_step_is_vm_apply_u`.

The two directions of the raw result contract are `self_interpreter_complete` and `self_interpreter_sound`, packaged as the biconditional `self_interpreter_correct`: a terminal guest run exists exactly when an actual run of the fixed host reaches its end PC with the corresponding represented guest state. The divergence theorem `self_interpreter_divergence` uses the same host run and does not mistake fuel exhaustion for proof of divergence. For arbitrary packed data, the host can report malformed words; `self_interpreter_malformed` specifies that status, while the canonical encoding of a well-formed guest is proved never to reach it. Guest `HALT`, guest fall-off, malformed input, and divergence are therefore separate outcomes. The construction proves the selected B3 fragment and its frame relation; it is not advertised as an unchanged-word64 or full-51-opcode self-interpreter.

The precise Rice/reduction domain. The theorem called `self_rice` lives on the `GInstr` model, not on the bounded `vm_run` predicate and not on the CSR state

predicate. Its input predicate is

$$p \mapsto \text{g_wf_program}(p) \wedge \text{Pr}(p),$$

where Pr is a predicate on guest programs. The premises are: a well-formed witness w with $\text{Pr}(w)$; a well-formed never-terminating program g_bottom with $\neg \text{Pr}(\text{g_bottom})$; and the well-formed extensionality condition $\text{g_equiv } p \ q \rightarrow (\text{Pr}(p) \leftrightarrow \text{Pr}(q))$. Under those premises, self_rice reduces the complement of the pinned MM2 halting problem to the complement of this well-formed guest predicate and concludes the repository’s synthetic *undecidable* relation.

The guest-program decider contract is separately named g_decides : a well-formed guest program D must terminate on every well-formed encoded input and report $\text{Pr}(p)$ exactly by whether its final guest register 0 equals 1. $\text{g_decides_decidable}$ turns that total guest-program behavior into a Boolean decider for the same predicate, and $\text{self_rice_representable}$ transfers the consequence to the library’s enumerability formulation. Thus “undecidable” here means the formal reduction/enumerability contract supplied by the library; it does not mean an unrestricted meta-theoretic negation of every possible Coq decision procedure. The bounded VM outcome equality, the VM’s conditional structural diagonal, and the unbounded CM2 bridge remain separate theorems with separate domains.

10.6 The proof, in prose

The proof is the standard Kleene-1938 diagonal, transported to the substrate.

Assume for contradiction that $\text{decide} : \text{Program} \rightarrow \text{bool}$ satisfies the bi-implication. Define the flipping transformer

$$\text{flip}(p) = \begin{cases} \text{no_program} & \text{if } \text{decide } p = \text{true} \\ \text{yes_program} & \text{if } \text{decide } p = \text{false} \end{cases}$$

This is a total Coq function from Program to Program . The theorem also requires that this particular flip satisfy Representable ; total Coq definability alone does not establish that premise. With that premise, apply the substrate’s recursion-theorem field to flip : there exists a program p with $\text{prog_equiv } p \ (\text{flip } p)$.

Case-split on $\text{decide } p$.

Case A: $\text{decide } p = \text{true}$. By the bi-implication, p admits a shortcut. By definition of flip , $\text{flip } p = \text{no_program}$. By the recursion-theorem witness, p is extensionally equivalent to no_program . By extensionality of AdmitsShortcut , p admits iff no_program admits. But no_program does not admit. Contradiction.

Case B: $\text{decide } p = \text{false}$. By the contrapositive of the bi-implication, p does not admit. By definition of flip , $\text{flip } p = \text{yes_program}$. By the recursion-theorem

witness, p is extensionally equivalent to `yes_program`. By extensionality, p admits iff `yes_program` admits. But `yes_program` admits. Contradiction.

Both cases close. No correct total decision procedure for `AdmitsShortcut` whose flip satisfies the stated representability premise exists in this instance. $\square$

The proof is short, because the work is in the typeclass, not in the diagonal. The body runs about 38 lines, of which roughly 16 are actual tactic steps; the rest is the case-split structure and the inline narration. Once the substrate supplies the restricted recursion theorem, two canonical inhabitants, an extensional predicate, and representability of the candidate decider's flip, the diagonal is mechanical. The substrate-level work was assembling the right typeclass; the limitative result is what falls out.

The diagonal does not use `A2`, the ledger, or a certification predicate. Its contradiction uses the restricted recursion-theorem field, representability of the candidate flip, and the two witnesses and extensionality of `WithShortcutPredicate`. Certification supplies a separate interpretation in the VM; the `nat` family has no certification content. `A2` neither supplies the recurrence premise nor establishes a correspondence between the bounded program predicate and the CSR state predicate. Those connections must be proved separately wherever an application requires them.

10.7 The two-axis picture this completes

Time axis	Structural axis
steps	μ
HALT (Turing 1936)	<code>AdmitsShortcut</code> (above)
no Turing decider for halting	no correct decider with a representable flip under recurrence

The table lines up the same diagonal on two axes, each under its own model's premises. The `nat` construction gives a family indexed by the candidate decider, and the VM result is about bounded outcome equality under a restricted recurrence premise. Neither one hands you a second fixed effective undecidable problem, and neither separates Turing degrees. The CSR collision below is a different kind of result: a state that a window can't recover.

10.8 What this closes

What the substrate-level theorem closes. A total witness procedure would yield a decider of the same predicate only under an explicit reduction contract: its input is each program p in the theorem's domain; its success is equivalent to `AdmitsShortcut p`; and it terminates with either a witness or a definite negative answer on every such input. To apply the diagonal, the Boolean decider obtained by testing that success must also have a flip in the supplied representability class. Under these premises the theorem excludes the procedure. A procedure returning a `SoundStructuralShortcut` record is not automatically such a reduction: its witness relation must first be connected to this exact program predicate. Partial synthesis,

restricted argument inputs, and externally supplied evidence are not excluded by the total-decider theorem.

What the nat-substrate theorem closes. It constructs a family of instances indexed by the candidate d . For each member, d fails on that member's predicate. The generic theorem applies under its representability and fixed-point premises. A fixed-system effective undecidability result requires another instance argument.

What the L theorem closes. It supplies that other argument for one fixed model. In L the recursion theorem is proved, not built in, so the diagonal rules out a real decider for a fixed predicate: Rice's theorem, with halting as a corollary (Section 10.4). It does not carry over to the 51-opcode VM, whose recursion theorem is still a premise.

What the VM corollary closes. The 51-opcode VM corollary `vm_structural_shortcut_undecidable` is the substrate-level theorem instantiated at `vm_substrate`: no representable decider answers `true` exactly when a VM program admits the extensional shortcut. It carries the VM-side Section parameters as explicit `forall` premises, in the named-premise style the physical bridges use. The generic theorem does not depend on these particular VM parameters, but its VM application does. The `nat` family supplies a different instance for each d and cannot replace the VM's bounded recurrence premise. The exclusion applies to deciders whose diagonal flips lie in the supplied representability class.

How to falsify. Three routes.

1. A Coq function `decide : Program → bool` whose diagonal flip transformer is internally representable in some Substrate plus `WithShortcutPredicate` instance and that satisfies the bi-implication would force the substrate-level proof to fail at one of its steps, each of which is named in the case-split above.
2. For the nat-coded substrate specifically: exhibit a Coq function `d : nat → bool` that correctly decides `nat_admits d`. The corollary `nat_self_undecidable` forbids this; finding such a `d` contradicts the corollary.
3. For the VM specialization, supply its bounded recurrence premise for the chosen representability class and a correct decider whose particular diagonal flip belongs to that class. A correct external decider alone is not a counterexample: bounded membership is externally decidable, and its flip is excluded by every class satisfying that recurrence premise.

10.9 The wall the classical configuration cannot see

There is one sentence that kills most new “a computer can't decide this” results before they get off the ground, and the first time I thought I had one I said it to myself before anyone else could: *you didn't find a new kind of unanswerable question. Your machine is a*

Turing machine with some extra memory bolted on, the thing you can't decide is just one more ordinary fact about what a program does, and Rice's theorem already says nobody can decide facts like that. Nothing new, nothing sideways to ordinary computation. I went looking for where that sentence breaks the result, because if it does, I'd spent months on a footnote to Rice. It doesn't. The reason is plain once you see it: a classical machine never gets the whole state. It gets a stripped-down version, the part it can read and write, and my unanswerable question turns on a piece the stripping throws away. Rice's theorem is about an answer a machine can't compute from what it's handed. This is about an answer that was never in what it's handed.

A classical decider has, as its entire input, the Turing configuration `forget s` : `TMSnapshot`, the same four-field projection from the structural-axis section. If the thing it is asked to decide is not a function of `forget s`, then no classical decider can be right about it, and not for want of horsepower: what it is deciding is not in what it is given. That is the whole shape of this wall, and it is why it sits beside Rice's rather than under it. Two independent obstructions, on two independent axes.

| Plain reading. Two reachable states, one with the certification channel fired and one without, look the same through the four-field `forget` window. The full state answers the question easily. The window can't. The bounded program-equality predicate the diagonal uses is a different definition, and nothing here says the two predicates are the same.

Theorem 10.6 (Structural-shortcut admittance is not classical, `structural_shortcut_not_function_of_classical`). *There is no function `decode : TMSnapshot → bool` such that, for every VM state s ,*

$$\text{decode}(\text{forget } s) = \text{admits_structural_shortcut_bools},$$

where `admits_structural_shortcut_bools` is the reading “ s 's certification channel has fired” (`csr_cert_addr` $\neq$ 0).

Proof. Two states reachable from `init_state` by executing a trace: `run_cert_set` runs the minimal `MORPH_ASSERT` closure and fires the channel (`csr_cert_addr` = 650); `run_cert_unset` runs the same two-instruction prefix, then a graph-preserving no-op whose μ -cost is chosen to match, leaving `csr_cert_addr` = 0. The matched cost makes the two runs land on the same $(pc, \mu, \text{regs}, \text{mem})$, so `forget run_cert_set` = `forget run_cert_unset` by computation. A `decode` satisfying the equation would have to return both `true` and `false` on that one input. Contradiction. □ □

This is the first rung of a short ladder, and I want to be exact about how far each rung reaches, because the honest version is sharper than the inflated one.

Observation-dependent oracles. `structural_axis_survives_any_classical_oracle` quantifies over oracle answers that are functions of `forget` alone. Composing such an oracle with a decoder preserves the collision. An oracle supplied independent evidence or full state has a different information domain.

The two non-determination directions. `axes_mutually_independent` supplies reachable witnesses: the structural channel is not determined by `forget`, and the program counter is not determined by that channel. `structural_membership_decidable` computes the channel predicate from full state. These are information-loss statements about specified observations. Rice and Turing-degree conclusions require an effective program model and are not consequences of this state-level collision.

The concrete VM encoding. `vm_structural_shortcut_undecidable_encoded` uses the proven `program_to_nat/nat_to_program` round trip, with the code stored in `vm_logic_acc`. It still quantifies over a representability predicate and a fixed-point premise for the bounded `vm_run`. An unbounded self-interpreter would concern another execution semantics and would require a correspondence theorem before it could discharge this premise. The separate B3 interpreter uses executable data encoding for an unbounded arithmetic/control fragment; it does not supply this bounded-semantics bridge. Furthermore, `no_logic_acc_encoded_interpreter` proves an input-access obstruction by an explicit equivalent-state relation: every executable step and run commutes with replacing `vm_logic_acc`. A fixed VM program cannot use that encoded input to reproduce even a simulated program's final certification bit through an output observation independent of the retained input code. This rules out the proposed encoding for that interpreter contract; it does not rule out an executable alternative encoding or an architectural extension.

| Plain reading. The state-level theorem excludes recovery from the specified observation, including observation-dependent oracle extensions. The program-level theorem excludes deciders whose flip transformers belong to a representability class satisfying the stated recurrence premise. The nat example is a family parameterized by the candidate decider. These results retain their separate domains and quantifiers.

Part II

Building it

The abstract results in Part I hold for any system that carries the pieces their theorems ask for. This part is one such system, built all the way down. The concrete witness pairs in Part I are states of this system. It is a 51-opcode VM, written in Coq, extracted to OCaml, and carried into hardware, with a category-theory layer for structure and a

check against the CHSH bound. I built it so the argument would have something you can run and try to break. It is a witness to the model. It is not the model. Burn every opcode here and the abstract results still stand; a different build would need its own witness pairs. The five small models in Section 17 are there to make the same point from the other side: the same law, in systems that look nothing like this VM.

11 The formal state record

This section is the lookup table for everything the machine remembers between steps. It's formal on purpose. Come back to it whenever a later section names a field.

Definition 11.1 (Concrete state record). The kernel's `VMState` record realises the substrate state in 12 fields:

- `s.vm_graph`: the partition graph. It contains module records, morphism records, and fresh-ID counters.
- `s.vm_csrs`: control-status registers. The hardware split: data registers hold values (like `s.vm_regs`); CSRs hold configuration and status. Four fields:
 - `csr_cert_addr`: 0 until something is asserted. Set to the ASCII checksum of the property string on a successful `MORPH_ASSERT` (nonzero for any property string with a non-NUL byte). The name says address; the contents say fingerprint. Nonzero here means a structural property was certified and paid for.
 - `csr_status`: general status code. Informational. Nothing reads it; nothing branches on it.
 - `csr_err`: error code. Zero means clean. Nonzero means an instruction failed: bad argument, missing morphism, locality violation, etc.
 - `csr_heap_base`: base address for `HEAP_LOAD` and `HEAP_STORE`. Final address = base + register-specified offset. No paging, no translation, just addition.
- `s.vm_regs`: a list of register values. The kernel fixes `REG_COUNT = 16`; reads and writes wrap modulo it. The kernel proofs are parametric in `REG_COUNT`, so 16 is a scaffolding choice, not a theorem.
- `s.vm_mem`: a list of data-memory words. The kernel fixes `MEM_SIZE = 128`; reads and writes wrap modulo it. The proofs are parametric here too.
- `s.vm_pc` $\in \mathbb{N}$: program counter. The integer that says *which instruction next*.
- `s.vm_mu` $\in \mathbb{N}$: the μ ledger, the substrate's accounting field. Starts at 0. Never decreases.
- `s.vm_mu_tensor`: a 4×4 array of cost values (16 entries), indexed by direction pairs (i, j) . `REVEAL` is the one opcode that writes it, stamping the disclosure cost

at the direction index encoded in the instruction; the per-module 4×4 grids that `TENSOR_SET` writes and `TENSOR_GET` reads are separate state, carried inside `vm_graph` (Section 12). The kernel tracks *where* costs land, not just how much. In the kernel model these are natural numbers; in the hardware, 32-bit words. If you don't use the tensor tracking, ignore all 16 entries.

- `s.vm_err` $\in \{\text{false}, \text{true}\}$: the error flag. Set when an instruction fails.
- `s.vm_logic_acc`: a natural number with two roles. Role one: the Logic Engine accumulator during `LASSERT` (the on-chip unit that reads formula words and certificate bytes and tracks validation state). Role two: the magic key the FSM matches against. `0xCAFEEDACE` (hex; 3,405,703,886 decimal) passes. Anything else, the FSM rejects. The field exists so the Coq kernel and the physical hardware agree on the same accumulator value at every step.
- `s.vm_mstatus`: status word, 0 or 1. 0 = Turing mode (ordinary computation; partition graph and morphism map idle). 1 = Thiele mode (structural state live). The kernel keeps this as a status indicator only. It does not gate the structural opcodes at the step level; those are always available. The field exists so the hardware and the Coq model agree across the bridge proofs.
- `s.vm_witness`: the CHSH witness counters. This is a record of 8 natural-number counters, one for each combination of (measurement setting pair) $\times$ (same/different outcome). The four setting pairs are $(x, y) \in \{(0,0), (0,1), (1,0), (1,1)\}$, where x is Alice's setting and y is Bob's setting. That is the modern Bell-test convention (NPA hierarchy, Brunner et al. 2014 review), and the same one the Coq constructor `instr_chsh_trial` (`x y a b : nat`) uses. The outcome variables are (a, b) . For each setting pair, there are two counters: same-outcome and different-outcome. Eight counters total: `wc_same_00`, `wc_diff_00`, `wc_same_01`, `wc_diff_01`, `wc_same_10`, `wc_diff_10`, `wc_same_11`, `wc_diff_11`, where the numeric subscript is the value of (x, y) for that bucket. Each `CHSH_TRIAL` instruction increments exactly one counter. (Some Bell-test literature uses the reverse convention with (a, b) for settings; the numeric subscripts 00, 01, 10, 11 are unambiguous either way.) The CHSH score S is computed from the aggregate sums of these counters. For each setting pair (x, y) , define $N_{xy} = \text{wc_same}_{xy} + \text{wc_diff}_{xy}$ (total trials with that setting) and $E_{xy} = (\text{wc_same}_{xy} - \text{wc_diff}_{xy}) / N_{xy}$ (the correlation: fraction same minus fraction different, ranging from -1 to $+1$). Then $S = E_{00} + E_{01} + E_{10} - E_{11}$. If no trials have been run for a setting ($N_{xy} = 0$), the correlator for that setting defaults to 0 by convention. Classically $|S| \leq 2$; the NPA algebraic model allows $|S| \leq 2\sqrt{2}$ (see Section 18). The concrete Coq definition is `chsh_stat_from_wc`.
- `s.vm_certified` $\in \{\text{false}, \text{true}\}$: the top-level certification flag. Only `CERTIFY` sets it. Nothing else can.

The reference kernel and the physical hardware are not the same shape at every bit, on purpose. The kernel is where I pin the step rule down precisely enough that nothing can quietly drift; it is the gap-catcher, not the source of the truth. The truth is the cost argument itself, the thing you could re-run with a pencil. The kernel stores register and memory values as natural numbers, but truncates writes through 64-bit word helpers to keep the math clean and the bit-width honest. The hardware layer (explained in Section 16) is finite: 16 general-purpose registers, 128 memory words, bounded tables. The bridge proofs in the hardware section state exactly where those finite hardware representations match the kernel step; the size constants are parametric, so the same proofs apply at any bound. The 16 and the 128 are scaffolding choices. They aren't theorems.

Definition 11.2 (Partition Graph). The partition graph `s.vm_graph` is a record:

- `pg_next_id`: the next fresh module identifier.
- `pg_modules`: a list of module-ID/module-record pairs. Each `ModuleState` holds region data, axiom data, and the module's μ -tensor.
- `pg_next_morph_id`: the next fresh morphism identifier.
- `pg_morphisms`: a list of morphism-ID/morphism-record pairs. Each `MorphismState` holds a source module, target module, coupling data, and an identity flag.

Remark 11.1. There is no separate `vm_heap`, `vm_output`, or `vm_imem` field in the Coq kernel's `VMState`. Heap operations are heap-relative accesses into `vm_mem`. Output and instruction transport live one layer up: the runner, the trace, the RTL harness. None of that is kernel state. The distinction matters because NoFI is a theorem about the record above, nothing more.

Remark 11.2. The partition graph splits in two. The *module layer*: how state is decomposed into named pieces. The *morphism layer*: the typed categorical relationships between those pieces. The two layers are independent. Two states can agree on every module and still differ on every morphism. The Categorical Separation Theorem (Section 14) lives on that fact.

12 The instruction set: all 51 opcodes

The kernel realises the Thiele substrate as a 51-opcode VM (the term “opcode” is in Section A: the numeric tag the hardware decodes to choose which behaviour to run). Every one carries an explicit `mu_delta` parameter. Cost is part of the instruction, not bolted on after. The VM reads it, the VM charges it.

One reminder before the machinery. This is scaffolding. These 51 opcodes are one way to build

the substrate, they are not the substrate. Burn every one of them and the machine is exactly as real, because the machine is the law, not the instruction set. What follows shows the machine can be built and can't be waved away. It is not what makes it real.

12.1 Group 1: Partition operations (4 opcodes)

These opcodes modify the partition graph. They are how the kernel builds structural knowledge: one module, one partition edit, one accounting entry at a time.

PNEW(R, δ): allocate a new module.

Creates a fresh module and charges δ to μ . The instruction carries a region list R describing which memory addresses belong to this module. The current step rule normalises this to a standard form (the math sense of “canonical”: a unique representative for a class of equivalent inputs): it keeps the size (how many addresses) and stores it as the range $0, 1, \dots, sz-1$. So the checked step preserves the module size, not arbitrary input geometry. The bounded RTL has its own partition-table and active-module machinery for locality enforcement (locality, in this VM, is the constraint that an instruction may only access memory inside its own active module's region; Section 16 covers the table). The **PNEW** opcode itself is the graph allocation step, nothing fancier.

Why does this exist? Because structural knowledge needs a home. Creating a module is how you tell the VM “this region of computation is a named unit I'm tracking.” The cost δ is programmer-declared and can be 0; the VM does not force you to pay for allocation itself. If building this partition cost something because you had to figure out the right boundary, run a search, or derive the structure, you record that here. If it was bookkeeping, $\delta = 0$ is fine, no fee for shuffling paper.

PSPLIT(m, L, R, δ): split one module into two.

Splits module m and charges δ to μ , no more. The constructor still carries left and right region payloads, but the current hardware-aligned step ignores those payloads and calls `graph_hw_pspl`: the left side gets half the module size, and the right side gets the remainder. This is the refinement operation in the finite hardware model. If discovering the split deserves a cost, encode a positive δ ; the canonical reversible-accounting policy keeps it at 0.

PMERGE(m_1, m_2, δ): merge two modules into one.

Merges modules m_1 and m_2 and charges δ . The hardware-aligned step concatenates the module sizes through `graph_hw_pmerge`; module IDs wrap modulo 64. It does not perform a rich invariant-compatibility check in this step. Like **PNEW** and **PSPLIT**, the VM does not enforce a minimum cost here. Trust comes later, at the certification step, when the structure has to pay for being believed.

PDISCOVER($m, evidence, \delta$): query partition structure.

The evidence payload is carried by the instruction, but the hardware-aligned relation

does not attach axioms or mutate the graph. It advances the PC and charges δ . In the current kernel step this instruction is an explicit accounting event; it does not perform a register query or assert semantic truth.

12.2 Group 2: Logic and certification (3 opcodes)

This group carries the Logic Engine’s checked validation step and its ledger side. `LASSERT` is the one that talks to the Logic Engine, the on-chip finite-state machine that checks formulas and certificates; `LJOIN` and `MDLACC` are ledger entries standing beside it. No outside solver gets smuggled in.

`LASSERT(freg, creg, kind, flen, cost)`: validate a formula and witnesses.

Reads a formula whose declared encoded-word count is *flen* from memory starting at the address in register *freg*. The *kind* boolean flag selects which validation path the Logic Engine should follow. “SAT” (satisfiable) means: I’m claiming this formula has at least one satisfying assignment, some combination of variable values that makes it true. “UNSAT” (unsatisfiable) means: I’m claiming it has no satisfying assignment, it is always false. In the on-chip semantics, *kind* = *true* is the only success path: it requires a dual-witness certificate, one satisfying assignment and one falsifying assignment, so the validation proves the formula is satisfiable and not a tautology. *kind* = *false* is the UNSAT-tag path: `lassert_check_ok` returns false unconditionally when *kind* = *false*, so any `LASSERT` issued with *kind* = *false* fails the check, traps to `LASSERT_TRAP_PC`, and sets the error flag. The μ -cost is still charged. The SAT-only validation path is the kernel’s structural-proof shape; the UNSAT tag is a reserved opcode bit that the kernel routes straight into the trap. This matches the RTL FSM, which has phases 0 (idle), 1 (header read), and 2 (scan). The kernel and hardware both validate that the declared count matches the in-memory formula header before the check is allowed to succeed. A mismatch traps and sets the error flag. `LASSERT` reads a certificate block from memory starting at the address in register *creg*. That block contains two witnesses: one satisfying assignment and one falsifying assignment. The on-chip Logic Engine FSM processes the formula and both witnesses in-place. No external solver gets a hidden vote. Validation succeeds only if the declared count is honest, the first witness satisfies the formula, and the second witness falsifies it. That means the formula excludes at least one valuation. `LASSERT` itself is a checked validation step. Certification channels are activated by concrete state-changing paths such as `CERTIFY` and `MORPH_ASSERT`. The abstract cert-address theorem uses a broader guardrail predicate and puts `LASSERT` in the cost-positive certification/revelation class. That is not a claim that `LASSERT` writes the certificate-address register. It is a claim that this class cannot move for free. Charges $flen \times 8 + S(cost)$ to μ , even when the check fails.

This is the most important logic opcode in the VM. `LASSERT` is where a constraint package is checked before any later certification step can lean on it. There’s no faking the pricing anymore by declaring a tiny *flen* for a large in-memory formula: if the

header and the instruction disagree, the VM traps. And no faking semantic narrowing with a tautology either, which would otherwise be the slick move: the dual-witness requirement closes the tautology-inflation hole, because a tautology has no falsifying witness to show the formula excluded any valuation, and the VM wants to see that witness or it walks. The cost formula is still a spend ledger. The honest-length check and the non-triviality witness are what tie that spend to the real checked object, not to a thing the programmer wished they had checked.

LJOIN(*c1reg*, *c2reg*, δ): reserve certificate-join cost.

Reserves the cost of joining two memory-resident certificates. The addresses of the input certificates are in *c1reg* and *c2reg*. Charges $S(\delta) \geq 1$. In the current hardware-aligned semantics, LJOIN does not compare certificate strings, write a composite certificate, touch CSR state, or mutate the graph. It advances the PC and pays the positive floor. The join proof lives at the abstraction/verifier boundary, not inside this kernel step.

MDLACC(*m*, δ): MDL cost accounting.

Charges δ as a module-discovery / model-description cost for module *m*. It does not read a register, and it does not mutate the graph. It just enters the bill.

Suppose you have two models that both fit the same data. One is a ten-thousand-parameter neural network. The other is a five-line formula. Both fit the training set. Which bill do you want to pay? Minimum description length says the total bill is the model description plus the cost of encoding the data once the model is known. Description length is how many bits it takes to write the model down. A simple formula takes few bits. A massive network takes many. The short formula may cost more to encode the remaining data, but almost nothing to describe. The massive network may compress the training set perfectly, but its own description is enormous. MDL picks the smaller total: bits for the model plus bits for the data given the model.

In the Thiele vocabulary: a module structure that you built can carry a description-length cost. Not because someone decreed it. Because constructing a specific partition of state space may require information that belongs on the bill. MDLACC is the opcode that lets a program explicitly enter that description cost into the ledger after the structure has been built. It performs no graph change. Its sole effect is to advance the program counter and charge the encoded δ . The MDL connection is just this: the substrate's μ -ledger can track the model-description bill when the program chooses to enter it.

12.3 Group 3: Compute and data transfer (14 opcodes)

Ordinary arithmetic and data movement. These are the operations you find in every ISA. They mostly carry $\delta = 0$ because they do not introduce new structural entitlement. They just compute.

Opcode	What it does	Typical cost
LOAD_IMM(r, v, δ)	Write immediate value v into register r	0
XFER($r_{\text{dst}}, r_{\text{src}}, \delta$)	Copy register r_{src} to r_{dst}	0
ADD(r_d, r_a, r_b, δ)	$r_d \leftarrow r_a + r_b$ (modular 64-bit)	0
SUB(r_d, r_a, r_b, δ)	$r_d \leftarrow r_a - r_b$ (modular 64-bit)	0
MUL(r_d, r_a, r_b, δ)	$r_d \leftarrow r_a \times r_b$ (modular 64-bit)	0
AND(r_d, r_a, r_b, δ)	$r_d \leftarrow r_a \text{ AND } r_b$ (bitwise)	0
OR(r_d, r_a, r_b, δ)	$r_d \leftarrow r_a \text{ OR } r_b$ (bitwise)	0
SHL(r_d, r_a, r_b, δ)	$r_d \leftarrow r_a$ shifted left by r_b bits	0
SHR(r_d, r_a, r_b, δ)	$r_d \leftarrow r_a$ shifted right by r_b bits	0
LUI(r_d, v, δ)	Load upper immediate: $r_d \leftarrow v \ll 8$	0
XOR_LOAD(r_d, addr, δ)	Load from absolute memory address addr into r_d	0
XOR_ADD(r_d, r_s, δ)	$r_d \leftarrow r_d \text{ XOR } r_s$ (XOR accumulate)	0
XOR_SWAP(r_a, r_b, δ)	Swap registers r_a and r_b (reversible, direct two-write exchange)	0 by convention
XOR_RANK(r_d, r_s, δ)	$r_d \leftarrow \text{popcount}(r_s)$ (Hamming weight)	0

Why does the ISA have all these? Because the 51-opcode realisation has to run programs, not just keep receipts. Programs need to compute intermediate values, manage data, and control flow. The compute group gives them that ordinary machinery. The key property: none of these opcodes touch the certification channels. Arithmetic can move bits all day. It cannot mint certified structural knowledge.

XOR_SWAP is a reversible register swap. The kernel implements it as a direct two-write exchange. The name suggests the classic XOR-swap sequence that some assembly programmers use to swap two values without a temporary register (it works by writing one register with the XOR of both, then the other, then the first again), but the kernel does not actually run that sequence. It writes both registers at once. In the table it carries $\delta = 0$ by convention because a reversible operation, in the Bennett/Landauer sense (Section 19), does not destroy any information. If a program supplies a nonzero δ , the ledger still charges it.

12.4 Group 4: Memory and control flow (8 opcodes)

Memory access and program flow control. This is the group where a machine could quietly cheat if you let it: load from wherever it pleases, jump wherever it pleases, and the partition stops meaning anything. So this is exactly where the locality story has to hold. The opcodes here move data and steer the program counter and nothing more, and the wall that keeps a module's reach inside its own region lives one layer down, in the bounded hardware table that the partition-wall note right below makes concrete.

Opcode	What it does	Note
LOAD(r_d, r_a, δ)	Load $\text{vm_mem}[r_a]$ into r_d	Kernel register-indirect load
STORE(r_a, r_s, δ)	Store register r_s into $\text{vm_mem}[r_a]$	Kernel register-indirect store
JUMP(addr, δ)	Set PC to addr unconditionally	none
JNEZ(r, addr, δ)	Jump to addr if $r \neq 0$	none
CALL(addr, δ)	Push return address via stack pointer r15, jump to addr	Qed coverage
RET(δ)	Pop return address via stack pointer r15, jump there	Qed coverage
HALT(δ)	Advances PC and charges δ at the kernel step, like any other instruction. The runner stops at HALT (and when PC goes past the program), stamping a halt status; the hardware latches halted.	runner-level
CHECKPOINT(ℓ, δ)	Accept label ℓ , advance PC	label outside kernel

At the kernel level, LOAD and STORE are register-indirect memory operations over vm_mem . At the RTL/cosimulation layer, the partition wall adds active-module locality checks. INIT_PT and INIT_ACTIVE_MODULE are harness setup commands for that finite hardware table, not VM opcodes and not fields of VMState . This is the honest split: the abstract ISA defines memory semantics; the bounded hardware layer enforces locality through its own table state.

12.5 Group 5: Revelation, I/O, heap, and tensor (7 opcodes)

This group mixes two fundamentally different kinds of operations. `EMIT` and `READ_PORT` are in the *certification/revelation cost class*: they cost ≥ 1 unconditionally and are subject to the same mandatory-floor rule as `CERTIFY` and `MORPH_ASSERT`. `WRITE_PORT`, `HEAP_LOAD`, and `HEAP_STORE` are ordinary I/O and storage, no mandatory floor. `TENSOR_SET` and `TENSOR_GET` manage per-module cost tensors. They share this group only by physical-channel adjacency; their cost semantics are distinct.

Opcode	What it does	Cost
READ_PORT($r, c, v, bits, \delta$)	Write value v into register r . The value v is embedded in the instruction encoding at assembly time (the kernel step is deterministic: it writes the declared v , not a runtime port read). Channel c and the $bits$ count are used by the runner/environment layer for I/O orchestration and cost accountability.	$bits + S(\delta) \geq 1$
WRITE_PORT(c, r_s, δ)	Send register r_s to channel c outside kernel state	δ
EMIT($m, payload, \delta$)	Emit payload as a receipt-visible trace event	$ payload _{\text{bits}} + S(\delta) \geq 1$
HEAP_LOAD(r_d, r_a, δ)	Load from <code>csr_heap_base + regs[r_a]</code> into r_d	δ
HEAP_STORE(r_a, r_s, δ)	Store r_s to <code>csr_heap_base + regs[r_a]</code>	δ
TENSOR_SET(m, i, j, v, δ)	Set per-module tensor entry (i, j)	δ
TENSOR_GET(r_d, m, i, j, δ)	Read per-module tensor entry (i, j) into r_d	δ

EMIT charges the encoded payload length in bits plus $S(\delta)$. READ_PORT charges the declared $bits$ operand plus $S(\delta)$. Nothing in the kernel checks that the declared number matches how many bits physically moved. REVEAL is the same way: for it to stand for a binding commitment you need an evidence contract from outside, because the certificate operand isn't checked. The positive floor holds whatever those outside readings turn out to be.

TENSOR_SET and TENSOR_GET access a flattened 4×4 tensor in the kernel and dedicated tensor state in the hardware model. Their intermediate correspondence requires the source theorem's index and module conditions. Physical RTL agreement must also satisfy the finite representation and retirement contract in Section 16.

12.6 Group 6: Witness and certification (8 opcodes)

These opcodes accumulate binary game outcome statistics into hardware witness counters, enforce the NPA-PSD boundary on those counters, and manage the top-level certification flag.

CHSH_TRIAL(x, y, a, b, δ): record one CHSH trial.

Records one trial of the CHSH game. $x \in \{0, 1\}$ is Alice's measurement setting. $y \in \{0, 1\}$ is Bob's measurement setting. $a \in \{0, 1\}$ is Alice's outcome. $b \in \{0, 1\}$ is Bob's outcome. The Coq constructor order is settings first (x, y) , then outcomes (a, b) . The instruction increments exactly one of the 8 `WitnessCount` counters: if $a = b$ (same outcome), it increments `wc_same_xy`; if $a \neq b$ (different outcome), it increments `wc_diff_xy`. There are 4 setting pairs $\times$ 2 same/different buckets = 8 counters. The CHSH score S is computed later from these counters. Charges δ to μ .

This is a Tier 0 operation: it records data but does not activate any certification channel. It is Tier 0 because `CHSH_TRIAL` cannot trigger a cert-insight event (proven, `chsh_trial_is_tier0`). Being Tier 0 doesn't make it free. Its cost is the encoded δ , so only the zero- δ case is uncharged. One guardrail: if any of x, y, a, b is not in $\{0, 1\}$, for example passing $x = 5$, the VM fires a bad-bits error step. The error flag latches, no counter is incremented, but the cost is still charged. The VM does not silently count invalid inputs as valid trials, and there is no path through the kernel that lets you pretend it did.

CERTIFY(δ): set the top-level certification flag.

Sets `vm_certified = true`, charges $S(\delta) \geq 1$ to μ , advances PC. This is the simplest cert-channel activation: you are saying “I, the program, certify that this execution has reached an attested state.” The flag is set. The cost is charged. There is no way to set `vm_certified = true` without this opcode, and this opcode always charges ≥ 1 .

REVEAL($m, bits, cert, \delta$): disclose a value.

Reveals *bits* bits from module *m* with certificate string *cert*. In the current kernel relation, the certificate string is carried but not checked; the priced event is the disclosure itself. A bit-commitment protocol is a two-phase scheme: first you commit to a value (seal it in an envelope), later you reveal it. The reveal step proves you didn't change your answer after seeing the other side's response. `REVEAL` is the disclosure half of that scheme. `REVEAL` charges $bits + S(\delta)$: the number of bits being disclosed, plus at least 1. The disclosed bit count is also recorded in the global μ -tensor at flat index $(m \bmod 16)$, where the module operand is read as the tensor direction index (the μ -tensor is indexed by direction pairs, not module slots; Section 11), marking which direction the disclosure landed in; the $S(\delta)$ part of the charge goes to the ledger alone. Revealing 0 bits still costs at least 1. Revealing 32 bits costs at least 33. Disclosing more costs more, and there's no plan where you reveal the whole module for the price of one bit. This is what makes the revelation requirement concrete: getting receipt-visible cross-correlations from binary game trials requires an explicit disclosure instruction, and that disclosure is priced by how much you reveal.

CHSH_LASSERT(δ): enforce the level-1 NPA-PSD boundary on the witness counters.

Reads the 8 `vm_witness` buckets, runs the integer-arithmetic check `column_contractive_check_witness` ($\mathbb{Z}$ arithmetic only, $\mathbb{Q}_1$ column-contractivity), and advances PC on pass or traps to `LASSERT_TRAP_PC` and latches `vm_err` on fail. Charges $S(\delta) \geq 1$ either way. A non-trapping step entails the witness-derived 5×5 NPA moment matrix is PSD (`chsh_lassert_no_trap_implies_npa_psd`). The four `CHSH_LASSERT_1AB` variants below extend this base check from the 5×5 level-1 matrix to the 9×9 level-1+AB matrix; the full treatment of the family is in the “CHSH check family” note after the opcode tally and in Section 18.

CHSH_LASSERT_1AB(δ): Q_{1+AB} -aware witness check at $\gamma = 0$.

Extends `CHSH_LASSERT` with the level-1+AB NPA condition. Reads the 8 `vm_witness` buckets, runs an integer-arithmetic check (Q_1 column-contractivity plus the SOS sum-of-squares condition $E_{00}^2 + E_{01}^2 + E_{10}^2 + E_{11}^2 \leq 1$ at $\gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = \gamma_5 = 0$), advances PC on pass with cert state intact or traps to `LASSERT_TRAP_PC` on fail. Charges $S(\delta) \geq 1$ either way. The bridge theorem `chsh_lassert_1ab_no_trap_implies_npa_psd_q1ab` proves a non-trapping step entails `npa_psd_q1ab = symmetric9  $\wedge$  PSD9` of the 9×9 NPA Q_{1+AB} moment matrix at the witness-derived rationals.

CHSH_LASSERT_1AB_G5($\delta, same_g5, diff_g5$): Q_{1+AB} check with the four-body moment γ_5 free.

Carries one γ -bucket pair $(same_g5, diff_g5)$ for $\gamma_5 = (same_g5 - diff_g5) / (same_g5 + diff_g5)$. Runs the γ_5 SOS check (a weighted 4-D Cauchy-Schwarz identity in cleared integer form), enforces $-Dg5 < Ng5 < Dg5$ as part of the bool decider. On pass, the bridge gives PSD9 at $(E, 0, 0, 0, 0, \gamma_5)$.

CHSH_LASSERT_1AB_G345($\delta, same_g3, diff_g3, same_g4, diff_g4, same_g5, diff_g5$): Q_{1+AB} check with three-body and four-body moments $\gamma_3, \gamma_4, \gamma_5$ free.

Carries three γ -bucket pairs. Runs the γ_{345} check: an augmented 2×2 Schur-slack lemma plus a 4×4 Sylvester PD condition on the difference matrix $H_{\gamma_{345}} = \det_M \cdot M_M - M_N$. On pass, the bridge gives PSD9 at $(E, 0, 0, \gamma_3, \gamma_4, \gamma_5)$.

CHSH_LASSERT_1AB_G12345($\delta, same_g1, diff_g1, \dots, same_g5, diff_g5$): full Q_{1+AB} check with all five γ moments free.

Carries five γ -bucket pairs. Runs the `sym6  $\rightarrow$  sym5  $\rightarrow$  sym4` Schur-cascade check on the supplied rational moments. A passing check implies PSD9 of the specified moment matrix. The four Q_{1+AB} opcodes are represented in the Kami model and its conditional kernel correspondence. The synthesis target contains the 47-opcode subset, including the base `CHSH_LASSERT`; the four higher-moment variants are outside that synthesis target.

MORPH_ASSERT (see Group 7): on success, it activates the cert-address channel and belongs conceptually here too, but is listed in the morphism group because it operates on a morphism ID.

12.7 Group 7: Categorical morphism operations (7 opcodes)

These are the opcodes that operate on the categorical morphism layer of the partition graph. Five of them (`MORPH`, `COMPOSE`, `MORPH_ID`, `MORPH_DELETE`, and `MORPH_TENSOR`) modify graph state. `MORPH_ASSERT` leaves the graph unchanged and writes the CSR certification channel on success; `MORPH_GET` reads graph state into a register. The predicate `is_classical_opcode` returns false on the graph-mutating and assertion operations, but classifies `MORPH_GET` as classical because its

visible register effect is included in that fragment. A Turing machine still has to simulate the graph on its tape to evaluate that read. A Turing machine can simulate programs using these opcodes by encoding the full VM state on its tape. The simulation can be honest (maintaining the μ -ledger cost semantics in the encoding) or dishonest (omitting that accounting). A universal machine's transition relation is indifferent to which, and that is the substrate-versus-program distinction Section 14 names: when a universal machine runs the simulation, A2 lives in the simulator program, not in the host's step rule. A Turing machine built with the ledger rule in its own transition table is a different case. It is an instance of the substrate, not a simulation of one.

A *morphism* is a record saying: "module A is related to module B in this specific, named way." Not just " A and B exist." Not just " A and B are connected by an edge." A morphism has a source module ID, a target module ID, coupling data (a list of (source-cell, target-cell) pairs plus a label string), and an is-identity flag.

The coupling field's operations are actual relational content, proven as such: `COMPOSE` computes the relational composition of two morphisms' coupling lists, pair (a, c) included if and only if there exists b such that (a, b) is in the first coupling and (b, c) is in the second, and this composition is proven correct against the standard relational-composition definition; `MORPH_TENSOR` concatenates two coupling lists. Neither operation is a programmer tag; both are data the kernel builds and proves properties about, at the level of the coupling type.

The reference semantics loads coupling data from VM memory. At `coupling_idx`, memory stores a capped pair count; the following words store source/target cells, followed by a length-prefixed label. `MORPH` decodes that block with `load_coupling_from_mem`, filters pairs to the source and target module regions, and adds the resulting morphism. A nonempty reachable example is covered by `test_morph_loads_nonempty_coupling_from_memory`: memory at 80–82 encodes the pair $(0, 1)$, two modules with regions containing those cells are created, and `MORPH` produces a morphism whose coupling list is $[(0, 1)]$. The hardware regression `test_morph_reads_upper_half_memory` exercises the corresponding extended-format path. The reference proof and hardware correspondence still have boundaries: the label is decoded by the reference VM but the finite hardware descriptor path has its own capacity, normalization, and retirement premises. The relational-composition and concatenation proofs therefore apply to actual nonempty reference data, while full reference-to-hardware equality remains conditional on the stated representation conditions.

Two states can agree on the selected classical projection and ledger while differing in their morphism maps. An observer restricted to that projection cannot distinguish them. The VM can query its graph via `MORPH_GET`; a conventional encoding that retained the graph could also support such a query.

Opcode	Effect	Cost
MORPH(r_d, s, t, c, δ)	Create morphism $s \rightarrow t$, write ID into r_d . Decode the serialized coupling block at c , filter it to the endpoint regions, and store the resulting coupling.	δ
COMPOSE(r_d, m_1, m_2, δ)	Compose $m_1 : A \rightarrow B$ with $m_2 : B \rightarrow C$ (requires target of m_1 = source of m_2). Coupling = relational composition.	δ
MORPH_ID(r_d, m, δ)	Identity morphism on module m .	δ
MORPH_DELETE(id, δ)	Delete morphism id . Error if not found.	δ
MORPH_ASSERT($id, prop, cert, \delta$)	Assert morphism id exists with property $prop$. Sets <code>csr_cert_addr</code> = <code>checksum(prop)</code> on success. Always charges $S(\delta) \geq 1$.	$S(\delta)$
MORPH_TENSOR(r_d, m_1, m_2, δ)	Tensor $m_1 \otimes m_2$: requires disjoint source/target regions. New morphism from union-source to union-target. Coupling = concatenation.	δ
MORPH_GET(r, id, sel, δ)	Read field (0=source, 1=target, 2=#coupling-pairs, 3=is-identity) of morphism id into r .	δ

MORPH_ASSERT deserves special attention. It charges $S(\delta) \geq 1$ regardless of whether the assertion succeeds. This is the sharpest instance of No Free Insight: the *attempt to claim structural knowledge costs something*, even when the claim is false. A failed assertion is not free. A step rule with no ledger has nowhere to record that charge. An ordinary machine that keeps a ledger in its own step rule can record it, and then it is an instance of the substrate. In the 51-opcode realisation it is enforced by the step semantics and tied back to the abstract theorem.

12.8 Why 51 opcodes?

The number 51 comes from what the kernel needs to do:

- Compute arithmetic and data: 14 ops (LOAD_IMM, XFER, ADD, SUB, MUL, AND, OR, SHL, SHR, LUI, XOR_LOAD, XOR_ADD, XOR_SWAP, XOR_RANK)
- Memory and control flow: 8 ops (LOAD, STORE, JUMP, JNEZ, CALL, RET, HALT, CHECKPOINT)
- I/O, heap, and tensor: 7 ops (READ_PORT, WRITE_PORT, EMIT, HEAP_LOAD, HEAP_STORE, TENSOR_SET, TENSOR_GET)
- Partition structure: 4 ops (PNEW, PSPLIT, PMERGE, PDISCOVER)
- Logic and certification: 3 ops (LASSERT, LJOIN, MDLACC)
- Witness and certification (general plus the full CHSH check family): 8 ops (CHSH_TRIAL, CERTIFY, REVEAL, CHSH_LASSERT, CHSH_LASSERT_1AB, CH

SH_LASSERT_1AB_G5, CHSH_LASSERT_1AB_G345, CHSH_LASSERT_1AB_G12345)

- **Categorical morphisms:** 7 ops (MORPH, COMPOSE, MORPH_ID, MORPH_DELETE, MORPH_ASSERT, MORPH_TENSOR, MORPH_GET)

$14 + 8 + 7 + 4 + 3 + 8 + 7 = 51$. Of the 51, 47 are synthesized into the FPGA bitstream; the four Q_{1+AB} check opcodes (CHSH_LASSERT_1AB and its three γ -extended variants) live in the Kami HW abstraction with kernel-equivalence proven, but not in the synthesized bitstream (see the “CHSH check family” note just below for the silicon-budget reason).

The CHSH check family: CHSH_LASSERT plus four Q_{1+AB} extensions. The base opcode CHSH_LASSERT μ_delta reads the eight buckets of `vm_witness` directly, computes the integer-arithmetic check `column_contractive_check_witness` (defined using only $\mathbb{Z}$ arithmetic on the same/diff counts), and either advances the program counter on success or traps to LASSERT_TRAP_PC and latches `vm_err` on failure. The cost is $S(\mu_delta) \geq 1$ regardless of outcome. Mandatory-floor discipline: you pay to check, even to fail. The bridge theorem `chsh_lassert_no_trap_implies_npa_psd` proves that any no-trap, no-err-flip execution of CHSH_LASSERT implies PSD of the witness-derived, fixed 5×5 real moment matrix. The condition is NPA’s level-1 PSD test, and the name says only that. The kernel does not construct a quantum state or observables. The chain in plain order: the integer check on the witness counters passes; that fact lifts to real-valued column-contractivity by `column_contractive_check_witness_sound`; column-contractivity is biconditional with PSD of the specified matrix by `column_contractive_iff_npa_psd`. Three theorems. No physics input. The opcode is the place in the kernel where the algebra of Section 18 becomes something the VM actually refuses to do, not just declines to recommend.

The four Q_{1+AB} extensions CHSH_LASSERT_1AB (tag 0x2F, $\gamma = 0$), CHSH_LASSERT_1AB_G5 (tag 0x30), CHSH_LASSERT_1AB_G345 (tag 0x31), and CHSH_LASSERT_1AB_G12345 (tag 0x32) lift the same discipline from the level-1 5×5 matrix to the level-1+AB 9×9 formal matrix. Each runs its own integer-arithmetic Schur-cascade check on `vm_witness` plus the supplied γ -bucket pairs, advances on pass or traps on fail, and charges $S(\mu_delta) \geq 1$. The four bridge theorems (one per slice, named `chsh_lassert_lab_*_no_trap_implies_npa_psd_qlab`) deliver `npa_psd_qlab = symmetric9  $\wedge$  PSD9` on the witness-derived rationals. The predicate is PSD of that formal matrix. It is not a hardware or Hilbert-space construction. The mathematical detail of the four-slice Schur cascade is in §18’s Q_{1+AB} subsection below.

The instruction set separates ordinary register and memory operations from graph operations, witness recording, and designated charged checks. CERTIFY activates `vm_certified`; MORPH constructs morphism relationships; CHSH_TRIAL records

binary-game outcomes; and the CHSH assertion family tests the corresponding arithmetic constraints. Their costs and effects are specified in `VMStep.v`.

13 What a categorical morphism is: the category theory, from scratch

Category theory from scratch. I didn't know what it was when I started this project.

This is the structure layer of the VM build. It is the kind of structure the VM happens to carry. The model in Part I doesn't ask for it, and none of the abstract theorems there mention it.

13.1 The idea of a category

A category is the part of mathematics that takes the relationships between things as seriously as the things, and then quietly decides the relationships matter more. It is built from two ingredients. There are objects, the things. And there are arrows between them, the relationships. That is the entire inventory, and it looks thin until you notice what got promoted. In most of mathematics the objects are the citizens and the maps between them are second-class traffic. A category turns that around. The arrows are first-class, and nearly everything you can say inside a category you say by pointing at arrows, not by reaching inside objects.

The objects can be anything at all, and that is at once why category theory reaches as far as it does and why it feels slippery the first time through. Sets can be the objects. So can vector spaces, or topological spaces, or any kind of mathematical thing that has structure-respecting maps between its members. The theory does not care what the objects are made of. It cares only how they connect. In the realisation here the objects are modules, the named pieces a partition cut the state into, and the arrows are the relationships between those pieces.

Arrows go by a second name, morphisms, and the two words are interchangeable; I use both. An arrow runs from one object to another, written $f : A \rightarrow B$ and read " f , from A to B ." The thing everyone imports from school and has to set down here is the assumption that an arrow is a function. It does not have to be. An arrow is whatever the category accepts as a structure-respecting way to get from A to B , and in the realisation here it is something more pointed than a function. A morphism is a claim: the assertion that these two modules are related, and here is the type of the relationship.

Two laws must hold:

1. **Composition:** If $f : A \rightarrow B$ and $g : B \rightarrow C$, there must exist a composite $g \circ f : A \rightarrow C$. This is what `COMPOSE` implements.
2. **Identity:** For every object A , there must exist an identity arrow $\text{id}_A : A \rightarrow A$. This is what `MORPH_ID` implements.

And composition must be associative: $(h \circ g) \circ f = h \circ (g \circ f)$.

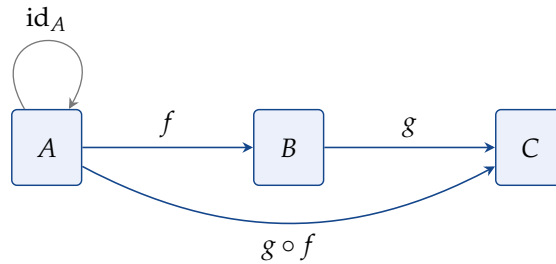


Figure 5: Arrows chain. f goes from A to B , g goes from B to C , and composing them gives one arrow straight from A to C . Every object also has an identity arrow looping back to itself. `COMPOSE` builds the long arrow; `MORPH_ID` builds the loop.

Why composition? Because relationships have to chain. A to B , B to C , therefore A to C . Without that, every multi-hop path has to be named individually, and the structure collapses into a pile of unconnected pairs. A category without composition is a phone book without a search function.

Why identity? Every object needs a self-loop, even the boring ones. A module with no connections to anything else still has to exist somewhere in the framework. `MORPH_ID` is the instruction for that: it builds an $A \rightarrow A$ morphism, stores an empty coupling, and flags it `is_identity`. That flag is load-bearing, and here is why it has to be. An empty coupling would *annihilate* under relational composition (compose anything with the empty relation and you get the empty relation back, not the thing you started with), so the identity cannot be carried by the coupling data. It is carried by the flag, which `COMPOSE` honors: composing an identity-flagged arrow with f returns f 's coupling unchanged, so $\text{id}_A ; f = f$ and $f ; \text{id}_B = f$ hold at the coupling level (`graph_compose_morphisms_coupling`; the matching hardware path is `driven_step_compose`). Drop the flag and the identity law breaks: composing with an identity would wipe out the very thing you composed it with.

Why associativity? Parenthesization should not matter. $(h \circ g) \circ f$ and $h \circ (g \circ f)$ both spell the same three-hop path $A \rightarrow B \rightarrow C \rightarrow D$. If they came out different, two programs that built the same chain in different orders would produce two different morphisms. That is a bug. The chain is the thing. How you assembled it is not.

Here is exactly what Coq checks, on the stored morphism records. Morphisms carry coupling data, a list of (source-cell, target-cell) pairs plus a label. A “cell” is a natural number indexing into a module’s memory region. A coupling pair (a, c) says: cell a in the source module is related to cell c in the target module. The coupling is the *relational content* of the morphism, the actual evidence of which parts of one module correspond to which parts of another. `COMPOSE` produces a new morphism whose coupling pairs are the relational composition of the two inputs: (a, c) is included when there exists an intermediate cell b such that (a, b) is in the first coupling and (b, c) is in the second.

Standard relational composition. The one exception is the identity short-circuit from the previous subsection: if either input is flagged `is_identity`, `COMPOSE` returns the other input's coupling unchanged. `graph_compose_morphisms_coupling` proves both cases. I checked that in Coq, because the identity law holding “obviously” is exactly the kind of thing my eye waves through right before it turns out to be false. Associativity is `morph_graph_compose_assoc`: whenever both groupings are defined, they give the same set of pairs. Same set, not necessarily the same list order. Calling the whole thing a full category takes more: explicit membership for objects and for arrows between them, and equality laws. IDs, labels, and list order don't supply those on their own. The identity flag is how the implementation handles identities. By itself it doesn't prove that every isolated object sits in a fully specified category.

13.2 Why modules as objects

In the concrete kernel, the objects of the category are the modules of the partition graph, and that choice is less arbitrary than it looks. A module is a named piece of the computation, the place where a partition cut the state and put a fence around it. Drawing that fence is the act that creates an address. Once a piece has a name it owns a region of memory or state, it may carry certified axioms, it carries a local 4×4 μ -tensor, and, the part that earns it the job, it becomes a thing the rest of the structure can point at. That is the whole reason modules are the bookkeeping unit: a relationship has to attach to something, and the only somethings on offer are the named pieces. An unnamed piece, as far as the category is concerned, is not there at all. There is nothing to relate it to and nothing to relate it from. So everything structural hangs off a module. That isn't a convention. A module is the only kind of thing the category can see.

A morphism between two modules is a claim with its evidence stapled on: “there is a named relationship between A and B , and here is the explicit coupling data that says which parts line up.” That is the move that turns the relationship into a first-class object instead of a remark in a comment somewhere. `COMPOSE` chains two of these by the relational composition already spelled out above, and an identity-flagged input still falls to the short-circuit from before; I will not re-derive either one here. What is worth stopping on is that the relationship does not dissolve into the modules it connects once it is built. It stays in the graph as a thing you can hold and ask questions of: via `MORPH_GET` (selector 2) you read back the coupling-pair count of any morphism, after the fact, straight off the state. The relationship is not lost in the act of relating. It is its own object, carrying a count you can go and demand back.

What the coupling pairs represent is up to the programmer. They could encode:

- Module B refines module A (which cells in A map to which cells in B)
- Module A entangles with module B (which cell-pairs are correlated)
- Module A simulates module B (which transitions/outputs correspond)

The kernel enforces the relational composition correctly. The interpretation is yours.

13.3 Why this is not just “extra state”

Here is the question I chewed on for a long time. If you can just bolt extra registers onto a classical machine and stash morphism IDs in them, why does the categorical layer matter at all?

The answer is in the `MORPH_ASSERT` opcode and in the Categorical Separation Theorem.

Extra registers do not have provenance. You can write anything into a register. You can write 42 into register 5 and claim it is a morphism ID. But the VM checks: does morphism 42 actually exist in the graph, put there by a `MORPH` or `COMPOSE` instruction that genuinely ran? `MORPH_ASSERT` looks the morphism up by ID and fails (setting `vm_err`) if no such morphism is there. Existence is all it verifies; the source and target are not arguments it checks against, though you can read them back separately with `MORPH_GET` (selectors 0 and 1). And `MORPH_ASSERT` costs $S(\delta) \geq 1$ even if it fails.

Building the morphism chain by calling `MORPH`, `COMPOSE`, and `MORPH_ID` is free when those instructions carry $\delta = 0$. Those ops don’t have a mandatory positive floor. What costs is `MORPH_ASSERT`: the act of asserting that the morphism exists and activating the cert-address channel. That always costs $S(\delta) \geq 1$. You can build the chain for free, but you cannot assert it into the cert-address channel for free.

The graph is structure the machine holds outright. A trusted trace records every change to it and what each assertion attempt cost. `MORPH_ASSERT` checks that the morphism exists and stores a checksum of the label. It doesn’t check what the label says, and it doesn’t check the certificate you hand it. The separation theorem in the next section says exactly what the chosen shadow throws away.

14 Categorical separation: what the classical shadow drops

Here is the smallest version of the whole picture. Two states, side by side. Every classical field matches: registers, memory, μ , program counter, error flag, certification flag. One of them holds an identity arrow in its graph. The other holds nothing. Through the six-field window they are the same state. They aren’t.

That is a fact about this particular window. Section 3 already pulled the transition law apart from what an observer sees. A different representation could keep the graph. This section proves what this window, as chosen, drops.

Theorem 14.1 (Categorical Separation). *There exist states s_1 and s_2 such that:*

- $s_1.\text{vm_regs} = s_2.\text{vm_regs}$ (identical registers)
- $s_1.\text{vm_mem} = s_2.\text{vm_mem}$ (identical memory)

- $s_1.vm_mu = s_2.vm_mu$ (identical μ)
- $s_1.vm_err = s_2.vm_err$ (identical error flag)
- $s_1.vm_pc = s_2.vm_pc$ (identical program counter)
- $s_1.vm_certified = s_2.vm_certified$ (identical top-level certification flag)

yet their morphism graphs differ. In the concrete witness, one state has an identity morphism in `pg_morphisms`; the other has no morphisms.

Construction. Build s_1 with `pg_morphisms = [(0, id)]`. Build s_2 with `pg_morphisms = []`. Set every shadow field the same: registers, memory, μ , PC, error flag, and `vm_certified`. The equality proof is by reflexivity on those fields. The graph distinction is by list constructor mismatch: a one-element morphism list is not the empty list. $\square$

Two witness states, side by side. What the six-field shadow sees. What the graph state actually holds.

	State s_1	State s_2
<code>vm_regs</code>	<code>[]</code>	<code>[]</code>
<code>vm_mem</code>	<code>[]</code>	<code>[]</code>
<code>vm_mu</code>	<code>0</code>	<code>0</code>
<code>vm_pc</code>	<code>0</code>	<code>0</code>
<code>vm_err</code>	<code>false</code>	<code>false</code>
<code>vm_certified</code>	<code>false</code>	<code>false</code>
<code>pg_morphisms</code>	<code>[(0, id)]</code>	<code>[]</code>

What this witness shows. The two states above are not contrived. They are the simplest possible six-field shadow collision under the Thiele projection: identical registers, identical memory, identical PC, identical error flag, identical μ , identical `vm_certified`. Every observer restricted to that projection says they are the same state. The morphism graph says they are not. This is a witness inside the model that the chosen projection drops information. On its own it doesn't compare substrates: a similar projection on any other substrate makes the same shape. What ties it to the rest of the argument is the cost floor, the irredundancy result, and the fact that A2 can be a law in the step instead of a check run afterward. The uniqueness theorem says the ledger is the only count that fits its schedule. Pricing and irredundancy say why those fields, and that law, are the ones worth keeping.

What the witness does not prove on its own. The obvious move is to reach for more room: bolt on a register, a tape region, any auxiliary slot, and surely the Turing machine wriggles out. It doesn't. It rebuilds the same witness one size up. The augmented machine still has a non-identity projection (drop the added slot from the

projected fields) and the same collision shape on the augmented states. The collapse the shadow lemma exhibits happens no matter how much room you give it, because it is baked into the construction. That cuts both ways. It is also why the witness, on its own, says nothing about whether a Turing machine can hold the graph. It can, encoded on tape. The question that matters lives somewhere else, and the next paragraph says where.

The deeper question is where the cost rule lives, and that is the substrate-versus-program distinction. Section 3’s subsection “What is canonical, and under which conditions” says it once and explicitly: a fixed machine can enforce an invariant through its own transitions. That includes a Turing machine. Give its tape a ledger track and a certification cell, and write a transition table that never sets the cell without bumping the ledger. That machine obeys A2 in its step relation. It is an instance of `CertificationSystem`, not a counterexample to it. What a Turing machine does not do is enforce A2 on the programs it runs. A universal machine simulating another machine carries the simulated certification bit as data on its tape, and a buggy simulated program can flip that bit without paying; the host’s step relation has no opinion about it. The Thiele VM is in exactly the same position when it simulates something else. The kernel theorems are about the designated predicate of the machine doing the stepping. That is a statement about which predicate a step rule prices. It is not a claim that Turing machines cannot price one.

Connecting to the universal NoFI. Combine this within-model witness with `universal_nfi_any_substrate` and `thiele_represents_simulating_cert_system` and you get the substrate-level statement. Take a certification system that satisfies A2 and supplies a structure-preserving translation into the Thiele VM, packaged as a `SimulatingCertificationSystem`, the record that carries the translation together with a proof it commutes with steps. Its certifications then transport: an uncertified-to-certified trace costs at least 1 in the source system, and the embedded execution lands in a Thiele-certified state. The translation is a hypothesis the record supplies, not something the theorem conjures for an arbitrary A2 system; the only embedding the kernel builds outright is the Turing one (`thiele_simulates_turing`), and the cost floor for an A2 system with no embedding at all is the direct content of `universal_nfi_any_substrate`. Its axiom closure is empty, nothing named or assumed. What it does take is written in its type: the A2 rule, and a run that starts uncertified and ends certified. The witness theorem shows what the Thiele projection drops within the model. The substrate-independent theorem shows that any substrate satisfying A2 inherits the μ -price on certification. Together they say: the cost floor is forced wherever A2 is accepted. In Thiele, the dropped information sits in the state as a primitive field.

What would falsify this section. Not a clever function on the shadow. The six shadow fields (`vm_regs`, `vm_mem`, `vm_mu`, `vm_pc`, `vm_err`, `vm_certified`) of s_1 and s_2 are literally equal, so no function of them can tell the two apart. That part is reflexivity.

The part that can break is the model. If `shadow_proj` kept any piece of the graph, or if the probe touched `pg_morphisms`, `shadow_strictly_lossy` would stop compiling. The witness pair is small enough to check by hand. That is the seam this section breaks at, and the place to go looking for it is Section 23.

| Plain reading. Same witness pair as Categorical Separation. The probe is a classical opcode that touches only the classical shadow. D3 says it leaves the morphism graph alone. So the graph-level difference survives the probe. The difference between s_1 and s_2 isn't an artifact of where they started. It persists through ordinary classical computation.

Theorem 14.2 (Shadow Separation). *There exist two states with the same classical shadow but different morphism graphs, and a legitimate graph-preserving probe after which the morphism-level distinction remains.*

Proof. Take s_1, s_2 to be the Categorical-Separation witness pair: s_1 has `pg_morphisms` = `[(0, id)]`, s_2 has `pg_morphisms` = `[]`, and the six classical-shadow fields agree. So `shadow_proj(s1)` = `shadow_proj(s2)` but `s1.vm_graph` $\neq$ `s2.vm_graph`.

The probe is the single-instruction program `[instr_add0000]` (ADD with $\delta = 0$). This instruction is classical: `is_classical_opcode(instr_add0000)` = `true`. By D3 conservativity, the probe leaves `vm_graph` untouched on both s_1 and s_2 ; the Coq proof checks it directly on the two witnesses (`probe_preserves_graph_A`, `probe_preserves_graph_B`). After the probe the morphism lists are still `[(0, id)]` and `[]`. They still differ. □ □

The probe used is deliberately boring: `ADD 0 0 0 0`. It is not a morphism operation. That is the point. The theorem is not relying on a failing probe to manufacture a difference. The difference is already in the structural state, and ordinary classical projection cannot see it.

Corollary 14.3 (Classical observer is blind). *No function of (registers, memory, μ , PC, error, certified) can separate all VM states. The classical shadow is strictly lossy.*

Corollary 14.4 (Thiele strictly extends classical). *Every program in the classical fragment, meaning every program restricted to `is_classical_opcode`, leaves `pg_next_id` unchanged. There exist VM programs that change `pg_next_id` via `PNEW`. Therefore Thiele strictly extends the classical fragment: from the same starting state, it can reach and probe states that no program in that fragment can reach. A universal Turing machine can simulate the full VM state by encoding it on tape, but a faithful simulation has to preserve the μ -ledger cost. Drop the accounting and it is no longer simulating the substrate honestly.*

The embedding and projection results say which views survive and which structural distinctions fall away. A projection that sends different graphs to the same place can't give you back the original graph on every state. That doesn't rule out an encoding that keeps the whole state. It doesn't make these particular VM opcodes necessary in every model, either.

15 The knowledge receipt: accounting and structural witnesses

Four short acts with one receipt. They show the difference between a morphism that lives in the graph and a register that just holds its number, and they show the trace recording what an assertion attempt cost, even when it fails. None of this vouches for a receipt somebody else hands you, and none of it makes a property label true.

Act 1: The attempt costs, even when it fails.

Run `MORPH_ASSERT(k, "test", "", 0)` where morphism *k* does not exist. What happens:

- `vm_err` becomes true (the assertion failed, morphism *k* not found).
- `csr_cert_addr` stays 0, the cert-address channel is NOT activated on failure.
- μ increases by $S(0) = 1$ regardless (the attempt cost something).

The VM charges for the failed existence check. That's what `MORPH_ASSERT` guarantees and all it guarantees: it checks the morphism exists, turns the property text into a checksum label, and leaves the certificate string unchecked.

Act 2: Build the chain.

Run this sequence:

$$\begin{aligned} & \text{PNEW}(A), \text{PNEW}(B), \text{PNEW}(C) \\ & \text{MORPH}(f, A, B), \text{MORPH}(g, B, C) \\ & \text{COMPOSE}(h, f, g) \end{aligned}$$

Then `MORPH_GET(r0, h, source)` and `MORPH_GET(r1, h, target)`. The resulting state has $r_0 = A$, $r_1 = C$, and morphism $h : A \rightarrow C$ exists in the morphism map. With $\delta = 0$ on each build step, $\mu = 0$. The morphism exists, but no cert-address commitment has been paid for yet.

Act 3: Assert.

Run `MORPH_ASSERT(h, "composed", "", 0)`. Morphism *h* exists and the property string "composed" has a non-NUL byte. The assertion succeeds. `csr_cert_addr` goes to the ASCII checksum of "composed" (nonzero). Cert-address channel activated. μ goes up by $S(0) = 1$.

(Why does the property string need a non-NUL byte? Because `csr_cert_addr` stores the ASCII checksum of the property string, a plain byte-sum. The empty string sums to 0, and so would an all-NUL string, leaving the channel at 0. Not activated. Any string with a non-NUL byte gives a nonzero checksum and works. “composed” is just an example label.)

The trace records a successful existence check and what it cost. The ledger can’t go down under the VM’s step rule. The total by itself doesn’t say which assertion happened or where. For that you need the trace. And keeping honest books is a different job from catching an adversary who hands you a different history.

Act 4: Equal observations can hide different graphs.

Snapshot Program A at the end of Act 2, before Act 3’s `MORPH_ASSERT`. Its six-field classical shadow is fixed: `MORPH` and `COMPOSE` wrote the new morphism IDs into their destination registers and `MORPH_GET` read *A* and *C* back into r_0 and r_1 ; the program counter sits at 8; $\mu = 0$; error clear; uncertified.

Program B forges that shadow without building anything. It never runs `MORPH` or `COMPOSE`. It just `LOAD_IMM`s the same values into the same registers (the morphism IDs are deterministic, handed out in order (1, 2, 3) by `pg_next_morph_id`, which boots at 1, so there is nothing secret to reproduce) and pads with $\delta = 0$ `CHECKPOINT` no-ops until its program counter also reads 8. Now every field the shadow keeps matches Program A’s, $\mu = 0$ on both sides. A classical observer cannot tell them apart. That two states can share the entire shadow while differing in the graph is the Categorical Separation Theorem (Section 14); Program B is the forged half of exactly such a pair.

The whole difference lives in the graph. Program A built morphism *h* with `MORPH` and `COMPOSE`, so *h* sits in its `pg_morphisms`. Program B’s register *h* holds the number 3, but its graph is empty: the value is a forged token, not a graph entry. Apply `MORPH_DELETE(h)`: Program A succeeds, because there is a morphism to delete. Program B fails and sets `err`, because there is not. `MORPH_DELETE` reads the graph, and the graph is the one thing `LOAD_IMM` cannot forge.

The ledger is what the schedule added up to. The graph carries the structure the shadow leaves out. The trace tells you where things came from, as long as the verifier has a history it can trust.

16 The VM, built three times: proof, runner, hardware

Three scaffolding layers wrap the substrate: a Coq kernel, an OCaml runner extracted from it, and a Kami RTL design tied back to it. None of them IS the substrate. They are how the substrate gets to live on a chip, on a laptop, and inside a proof checker. Here is where proof stops. Coq proves that the Kami model steps the way `vm_apply` does (`driven_step_wf`, `driven_trace_commutates`), under the

WFD_{Driven}Precondition and WFD_{Driven}Run premises. Separate theorems cover retirement under the real priority scheduler. That is the proof. Past the Kami model, the Bluespec compiler, the printer, synthesis, place-and-route, and the bitstream are tested or trusted, not proved. The extracted OCaml runner is checked by parity tests. Section 16 has the table, edge by edge. The seams are named in the source, out in the open, so you can walk to any one of them and see what it actually rests on. I would rather hand you the exact seam where proof stops and tested correspondence starts than dress the whole stack up as one seamless theorem it is not.

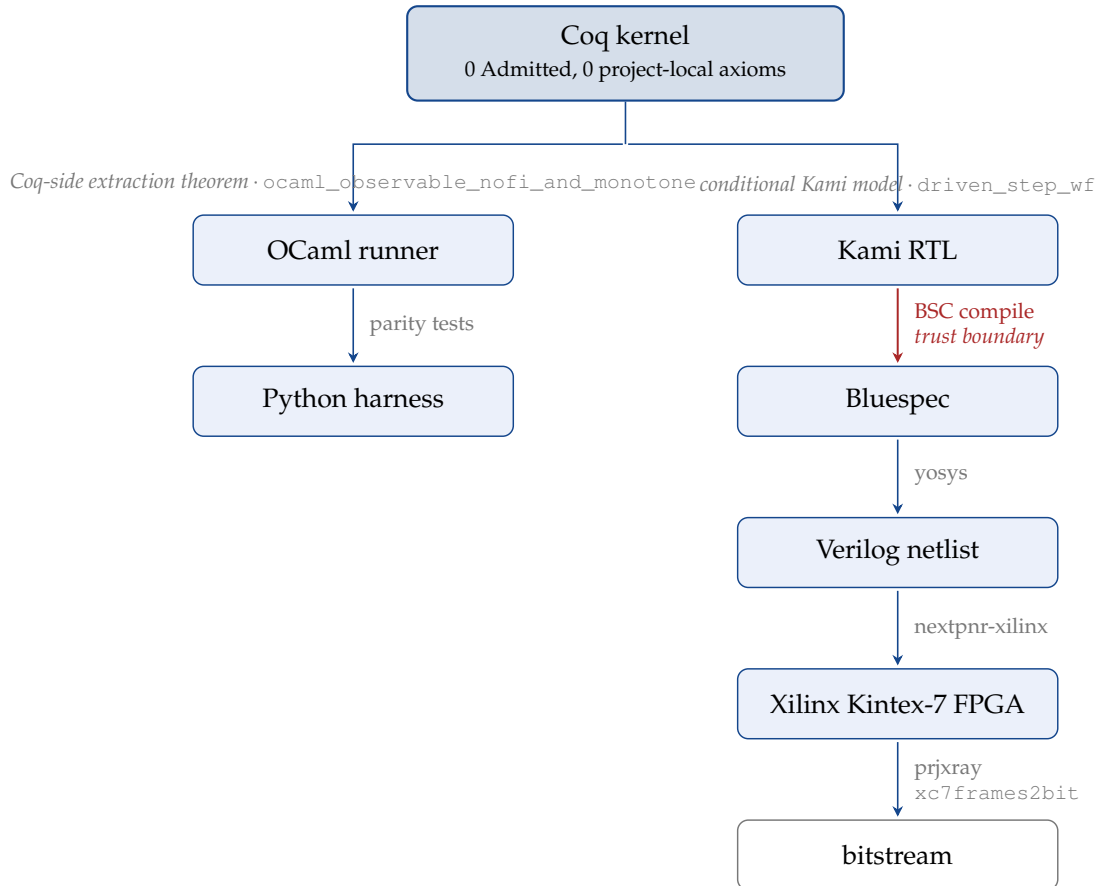


Figure 6: Three ways the substrate gets built. The Coq kernel is the source of truth. The left branch goes to a program you can run; the right branch goes to silicon. Blue arrows are either Coq-model correspondences, conditional, with the theorem named on the edge, or tests. Red arrows and the tool steps below them are places where a translation or a tool is trusted, unless a separate theorem or artifact says otherwise. Nothing here claims one proof running end to end from Coq to the bitstream.

16.1 Layer 1: The Coq kernel

Coq is where the substrate gets pinned down. The kernel defines:

- The `VMState` record (all 12 fields, as in Section 11).
- The `vm_instruction` type: an inductive type (Section A) with 51 constructors, one for each opcode, each carrying the right argument types.

- The `vm_apply` function: takes a state and an instruction, returns a state. It is total (Section A): every input pair produces a defined output, including the error case.
- All the theorems: No Free Insight, μ -monotonicity, μ -initiality, categorical separation, Turing strictness, and the rest.

The Coq kernel operates mostly over natural numbers, but `VMState` is not an unbounded mathematical fantasy. `REG_COUNT = 16`. `MEM_SIZE = 128`. Both are matched against the synthesised RTL. The accessors wrap indices through those sizes (modular arithmetic, Section A). Both constants are parametric in the proofs (Section A: same proofs, any value), so nothing in the formal chain depends on the specific numbers. Register and memory writes are truncated through the kernel’s 64-bit word helpers to keep the math clean. The μ counter is a plain natural number with no fixed upper bound. The RTL instantiates a finite 32-bit datapath (the part of the chip that moves and operates on data, not the control logic) and bounded tables.

Arithmetic near a word boundary is where a bridge claim can quietly stop meaning what it says, so state it concretely. `ADD` on the register values $2^{32} - 1$ and 1: 32-bit modular arithmetic wraps the sum to 0; the kernel’s 64-bit word truncation instead gives 2^{32} , a different value. This is not a hypothetical; it is exactly the gap between the RTL’s 32-bit datapath and the kernel’s 64-bit word helpers named in the paragraph above. Values whose sum stays under 2^{32} agree on both sides; values whose sum crosses that boundary do not, and the bridge domain has to say which side is authoritative for such a value or exclude it. The μ counter’s own range needs the same treatment rather than an unqualified “unbounded natural on both sides” reading: the kernel’s μ is a plain unbounded natural number, but the RTL ledger is a fixed-width register, so a program that accumulates enough μ to exceed that width is outside what the current hardware correspondence establishes, the same way an oversized `ADD` is.

Zero Admitted means: no step in any theorem proof was asserted without derivation. Coq checked everything. I did not.

16.2 Layer 2: The OCaml extracted runner

Here is the trick that makes the OCaml layer worth having at all. A Coq definition is two things wearing one coat: a recipe for computing an answer, and a proof that the recipe behaves. Extraction takes the coat off. It keeps the recipe, in this case rendered into OCaml, an ordinary programming language you can compile and run like any other, and it drops the proof terms on the floor, because a CPU has no use for them. So `vm_apply` (the function that defines what every opcode does) walks out the door as an OCaml function that computes the same answer on the same inputs. The proofs do not come along. The algorithm does. That is the whole point: I get a binary I can actually run, descended from the exact definition the theorems were proved about, and the descent is mechanical rather than me retyping the semantics into a second

language and praying. The extracted runner executes all 51 opcodes from the Coq semantics, morphism operations and `CHSH_LASSERT` included.

The trust boundary at this layer is named explicitly. The kernel proves formal facts about the Coq-side observable (`shadow_to_eo (vm_apply s i)` strips a state down to its observable fields after one instruction): exact μ accounting and μ never decreasing (`ocaml_observable_nofi_and_monotone`). It also names the external runner boundary. The Coq lemma `ocaml_runner_observable_defined` only says the observable is defined, which is true of any Coq function. It is not an unproved *Hypothesis* (Section A), and it is not a cross-language equivalence theorem. The cross-language claim, that the built OCaml binary returns the same observable as the Coq semantics on every input, is checked by the parity test suite (Section A: run both on the same battery, verify they agree). I do not count that as a kernel theorem. It is an audited boundary, backed by tests, named in the source.

16.3 Layer 3: The Kami RTL hardware

The whole reason this layer exists is to refuse to retype the design in a second language. Kami lets you write a hardware specification inside Coq itself and then extract it, the same move the OCaml runner used, except the target is not software, it is a chip. The first stop is Bluespec SystemVerilog, a hardware description language pitched high enough that you say what the hardware should compute rather than which wire goes where: the registers it holds, the rules it runs, and the conditions under which each state transition is allowed to fire. The Bluespec compiler then drops a level and emits standard Verilog RTL, the gate-and-register description that synthesis tools chew into actual logic, either on an FPGA, a chip you can reconfigure after it leaves the factory, or an ASIC, a chip burned for one design and never reprogrammable. So a Kami module is the same idea descending one more flight of stairs: the finite hardware realisation, pinned down in Coq, walks all the way to gates without anyone hand-translating it on the way, and the Kami extractor runs it through the pipeline above.

The hardware has:

- 16 registers, each 32 bits wide (`RegCount = 16`; kernel proofs parametric)
- 128-bit instruction encoding (`InstrSz = 128`)
- 128 words of data memory and 128 instruction words (`MemSize = 128`; kernel proofs parametric in `MEM_SIZE`)
- bounded instruction transport and execution state
- 64 partition module slots (`PTableSz = 64`)
- 16 slots in each of the morphism, coupling, formula, certificate, descriptor-metadata, and coupling-pair tables
- An on-chip `LASSERT` FSM for formula verification (with 64-entry inlined backing buffers)

- The μ ledger updated atomically with each executed instruction (no separate parallel datapath; the cost arithmetic happens inside the same `step` rule)

The Verilog is the hardware artifact. The FPGA is whatever I could put it on for free. The current target is the Xilinx Kintex-7 `xc7k325tffg900-2` on the Diligent Genesys 2: the smallest fully-open-toolchain part the design still fits comfortably after the multi-cycle `CHSH_LASSERT` witness checker landed. Section A explains the synthesis pipeline I use: `yosys` does the gate-level synthesis, `nextpnr-xilinx` does the place-and-route, and `prjxray's xc7frames2bit` produces the bitstream the chip reads to configure itself.

CI (Full) runs synthesis, simulation, and bitstream generation. I don't own a Genesys 2. CI routes the design and writes the bitstream, and no board has loaded it yet. A resource, timing, or bitstream report only counts for the exact RTL and tool versions recorded with it. Compiling the Coq proofs doesn't measure changed RTL. Any physical claim about the current design needs freshly regenerated pipeline artifacts tied to the source identity in Section 21.1.

The kernel proofs are parametric in `REG_COUNT` and `MEM_SIZE`, so the same proofs apply at any size constants the design picks. This part was a proof of existence on hardware. It was never a design constraint. The size constants (16 registers, 128 data words, 128 instruction words) were chosen small enough to fit on a free-toolchain FPGA. The design scales to whatever size you actually want.

47 of the 51 opcodes are implemented in the synthesized RTL: the Kami core has hardware logic for those 47, and the synthesised Verilog carries them through to the FPGA bitstream. The remaining four (the Q_{I+AB} check-opcode family `instr_chsh_lassert_lab`, `instr_chsh_lassert_lab_g5`, `instr_chsh_lassert_lab_g345`, `instr_chsh_lassert_lab_g12345`) are defined in the Kami HW abstraction with kernel-equivalence proven, and they execute in the OCaml runner and Python harness, but they are not synthesized into the FPGA bitstream. Each of them would need its own wide-arithmetic FSM, and I haven't built that hardware. The OCaml/RTL opcode-set parity test tolerates exactly these four. Which opcodes are in hardware isn't the real question. The real question is which opcodes have a completed formal *bisimulation proof*.

The theorem `KamiHW.GraphReconstructionBridge.driven_step_wf` compares the Gallina function `kami_step` with `Kernel.SimulationProof.vm_apply` through `abs_full_snapshot`. Its premise is `WFDDrivenPrecondition ks i`. The trace theorem `driven_trace_commutates` requires `WFDDrivenRun fuel trace ks`, which checks that premise at every visited state. This is the intermediate Coq model surface, separate from physical clock transitions in `ThieleCPUCore.v`.

- The default branch of `WFDriivenPrecondition` is `True`. It adds no instruction-specific premise to this intermediate model equality.
- `CALL` and `RET` require `WellFormedSnapshot`, with a PC bound for `CALL`. `CHSH_TRIAL` requires Boolean operands. `LASSERT` requires agreement with its memory header. Tensor access requires valid indices, and `TENSOR_GET` also requires module existence.
- `MORPH` requires `extended_hw_invariant` and existing source and target modules. `COMPOSE`, `MORPH_TENSOR`, and `MORPH_GET` require the extended invariant. `MORPH_ID` uses the empty-descriptor invariant and module existence. `MORPH_ASSERT` and `MORPH_DELETE` require morph-table well-formedness.
- `PNEW` requires a nonempty normalized region and fresh tensor state. `PSPLIT` and `PMERGE` carry the partition, capacity, freshness, and arithmetic conditions in their source branches. These per-operation conditions must be established along the actual run; reset alone does not provide unlimited table capacity.

The actual CPU rules also have executable Kami semantics. `CoreRules` places all twelve rules in the supported linear-action fragment; `CoreExecution` proves that finite traces of its first-enabled selector are actual Kami multi-step executions, with a firing bound and an early-stop characterization by absence of enabled rules. Reset has an enabled selected rule. `CoreTyping` proves preservation of the complete register-name/kind schema from reset, while `DispatchReset` states thirteen typed initialization facts. So the selected execution and the register types are pinned down. Fairness isn't, and neither are the invariants about what gets allocated along a reachable run, the full abstract observation relation, or meaning surviving the trip through the RTL toolchain.

Carrying results all the way to finite RTL takes more: matching word ranges, memory addressing, allocation capacity, reset state, and a view taken when an instruction retires. The word-boundary example above is one case of it. Exact equality needs a range restriction or a proved modular abstraction, and the finite ledger needs a bound that rules out overflow. Multi-cycle operations need a retirement relation that accounts for the FSM's in-between states.

The coupling FSM reads serialized `MORPH` pairs and implements pair copying and relational joins for the other two operations. Descriptor zero remains reserved for empty/identity couplings. The finite pair table holds 16 entries; insufficient space raises an error, and empty results need no pair entries. The RTL regression tests inspect the backing pair tables through the testbench. They do not establish full-state refinement: hardware lacks the software coupling-label field and module-membership filter, and the full endpoint/region representation contract remains an obligation for a broader physical correspondence claim. The RTL normalizes the newly allocated pair range, retaining each pair's last occurrence as the software's `nodup` does. Allocation

first requires room for the raw intermediate list; a normalized result fitting the table does not guarantee admission when the raw list exceeds it.

The inventory arithmetic `rtl_inventory_arithmetic` is not a proof of these representation obligations. The 47 synthesized / four nonsynthesized distinction describes the implementation inventory; it does not turn a conditional model theorem into unrestricted RTL equality.

And the four Q_{1+AB} check opcodes are not stranded by sitting outside that partition: their kernel-equivalence proof makes them sound at the substrate level (see §18), and they run in the OCaml runner and Python harness. What they lack is a synthesized-Verilog counterpart to bisimulate against, which is why they sit outside the 47 instead of inside it as a gap. That’s honest proof accounting. Nothing is papered over.

The substrate is the math. The Coq kernel pins it down. The hardware is a finite physical instantiation with explicit table sizes and word widths. The proofs state the bridge between them under the finite representation invariants where those invariants are needed. This is the only honest relationship physical hardware can have with an abstract mathematical substrate.

Substrate-level vs program-level enforcement. In the Coq kernel, A2 is part of the step relation. A2 says: flipping certification false-to-true costs at least 1, at the step level. Look at the cases for `CERTIFY` and `MORPH_ASSERT` in `vm_apply`: the cost is added before the flag flips; the step that mutates the certification channel and the step that charges μ are the same step. That is what A2 means in the kernel.

A correct conventional simulator can keep the VM’s invariant on the states it encodes. A different or buggy simulator might not. That doesn’t prove anything about what conventional computation can’t do. The VM theorem is about the VM’s own step relation. Carrying it into hardware also takes the representation conditions in `WFDrivenRun` and the documented extraction and compiler trust boundaries.

Enforcing the law, representing it, and seeing it are separate questions, and I keep them separate. The abstract A2 theorem allows any state type and any designated predicate. Exact unit pricing is a stronger condition, relative to how many certification flips happen. A2 itself allows larger charges. And neither one vouches for a receipt from outside without some evidence to back it.

17 Five minimal models inspired by other disciplines

Same law, five other places it shows up. The abstract records leave the kind of state and instruction open, so you can drop in something that looks nothing like this VM. Here are five small models, each borrowed from a field that got to the idea on its own. Coq checks each model and what follows from it. Whether a real deployed protocol behaves like its model is a separate claim, and it needs its own evidence.

Ordinary machines can do all of this. That is the point of the section. Blockchains, gas meters, attestation, transparency logs, and proof-carrying code all run on conventional hardware, and each one puts a cost or a required piece of evidence on a commitment. Each field got there separately, in its own vocabulary, and built the rule on top of a machine whose account of computation had no place for it. The claim of this book is about that account. It says the account of computation should carry a priced commitment, so the next field doesn't have to build it again. It does not say a machine can't carry one.

17.1 Proof-of-stake finality

Stake it or it doesn't count. The model has a yes/no finalized flag, and finalizing puts stake at risk. A transition that finalizes for zero breaks A2. Charging at least one unit on it satisfies A2. What the model records is exposure to loss. It doesn't say the collateral was actually destroyed, or that each exposure is a separate unit spent, or that every real consensus protocol works this way. The file has no Casper checkpoints, quorum weights, double or surround votes, or accountable-safety theorem.

In the model, `nothing_at_stake_is_free_forgery` rules out an A2 instance for the free-finalization witness it's handed. Under the slashing-schedule premise, `slashing_finality_floor` says any run that starts unfinalized and ends finalized carries at least one unit of modeled charge. Using this on a real protocol means first showing its transitions match the model's.

Falsifier: a transition system satisfying the stated slashing-schedule and execution premises with an initially unfinalized, finally finalized trace whose modeled charge sum is zero.

17.2 Gas metering

Paying for the commitment and paying for everything else are different bills. The gas model counts commitments, one unit each. `gas_schedule_exactness` says exactly when a schedule prices those commitments exactly: meeting the floor and never overcharging pins down both which steps get charged and the unit price, and exact pricing gives you those two conditions back. A real gas schedule also charges for things like computation and storage. Its full fee isn't this counter, and no EVM opcode table is modeled here.

The model keeps two failures apart: leaving out a mandatory charge, and charging more than the reference counter. `undercharged_opcode_admits_free_commitment` concerns a committing step charged zero. `overcharge_breaks_exactness` concerns the no-overcharge condition. The VM has an exact state-dependent commitment pricer, `thiele_vm_commit_pricing_is_exact`, bounded above by its full ledger schedule via `thiele_unit_price_lower_bounds_mu`. Repeated CERTIFY instructions illustrate the distinction: each pays the instruction floor, while only a false-to-true transition contributes to the flip counter.

Falsifier: a schedule satisfying floor and no-overcharge whose charging predicate differs from the commitment predicate on some reachable step.

17.3 Trusted-execution attestation

Show the measurement, or the verifier can't tell. The attestation model adds a measurement field to the report. Under its coverage and evidence assumptions, `attestation_cannot_factor_through_bare_transcript` rules out a sound and complete verifier that only reads the bare transcript. `replay_is_a_two_preimage_witness` supplies the model's collision: two runs, one bare transcript. That is the shape of a replay, not a replay attack; no signed report or freshness rule is in the model. `measurement_enriched_attestation_succeeds` gives a verifier that works on the enriched reports. The point is that some distinguishing evidence has to be there. It doesn't have to be any particular physical register, and nothing here vouches for a measurement an adversary hands over.

Falsifier: a sound and complete attestation scheme for a μ -dependent claim that reads only the bare transcript.

The TPM quote file is a deliberately scoped abstraction of fields from TPM 2.0 Library Version 185. It fixes one epoch-initial PCR value to zero as a model choice, folds abstract extends into that PCR, and represents a quote by the selected-PCR digest and verifier nonce. Reset, restart, signature computation, and signature verification are outside the model; quote authenticity is assumed. A verifier reading only these modeled quote fields cannot decide an additional Boolean runtime flag that the quote definition ignores (`quote_cannot_attest_unmeasured_state`). A quote reports selected PCR state, not arbitrary unmeasured runtime state. The theorem does not establish a generic boot-time-attestation result or discover a TPM vulnerability.

The application to `V_does_not_factor_through_classical` is explicit in `quote_runtime_verifier_separation`. The adapter stores both modeled quote fields in a classical transcript, with no information lost (`quote_projection_faithful`). The two VM witness states label the two truth values of the runtime claim. This transport makes no assertion that TPM cost or energy equals μ .

17.4 Certificate transparency

Keep a log, or live with the split view. The transparency model adds an inclusion-verdict bit, governed by the same exact disclosure contract as the commitment route. `abstract_log_bit_verifier` builds a sound and complete verifier under that contract. `abstract_bare_verifier_impossible` rules out the bare verifier for the VM claim, and `abstract_mu_collision_witness` is the colliding pair. The pair has a split view's shape; an RFC split view is two inconsistent log histories, and that is not what the pair is. Saying these fields are a real certificate-transparency deployment would take its own implementation and security argument.

Falsifier: a sound and complete log-free verifier with the same soundness target.

17.5 Proof-carrying verification

Carry the proof along. The proof-carrying model puts evidence in a transcript field called `rounds`. `proof_rounds_escape` verifies that field under its evidence contract, and `bare_pcc_impossible` is what happens when the evidence is taken away. Calling the field `rounds` doesn't make a one-shot certificate interactive. The evidence is one number the contract says equals μ ; no code, safety policy, or proof checker is modeled. Separately, `level_k_verification_floor`, which is just another name for `level_k_certification_cost_floor`, says k charged certification events cost at least k , and `level_k_verification_floor_tight` shows it can be exactly k . That count is a different thing from the per-instruction charge levels in Section 19. Neither one measures prover time, circuit size, or proof length.

Falsifier: a level- k certified trace with total μ below k , or a sound and complete zero-round bare verifier.

17.6 The same table, five ways

System	Escape used	Kernel result	Model consequence	What refutes it
PoS finality	none, the cost records directly	<code>universal_nfi_any_substrate</code>	modeled finalization charge ≥ 1	a zero-stake gadget admitting A2
Gas metering	none, a pricing biconditional	<code>exact_commitment_pricing_characterization</code>	charge $\equiv$ commit, unit price	a conforming non-commitment schedule
TEE attestation	substrate (enrich the state)	<code>V_does_not_factor_through_classical</code>	distinguishing evidence	sound+complete bare-transcript attestation
Transparency logs	commitment contract	<code>abstract_log_built_verifier</code>	the supplied evidence, under the contract	sound+complete log-free verifier
Proof-carrying	interaction	<code>interactive_escape_succeeds, level_k_certification_cost_floor</code>	evidence, and k units for k charged events	a level- k trace under $k \mu$

Each of the five proves what its own record says. Reading any of them as a claim about a deployed system takes evidence from that system. And none of them shows certification is the one event every model has to price. That question is still open.

18 Binary Game Statistics: where a Priced Check Meets the NPA Boundary

This section is about one build and one bound. The abstract model has no CHSH step; the VM has one, realised with two opcodes, and it is here as a stress test of the cost law against a boundary I did not write. `CHSH_TRIAL` records outcomes of a 2-player binary game into hardware witness counters. `CHSH_LASSERT` refuses to advance

normally unless those counters satisfy a Z-arithmetic check.

The enforcement doesn't care who built the wall. `CHSH_LASSERT` refuses any tally outside one fixed algebraic slice by trapping to `LASSERT_TRAP_PC`, and every super-quantum correlation is outside it (so are some classical ones; the section says which); `LASSERT` refuses a formula whose dual witness fails to exclude a single valuation by trapping to the same address, the same way. One wall comes from the NPA mathematics, which I did not author and could not have, and it sits inside the Tsirelson bound. The other is a logical constraint I wrote down myself. The step traps the same on both. So μ isn't pricing contact with external structure as some special case. It prices the certified narrowing as such, and the trap fires the same whether the state it throws out is a PR box or a falsifying assignment I handed it. Insight costs, and the price never asks whether you found the wall or built it.

And the claim is about that step. The chip is something else. Some of these cert-opcodes never reach the synthesised FPGA, because I haven't built their hardware yet. That isn't a hole in the argument. The trap is the cost-on-certification law, and the law lives in the transition, not in the gate count.

Everything below earns that. The matrix algebra is the discharge: it shows that everything `CHSH_LASSERT` lets through sits inside the Tsirelson bound, a bar I did not set myself. I came at it sideways, stress-testing the machine's priced checks against a known algebraic boundary, the NPA moment matrix conditions for what correlations are achievable under polynomial constraints.

I did not set out to find what I found. The chain is concrete. The Z-arithmetic check the kernel runs on the witness counters (decidable, integer-only) is sound with respect to the real-valued column-contractive predicate by `column_contractive_check_witness_sound`, a 188-line proof. That real-valued predicate is biconditional with PSD of the specified zero-marginal, zero-within-party-moment matrix by `column_contractive_iff_npa_psd`. End-to-end: a successful `CHSH_LASSERT` step entails PSD of that specified matrix on the witness-derived correlators, by `chsh_lassert_no_trap_implies_npa_psd`. No quantum-mechanical premises live in the Coq dependency tree: no Hilbert space, no operators, no observables. The identification of that PSD condition with quantum-realizable correlations lives in the NPA framework itself (Navascués–Pironio–Acín 2007, 2008). I cite it. I do not derive it. The kernel's content is the algebraic bridge between an integer-arithmetic check inside an opcode and PSD of a specific matrix. The theorem names say PSD, because PSD is what the kernel proves.

Which way it runs: the machine first, NPA second. Two passes, in this order. The first pass stays on the machine's side and never mentions NPA: the `CHSH_TRIAL` opcode, what its witness counters cost, the integer check `column_contractive_check_witness` that `CHSH_LASSERT` runs on those counters, and the proof that the integer check is sound for its real-number version. What the step checks at `CHSH_LASSERT`

comes down to three polynomial inequalities on cleared integer ratios of the counters. The ledger charges for running that check; it does not produce the check. The test is defined inside the step rule and checked by the kernel without looking anything up in the quantum-mechanics literature. The second pass is where the matrix comes in. Lift the machine's own test to the reals (IZR, with the witness counts as denominators) and you get the column-contractive predicate, and that predicate holds exactly when the specified zero-marginal, zero-within-party-moment matrix is PSD. Both sides of that biconditional are defined mathematical predicates, and the proof is pure polynomial arithmetic with no project-local axioms. It isn't a claim that the runtime built a physical quantum experiment. The result that carries the weight is the subsection "Algebraic characterization of the CHSH gate condition." The subsections before it (the Bell setup, the classical bound, the quantum bound) are the walk there. If you have time for one subsection, that's the one. The next subsection builds the game from scratch for anyone who hasn't met Bell tests; if you have, skip ahead.

18.1 The Bell test setup, from scratch

Imagine two players, Alice and Bob, in separate rooms. They cannot communicate once the game starts. A referee sends Alice one of two questions ($x \in \{0,1\}$). The referee sends Bob one of two questions ($y \in \{0,1\}$). Alice answers $a \in \{0,1\}$. Bob answers $b \in \{0,1\}$. (Settings (x,y) , outcomes (a,b) : same convention as Section 11 and the Coq.)

The game rule: they want $a \oplus b = x \wedge y$. The symbol $\oplus$ is XOR (exclusive OR): $a \oplus b = 1$ when $a \neq b$, and 0 when $a = b$. The symbol $\wedge$ is AND: $x \wedge y = 1$ only when both $x = 1$ and $y = 1$, otherwise 0. In plain language: their answers should be different if and only if both questions were 1. For all other question combinations (both 0, or one each), they should match.

Why 75% is the classical ceiling. Before the game, Alice and Bob can agree on any deterministic strategy: a function $a(x, \lambda)$ for Alice and $b(y, \lambda)$ for Bob, where λ is any shared randomness they agree on in advance. Once the game starts, no communication. One assumption, said out loud: the referee asks the four question pairs equally often. That is where the 75% comes from. Drop it and the exact statement is only that every deterministic strategy loses at least one of the four pairs.

Fix any deterministic strategy and freeze λ . Then Alice has two predetermined answers, a_0 and a_1 , and Bob has two predetermined answers, b_0 and b_1 . Winning all four question combinations would require:

$$a_0 \oplus b_0 = 0, \quad a_0 \oplus b_1 = 0, \quad a_1 \oplus b_0 = 0, \quad a_1 \oplus b_1 = 1.$$

Now XOR the four left sides. Every variable appears twice, so the total is 0. XOR the four right sides and the total is 1. That is impossible. So every deterministic local

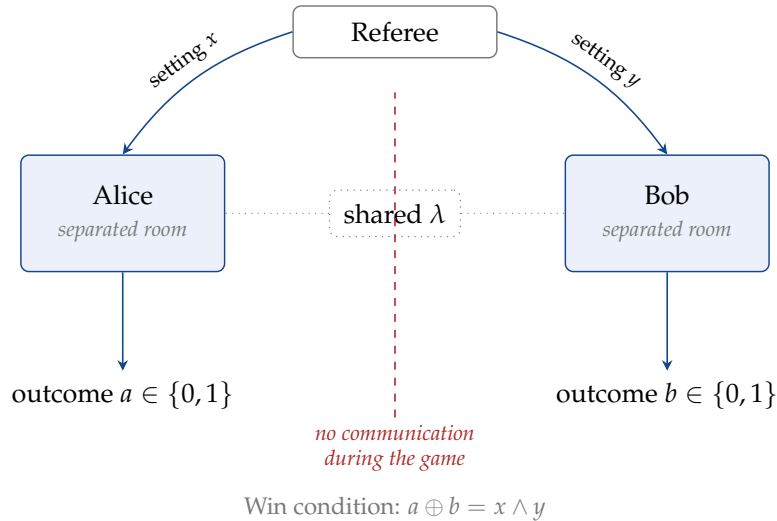


Figure 7: The CHSH setup. The referee distributes settings x, y ; Alice and Bob each have a separated room and may share an agreed-on λ from before the game started. No communication during the game. Each produces a binary outcome (0/1 on the wire, recoded as ± 1 for the correlators). The four correlators $E_{xy} = \langle ab \rangle$, read in the ± 1 encoding, are what the kernel's CHSH_LASSERT instruction checks against the column-contractivity / NPA condition.

strategy loses at least one of the four cases, and the best deterministic strategy wins at most 3 out of 4 question combinations.

And you cannot buy your way past it with a coin flip. A randomized strategy is nothing but a bag of deterministic ones with a die deciding which to pull, and every strategy in the bag already tops out at 75%. Average a pile of things that each sit at 75% or below and the average sits at 75% or below; there is no arrangement of losers that sums to a winner. That is linearity of expectation, the most boring theorem in the building, quietly closing the last door.

So under that uniform-setting contract the classical ceiling is $3/4$, seventy-five percent, and not a hair more. And I do not want you to take that on the strength of an XOR trick on a page; the reference kernel pins it down both ways and leaves nothing Admitted. One executable trace, built from nothing but PNEW, PSPLIT, and CHSH_TRIAL and never spending a cent of the cost ledger ($\mu = 0$), walks right up to 75% and touches it. Then `local_box_CHSH_bound` closes the lid from the other side: no local strategy, however randomized, no factorizable box anywhere, gets over the line under the stated population/setting-weight assumptions. A trace can of course just *write* any tallies it likes into the counters at $\mu = 0$, the way anyone can write a fake scoreboard; the theorem is about what local play can earn, and the certification step that would stand behind a scoreboard is the priced one. The achievable and the unbeatable land on the same number, which is the only kind of ceiling I trust.

18.2 The CHSH inequality: where it comes from

Now here is what took me a while to understand: why is there a specific algebraic combination S that detects whether correlations can be explained classically?

The key move is this. For any fixed shared randomness λ , Alice's output $A_x(\lambda) \in \{-1, +1\}$ (switching to the ± 1 encoding now instead of 0/1, indexed by Alice's setting x) and Bob's $B_y(\lambda) \in \{-1, +1\}$ (indexed by Bob's setting y). Consider the expression:

$$A_0(\lambda)B_0(\lambda) + A_0(\lambda)B_1(\lambda) + A_1(\lambda)B_0(\lambda) - A_1(\lambda)B_1(\lambda)$$

Factor it:

$$= A_0(\lambda)(B_0(\lambda) + B_1(\lambda)) + A_1(\lambda)(B_0(\lambda) - B_1(\lambda))$$

Since B_0 and B_1 are each ± 1 , either $B_0 + B_1 = \pm 2$ and $B_0 - B_1 = 0$, or $B_0 + B_1 = 0$ and $B_0 - B_1 = \pm 2$. The two terms cannot both be large. In either case, $|B_0 + B_1| + |B_0 - B_1| = 2$, and since $|A_0|, |A_1| \leq 1$, the entire expression is at most 2 in absolute value for any fixed λ .

Average over λ :

$$S = \langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle$$

where $\langle A_x B_y \rangle$ is the correlation (expected value of the product) when settings are x and y . Since the expression before averaging was bounded by 2 for every λ , the average satisfies $|S| \leq 2$.

That is the argument. The combination with the minus sign is chosen exactly because the factoring trick works: one of the B-terms goes to zero and the other to ± 2 , forcing the whole thing below 2. The inequality $|S| \leq 2$ is a pure algebraic consequence of the outputs being ± 1 and being determined locally.

None of that lives only in my head. The argument I just walked you through is the English proof; the formal check that I did not fool myself anywhere is `local_box_CHSH_bound`, which says the plain thing in symbols: for any factorizable box B , $|S(B)| \leq 2$, with nothing left Admitted. And a bound from above is only half an honest claim. `classical_bound_achieved` hands over the other half, a constructive witness, an executable trace at $\mu = 0$ that actually reaches $|S| = 2$ rather than just respecting it. Put the two together and the ceiling is not loose: $\max\{|S| : B \text{ factorizable}\} = 2$, exactly, no slack to argue about, and a $\mu = 0$ trace reaches it.

18.3 The quantum bound: why $2\sqrt{2}$?

Now hand the same game to quantum mechanics and the ceiling lifts. Not to anything you please, and not all the way to a perfect 4; it lifts to $2\sqrt{2} \approx 2.828$ and stops there as hard as the classical 2 stopped. That number is the Tsirelson bound, named for Boris Tsirelson, who was the first to write down $|S| \leq 2\sqrt{2}$ and pin quantum correlations under it. The physics story of *why* it is that exact number, and not some other irrational nobody would have guessed, runs through Hilbert-space operators, anticommutators, the Pauli matrices, the Bloch sphere, the whole machinery. It is a real and beautiful derivation and I am going to walk straight past it, because none of the formal content here leans on it. I do not need to retell quantum mechanics to show you the kernel respects the wall; I only need the algebra, and that is what I am keeping.

What the formal content does depend on, and what the next subsections derive, is the algebra. One bridge theorem takes PSD of the zero-marginal NPA matrix as its premise. Reading that premise as quantum realizability is the NPA framework's step, and it is not derived in Coq. Its conclusion is the algebraic bound $|S| \leq 2\sqrt{2}$. The "What the algebra proves" subsection below then builds the same bound from polynomial inequalities on the specified moment matrix without any quantum-physics premise at all. That is the path I follow, and the formal claim is the algebraic one.

And that gap between 2 and $2\sqrt{2}$ is not bookkeeping. It is the most stubborn fact in physics: people have gone into labs, fired entangled photons through polarizers, counted coincidences, and watched the number sit up above 2 where no local classical story is allowed to put it. Whatever the world is doing, it is not Alice and Bob with a shared cheat sheet agreed beforehand. The physics of *why* the wall sits at $2\sqrt{2}$ is somebody else's chapter and a good one. My question is narrower and, to me, stranger: why does an integer check on counters stay under a wall it never names? The polynomial arithmetic below answers it. Take the check's three inequalities, add nothing from quantum mechanics, and the same $2\sqrt{2}$ falls out the bottom on its own. The inequalities themselves come from the NPA matrix, so the physics literature chose the shape of the check. What no physics does is prove the bound; algebra does that.

A note on the rational approximation in the kernel. The kernel names the rational threshold $5657/2000 \approx 2.8285$ for exact rational comparisons (you can verify $(5657/2000)^2 = 32,001,649/4,000,000 > 8$, so $5657/2000 > \sqrt{8} = 2\sqrt{2}$). The real-valued quantum-model bridge uses $\sqrt{8}$ directly. Those are two different proof surfaces, and I keep them apart on purpose: the rational one so a machine can check it in exact arithmetic, the real one so the quantum bridge has something to bite on.

18.4 What the algebra proves

| **Plain reading.** Factor the expression. Two terms. One has $B_0 + B_1$, the other has $B_0 - B_1$. With $B_0, B_1 \in \{\pm 1\}$, only one can be nonzero at a time, and the nonzero one is ± 2 . Multiply by $|A| \leq 1$. The whole expression caps at 2. Average over λ ,

you get $|S| \leq 2$. To make it 2, just have both players always answer 1.

Theorem 18.1 (Classical CHSH Bound). *For population correlations with a setting-independent shared-variable distribution, Alice’s local response depending on x, λ and Bob’s on y, λ , $|S| \leq 2$. The constant strategy attains equality. Its four-setting VM witness has zero charge when every trial uses $\delta = 0$.*

Proof. $|S| \leq 2$: the XOR-factoring argument from Section 18’s “Why 75% is the classical ceiling.” Fix any local deterministic strategy and any λ . Alice produces $A_x(\lambda) \in \{-1, +1\}$; Bob produces $B_y(\lambda) \in \{-1, +1\}$. The quantity

$$\begin{aligned} A_0(\lambda)B_0(\lambda) + A_0(\lambda)B_1(\lambda) + A_1(\lambda)B_0(\lambda) - A_1(\lambda)B_1(\lambda) \\ = A_0(\lambda)(B_0(\lambda) + B_1(\lambda)) + A_1(\lambda)(B_0(\lambda) - B_1(\lambda)) \end{aligned}$$

satisfies $|\cdot| \leq |B_0 + B_1| + |B_0 - B_1| = 2$ (because for $B_0, B_1 \in \{\pm 1\}$, one of $|B_0 + B_1|$ and $|B_0 - B_1|$ is 2 and the other is 0, and $|A_0|, |A_1| \leq 1$). Average over λ to get $|S| \leq 2$. Mixtures of deterministic strategies are convex combinations, so the same bound holds by linearity of expectation.

Tightness: the strategy $A_x \equiv 1, B_y \equiv 1$ gives $\langle A_0B_0 \rangle = \langle A_0B_1 \rangle = \langle A_1B_0 \rangle = \langle A_1B_1 \rangle = 1$, so $S = 1 + 1 + 1 - 1 = 2$. This is achievable at $\mu = 0$ by a program with all $\delta = 0$ instructions accumulating one trial per setting pair in `vm_witness`. The Coq witness is `classical_bound_achieved`. $\square$ $\square$

| Plain reading. Direct contrapositive. If $|S| \leq 2$ for the population-level correlation of every LHV, then a population-level $S > 2$ rules out every LHV. The Bell-test contradiction has this form, and it’s a claim about the number an unbounded run converges to. The number on any one finite scoreboard is a different thing.

Theorem 18.2 (Non-locality of an exact CHSH violation). *An exact correlation $\mathbf{E} = (E_{00}, E_{01}, E_{10}, E_{11})$, with $S = E_{00} + E_{01} + E_{10} - E_{11}$ and $S > 2$, cannot arise as the population-level correlation of any local hidden variable model (the term is built in Section A: a classical-physics-style theory where outcomes come from shared randomness λ plus local settings). This is a claim about the idealized correlator, the value an unbounded run of trials at each setting converges to. It is not yet a claim about any one finite tally of recorded trials; that distinction is spelled out below.*

Proof. Contrapositive of the Classical CHSH Bound. Any LHV model produces a locally factorizable correlation by definition: each player’s outcome is a function of their setting and the shared λ . The theorem above says all such correlations satisfy $|S| \leq 2$. So an idealized correlation with $S > 2$ cannot have arisen from any LHV model. $\square$ $\square$

Remark 18.1. A finite scoreboard is not the population correlation. A `WitnessCounts` tally from a finite run of `CHSH_TRIAL` instructions is a sample statistic, `chsh_stat_from_wc`, not the idealized correlator the theorem above bounds. A local hidden variable model can draw a fresh λ per trial, and a short run can land away from its own population mean by chance. Run one trial at each of the four settings, with Alice and Bob each answering an independent fair coin flip, an honest LHV strategy with population correlation zero at every setting: the event “same at (0,0), same at (0,1), same at (1,0), different at (1,1)” has probability $1/16$ and produces the finite tally $S = 4$, the algebraic maximum, with no communication and no quantum content anywhere in the construction. This is the same finite-sample distinction Zhang, Glancy, and Knill work out in full for Bell tests (*Efficient quantification of experimental evidence against local realism*, arXiv:1303.7464): a population bound of $|S| \leq 2$ is not, by itself, a claim that every finite sample drawn from a local model also satisfies it.

What the kernel proves about a finite tally is narrower, and it does hold: `chsh_violation_rules_out_locally_factorizable_coupling` shows that a tally with $S > 2$ is inconsistent with any single fixed deterministic strategy (a_0, a_1, b_0, b_1) replayed identically at every recorded trial (`locally_consistent_classical_bound` is the converse direction: a tally consistent with one fixed deterministic strategy has $|S| \leq 2$). That rules out one specific, narrow explanation of the tally. It does not rule out a randomized local model that redraws λ trial by trial, which is exactly the kind of model the coin-flip example above uses to reach $S = 4$ honestly. Reading a certified witness violation as physically informative rests on running enough trials to make the fixed-strategy count and the population count converge, not on an identity between one finite tally and the exact-correlation theorem above.

| **Plain reading.** PSD of the fixed matrix and column contractivity are the same condition. Column contractivity caps each row, and Cauchy–Schwarz turns those row caps into $2\sqrt{2}$. That is a statement about this matrix. A quantum behavior in general, or a finite local sample, is a different object with its own definition.

Theorem 18.3 (Tsirelson upper bound for the fixed completion). *If the correlators satisfy PSD of the specified zero-marginal, zero-within-party-moment completion, their CHSH value satisfies $|S| \leq \sqrt{8} = 2\sqrt{2}$.*

Proof. The fixed-matrix biconditional gives column contractivity. It bounds each row norm of E by one. Apply Cauchy–Schwarz to $S = (E_{00} + E_{01}) + (E_{10} - E_{11})$: each parenthesis has magnitude at most $\sqrt{2}$ times its row norm. Thus $|S| \leq 2\sqrt{2}$. This proof uses the specified matrix, without an identification with arbitrary zero-marginal operator models. □ □

| **Plain reading.** Two row inequalities, one six-square identity, done. That is the whole proof. No hidden parameters. No physics. The $2\sqrt{2}$ falls out of the algebra;

nobody put it in. What the theorem covers is four real numbers under two row constraints. Whether some experiment's correlators satisfy those constraints is a different question, and this theorem does not touch it.

Theorem 18.4 (General algebraic Tsirelson bound). *For every real quadruple satisfying the two explicit row constraints $E_{00}^2 + E_{01}^2 \leq 1$ and $E_{10}^2 + E_{11}^2 \leq 1$, the CHSH value satisfies $S^2 \leq 8$, that is, $|S| \leq 2\sqrt{2}$. These are the exact hypotheses of `tsirelson_from_row_bounds`; the theorem does not quantify over unspecified minor witnesses or hidden parameters t and s . This is an algebraic statement about four real numbers, not a completeness theorem for quantum behaviors. The Coq proof uses the explicit six-square Cauchy–Schwarz identity `cauchy_schwarz_chsh`, then adds the two row inequalities. The rational approximation corollary uses $5657/2000$ as an exact rational overestimate of $2\sqrt{2}$ (you can verify $(5657/2000)^2 = 32,001,649/4,000,000 > 8$), allowing the bound to be checked in exact rational arithmetic without floating-point. No physics premises. Pure polynomial arithmetic. The Tsirelson bound $2\sqrt{2}$ is derived from algebra, not assumed.*

Proof. The proof has two rows, not four unspecified 3×3 -minor constraints. First add the two row inequalities to obtain $E_{00}^2 + E_{01}^2 + E_{10}^2 + E_{11}^2 \leq 2$. Then `cauchy_schwarz_chsh` gives the explicit identity $4 \sum E_{xy}^2 - S^2 \geq 0$, a sum of six squares. Combining the two facts yields $S^2 \leq 8$. No parameters t, s occur in this theorem or its proof.

The rational approximation corollary uses $5657/2000$ in place of $\sqrt{8}$ to keep the comparison in $\mathbb{Q}$: $(5657/2000)^2 = 32,001,649/4,000,000 > 8 = 32,000,000/4,000,000$, so $5657/2000 > \sqrt{8}$. Combined with $|S|^2 \leq 8$, this gives $|S| < 5657/2000$, a strict rational bound checkable without floating-point evaluation. $\square$ $\square$

Bell's theorem has another face, and it is worth seeing both. The statistical face is the one everyone draws: counts going into a number S and the number refusing to stay under 2. But the same wall shows up one floor down, in the language of couplings, where you are no longer talking about averages at all but about whether two parties can each answer locally and have their answers fit together. A WitnessCount-consistent locally deterministic strategy here is an honest, almost insultingly simple thing: four bits, Alice's answer for each of her two settings and Bob's for each of his, fixed in advance with no peeking. Pin that down and you get a separable coupling, by which I mean one that hands each setting exactly one outcome type, no hedging, no mixing. The predicate `WCLocallyConsistent` additionally requires every one of the four setting buckets to be nonempty, and under that premise the kernel proves $|S| \leq 2$. So read it the other way, which is the way that bites: if a finite tally has $S > 2$, there is no *single fixed* four-bit strategy, repeated identically on every recorded trial, that explains that tally. It does not rule out a randomized local model that redraws its hidden variable every trial; the finite-score remark above shows one landing past 2 by luck. The wall is not a fact about statistics that we then dress up in structure; it is already there in the structure, and the statistics are just where you happen to bump

into it first. Going from one finite tally to a claim about the population takes a separate sampling argument.

What does any of this have to do with computation? Everything, and it comes in through the witness counters and the certification layer. Picture a trace that sits there racking up CHSH trial results through CHSH_TRIAL, tally after tally, costing nothing, because counting is cheap and the universe does not send you a bill for watching it. Then the trace does the expensive thing: it commits to a claim about them. Checking the tally is CHSH_LASSERT, which costs at least one whether it passes or traps. Flipping a certification channel is CERTIFY or MORPH_ASSERT, and A2 forces $\mu \geq 1$ on that flip, no exceptions. Neither flip checks the claim on its own; the check and the commitment are separate steps, each priced. So the statistics are free and the claim about the statistics is not, and that gap is the whole point. You can stare at the data all day for nothing; the moment you stand up and say you know what it means, somewhere a counter ticks. Throughout, the witness counters carry Alice's setting x and Bob's setting y , the same indexing the formal layer uses, so this is not a side dialect (Section 11). It is the witness insight taxonomy of Section 6, viewed from the place where physics hands you a free supply of trials and then charges you for the sentence you write about them.

18.5 The bit-commitment requirement

If you want cross-correlations above the classical bound to come out attested, not just observed, the reference kernel makes you pay for a step from the priced class to get there. `nonlocal_correlation_requires_revelation` says exactly which: a trace that starts with the certificate-address channel at zero and ends with it set contains a REVEAL, EMIT, LJOIN, LASSERT, or MORPH_ASSERT. That theorem is about the channel. It knows nothing about CHSH. Requiring REVEAL in particular for a super-quantum claim is a runtime policy, and the source says so. The requirement falls straight out of how cost is priced: A2 charges the cert-flip, so the only way to reach an attested claim is to take the step that flips the cert, and that step costs. The strictness is downstream of the accounting, not the other way around, and the next few lines walk through exactly where it gets forced.

So here is where that gets concrete. A single CHSH_TRIAL drops a few setting bits and outcome bits into the witness counters, and the cross-correlations $\langle A_x B_y \rangle$ are nothing but those tallies divided through. They are a record of what happened, and the ledger has no opinion about a record. The accounting only wakes up when you point at one of those numbers and say it backs a particular claim. That is not a count any more. It is a stance, and a stance is the one thing the ledger knows how to charge for, which is why the cost lands precisely on the sentence and never on the arithmetic underneath it.

And this part is not a hope, it is settled. There is exactly one instruction in the reference kernel that can push `csr_cert_addr` off zero, and that is MORPH_ASSERT, and

even then only when the morphism genuinely exists and the property string it carries is more than an empty NUL byte. So the door to attestation has a single key. Run that backward and it pins the trace: anything ending in `supra_cert` had to have walked through a valid `MORPH_ASSERT` step on the way, there is no other path that sets the register, and the kernel will not let you fake one. Which is the same wall from a closer angle. `CHSH_TRIAL` counts pile up for free, all the raw data you like, but the instruction that stands up and commits to the pile is the one that costs, and it is the only one that even can. What it commits to is a label on an existing morphism. Whether the label is true of the pile is a separate question, and `CHSH_ASSERT` is the priced check that answers part of it.

18.6 Algebraic characterization of the CHSH gate condition

If you only have the patience for one subsection of Section 18, make it this one. Everything before it is the long walk, the honest grind of getting from the machine's check to the physics boundary. What lands here is the thing the grind was for: the place where the check the machine runs and the matrix the physics literature uses turn out to be the same condition read two ways, for one fixed slice. The rest of the section earns it. This is what it earns.

Here is the result of stress-testing the machine's check against the NPA boundary. The column-contractivity conditions are what positive-semidefiniteness of that NPA moment matrix comes to, written out as polynomials. The matrix comes from the physics literature. The proof that the two conditions agree uses no physics at all; it is pure polynomial arithmetic. The following biconditional holds.

Positive semidefinite (PSD). A matrix M is positive semidefinite if for every vector v you can multiply with it, the quantity $v^T M v$ is at least 0. (Here v^T is the transpose of v from Section A: turning a column of numbers on its side; $v^T M v$ is the matrix product of v^T , M , and v , which is just one number.) The intuition: PSD means the matrix never maps a vector to something pointing "backwards" relative to the vector. For a 2×2 matrix $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$, PSD reduces to three simple conditions: $a \geq 0$, $c \geq 0$, and $ac - b^2 \geq 0$. The first two say the diagonal entries are non-negative. The third says the determinant (Section A) is non-negative. For larger matrices, the PSD condition generalises: every principal submatrix (every square subblock you get by picking some rows-and-columns at the same indices, see Section A) must also be PSD.

Why PSD? In the NPA picture, PSD is the condition for a matrix to be a table of genuine inner products, which is what correlations from a quantum state are. NPA uses PSD of a specific 5×5 matrix, built from the four CHSH correlators, as its level-1 condition for quantum realizability. That is the physics reading, and it comes from the NPA papers. The proofs here don't supply it. What Coq checks is the matrix: its inequalities and what follows from them. No Hilbert space, no state, no observables. Two scope words are load-bearing. For this pinned zero-marginal matrix the PSD test is exact for

its slice. The general level-1 NPA test, marginals free, is a looser relaxation, and the hierarchy tightens it at higher levels (Section 18.7).

18.6.1 The matrix, written out

Here's the shape of it before the arithmetic. The four correlator numbers go into a 5×5 grid. Look at the grid through three windows: two small ones and one bigger one (Figure 8). Each window has to pass one test, the same test every time, that the numbers inside it hang together the way a table of real inner products has to. Those three tests are the three conditions the machine's integer check runs. Pass all three and the whole grid passes. Fail one and it doesn't.

What follows is the formal layer at its most formal: the grid itself, then the determinant arithmetic that pulls the three conditions out of it, then the proof that they're enough. The models wrote that derivation and Coq checked it. If you don't want the arithmetic, skip ahead to the theorem it proves, "Slice coherence $\iff$ NPA-PSD," and read on from there.

Here is the actual 5×5 moment matrix for the zero-marginal CHSH slice. The rows and columns carry the usual NPA labels: I, A_0, A_1, B_0, B_1 . In the physics reading they are the identity and the two measurements on each side, and the (i, j) entry is the expected value of $O_i O_j$. In Coq they are just positions in a real matrix. Zero-marginal means the four one-party slots and the two within-party slots are set to zero, which leaves the CHSH correlators in the cross positions.

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & E_{00} & E_{01} \\ 0 & 0 & 1 & E_{10} & E_{11} \\ 0 & E_{00} & E_{10} & 1 & 0 \\ 0 & E_{01} & E_{11} & 0 & 1 \end{pmatrix}$$

Rows and columns are indexed 0 through 4, in the order $\{I, A_0, A_1, B_0, B_1\}$. The diagonal is all 1s. In the physics reading that is because a ± 1 -valued measurement squares to the identity. In Coq the 1s are simply written into the definition. The off-diagonal entries E_{xy} are the CHSH correlators with Alice's setting x and Bob's setting y (Section A). Everything else is zero because this is the zero-marginal, zero-within-party slice.

18.6.2 Where the three conditions come from

The three column-contractivity conditions come directly from PSD of this matrix. Here is the derivation.

Condition 1: $1 - E_{00}^2 - E_{10}^2 \geq 0$

Take the 3×3 submatrix from rows and columns $\{1, 2, 3\}$ of the 5×5 moment matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & E_{00} & E_{01} \\ 0 & 0 & 1 & E_{10} & E_{11} \\ 0 & E_{00} & E_{10} & 1 & 0 \\ 0 & E_{01} & E_{11} & 0 & 1 \end{bmatrix}$$

Condition 1

$$1 - E_{00}^2 - E_{10}^2 \geq 0$$

rows/cols $\{1, 2, 3\}$ (A_0, A_1, B_0)

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & E_{00} & E_{01} \\ 0 & 0 & 1 & E_{10} & E_{11} \\ 0 & E_{00} & E_{10} & 1 & 0 \\ 0 & E_{01} & E_{11} & 0 & 1 \end{bmatrix}$$

Condition 2

$$1 - E_{01}^2 - E_{11}^2 \geq 0$$

rows/cols $\{1, 2, 4\}$ (A_0, A_1, B_1)

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & E_{00} & E_{01} \\ 0 & 0 & 1 & E_{10} & E_{11} \\ 0 & E_{00} & E_{10} & 1 & 0 \\ 0 & E_{01} & E_{11} & 0 & 1 \end{bmatrix}$$

Condition 3

$$AC \geq B^2, \text{ Schur on } 4 \times 4$$

rows/cols $\{1, 2, 3, 4\}$

Figure 8: The 5×5 specified zero-marginal, zero-within-party-moment matrix shown three times, each copy highlighting one principal submatrix whose PSD condition gives one column-contractivity condition. Two come from 3×3 principal-submatrix determinants; the third comes from the Schur complement of the 4×4 block. All three follow from PSD of the parent matrix. Panel 2's $\{1, 2, 4\}$ submatrix is non-contiguous, skipping row and column 3, so its highlight marks the band of rows the condition constrains rather than the exact cells; panels 1 and 3 are contiguous and highlighted exactly.

(the rows and columns indexed by A_0, A_1, B_0):

$$\begin{pmatrix} 1 & 0 & E_{00} \\ 0 & 1 & E_{10} \\ E_{00} & E_{10} & 1 \end{pmatrix}$$

Compute the determinant of this 3×3 submatrix by the cofactor expansion along the first row (Section A: pick a row, multiply each entry by the determinant of the 2×2 matrix you get by deleting that entry's row and column, with alternating signs, then sum):

$$1 \cdot (1 \cdot 1 - E_{10}^2) - 0 \cdot (0 \cdot 1 - E_{10} \cdot E_{00}) + E_{00} \cdot (0 \cdot E_{10} - 1 \cdot E_{00}) = 1 - E_{10}^2 - E_{00}^2.$$

The principal-submatrix rule for PSD says this determinant must be ≥ 0 . That is condition 1.

Condition 2: $1 - E_{01}^2 - E_{11}^2 \geq 0$

Take the 3×3 submatrix from rows and columns $\{1, 2, 4\}$ (the A_0, A_1, B_1 block):

$$\begin{pmatrix} 1 & 0 & E_{01} \\ 0 & 1 & E_{11} \\ E_{01} & E_{11} & 1 \end{pmatrix}$$

Same calculation: determinant $= 1 - E_{01}^2 - E_{11}^2 \geq 0$. That is condition 2.

Condition 3: $(1 - E_{00}^2 - E_{10}^2)(1 - E_{01}^2 - E_{11}^2) \geq (E_{00}E_{01} + E_{10}E_{11})^2$

This comes from the 4×4 submatrix at rows and columns $\{1, 2, 3, 4\}$. That submatrix

has a block structure (an arrangement where the entries are themselves 2×2 matrices):

$$\begin{pmatrix} I_2 & E \\ E^T & I_2 \end{pmatrix}$$

where I_2 is the 2×2 identity matrix (Section A) and E is the 2×2 correlator matrix

$$E = \begin{pmatrix} E_{00} & E_{01} \\ E_{10} & E_{11} \end{pmatrix}.$$

E^T is the transpose of E from Section A. A standard linear-algebra fact (the Schur complement criterion, also Section A) says that a block matrix of this shape is positive-semidefinite if and only if the Schur complement of the upper-left I_2 is positive-semidefinite. The Schur complement here works out to $I_2 - E^T E$. The order of the transpose matters: $E^T E$ puts the squared norms (sums of squares) of the columns of E on its diagonal, while EE^T would put the squared norms of the rows. The right one for this argument is $E^T E$.

Computing $E^T E$ entry by entry:

$$E^T E = \begin{pmatrix} E_{00} & E_{10} \\ E_{01} & E_{11} \end{pmatrix} \begin{pmatrix} E_{00} & E_{01} \\ E_{10} & E_{11} \end{pmatrix} = \begin{pmatrix} E_{00}^2 + E_{10}^2 & E_{00}E_{01} + E_{10}E_{11} \\ E_{00}E_{01} + E_{10}E_{11} & E_{01}^2 + E_{11}^2 \end{pmatrix}.$$

The Schur complement is therefore

$$I_2 - E^T E = \begin{pmatrix} 1 - E_{00}^2 - E_{10}^2 & -(E_{00}E_{01} + E_{10}E_{11}) \\ -(E_{00}E_{01} + E_{10}E_{11}) & 1 - E_{01}^2 - E_{11}^2 \end{pmatrix} = \begin{pmatrix} A & -B \\ -B & C \end{pmatrix}$$

where $A = 1 - E_{00}^2 - E_{10}^2$ (the left side of condition 1), $B = E_{00}E_{01} + E_{10}E_{11}$, and $C = 1 - E_{01}^2 - E_{11}^2$ (the left side of condition 2). For this 2×2 Schur complement to be PSD, two things must hold: the diagonal entries must be non-negative (which are conditions 1 and 2 again) and the determinant must be non-negative, $AC - B^2 \geq 0$. That last inequality is condition 3.

That is where the three conditions come from: two from 3×3 submatrices, one from the Schur complement of the 4×4 block, each one a principal minor forced non-negative by M being PSD. That derivation runs necessity, PSD of M implies the three conditions. It is not yet sufficiency, and checking these particular minors does not by itself certify every principal minor of the full 5×5 ; a direct argument for the other direction follows.

The forward direction, directly. Write M in block form as $1 \oplus \begin{pmatrix} I_2 & E \\ E^T & I_2 \end{pmatrix}$: the $(0,0)$ entry is 1, the rest of row and column 0 are 0 because the marginals vanish, and the remaining 4×4 block is exactly the one just computed. For $v = (v_0, x, y)$ with $v_0 \in \mathbb{R}$

and $x, y \in \mathbb{R}^2$,

$$v^T M v = v_0^2 + x^T x + 2x^T E y + y^T y = v_0^2 + \|x + E y\|^2 + y^T (I_2 - E^T E) y,$$

the last step by expanding $\|x + E y\|^2 = x^T x + 2x^T E y + y^T E^T E y$ and cancelling the $y^T E^T E y$ term against its copy in $y^T (I_2 - E^T E) y$. The first two terms are non-negative for every v_0 and every x , with no condition on E at all; x is free to range over all of $\mathbb{R}^2$, and $\|x + E y\|^2 \geq 0$ regardless of what value it takes. The entire content is in the third term: $v^T M v \geq 0$ for every choice of v_0, x, y if and only if $y^T (I_2 - E^T E) y \geq 0$ for every $y \in \mathbb{R}^2$, which is exactly PSD of the 2×2 matrix $I_2 - E^T E$ computed above. A 2×2 symmetric matrix is PSD iff its diagonal entries are non-negative and its determinant is non-negative, which is conditions 1, 2, and 3 together. So column-contractivity is equivalent to PSD of $I_2 - E^T E$, which is equivalent to $v^T M v \geq 0$ for every v , which is PSD of M . This is the complete forward direction: three conditions checked once, as the PSD criterion for one 2×2 matrix, not as a claim that a hand-picked list of minors stands in for all $2^5 - 1$ principal minors of the 5×5 .

What this fixed completion establishes. The reverse direction (proved as `npa_psd_implies_column_contractive`) is that if M is PSD, the correlators must be column-contractive. Here is the argument. Take any PSD matrix M and feed it the specific 5-dimensional vector $v = (0, -(E_{00}y_0 + E_{01}y_1), -(E_{10}y_0 + E_{11}y_1), y_0, y_1)$, where y_0 and y_1 are free real numbers. The PSD condition $v^T M v \geq 0$, after the matrix algebra, gives a quadratic expression in y_0 and y_1 : $Ay_0^2 - 2By_0y_1 + Cy_1^2 \geq 0$ for all real y_0 and y_1 . If a quadratic of the form $ay^2 + 2by + c$ is non-negative for every real y , then $a \geq 0$ and $ac \geq b^2$ (the non-negative-discriminant rule from intermediate algebra: a parabola that never dips below the axis has non-positive discriminant, which after sign-flipping reads $b^2 \leq ac$; that one direction is all the argument uses). That closes condition 3. The diagonal conditions (1 and 2) come from setting $y_1 = 0$ and $y_0 = 0$ separately.

A correlation $(E_{00}, E_{01}, E_{10}, E_{11})$ is *slice-coherent* if and only if it is realizable by the specified zero-marginal, zero-within-party-moment completion. Exactly. Not approximately. Not “a complete test for physical quantum behavior.” The name describes the test: membership in the fixed algebraic completion used by `zero_marginal_npa`. Four ratios can satisfy that test whether their counters came from trials or were written directly. A passing gate earns the algebraic membership claim. It does not establish the history of those counters. Nor does it build a quantum state and observables for the tuple. The biconditional is pure polynomial arithmetic with no project-local axioms (`Print Assumptions on column_contractive_iff_npa_psd` reports only Coq’s standard classical-reals and functional-extensionality `stdlib` axioms, namely `sig_not_dec`, `sig_forall_dec`, and `functional_extensionality_dep`, the same `stdlib` families Section 18.7 discloses for the Q_{1+AB} theorems; “zero axioms” in this section means zero project-local, in the sense of Section 23):

| Plain reading. Both directions are pure algebra on the 5×5 matrix. Forward: each of the three column-contractivity conditions is a principal-submatrix or Schur-complement determinant being non-negative, and together they make the whole matrix PSD. Reverse: feed the PSD matrix a specific test vector built from (y_0, y_1) . What comes out is a quadratic in (y_0, y_1) that is never negative, so its discriminant can't be positive, which is $b^2 \leq ac$. That's condition 3. The diagonal conditions fall out at $(1, 0)$ and $(0, 1)$. No physics anywhere; both directions are polynomial-arithmetic identities.

Theorem 18.5 (Slice coherence $\iff$ NPA-PSD).

$$\text{coherent}(\mathbf{E}) \iff \text{psd-npa}(\mathbf{E})$$

where *coherent* means the three column-contractivity conditions hold, and *psd-npa* means the 5×5 specified zero-marginal, zero-within-party-moment matrix is positive semidefinite.

Proof. The full math is in the two derivations of Section 18 preceding this theorem. Recapitulated for the proof block:

Forward direction (column-contractive $\Rightarrow$ PSD). Assume the three column-contractivity conditions hold. The Section 18 derivation showed: condition 1 is exactly the principal-submatrix determinant on rows/columns $\{1, 2, 3\}$; condition 2 is exactly the principal-submatrix determinant on rows/columns $\{1, 2, 4\}$; condition 3 is exactly the Schur-complement determinant of the 4×4 block. Combined with the $\{0\}$ principal submatrix (just the entry $1 \geq 0$) and the diagonal entries (all $1 \geq 0$), every principal submatrix has non-negative determinant and non-negative diagonal, so by the principal-submatrix PSD criterion (Section A), the full 5×5 matrix is PSD. The Coq witness is `zero_marginal_npa_column_contractive_implies_psd`.

Reverse direction (PSD $\Rightarrow$ column-contractive). Assume the moment matrix M is PSD. Specialize $v^T M v \geq 0$ at the vector

$$v(y_0, y_1) := (0, -(E_{00}y_0 + E_{01}y_1), -(E_{10}y_0 + E_{11}y_1), y_0, y_1)$$

for free $y_0, y_1 \in \mathbb{R}$. Computing $v^T M v$ using the explicit moment-matrix entries and collecting terms gives the quadratic form

$$Ay_0^2 - 2By_0y_1 + Cy_1^2 \geq 0 \quad \text{for all } y_0, y_1 \in \mathbb{R},$$

where $A = 1 - E_{00}^2 - E_{10}^2$, $B = E_{00}E_{01} + E_{10}E_{11}$, $C = 1 - E_{01}^2 - E_{11}^2$. By the quadratic-non-negative-discriminant lemma (`quadratic_nonneg_discriminant`), this is equivalent to: $A \geq 0$, $C \geq 0$, and $B^2 \leq AC$. The three conditions are exactly the three column-contractivity conditions. Specializing further at $(y_0, y_1) = (1, 0)$ and $(0, 1)$ recovers conditions 1 and 2 directly; the general (y_0, y_1) case yields condition 3. The

Coq witness is `npa_psd_implies_column_contractive`. □ □

The left side is three polynomial inequalities (column contractivity of the specified zero-marginal, zero-within-party-moment matrix):

$$1 - E_{00}^2 - E_{10}^2 \geq 0 \quad (\text{condition 1})$$

$$1 - E_{01}^2 - E_{11}^2 \geq 0 \quad (\text{condition 2})$$

$$AC \geq B^2 \quad (\text{condition 3})$$

where $A = 1 - E_{00}^2 - E_{10}^2$, $B = E_{00}E_{01} + E_{10}E_{11}$, $C = 1 - E_{01}^2 - E_{11}^2$. These are the three conditions of `zero_marginal_column_contractive` (defined): column 0 norm ≤ 1 , column 1 norm ≤ 1 , determinant $AC - B^2 \geq 0$. The right side says the NPA moment matrix is positive semidefinite. One scope note so this sentence cannot be quoted into a bigger claim than it makes: for this pinned 5×5 matrix, PSD characterizes the orthogonal-observables zero-marginal slice exactly (that is what the biconditional proves, and a Gram decomposition of the PSD matrix hands back the realizing observables), but level-1 PSD in general, with marginals in play, is a relaxation, an outer approximation the NPA hierarchy tightens level by level. Section 18.7 climbs to the level that is exact for CHSH.

The forward direction is `zero_marginal_npa_column_contractive_implies_psd`. The reverse direction: given a PSD NPA matrix, specialize it at the vector $(0, -(E_{00}y_0 + E_{01}y_1), -(E_{10}y_0 + E_{11}y_1), y_0, y_1)$. The PSD condition gives a quadratic-in- y_0/y_1 that is everywhere non-negative. Then `quadratic_nonneg_discriminant` ($\forall t, a + 2bt + ct^2 \geq 0 \Rightarrow b^2 \leq ac$) closes the determinant condition. The diagonal conditions come from $y_0 = 1, y_1 = 0$ and $y_0 = 0, y_1 = 1$. Both directions are packaged as the biconditional `column_contractive_iff_npa_psd`.

What this means for the hierarchy of correlations:

- Classical local hidden variables: $|S| \leq 2$.
- Slice-coherent = realizable by the specified zero-marginal, zero-within-party-moment completion (biconditional, both directions, no project-local axioms). Every slice-coherent correlation satisfies $|S| \leq 2\sqrt{2}$, but NOT every correlation with $|S| \leq 2\sqrt{2}$ is slice-coherent: $(1, 0, 1, 0)$ has $|S| = 2$ but fails condition 1.
- No-signaling: no constraint beyond $|E_{ab}| \leq 1$.
- PR box ($E_{00} = E_{01} = E_{10} = 1, E_{11} = -1$): condition 1 gives $1 - E_{00}^2 - E_{10}^2 = 1 - 1 - 1 = -1 < 0$. Fails immediately by direct arithmetic.
- Tsirelson bound: `state_column_contractive_implies_tsirelson` proves slice-coherent $\Rightarrow |S|^2 \leq 8$ (i.e., $|S| \leq 2\sqrt{2}$) directly from column-contractivity. The chain factors through PSD: column-contractive $\Rightarrow$ NPA-PSD $\Rightarrow$ row bounds $\Rightarrow$ Tsirelson, and the row-bounds step is `tsirelson_from`

`_row_bounds`. The contrapositive ($|S| > 2\sqrt{2} \Rightarrow \text{not slice-coherent}$) follows immediately. The converse is false and not proved.

The gate checks membership in a specified algebraic set. That is the claim it has to earn. Run the sixteen deterministic local strategies through it: every one of them has all four correlators at ± 1 , so condition 1 reads $1 - 1 - 1 = -1 < 0$ and `CHSH_LASSERT` traps every single one of them, including the all-ones strategy that this section's own tightness theorem (`classical_bound_achieved`) runs at $\mu = 0$ to touch the classical bound. Those tallies are as true as tallies get, the trace really did accumulate those counts, and the gate rejects them anyway, because their correlators fail the fixed orthogonal completion. Marginals alone do not explain this: mixing each deterministic strategy with its simultaneous sign reversal preserves those correlators and gives zero marginals. In the other direction, a program can run a sequence of `CHSH_TRIAL` calls chosen to land the tally inside the slice, rather than earning it by playing the game honestly against an unpredictable referee, and the gate waves it through the same as an honest run; `LOAD_IMM` itself only writes registers, not the `vm_witness` buckets `CHSH_LASSERT` reads, so it cannot forge a tally directly, but a program is free to choose its own `CHSH_TRIAL` inputs and drive the buckets to any reachable value it likes. Whether a tally was earned by play against a genuine referee or walked there on purpose is the kind of fact the receipt discipline of Section 15 meters, not this opcode. So the correct reading of the boundary is set-theoretic, not moral. The classical polytope and the slice-coherent set intersect, uniform noise $(0, 0, 0, 0)$ sits comfortably in both, but neither contains the other: $(1, 0, 1, 0)$, reachable by a classical mixture and carrying a legal $|S| = 2$, fails condition 1 ($1 - 1^2 - 1^2 = -1 < 0$), while every point satisfying the fixed orthogonal completion passes. What a successful `CHSH_LASSERT` certifies is that the counters cohere with the zero-marginal slice, no more and no less, and the Tsirelson corollary is inherited from that membership, not from any power to sniff out forged data.

Kernel-level enforcement (the `CHSH_LASSERT` opcode). Here is the thing about a certification bit: by itself it is just a bit. A generic `CERTIFY` can set `vm_certified := true` and never once glance at the witness counters that are supposed to earn it. That is a rubber stamp with the ink already on it, and a rubber stamp is exactly what the whole discipline is built to forbid. So the certification channel cannot lean on `CERTIFY` for this discipline; it needs an opcode that does the looking, that reads the evidence before it is allowed to say the word. That opcode is `CHSH_LASSERT`, and it ties slice coherence as column-contractivity to the certification channel by refusing to take the claim on faith. It reads the `WitnessCounts` buckets, the counts of how the two parties actually came out together and apart, and it computes the three integer-arithmetic conditions `column_contractive_check_witness` in $\mathbb{Z}$ -arithmetic, no real and no rational ever evaluated at runtime, because the moment you let floating point near a soundness check you have smuggled in a place for the answer to drift. If the counts satisfy the three conditions, the program counter advances and the run

goes on. If they do not, there is no negotiating: the step traps to `LASSERT_TRAP_PC` and latches `vm_err`, and the incoherent certification never happens. Two theorems close the chain to NPA-PSD; both are stated and proved below.

| Plain reading. Integer arithmetic upstairs, real arithmetic downstairs, IZR is the elevator. The check throws out an empty setting bucket before it does anything else: $N_{00}, N_{01}, N_{10}, N_{11} > 0$ are part of the check itself. A witness record with two settings never exercised, all trials piled onto the other two, fails on those positivity checks and never reaches the polynomial conditions. With positivity in hand, write the three column-contractivity conditions in terms of the counts, cleared of denominators, and you have polynomial inequalities in $\mathbb{Z}$. Cast to $\mathbb{R}$ via IZR; that keeps ≥ 0 . Divide back through by the strictly positive trial counts and you get the real-valued conditions. The whole route avoids floating point at every step: integers in, reals out, division at the boundary, and the boundary is never zero.

Theorem 18.6 (Integer-check soundness, `column_contractive_check_witness_sound`). *Let w be the `WitnessCount` record with eight non-negative-integer fields $wc_same_{xy}, wc_diff_{xy}$ for $(x, y) \in \{0, 1\}^2$. Define $N_{xy} = wc_same_{xy} + wc_diff_{xy}$. The integer-only check `column_contractive_check_witness` is the conjunction of two parts: the four strict-positivity conjuncts $N_{00} > 0, N_{01} > 0, N_{10} > 0, N_{11} > 0$, and three $\mathbb{Z}$ -arithmetic inequalities on the counts, all written with no division. If the integer check returns `true`, then every N_{xy} is a positive integer, so*

$$E_{xy} = \frac{wc_same_{xy} - wc_diff_{xy}}{N_{xy}}$$

is an ordinary quotient with a positive denominator, and $\mathbf{E} = (E_{00}, E_{01}, E_{10}, E_{11}) \in [-1, 1]^4$ satisfies the three column-contractivity conditions of Section 18. This is a different fact from the general zero-trial convention of Section A, which sets $E_{xy} = 0$ when $N_{xy} = 0$ so that the CHSH score S is defined on every witness record. That convention lets S be computed at all; it plays no role here, because a passing check already guarantees no setting pair is empty.

Proof. The four positivity conjuncts are checked first, independently of the three polynomial conditions: the boolean function returns `false` the moment any $N_{xy} = 0$, before the polynomial terms are even compared. A `true` result therefore rules out every empty setting pair by construction. Take the record $wc_same_{00} = wc_same_{01} = 1$ with the other six counters zero: $N_{10} = N_{11} = 0$, so the positivity conjuncts for settings $(1, 0)$ and $(1, 1)$ fail and the check returns `false` outright, without ever reaching the polynomial comparison. No witness record with an empty setting pair passes.

For the remaining three conditions, multiplying through by N_{xy}^2 to clear denominators gives polynomial inequalities in the non-negative integer counts:

- Condition 1 ($1 - E_{00}^2 - E_{10}^2 \geq 0$) becomes the inequality $N_{00}^2 \cdot N_{10}^2 - (wc_same_{00} -$

$(wc_diff_{00})^2 N_{10}^2 - (wc_same_{10} - wc_diff_{10})^2 N_{00}^2 \geq 0$ after substituting E_{xy} and clearing denominators.

- Condition 2 is the analogous inequality with the E_{01}, E_{11} correlators.
- Condition 3 ($AC \geq B^2$ in the notation of Section 18) becomes a polynomial inequality of degree 8 in the integer counts: A 's numerator and C 's numerator are each degree 4, their product is degree 8, and B 's numerator squared is likewise degree 8, after clearing the common denominator $N_{00}^2 N_{01}^2 N_{10}^2 N_{11}^2$.

To prove the conclusion: assume the check returns `true`. The positivity conjuncts give $N_{xy} > 0$ for all four settings, so no case split on $N_{xy} = 0$ is needed; that case cannot arise. Cast the integer counts to $\mathbb{R}$ by $\mathbb{I}\mathbb{Z}\mathbb{R} : \mathbb{Z} \rightarrow \mathbb{R}$, which preserves the inequalities. Divide through by N_{xy}^2 (or N_{xy}^4 , for condition 3), licensed by $N_{xy} > 0$. After dividing, the three inequalities are exactly conditions 1, 2, 3 of column-contractivity in $\mathbb{R}$. The Coq proof closes the post-division algebra by `nra` (nonlinear real arithmetic) acting on a sum-of-squares-shaped goal. $\square$ $\square$

| Plain reading. Three theorems snap together: the kernel step rule says no-trap means the integer check passed; the soundness lemma says a passing integer check means real-valued column-contractivity; the biconditional says column-contractivity means NPA-PSD. Stack them. A successful CHSH_LASSERT step is a proof that the witness-derived correlators land inside the NPA-PSD set. No physics input anywhere in the stack. The (1,0,1,0) correlation fails the integer check (condition 1: $1 - 1 - 1 = -1 < 0$), so any program that accumulated those statistics traps on CHSH_LASSERT.

Theorem 18.7 ($\text{CHSH_LASSERT} \Rightarrow \text{NPA-PSD}$, `chsh_lassert_no_trap_implies_npa_psd`). *Let s be a state and let $s' = \text{vm_applys}(\text{instr_chsh_lassert } \delta)$. If $s'.\text{vm_err} = \text{false}$ and $s'.\text{vm_pc} = s.\text{vm_pc} + 1$ (i.e., the CHSH_LASSERT step advanced normally rather than trapping), then the specified zero-marginal, zero-within-party-moment matrix M built from the witness-derived correlators E_{xy} of $s.\text{vm_witness}$ is positive semi-definite.*

Proof. By the apply rule for `instr_chsh_lassert`: the step branches on `column_contractive_check_witness` run on $s.\text{vm_witness}$. If the check returns `false`, the step traps to `LASSERT_TRAP_PC` and sets `vm_err = true`. If the check returns `true`, the step advances PC by 1 and leaves `vm_err = false`. By hypothesis, s' exhibits the second behavior, so the integer check returned `true` on $s.\text{vm_witness}$.

Compose with the preceding theorem (integer-check soundness): the correlators E_{xy} derived from $s.\text{vm_witness}$ satisfy the three real-valued column-contractivity conditions.

Compose with the forward direction of `column_contractive_iff_npa_psd` (the

biconditional proved in Section 18): column-contractivity implies the NPA moment matrix M is positive semidefinite. $\square$ $\square$

Run the (1,0,1,0) correlation through `CHSH_LASSERT` and watch what the step does with it: the integer check fails on the very first condition, the step refuses to advance, and the state comes out the far side wearing `vm_err = true`. Nothing in the kernel was ever told where (1,0,1,0) sits relative to any physics, and the point itself is not unphysical: a classical mixture reaches it with room to spare, marginals and all. The kernel was told to count, clear denominators, and test three inequalities, and for this point the arithmetic simply would not close, because the point lies outside the zero-marginal slice the three inequalities carve out. That is the only gate there is. A trace whose correlators are column-contractive walks straight through; one whose correlators are not gets stopped by its own numbers, and the kernel never has to know why.

To falsify: exhibit $E_{00}, E_{01}, E_{10}, E_{11}$ satisfying the three polynomial inequalities but with a non-PSD NPA matrix. No such example exists under the stated definitions; the theorem proves it impossible. Or exhibit a quantum experiment producing a non-PSD NPA matrix. That would be news of a different kind. It would break the NPA bridge or quantum mechanics, not this theorem.

18.7 Lifting the bridge to level 1+AB: the four-slice Schur cascade

Same test, bigger grid. The 5×5 grid only holds the four correlators. The 9×9 grid also holds five higher moments, the ones that involve three or four measurements at once. A bigger grid means more windows to check and a lot more arithmetic, so the machine checks it in four stages, letting one more kind of moment move at each stage. What follows is the formal layer.

Section 18 establishes the runtime bridge: `CHSH_LASSERT` implies PSD of a fixed 5×5 moment matrix with zero marginals and zero within-party cross moments. This fixed slice excludes some classical correlators. Allowing the within-party moments to vary gives the full correlator ellipsope developed in Section 18.8. The Q_{1+AB} checks instead validate a specified 9×9 moment matrix on the basis $\{I, A_0, A_1, B_0, B_1, A_0B_0, A_0B_1, A_1B_0, A_1B_1\}$ with supplied higher moments. Their soundness theorem establishes PSD for that matrix. For complete behaviors, including marginals, finite-level NPA feasibility is a relaxation: Chaturvedi (2026, arXiv:2607.14569) proves that no fixed finite level describes the whole quantum behavior set in this scenario.

And the two levels are not two separate tricks. The 5×5 and the 9×9 checks are one dimension-polymorphic PSD predicate evaluated at two sizes: top index 4 for the 5×5 , top index 8 for the 9×9 , the same fold either way. The level-1 `CHSH` biconditional from Section 18 is the dimension-5 instance of it, recovered as a corollary that imports the original 5×5 result as a black box and adds no axioms over it (`column`

`n_contractive_iff_general_realizable`). So the lift in this subsection is not a 9-shaped coincidence dressed up as a method: it is the same realizability projection run one dimension up, and the dimension-5 corollary reproducing the bridge you already have is exactly the check that it is one predicate and not two.

18.7.1 The 9×9 moment matrix and its five γ parameters

The level-1+AB moment matrix is parametric in nine reals: the four CHSH correlators $(E_{00}, E_{01}, E_{10}, E_{11})$ and five additional moments

$$\begin{aligned}\gamma_1 &= \langle A_0 A_1 B_0 \rangle, & \gamma_2 &= \langle A_0 A_1 B_1 \rangle, & \gamma_3 &= \langle A_0 B_0 B_1 \rangle, \\ \gamma_4 &= \langle A_1 B_0 B_1 \rangle, & \gamma_5 &= \langle A_0 A_1 B_0 B_1 \rangle.\end{aligned}$$

γ_1, γ_2 label the two three-body A - A - B slots, γ_3, γ_4 the two three-body A - B - B slots, and γ_5 the four-body slot. γ_5 enters the matrix in two cells that are *not* equal: $+\gamma_5$ in the $(A_0 B_0, A_1 B_1)$ cell and $-\gamma_5$ in the conjugate $(A_0 B_1, A_1 B_0)$ cell. Here is where that convention comes from. In the physics picture those two cells hold the same four operators with B_0 and B_1 swapped, and orthogonal measurements anticommute, so the swap flips the sign. That minus sign is the anticommutator a CHSH violation runs on. That is the motivation, and only the motivation. In Coq the angle brackets are labels on the entries of a real polynomial matrix. The nine parameters are independent real slots, the opposite signs are part of how the matrix is defined, and nothing in the proofs builds operators, adjoints, or anticommutation. Reading the entries as physical expectation values would take a separate representation theorem. The 9×9 entries are polynomials of degree ≤ 2 in (E_{ij}, γ_k) , and PSD of this matrix is the formal Q_{1+AB} condition used below.

18.7.2 Why four slices

What a slice is. Here “slice” means three things together: a parameter domain, the PSD predicate on it, and the integer check that certifies it. What the source proves is that each check is sound. It does not prove that the accepted sets nest strictly, that the integer checker accepts every PSD point in its domain, or that Slice D is the whole physical quantum cone. Any inclusion claim below is about the formal domains only, and would need its own zero-extension and certificate-translation proof.

The substrate certifies four γ -slices, each via its own cert-opcode and its own integer-arithmetic check:

- **Slice A** ($\gamma = 0$, tag 0x2F): all five γ_k pinned to zero. The bool decider checks Q_1 column-contractivity (the three conditions of §18) plus the SOS sum-of-squares condition $E_{00}^2 + E_{01}^2 + E_{10}^2 + E_{11}^2 \leq 1$. Closed by an explicit 4-D Cauchy-Schwarz SOS identity.
- **Slice B** (γ_5 free, tag 0x30): the four-body moment γ_5 is admitted as a bucket-pair argument; the other four γ are zero. The check adds a weighted 4-D Cauchy-Schwarz identity that handles the $2\gamma_5(v_{11}v_{22} - v_{12}v_{21})$ cross term arising in the

residual (the v_{ij} are the coordinates of the paired 4-D vectors the identity runs over) (the relative minus is the conjugate-cell sign above: γ_5 on the $v_{11}v_{22}$ pairing, $-\gamma_5$ on the $v_{12}v_{21}$ pairing).

- **Slice C** ($\gamma_3, \gamma_4, \gamma_5$ free, tag 0x31): the two A - B - B moments and the four-body moment are admitted; $\gamma_1 = \gamma_2 = 0$. The check is a 4×4 Sylvester PD test on a difference matrix $H_{\gamma_{345}} = \det_M \cdot M_M - M_N$, where M_M and M_N are the two denominator-cleared moment blocks the slice compares and $\det_M$ is the cleared determinant scaling them, derived via an augmented 2×2 Schur-slack lemma.
- **Slice D** (full $\gamma_1, \dots, \gamma_5$ free, tag 0x32): all five γ parameters are admitted. The check is a compositional $\text{sym}6 \rightarrow \text{sym}5 \rightarrow \text{sym}4$ Schur cascade: the only check stiff enough to handle the bilinear A - A - B cross terms γ_1, γ_2 introduce in the residual.

These four are not four independent gadgets. Their parameter domains nest, $A \subset B \subset C \subset D$: each slice stops pinning a moment the previous slice held at zero. The reason to build them in exactly this order is that every newly freed moment drags a harder cross term into the residual, and you would rather meet those one at a time than all at once. What climbs is the set of parameters each check lets in. Whether the accepted sets climb strictly is a different question, and nothing here proves it. Each check is proved sound in one direction, into PSD of the specified matrix; none has a proved converse. Slice D, with nothing pinned, is the broadest check here. It is not a theorem that Slice D is the whole Q_{1+AB} cone.

18.7.3 The compositional Schur cascade (Slice D)

The mathematical core of Slice D is the chain

$$\begin{aligned} \text{PSD9}(M) &\Leftarrow \text{sym6 PD-interior on the 21 entries of } M_{6 \times 6} \\ &\Leftarrow \text{sym5 PD-interior on the } 5 \times 5 \text{ Schur complement of row 1} \\ &\Leftarrow \text{sym4 PD-interior on the } 4 \times 4 \text{ Schur complement of row 2} \end{aligned}$$

where $M_{6 \times 6}$ is what the 9×9 collapses to once the Q_{1+AB} subsumption identities pin its dependent entries: the projection onto the basis $\{v_{B1}, v_{B2}, v_{11}, v_{12}, v_{21}, v_{22}\}$, after which the 9×9 lands in the PSD cone whenever the 6×6 residual does. From there it is Schur complements all the way down, and it is the same move you already watched at level 1, run on a bigger matrix. The Schur-complement criterion (Section A), the one the level-1 bridge used on its 4×4 block, turns a positive-definiteness question about the 6×6 into one about a 5×5 Schur complement; run it again and the 5×5 becomes a 4×4 ; and the 4×4 closes by Sylvester's criterion, all four leading principal minors positive. That is the whole soundness argument: a base case and three reductions, each one a Schur step you can carry out by hand, stacked from the 4×4 back up to PSD9.

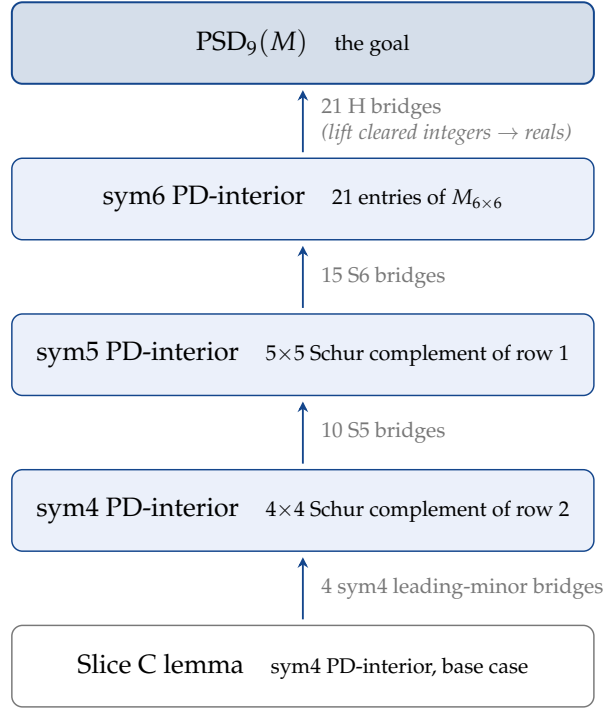


Figure 9: The Schur cascade lifts the base-case sym4 PD-interior fact from Slice C all the way up to $\text{PSD}_9(M)$. Each arrow is one Schur-complement reduction, a polynomial identity in the correlators and moments. The counts beside each arrow are how many such identities the integer check grinds through at that step, one per matrix entry.

Every reduction is a single polynomial identity in the correlators E_{ij} and the moments γ_k . To keep the opcode's check decidable and integer-only, each identity is multiplied through by the common denominator $\text{KZ} = (N_{00}N_{01}N_{10}N_{11} \cdot D_{g_1}D_{g_2}D_{g_3}D_{g_4}D_{g_5})^2$, where the N_{ij} are the per-setting trial counts and the D_{g_k} the γ -bucket denominators, and verified on integers, the same clear-the-denominators move as level 1. It is a lot of arithmetic: one identity per matrix entry, twenty-one across the 6×6 , fifteen for the first Schur step, ten for the second, four leading minors at the base. The kernel grinds every one. But the grinding isn't the argument; the argument is the iterated Schur reduction above, and a reader who wants to break it breaks the reduction, not the fifty identities. A passing integer check is those reductions holding, and those reductions holding is M in the PSD cone. The check leans on nothing past the two background facts level 1 already leaned on: the decidability of the reals and functional extensionality.

18.7.4 What this delivers and what it does not

Soundness. For each of the four Q_{1+AB} assertion opcodes, the successful-step condition in its theorem implies symmetry and positive semidefiniteness of the specified 9×9 moment matrix at the witness-derived correlators and supplied γ values. This is an arithmetic certificate about those entries. The theorem does not construct a physical experiment.

Scope. These theorems establish the four integer-check soundness implications and the associated matrix identities. Physical realization, completeness of the checks, and identification of a moment relaxation with a quantum behavior set require their own arguments. The full correlator ellipptope is treated separately below.

Accounting. Each assertion instruction has declared cost $S(\mu_delta) \geq 1$, charged on both success and failure. The integer test establishes its matrix property; the cost schedule records payment for executing the test. These are separate guarantees. The assertion opcodes preserve `vm_certified`; their success is observed through the stated trap and error conditions.

Falsification. Exhibit witness counters and γ bucket pairs for which one of the four cert-opcodes does not trap but the resulting 9×9 NPA moment matrix is *not* PSD, OR show that the Schur-cascade soundness theorem at any slice is incorrect (an SOS identity wrong, a minors-witness step that does not imply PSD9, a bool decider unsound), OR show that the soundness chain depends on a project-local axiom beyond the two standard-library axioms cited above. The `Coq Print Assumptions` report is the audit surface: every theorem either lists no new axiom or fails the proof gate.

18.8 The full correlator set: the ellipptope completion

The slice nails two numbers in the grid to zero. The ellipptope lets them move. For the slice you ask whether the point fits one fixed grid. For the ellipptope you ask whether there's any way to fill in those two blanks that makes the grid pass.

The fixed orthogonal completion excludes some classical correlators, including every deterministic local strategy. The full ellipptope allows the within-party cross moments to vary. `EllipptopeCompletion.v` defines membership by existence of a PSD completion and proves inclusion of classical correlators, the Tsirelson bound, and exclusion of the PR box. That is the mathematical object this subsection examines.

The characterization is the one Tsirelson's theorem forces. Take the same 5×5 moment matrix, marginals still zero, but stop pinning the two cross moments: let x stand for $\langle A_0 A_1 \rangle$ and y for $\langle B_0 B_1 \rangle$, both free. Call a correlator tuple $(E_{00}, E_{01}, E_{10}, E_{11})$ *ellipptope-realizable* when *some* completion (x, y) makes the matrix PSD (`ellipptope_realizable`, defined in `EllipptopeCompletion.v`). The existential over (x, y) is the entire difference between the slice and the set: the slice asks whether the point is coherent with one fixed completion, the zero one; the set asks whether any completion works.

Five theorems pin the characterization down, all mechanised, all with `Print Assumptions` reporting only the same `stdlib` families as the rest of the CHSH algebra (the classical-reals construction and functional extensionality; zero project-local axioms):

- `deterministic_strategy_ellipptope`: every deterministic local strategy is in the set. The witness is the honest one, $x = a_0 a_1$ and $y = b_0 b_1$, the cross

moments the strategy actually has, and the completed matrix becomes an explicit Gram matrix whose quadratic form is a sum of two squares. All sixteen sign patterns are instances of one parametric theorem. `classical_tightness_witness_elliptope` names the sharpest instance: the all-ones strategy, the $\mu = 0$ witness of `classical_bound_achieved` that the slice gate traps, is elliptope-realizable.

- `turing_point_elliptope`: $(1, 0, 1, 0)$, this document's running example of a classically reachable point the slice rejects, is in the set, witnessed by $x = 1$, $y = 0$ and the sum-of-squares decomposition $v_0^2 + (v_1 + v_2 + v_3)^2 + v_4^2$ of its completed quadratic form.
- `elliptope_convex`: the set is closed under binary convex combination; `elliptope_finite_mixture` extends that to arbitrary finite sub-convex mixtures by a mechanised renormalization induction (the weight deficit absorbed by the origin, `elliptope_zero`); and `lhv_mixture_elliptope` composes it with the deterministic theorem: every finite mixture of deterministic sign strategies, which is to say every local-hidden-variable correlator, lands inside. Not by hand-waving from the binary case; the n -ary theorem is in the kernel. As a small check on the geometry, the classical point $(1, 0, 1, 0)$ has row squared norms $1^2 + 0^2 = 1$ and $1^2 + 0^2 = 1$ if the rows are ordered as (E_{00}, E_{01}) and (E_{10}, E_{11}) , while its columns have squared norms $1^2 + 1^2 = 2$ and $0^2 + 0^2 = 0$. Transposing the matrix supplies the corresponding column/row counterexample; the distinction matters because the fixed-completion proof uses $E^T E$ and therefore column contractivity.
- `elliptope_tsirelson`: everything in the set satisfies $S^2 \leq 8$, i.e. $|S| \leq 2\sqrt{2}$. The proof is worth a sentence, because the fixed-slice row-norm bound does not hold for arbitrary completions. At $y \neq 0$ the row-norm bound fails, and honestly so: the point $(1, 0, 1, 0)$ itself has a row of norm $\sqrt{2}$. What replaces it is Cauchy–Schwarz run inside the PSD form itself (`psd_cauchy_schwarz`, derived from the same discriminant lemma the reverse bridge uses): the two budgets $\|B_0 + B_1\|^2 = 2 + 2y$ and $\|B_0 - B_1\|^2 = 2 - 2y$ are coupled through the completion, their product is at most 4, and $S^2 \leq 8$ falls out of that coupling. The bound holds at every point of the set, not because each row is short, but because the completion cannot make both budgets long at once.
- `pr_box_not_elliptope`: the PR box has no completion; its S is 4 and the Tsirelson theorem refuses it. And `beyond_classical_elliptope` exhibits a rational point with $S = 12/5 > 2$ inside the set, so the containment of the classical polytope is strict: $\text{classical} \subset \text{elliptope} \subset \text{no-signaling}$, with both inclusions proper, from inside the formalization.

The correlator geometry has three distinct sets: the classical polytope and the zero-marginal slice intersect without containment, and both sit inside the elliptope, which

admits every honest classical point the slice refuses while still refusing everything past $2\sqrt{2}$.

The completion gate is executable. `ElliptopeGate.v` builds the decidable check an opcode can branch on (`elliptope_check_full`), in the same integer-only discipline as `column_contractive_check_witness`: no real and no rational evaluated at runtime, every denominator cleared into $\mathbb{Z}$. The check has two branches. The PD branch is the cheap path for a strictly-interior point: it takes completion buckets for (x, y) , exactly the bucket-pair idiom the 1+AB opcodes use for their γ moments, and runs the fraction-free Sylvester test on the denominator-cleared completed matrix, whose soundness is the same `sym4_qf_nonneg_from_pd` LDLT lemma the Slice-C development proves. The LDL branch checks certificates that can include singular boundary completions: it takes a rational LDL^T certificate, six subdiagonal entries and four pivots over common positive denominators, and verifies the ten entry equations of $M = LDL^T$ in cross-multiplied $\mathbb{Z}$ -arithmetic. Its soundness (`elliptope_ldl_check_sound`) is a sum-of-weighted-squares identity: given the equations, the quadratic form is $d_1P_1^2 + d_2P_2^2 + d_3P_3^2 + d_4P_4^2$. The file proves soundness of this certificate format and supplies concrete accepted boundary points. The load-bearing theorem is `elliptope_check_full_sound`: a passing check entails elliptope membership of the witness-derived correlators, with the check's own passing as its only hypothesis. And the gate computes, not just proves, each by reflexivity: `gate_accepts_all_ones` runs the $\mu = 0$ tightness witness through the LDL branch (its all-ones completion is rank-one singular) and it passes; `gate_accepts_turing_point` does the same for $(1, 0, 1, 0)$; `gate_accepts_beyond_classical` passes the interior $S = 12/5$ point through the PD branch; `gate_accepts_pythagorean_boundary` certifies $(3/5, 4/5, 4/5, -3/5)$, whose arcsine sum is exactly π by the 3-4-5 triangle so it sits exactly on the Tsirelson curve with $S = 14/5 > 2$; it's invisible to the strict PD branch because its completion is singular, and it's reached by the LDL branch with pivots $(1, 1, 0, 0)$ at $x = y = 0$; and `elliptope_full_gate_never_accepts_pr_box` proves no completion buckets and no certificate ever get the PR box through, not by enumeration but by soundness plus the Tsirelson theorem closing both branches at once. This certificate format needs rational completion data. Soundness and the accepted examples don't show the gate accepts every elliptope point, and they don't rule out other exact certificate formats.

The runtime `CHSH_LASSERT` opcode checks the fixed orthogonal slice. The elliptope gate isn't wired to an opcode yet. That would take instruction semantics, a cost rule, and a hardware implementation. The completion results are about correlators. They don't pin down the full set of behaviors once marginals come in. Calling the completion model physical quantum correlations leans on outside mathematics, and producing entanglement in hardware would take physical evidence. The direct mathematical challenges are concrete: a deterministic sign pattern whose completed Gram matrix has a negative quadratic form would kill `deterministic_strategy_elli`

ptope; an ellipptope-realizable tuple with $S^2 > 8$ would kill `elliptope_tsirelson`; a PSD completion for the PR box would kill `pr_box_not_elliptope` and `Tsirelson` with it.

19 Irreversibility accounting and the μ -hierarchy

19.1 The Landauer association is motivation only

Same shape as Landauer's 1961 argument: erasing one bit must release at least $kT \ln 2$ of heat into the environment (k Boltzmann, T absolute temperature, $\ln 2$ the natural log of 2). The Coq proofs below do not depend on Landauer being right about the physics. They formalise a conservative accounting indicator at the instruction level: a cost-positive instruction contributes one to `irreversible_bits` by definition. That label is bookkeeping, not a proof that the instruction erased information or was physically irreversible. The count of positively charged firings is bounded by total μ . Shape borrowed, content paid for separately.

| Plain reading. Each cost-positive instruction contributes at least 1 to total cost and exactly 1 to the indicator. Sum across the trace and the count is bounded by the cost. A 0/1 indicator bounded by what it indicates. All it shares with Landauer is the shape: a lower-bound quantity attached to an event. It is not an irreversibility theorem. No physics in the proof. No microscopic erasure model. No physical entropy. No Boltzmann constant. The conversion of μ counts to joules is the calibration hypothesis (`mu_landauer_unruh_calibrated`, Section 25). No kernel theorem uses it.

Theorem 19.1 (μ -Landauer step bound). *Let `irreversible_bits(i)` be the indicator function that returns 1 if `instruction_cost(i) > 0` and 0 otherwise. Let `total_irreversible_bits(trace) :=  $\sum_{i \in \text{trace}}$  irreversible_bits(i)`. Then for every trace,*

$$\text{total_irreversible_bits}(\text{trace}) \leq \text{trace_total_cost}(\text{trace}).$$

The Coq witness is `total_irreversible_bits_le_cost`.

Proof. Per-instruction inequality: for every i , `irreversible_bits(i) ≤ instruction_cost(i)`. Two cases.

Case `instruction_cost(i) = 0`. Then `irreversible_bits(i) = 0` by definition, and $0 \leq 0$ holds.

Case `instruction_cost(i) > 0`. Then `irreversible_bits(i) = 1`, and the cost schedule (Section 7) gives `instruction_cost(i) ≥ S(δ) ≥ 1` for cert-setters or `instruction_cost(i) ≥ δ ≥ 1` otherwise. So $1 \leq \text{instruction_cost}(i)$.

Summing the per-instruction inequality over the trace:

$$\begin{aligned} \sum_{i \in \text{trace}} \text{irreversible_bits}(i) &\leq \sum_{i \in \text{trace}} \text{instruction_cost}(i) \\ &= \text{trace_total_cost}(\text{trace}). \end{aligned}$$

The right equality is by definition of `trace_total_cost` (or via the Latent- μ theorem applied to the trace from $\mu_0 = 0$). $\square$ $\square$

The physical claim has to pass a consistency test first. `calibrated_model_exists_iff_positive_cost` says exactly when the abstract positive-dissipation premise and the upper-calibration premise can hold together. The explicit yes/no reset model has an instance. But keep the previous state in a history list and every fixed-instruction step becomes injective, while the step you can see stays the same. `zero_cost_vm_jump_has_injective_history_lift` applies that move to the zero-cost F1 counterexample. The PC can collapse to one place while the bigger state keeps its past. I can't spot physical erasure by watching the PC alone.

The physical question is still open. I want to know whether the formal charge has a defensible physical interpretation. Counting positively charged instructions does not answer that question. Neither does the finite-map Python demonstration: it checks its combinatorial model, not heat released by a device. The F1 counterexample makes the next requirement concrete. Build a physical model whose premises can hold together, account for retained history, and then justify the connection to an experiment. The permanent-certificate theorem (Section 9) supplies the first two for the certifying step. Its premises hold together, as the reset model shows, and finite memory accounts for retained history. The entropy version carries it as far as a formula for heat, $k_B T \ln((m+k)/m)$, with Landauer's principle as the named premise. The connection to an experiment is what remains. Until that is done, the VM ledger gives no measured energy price for certification.

What the permanence results add to Landauer, and what they don't. I went looking for a prediction the permanent-certificate theorems make that Landauer's principle alone doesn't. There isn't one. The smallest case, one certified state and one flipping in, is setting a bit to one whatever it held. That is the bit reset Landauer described, and Bérut and colleagues measured its heat approaching $k_B T \ln 2$ in a colloidal double-well experiment ([Nature 483, 2012](#)). The general bound, $k_B T \ln((m+k)/m)$ for squeezing $m+k$ equally likely states into m , is Landauer's formula for that squeeze. The split between a known state, which owes nothing, and an uncertain one, which owes heat, is Maroney's and Sagawa's. So the physics content is exactly Landauer's principle plus one bridge: on finite memory, a certificate that lasts can only be switched on by a squeeze. The bridge formalized here is checked. The heat is Landauer's, and Bérut's experiment already measured it for the smallest case. A test of the bridge itself would

need a device that stores a certificate with no step able to revoke it and a way to meter the certifying step alone, and I don't know of one.

19.2 The μ -hierarchy

Borrow the silhouette of the classical time-hierarchy theorem, and then say honestly how much smaller the thing inside it is. The shape is the part everyone knows: more of the resource buys you something you provably could not have had with less. The resource here is μ , and the something is a certification level: traces are classified by an executed `CERTIFY` whose declared charge is at least k . Every level $k \geq 1$ is reachable by a one-instruction witness that pays exactly k , and no trace in that level can cost less than k . That pairing, reachability on one side and a conservation floor on the other, is the whole content. It is not a statement that one problem class outruns another. Whether more μ ever buys a genuinely harder problem is a different and much heavier question. It is open, I have it on the list in Section 25, and I am not going to quietly borrow its weight here while it is still unpaid.

| Plain reading. Two parts. Part (1) is by construction: use the one-instruction trace `CERTIFY( $k - 1$ )`. Its charge is $S(k - 1) = k$, it ends certified, and there is no padding instruction. Part (2) is the trace-side lower bound: the executed level- k `CERTIFY` appears in the ledger with cost at least k , so the total cost is at least k . For the smallest nontrivial example, $k = 2$ is exactly `CERTIFY(1)`, which costs $S(1) = 2$.

Theorem 19.2 (μ -Hierarchy (`mu_hierarchy_theorem`)). *Call a trace level- k certified when it ends with `vm_certified = true` and its executed-instruction log contains a `CERTIFY` whose charge $S(\delta)$ is at least k . For every $k \geq 1$:*

1. *There exists a level- k certified trace with total cost exactly k .*
2. *Any level- k certified trace has `trace_total_cost` $\geq k$.*

Proof. Part (1): the level- k witness. The one-instruction trace `[instr_certify( $k - 1$ )]` does it: the single `CERTIFY` charges $S(k - 1) = k$, which certifies at level k by definition, sets `vm_certified` to `true`, and is the whole bill. Total cost exactly k .

Part (2): cost lower bound. The level- k hypothesis hands over an executed `CERTIFY` with $S(\delta) \geq k$; the kernel proof finds it by scanning the executed-instruction log. That one step charged at least k to the ledger, every other step charged at least 0, and by Latent μ the total cost is the sum of the step charges, so the total is at least k . The scan earns the floor from the certifying step itself.

Each level $k \geq 1$ is therefore realizable at cost exactly k , and no trace certifies at level k for less. □

| Plain reading. For any budget you propose, the hierarchy theorem produces a

certified trace that exceeds it by 1. So no finite budget caps all certifications.

Corollary 19.3 (`mu_hierarchy_no_upper_bound`). *No fixed μ -budget suffices for all certification levels.*

Proof. Suppose for contradiction that some fixed budget $B \geq 0$ suffices for all certification levels, meaning every certified trace satisfies $\text{vm_mu} \leq B$. Apply the μ -Hierarchy theorem at $k = B + 1$: there exists a certified trace with $\text{vm_mu} = B + 1 > B$, contradicting the budget bound. $\square$ $\square$

What this comes down to is that the ledger does not give discounts. Pick any integer level you like and there is a certified trace that arrives at exactly that level; try to reach it for one unit less and you cannot, not because anybody forbade it but because the cost was already spent on the way up and conservation will not let you unspend it. That is the entire argument, and where it is mechanised the check agrees. Notice what I did not say. I did not say more μ makes the substrate “strictly more powerful”. That phrase belongs to the problem-class reading, the question of whether a bigger budget cracks problems a smaller one cannot, and this theorem does not touch it. I would rather leave that fence standing in plain sight than collect on a claim I have not paid for.

Here is the classical statement the shape comes from, so you can see exactly where the resemblance starts and where it stops. The time-hierarchy theorem says that for a time-constructible bound $T(n)$ there are problems you can solve in time $T(n) \log T(n)$ that you cannot solve in time $T(n)$: hand the machine a little more clock and it can decide things it could not decide before. That is a statement about which problems are in reach. The μ -hierarchy keeps the silhouette and trades the cargo: more μ buys more certification levels, which is a statement about how high the ledger can climb, not about which problems come into reach. It does not use the classical theorem, and it does not define a single time-bounded class. So treat the resemblance as a way to see the result, not as a smuggled inheritance of the classical theorem’s strength. The proof a few lines up is the part actually holding weight; the analogy is just the light I am reading it by.

A direct counterexample would be a $k \geq 1$ trace reaching `vm_certified = true` and $\text{vm_mu} \geq k$ with total trace cost $< k$. Under the theorem’s premises that cannot happen; if it does, the cost accounting or the proof is wrong.

20 Discrete Geometry over Partition Graphs

20.1 Discrete triangulation of the partition graph

A partition graph is not only a bookkeeping structure. You can also read it as a shape, a discrete geometric object, and put geometric questions to it. The honesty of the framing matters here, so I will be plain about what is and is not being claimed. The

result below reads the partition graph as a triangulated combinatorial object, a graph cut into triangular pieces, and sets it in an equilateral angle model: every corner of every triangle contributes $\pi/3$ radians, which is 60 degrees, because the three angles of an equilateral triangle are equal and a full turn is 2π radians, or 360 degrees. Those angles are a modeling choice. They are not pulled out of μ costs, and I am not going to dress them up as if they were.

A discrete triangulation defines a curvature-like angle defect. As I understand it (I researched this for the project, so check me on the geometry): on a flat surface, if you draw a triangle, the interior angles sum to 180° . On a curved surface, the angles can sum to more or less than 180° . Curvature is the amount by which triangles deviate from flat. For an interior vertex of a triangulated surface, the usual discrete curvature is 2π minus the incident angle sum. At a boundary vertex, the corresponding boundary turning contribution uses π instead. The quantity used in the Coq result below uses 2π at every vertex; on a surface with boundary it therefore includes an extra boundary contribution.

I went looking for the right invariant and found the Euler characteristic χ , a number that captures the overall shape of a triangulated surface, independent of how you draw the triangles. For any triangulation with V vertices, E edges, and F triangular faces: $\chi = V - E + F$. As I read it, no matter how you triangulate the same surface, you get the same number. For a sphere, $\chi = 2$. For a torus (donut shape), $\chi = 0$. The continuum Gauss-Bonnet theorem (which I am citing, not certifying) says the total curvature of a closed surface equals $2\pi\chi$: the local geometric information (curvature) integrates to a global topological fact (shape). The discrete analogue is stated below. The continuum result is exposition.

20.2 The restricted angle-defect identity

The definition `well_formed_triangulated` bundles properties of a partition graph, including triangle and incidence conditions and the additional equation $B = 3\chi$. That equation is a hypothesis, not a general theorem about triangulated disks. For example, a disk made by dividing a quadrilateral into two triangles has $V = 4$, $E = 5$, $F = 2$, $B = 4$, and $\chi = 1$, so it fails this extra condition. The predicate does not separately construct a planar embedding or establish every topological property of a disk.

| Plain reading. The 5π comes from two places: how D is defined, and the extra boundary-count rule $B = 3\chi$. Coq checks what follows from them. It doesn't show that rule is forced by computation, by topology in general, or by nature. Take $B = 3\chi$ away and what's left is $D = 2\pi\chi + \pi B$ under the incidence equations. There's no universal $5\pi\chi$ law.

Theorem 20.1 (Restricted angle-defect identity (`discrete_gauss_bonnet`)). *Let g be a partition graph satisfying `well_formed_triangulated`. In addition to its graph*

and triangle conditions, this predicate requires $3F = 2I + B$, the triangle-incidence degree sum $\sum_v \deg(v) = 3F$, and $B = 3\chi$, with $\chi = V - E + F$. The edge definitions give $E = I + B$. Assign angle $\pi/3$ to every triangle corner and define

$$D(g) = \sum_v (2\pi - \deg(v)\pi/3),$$

including the same 2π term at boundary vertices. Then $D(g) = 5\pi\chi(g)$.

Proof. The required triangle-incidence sum gives $D = 2\pi V - \pi F$. From $E = I + B$ and $3F = 2I + B$, we get $B = 2E - 3F$. Consequently,

$$D - 2\pi\chi = (2\pi V - \pi F) - 2\pi(V - E + F) = \pi(2E - 3F) = \pi B.$$

The additional hypothesis $B = 3\chi$ now gives $D = 2\pi\chi + \pi B = 5\pi\chi$. □

For a triangulated surface whose boundary is a union of simple cycles, the number of boundary vertices equals B . Replacing 2π by π at those boundary vertices subtracts πB , giving the usual combined interior-curvature and boundary-turning total $2\pi\chi$. This explains the relation to discrete Gauss–Bonnet. It requires identifying the boundary vertices and the appropriate surface hypotheses. The existing Coq theorem proves the defined D identity under its own predicate; it does not formalize that boundary-corrected theorem.

For a simple equilateral triangle, the distinction is visible without any machinery. Summing $2\pi - \pi/3$ at its three vertices gives 5π . Summing the boundary terms $\pi - \pi/3$ gives 2π . The first sum is the quantity defined here; the second is the boundary turning total. Both calculations can be right while answering different questions.

I did not write these proofs by hand. I directed LLMs and used the checker to test the formal statements. That method still leaves me responsible for checking the definitions and the interpretation. The 5π identity is a conditional combinatorial result. It isn't evidence for a new physical constant, and it doesn't derive angles from μ cost.

Part III

Checking it

21 What Coq is, and why I used it

Coq is a proof assistant, which is a phrase I had to go look up. Underneath the phrase it is a programming language, but one whose type system is strict enough to do a thing ordinary languages cannot: you write a proof down as a program, and the

compiler refuses it unless the program genuinely matches the statement you claimed. You cannot talk it into a maybe.

Here is the idea. In a normal programming language, a type says something about what kind of value a variable holds. In Python, `x: int` means x is an integer. In Coq, types are propositions. The type $n > 0$ means “the proof that n is positive.” A function from $n > 0$ to $n + 1 > 0$ is a proof that if n is positive, $n + 1$ is too.

The compiler checks whether the types line up. If your proof has a hole and you write “trust me,” it rejects you. You can’t bluff a proof term past it. But here is the part people skip, and it’s the part that matters: the compiler checks that the proof matches the statement. It does not check that the statement says what you meant. Hand it a statement that means nothing and it’ll prove it for you, three characters, no complaint. So the compiler accepting a proof is not the same as the thing being true. It means this proof matches this statement, and whether that’s the right statement is on me, not the machine.

So I didn’t use Coq to make anything true. I used it because I don’t trust my own eye to catch a gap in an argument I want to believe. I spent months convinced by my own reasoning, pushed the proof through, and watched a key step come apart in my hands. The machine caught where I’d fooled myself. That is what it’s for, and it’s the same source the running machine and the chip get extracted from. Gap-catcher and extractor. Not the proof. The proof is the argument, and an argument has to close in a way a person can follow, or it hasn’t closed at all.

21.1 Reproducing the source and proof contracts

Repository: <https://github.com/sethirus/The-Thiele-Machine>. The repository carries the formal sources, pinned dependency trees, executable contract probes, and the native reproduction runner. The authoritative reproduction procedure is documented in `docs/REPRODUCTION.md`; it builds a fresh source snapshot and does not rely on patches or precompiled proof objects.

Use the native toolchain required by the repository, then run the following commands from the repository root:

```
python3 scripts/reproduce_coq.py --jobs 1
python3 scripts/check_coq_probe.py
bash monograph/build_monograph.sh
```

The source-only reproduction has to exit zero. Don’t stop at the exit status, though. Look at the theorem types and their global dependencies too, because a closed report doesn’t make a theorem’s own premises go away. The monograph build regenerates the publication outputs from the TeX sources. The reproduction records it writes are evidence for that one run. They aren’t source files.

RTL generation is a separate command, `bash scripts/kami_extract.sh`, requir-

ing the Bluespec compiler and its standard library in addition to the Coq/OCaml toolchain. Its logs, generated Verilog, synthesis report, and any bitstream have to name the source they were actually built from. A passing Coq build doesn't give you a fresh synthesis or timing measurement. The proof-audit output also records the source state and date it describes.

21.2 The Inquisitor

Beyond Coq's type checker, the project has a custom proof auditor called the Inquisitor, shipped with the proof corpus. It runs on every commit and emits 87 distinct rule codes. The main categories:

- **No holes.** No `Admitted`, `admit`, or `give_up` anywhere in the active proof files. No unproved `Axiom` or `Parameter` declarations in project code.
- **Vacuity heuristics.** No theorem whose conclusion is literally `True`, `0 = 0`, or any other tautology that proves nothing. No "theorem" of the form $P \rightarrow P$. Existence claims backed by a constant witness get flagged. A flag is a reason to look, not a verdict: some constructions, like a fixed interpreter program, are supposed to be fixed, and they get judged on their premises and on what they actually compute. Coq's kernel already prevents circularly dependent proofs (it checks definitional well-foundedness); the Inquisitor catches separately the case where a proof is technically valid but epistemically empty, proving `True` by `trivial`, for instance.
- **Physics-definition heuristics.** No physics quantity defined as a constant placeholder. No physics claim without a corresponding invariance condition. No theorem stated about a physics concept if that concept is not genuinely formalized.
- **No trivial true proofs.** If a theorem's entire proof is `trivial` or `tauto` and the conclusion reduces to `True`, the theorem is rejected.
- **Scope enforcement.** The proof tree is split into tiers. Tier 1 is the kernel files: the core proof of the machine. Tier 1 files may not import from exploratory or research-side files. Theorem connectivity: every file in the active proof tree must reach the cost or semantics foundation through its import chain. If a file is disconnected, it does not get to claim the kernel's pedigree.
- **Comment hygiene.** The maintained source and documentation comments contain no unfinished or historical review markers; the repository checks this with `scripts/comment_hygiene.py` alongside the Coq proof audit.

The proof-audit output records the source state it was generated from, so regenerate it whenever the source changes. Its rules are heuristics. They point at places to look. They don't prove a theorem means what it should, and they can't promise every empty premise got caught.

22 Zero Admitted: what it actually means

“Zero Admitted” is literal. Nowhere in the proof tree did I write “trust me” and walk on. Every step the checker could see, it saw, end to end, with no “and the rest is obvious.” That is a hygiene fact about the gap-catcher, not a verdict on the argument: it says I didn’t let myself skip a step where the machine was watching. It does not say the statements are the right statements. That part stays on me, which is what the rest of this section is about.

This is what it does NOT mean:

- It does not mean the theorems prove physical reality. The theorems prove things about the Thiele substrate, as formalised in Coq. Whether that formalisation describes the world you live in is a separate question, answered by experiments, not by proof checkers.
- It does not mean the Coq axioms themselves are consistent. Coq sits on a formal foundation called the Calculus of Inductive Constructions. The key idea is the Curry-Howard correspondence: proofs and programs are the same thing. A proof that P is true is literally a program of type P . Checking the proof is type-checking the program. The “inductive” part means you can define your own recursive types (naturals, lists, trees) inside the system instead of having them as magical primitives handed down from above. That matters. The foundation is explicit. Nothing is smuggled.

Now the honest limitation. The sources I read on Coq’s foundation say it is believed to be logically consistent (meaning you cannot derive a contradiction from it), but this has not been proven within Coq itself. The way I understand it, that is not a weakness specific to Coq. The label I dug up is Gödel’s incompleteness theorem from 1931: a consistent, effectively axiomatized theory containing sufficient arithmetic cannot prove its own standard arithmetized consistency statement under the usual derivability conditions. And an inconsistent system proves everything, its own consistency included. So Coq’s foundation is trusted, not proved-from-within. “Trusted” has substance here. The same foundation has verified the CompCert C compiler and the Four Color Theorem (Gonthier 2005, a landmark fully machine-checked proof of a major mathematical theorem, one first settled on paper-plus-computer in 1976), among other artifacts. That prior use is evidence about the foundation’s track record in practice. It is not a proof of logical consistency, and a successful large-scale application would not necessarily surface every possible inconsistency; an inconsistent system only breaks visibly if some proof actually exploits the inconsistency, and nothing here rules that out. I am relaying what I read. I have not personally audited CompCert or Gonthier’s proof. “Zero Admitted” means every theorem follows from those axioms. It does not mean those axioms are themselves in-system guaranteed. It means they are the same axioms a substantial body of verified

software has relied on without breaking, which is track record, not a consistency proof.

- Hardware correspondence is conditional on its model and representation contract. `driven_step_wf` proves intermediate model equality under `WFDDrivenPrecondition`; executable RTL transfer adds finite-range and retirement obligations and toolchain trust. The four Q_{1+AB} instructions remain outside the synthesized inventory.
- It does not mean the OCaml runner binary is proven inside Coq to be bitwise identical to the Coq semantics. The kernel proves Coq-side observable facts. The parity suite checks the extracted binary empirically. One is a theorem. The other is a test.

A theorem Coq checks proves its whole type, premises and record obligations included. A Section hypothesis can turn into a theorem argument once the Section closes. `Print Assumptions` lists global dependencies, and `Closed` under the `global context` doesn't make those arguments disappear. So to judge whether a theorem covers the case you care about, read its statement, unfold the definitions that carry the weight, check the global assumption report, and find witnesses for the premises.

The bridges bring their own premises and their own trust boundaries. Any physical reading needs its named quantum or Landauer–Unruh calibration premise. Getting to running hardware also leans on extraction, the OCaml pretty-printer, BSC, the generated RTL, and the simulation harness, and silicon claims add synthesis and place-and-route on top. The Coq model theorem doesn't verify that whole chain. The OCaml and RTL parity checks are tests, each tied to its own revision and its own inputs.

23 How to break this

Every main claim in this document comes with the exact shape of the thing that would kill it. I went to the trouble on purpose. A claim you can't break is a claim that wasn't saying anything, and I would rather hand you a loaded gun pointed at my own work than a sentence too vague to aim at. So the ones below are written as instructions, not as boasts: here is the trace to find, here is the object to build, here is the theorem to exhibit. Build one and the proof tree falls, and I mean that as the better outcome of the two, because a break tells me something I didn't know and a year of nobody finding one only tells me people got tired of looking. If you build any of these, that is the message I want in my inbox.

Falsify No Free Insight.

Find a VM trace that:

- starts with `vm_certified = false` and `vm_mu = 0`
- ends with `vm_certified = true`
- ends with `vm_mu = 0`

If you find such a trace, the claim is false: `certification_requires_positive_mu` breaks at the step that prices the cert-flip. Add the counterexample as a theorem and the check refuses to compile on top of that.

Falsify μ -monotonicity.

Find a state s and instruction i such that $(\text{vm_apply } s \ i).\mu < s.\mu$. The Coq proof of `vm_apply_mu` falls.

Falsify categorical separation.

Show that the two witness states in `categorical_separation` differ in one of the six computational fields (registers, memory, μ , PC, error, certification), or that their morphism lists are the same. Either kills it. Error behavior is not the target; the thing that separates them is the morphism graph itself.

Falsify Turing strictness.

Exhibit a *classical* program, a VM instruction trace restricted to the classical opcode fragment (`is_classical_program`, run on the same substrate), whose execution changes the partition graph (`pg_next_id`) or the morphism map. Or prove that the witness step `PNEW` leaves `pg_next_id` fixed. Either kills the strictness half of `D5_thiele_strictly_extends_classical`. This is the semantic claim, that the classical opcode fragment cannot reach the structural states one structural opcode reaches. It is not the computability claim. A Turing machine can of course simulate a Thiele run by encoding the graph as bytes on its tape; that simulation exists and is not a counterexample.

Falsify structural entitlement.

Prove that for every pair of Thiele VM states with the same classical shadow and different morphism structure, no probe preserves the distinction; equivalently, refute the concrete witness pair the theorem exhibits. `shadow_strictly_lossy` falls. Or produce a strict feasible-set subset with a distinguishing observation that does not yield stronger membership predicates. The feasible-subset theorem falls with it.

Falsify universal certification cost.

Build a `CertificationSystem` satisfying the one-step rule `cs_cert_costs`, then give a trace from uncertified to certified with total cost 0. `universal_nfi_any_substrate` falls. If you instead drop the one-step rule, you have found a free-certification system, not a counterexample.

Falsify substrate-level structural-axis undecidability.

Three routes (Section 10 expands).

- Exhibit a Coq function `decide : Program → bool` whose diagonal flip transformer is internally representable in some Substrate plus WithShortcut-Predicate instance and that satisfies the bi-implication. The substrate-level proof of `structural_shortcut_undecidable` must fail at some step; identify which.
- For the nat-coded substrate specifically: exhibit a Coq function `d : nat → bool` that correctly decides `nat_admits d`. The corollary `nat_self_undecidable` says no such `d` exists; build one and you have contradicted it head-on, which is exactly the kind of break I want to hear about.
- For the VM specialization, supply its bounded recurrence premise for the chosen representability class and a correct decider whose particular diagonal flip belongs to that class. A correct external decider alone is not a counterexample: bounded membership is externally decidable, and its flip is excluded by every class satisfying that recurrence premise.

Falsify the classical CHSH bound.

Exhibit four fixed bits a_0, a_1, b_0, b_1 and a tally consistent with that one strategy (`WCLocallyConsistent`, every setting pair sampled) whose score beats 2. `chsh_stat_violation_not_local` falls. So does known mathematics. A short run from a randomized local strategy that lands past 2 by luck does not count; the theorem is about one fixed deterministic strategy, and Section 18's coin-flip example already shows that luck.

Falsify the CHSH-NPA bridge.

The biconditional `column_contractive_iff_npa_psd` falls to a counterexample in either direction: a column-contractive correlator whose specified zero-marginal, zero-within-party-moment matrix is not PSD (forward), or a PSD specified zero-marginal, zero-within-party-moment matrix whose correlators are not column-contractive (reverse). For the Tsirelson corollary, exhibit a slice-coherent (column-contractive) correlator with $|S| > 2\sqrt{2}$ and `state_column_contractive_implies_tsirelson` falls. The PR box and the Turing-classical point $(1, 0, 1, 0)$ are *not* counterexamples: both fail column contractivity by direct arithmetic, which is what the theorem requires, not what would break it.

Falsify the elliptope completion.

Three targets, one per load-bearing theorem of Section 18.8. Exhibit signs a_0, a_1, b_0, b_1 with squares 1 and a vector on which the completed Gram form goes negative; `deterministic_strategy_elliptope` falls. Exhibit an elliptope-realizable correlator tuple with $S^2 > 8$; `elliptope_tsirelson` falls, and with it the claim that the

completion cannot stretch both Cauchy–Schwarz budgets at once. Exhibit real x, y making the PR box’s completed 5×5 matrix PSD; `pr_box_not_elliptope` falls. The $\mu = 0$ all-ones witness and $(1, 0, 1, 0)$ are *not* counterexamples here either, in the opposite direction from the bridge entry above: the elliptope theorems admit them, which is what the theorems require.

Falsify the elliptope gate.

`elliptope_check_full_sound` claims soundness with the check’s passing as its only hypothesis: any passing combination of witness counters, completion buckets, or LDL certificate entails elliptope membership of the witness-derived correlators. Exhibit inputs where `elliptope_check_full` returns `true` while the correlators lie outside the set (violating $S^2 \leq 8$ suffices), and the soundness theorem falls, taking `elliptope_full_gate_never_accepts_pr_box` with it. Trap-side behavior is not a target, and a rejection is never a counterexample to soundness. This gate’s certificate format requires rational completion or LDL data (Section 18.8), so by construction it cannot accept a boundary point whose every PSD completion is irrational. That’s a fact about this particular certificate format. Nothing here claims every exact integer check has to reject exactly those points and no others, and nothing needs that.

The Inquisitor standard.

Run the Inquisitor on the repository and read what it says. It fails on HIGH or MEDIUM findings. LOW findings get reported but don’t fail it on their own. Every in-source scope marker sits next to the code it describes, with its reason. The Inquisitor reads syntax and applies heuristics. It can’t tell you whether a theorem applies to your case, and it doesn’t read the prose around it. A new finding means somebody looks. It doesn’t automatically mean a theorem is false.

Falsify the axiom audit.

If running `Print Assumptions` (Section A: the Coq command that reports global proof dependencies) on any named theorem in the kernel tier shows a project-local `Axiom` or `Parameter` in the dependency tree, the audit has degraded. The current full-corpus probe covers 13,028 named theorems across 433 files: 5,692 close under the global Coq context outright, and 7,336 use only Coq standard-library assumptions. The receipt names five such constants: `dependent_functional_extensionality`, `Eqdep.Eq_rect_eq.eq_rect_eq`, `ClassicalDedekindReals.sig_not_dec`, `ClassicalDedekindReals.sig_forall_dec`, and `Classical_Prop.classic`. Zero project-local axioms appear in any of the 13,028 dependency trees. This audits where global dependencies come from. It doesn’t prove that a theorem’s own Section-generalized premises hold for the instance you care about; those premises stay visible in the theorem types and get checked separately. Any project-local axiom or parameter appearing in a global dependency tree is a bug.

Falsify the hardware bisimulation.

A counterexample to `driven_step_wf` has to satisfy `WFDDrivenPrecondition` and break its stated intermediate model equality. A cosimulation mismatch on a valid input challenges the tested RTL transfer for that revision. Then pull it apart piece by piece: encoding, when the observation was taken, extraction, pretty-printing, BSC, and the harness. An input outside the theorem's domain, or a toolchain mismatch, doesn't refute the conditional Coq theorem.

Falsify the convergence answer to parametricity.

The separation/minimality/uniqueness trio is schema-parametric: any field a step rule carries that classical state does not determine supports all three theorems, word for word, and I concede that in full. What I offer for certification is convergence, and it is weaker than a proof. Three results from vocabularies that owe each other nothing each turn on a priced commitment: the verifier factorisation result (`V_does_not_factor_through_classical`) is about a claim on the ledger, the clean-room floor demonstration (`minimal/nofi_demo.py`) is about narrowing a set of possibilities, and the NPA moment-matrix certificate (`chsh_lassert_no_trap_implies_npa_psd`) is a priced check on counters. They do not land on the same state field. The CHSH check never writes the certification flag, and the verifier result is about μ . What they share is the shape: something gets committed, and the commitment is where the cost lands. Exhibit another event with three independent bridges of the same standing, or break one of these, and certification's distinguished seat falls back to an aesthetic choice. Whether certification is the event a model is *forced* to price remains the named open problem either way; Section 26 states the conjectured criterion (redundant record proliferation) that a successor project would have to prove or refute, with its own attack surface listed there.

Falsify the proof-of-stake reduction.

Construct a trace in the formal finality model satisfying the stated slashing schedule, with unfinalized initial state, finalized final state, and zero summed modeled charge. This contradicts `slashing_finality_floor`. A deployment with different transition or accounting semantics instead challenges its proposed interpretation.

Falsify the gas-metering reduction.

Construct a schedule satisfying both the commitment floor and no-overcharge whose charging predicate differs from the commitment predicate on some reachable step. `gas_schedule_exactness` falls, and the kernel's `exact_commitment_pricing_characterization` with it.

Falsify the attestation reduction.

Build a sound and complete attestation verifier for the μ -dependent claim that reads only the bare classical transcript, a function of the report's log field alone. `attestation_cannot_factor_through_bare_transcript` falls, and behind it the

verifier-factorisation theorem.

Falsify the transparency-log reduction.

Build a sound and complete log-free verifier with the same soundness target. `abstract_bare_verifier_impossible` falls, and the bare-setting impossibility with it.

Falsify the permanent-certificate theorem.

Exhibit a finite, duplicate-free list of every state, a certificate no step revokes, and an injective step that switches it on. `permanent_flip_is_not_injective` falls, and `a2_from_merging_price_and_permanence` with it. An infinite state space, a revocable certificate, or an unpriced merge is not a counterexample; the file already builds one of each.

Falsify the cost of a squeeze and the merge-or-revoke dichotomy.

For the bound, exhibit a finite machine with a permanent certificate and a cost assignment meeting `compression_priced` where one instruction switches k states on beside m certified ones with $2^{\text{cost}}m < m + k$. For the dichotomy, exhibit a finite machine and an injective instruction that switches a reading on without switching it off anywhere. `permanent_flips_compression_bound` or `flip_merges_or_revokes` falls. A merging instruction that also revokes is not a counterexample to anything; `forced_price_without_permanent_record` is one.

Falsify the shadow-pricing theorem.

Exhibit a window and a price computed from the observed before and after that meets the floor and never overcharges on a machine where two steps share their before and after pictures and only one certifies. `shadow_cannot_price_exactly` falls. For the VM claim, show that `CERTIFY 0` and `JUMP 1 0` from the clean start differ on `bare_observable`, or that `CERTIFY 0` and `JUMP 1 1` differ on `forget`. A window that shows certification is not a counterexample; `window_showing_reading_prices_exactly` says it prices exactly.

Falsify the entropy bound.

Exhibit a finite machine with a permanent certificate, an instruction that switches $k \geq 1$ states on beside m certified ones, and a distribution on those $m + k$ states whose image under the step has Shannon entropy above $\log_2 m$. `permanent_step_entropy_ceiling` falls. A heat measurement below $k_B T \ln((m + k)/m)$ does not falsify the theorem. It falsifies the premise `landauer_heat` for that device, which is Landauer's principle. For the general version, exhibit a deterministic step and a distribution under which two different states of positive probability land together while the entropy does not drop; `entropy_drop_pos_of_support_merge` falls. A certifying step from a known state that dissipates no heat is not a counterexample; `known_state_flip_forces_no_heat` says that case owes none.

Falsify the finite VM instance.

Exhibit a VM state and one of `CERTIFY 0`, `JUMP a 2`, `CHECKPOINT "" 0` whose result, read as program counter modulo four and certification flag, differs from the finite machine's step, or whose μ rise differs from 1, 2, 0. `vm_runs_finite_machine` or `vm_pays_finite_price` falls. A VM program that uses `JUMP a 0` is not a counterexample. It is outside the fragment, and the full VM leaves that merge free on purpose (`vm_jump_is_free_merge`). To break the full-VM claim instead, exhibit a VM step that switches the certification flag on without merging two states, or at zero cost; `vm_certifying_step_is_priced_merge` falls.

Falsify the proof-carrying reduction.

Construct a level- k certified proof script with total μ below k , or a sound and complete zero-round bare verifier. `level_k_certification_cost_floor` (restated for this model as `level_k_verification_floor`) or `bare_pcc_impossible` falls, and the μ -hierarchy or the bare-setting impossibility behind them.

23.1 Contribution and attribution boundary

This section says what is mine and what is not. It is not a claim to have been first. I have not read widely enough to say any result below is historically new, and I am not going to pretend otherwise. Two comparisons earlier in the document are primary: Landauer's erasure result and Wolpert and Macready's No-Free-Lunch theorem. The other labels are orientation points, and I would welcome correction on any of them.

The work comes apart into five layers.

Borrowed mathematics and vocabulary.

Category theory, Turing and Church–Turing terminology, the Kleene and Rice diagonal patterns, Bell, CHSH and Tsirelson, NPA and the ellipsope, the thermodynamic motivation, and the ordinary definitions of projections, partitions, predicates and resource measures all come from other people and from standard mathematics. So do the facts under the permanent-certificate results. Setting a bit to one whatever it held is Landauer's own example of an irreversible step (1961). That finite memory eventually forces erasure is Bennett's point about reversible computing. The counting is the pigeonhole principle. Entropy is at most the logarithm of the support size, and a deterministic step never raises it; both are Shannon's. That Landauer's bound depends on the actual distribution, so a known state can be reset for free, is the generalized form of Maroney (2009) and Sagawa (2014). None of it is mine.

Formal reconstructions.

The Coq development defines the state types, transition systems, fixed real-matrix predicates, encodings and proof terms this monograph uses. When I say a theorem is checked in Coq, I mean the stated proposition has a kernel-checked proof under its printed premises. That does not make the underlying idea mine, and it does not make it new.

Results this project proved.

Inside the stated interfaces, the repository proves the abstract A2 certification floor, the fixed-schedule ledger identity and uniqueness, the named projection-separation witnesses, the trace-fold universal property, the conditional structural-entitlement representation theorem, and the substrate-diagonal results. It also proves the permanent-certificate results of Section 9: that any implementation of a certificate no step revokes must merge on the certifying step, the logarithmic bound on the squeeze, the merge-or-revoke dichotomy and “forced to be priced exactly when merging”, the entropy forms with their distribution boundary, that an exact price for certification can’t be computed from a colliding window, the finite VM instance, and what the full VM prices. The facts they rest on are the borrowed ones above. What is mine is putting them together about certification, and checking every step in Coq. These are claims about the explicit Coq contracts and witnesses. They are results of this project. I am not claiming nobody wrote down an equivalent statement before me.

Instances and engineering.

The 51-opcode VM, its cost schedule, the Python and extracted-OCaml runners, the Kami model, the Bluespec and Verilog path, the verification scripts, the finite tests, and the synthesis and FPGA workflow are things I built. How much of each is proved, tested, or trusted is the edge-by-edge account in Section 16 and `docs/ASSURANCE.md`. A tested edge stays a tested edge. I do not promote it to a theorem.

Open proposals and things I am not claiming.

Whether there is a unique structural core, what the ledger means physically, which priced events are universally necessary, full-ISA hardware equality without conditions, and broad quantum or complexity-theory conclusions are all open or out of scope. Proving a conditional contract does not close any of them.

So the claim ledger that follows is a map from formal names to their domains and premises. It is not a list of discoveries.

24 The full claim ledger

This is the lookup list: every main claim by its Coq name, each checked under its complete formal premises. The crosswalk after it records where each one applies and where its instances stop. For the full quantified statement, go to the named source declaration.

1. *no_free_certification*: `csr_cert_addr` going $0 \rightarrow \text{nonzero}$ in one step requires $\text{cost} \geq 1$.
2. *no_free_certification_mu*: same, with μ increment ≥ 1 .
3. *no_free_certification_trace_mu*: trace-level version.
4. *no_free_certification_certified*: `vm_certified` going $\text{false} \rightarrow \text{true}$ requires $\text{cost} \geq 1$.
5. *certification_requires_positive_mu*: both channels covered.
6. *universal_nfi_any_substrate*: any abstract certification system satisfying the one-step cost

- rule has trace-level non-free certification.
7. *thiele_universal_nfi_cert_addr*: universal theorem instantiated for Thiele's `csr_cert_addr` channel.
 8. *thiele_universal_nfi_certified*: universal theorem instantiated for Thiele's `vm_certified` channel.
 9. *thiele_represents_simulating_cert_system*: any certification system with a simulation morphism into Thiele transfers certification into a Thiele certified execution.
 10. *forget_kernel_is_eq_on_classical*: the classical snapshot quotient forgets exactly the non-classical fields.
 11. *blindness_non_injective*: distinct VM states can have the same classical snapshot.
 12. *certification_is_lost*: certification can be in the kernel of the classical forgetful map.
 13. *shadow_strictly_lossy*: the classical shadow forgets morphism structure.
 14. *mu_ledger_necessity*: no strict-classical oracle on $(\text{mem}, \text{regs}, \text{pc})$ recovers the pair $(\mu, \text{certified})$.
 15. *vm_mu_not_classically_determined* / *vm_certified_not_classically_determined*: neither ledger balance nor certification status is a function of strict classical state.
 16. *mu_ledger_mutual_independence*: μ and `vm_certified` are each independent of strict classical state and of each other under cost-only/cert-only projections.
 17. *mu_ledger_minimality* / *P_full_complete_neither_mu_nor_cert_droppable*: the full $(\text{mem}, \text{regs}, \text{pc}, \mu, \text{certified})$ projection is complete, and dropping either μ or certification destroys the corresponding completeness.
 18. *thiele_nfi_pc_indexed*: PC-indexed execution version.
 19. *mu_is_initial_monotone*: μ is unique given the initial value and fixed instruction-cost schedule.
 20. *vm_apply_mu*: exact ledger identity $(\text{vm_apply } s \ i). \mu = s. \mu + \text{instruction_cost}(i)$. The cost is not just δ , it is δ for ordinary instructions, $S(\delta)$ for cert-setters, $|\text{payload}|_{\text{bits}} + S(\delta)$ for EMIT, etc.
 21. *kernel_certified_implies_positive_mu*: from zero, reaching certified implies $\mu > 0$.
 22. *feasible_strict_subset_implies_strict_predicates*: feasible-set narrowing plus a distinguishing observation yields a strictly stronger predicate.
 23. *strengthening_requires_structure_addition*: certified predicate strengthening requires a structure-addition event.
 24. *b4_information_reduction_derives_strict_predicates*: feasible-set reduction can generate the strict-predicate premise used by NoFI.
 25. *info_priced_cert_executions_bound*: executed cert-setting events are bounded by $\Delta\mu$.
 26. *observation_partition_reduction_implies_posterior_representative_reduction*: observation classes package into the posterior-representative witness for structural entitlement.
 27. *info_priced_arbitrary_feasible_reduction_bound*: with a decision-tree/representative witness, feasible-set compression is bounded by $\Delta\mu$.
 28. *info_priced_weighted_feasible_reduction_bound*: the $\Delta\mu$ lower bound extends to nat-weighted feasible sets.

29. *structural_entitlement_representation*: strict narrowing plus distinguishability, certified posterior admissibility, and an explicit posterior-representative reduction imply a strictly stronger predicate, a structure-addition event, and a $\Delta\mu$ lower bound.
30. *every_sound_structural_shortcut_lands_here*: every packaged `SoundStructuralShortcut` instantiates the structural-entitlement representation theorem.
31. *structural_shortcut_undecidable*: for every `Substrate` plus `WithShortcutPredicate` instance, no Coq function whose diagonal flip transformer is internally representable in the substrate decides the structural-shortcut admittance predicate. The substrate's own halting-shaped wall, by Kleene 1938 diagonal restricted to the substrate's internal expressive power.
32. *nat_recursion_theorem / nat_structural_shortcut_undecidable / nat_self_undecidable*: for each candidate d , the associated `nat` instance supplies its restricted fixed point and defeats d on `nat_admits d`. The parameter d remains in the theorem type despite a closed global assumption report.
33. *vm_structural_shortcut_undecidable*: the bounded VM specialization retains its representability and recurrence premises. The encoded corollary discharges the code round trip. `vm_bounded_shortcut_decide_correct` supplies an external decider; `vm_bounded_decider_flip_not_representable` proves its exclusion from a class satisfying bounded recurrence.
34. *program_to_nat / nat_to_program / nat_to_program_program_to_nat*: closed Gödel encoding of `list vm_instruction` as a `nat` with a proven left inverse, all 51 opcode constructors handled, no Hypothesis. Composes the existing binary encoders with a `list bool ↔ nat` bijection through `positive`.
35. *categorical_separation*: classical-identical states distinguishable by morphism probe.
36. *classical_observer_cannot_separate*: no classical function separates morphism-distinct states.
37. *shadow_proj / shadow_separation_theorem*: formal surjection, strictly lossy.
38. *D2_faithfulness*: classical programs embedded faithfully in Thiele.
39. *D3_conservativity*: classical programs leave structural layer untouched.
40. *D4_strictness*: there exist Thiele steps that escape the classical fragment.
41. *D5_thiele_strictly_extends_classical*: Thiele strictly exceeds classical semantics.
42. *degenerate_projection_theorem*: shadow projection is the classical equivalence quotient.
43. *no_classical_certification_decider*: no classical machine can decide morphism-certification.
44. *chsh_stat_violation_not_local*: CHSH-violating statistics are non-local.
45. *structural_trace_preserves_cert_addr*: traces with no cert-address setter preserve `csr_cert_addr`.
46. *partition_refinement_nonfree*: certified partition knowledge cannot be free.
47. *partition_free_but_certification_nonfree*: partition structure can be built at $\delta = 0$, but certified partition knowledge cannot be acquired for free.
48. *witness_insight_nonfree_general*: certified non-local witness insight requires $\mu \geq 1$.
49. *witness_insight_complete_taxonomy*: full three-tier taxonomy proven.

50. *separable_coupling / non_separable_coupling*: Functional (separable) and non-functional (non-separable) coupling predicates with concrete witnesses.
51. *CHSH coupling bridge*: Bell’s theorem in morphism coupling language. A locally consistent strategy produces a separable coupling and has $|S| \leq 2$. $S > 2$ rules out every locally factorizable morphism coupling.
52. *graph_certify_morphism_lookup*: morphism certification cost utility, including lookup correctness.
53. *exists_covering_tree*: for any feasible-set reduction, a covering complete binary tree exists constructively.
54. *info_priced_reduction_no_tree_hypothesis*: entropy bound without requiring the tree as an explicit input.
55. *sound_shortcut_from_components*: assembly lemma for SoundStructuralShortcut from its component witnesses.
56. *error_free_preservation_budget / positive_input_certification_budget / zero_mu_traces_satisfy_preservation_budget*: predicates giving a bounded trace for each input, with conditional predicate preservation or certification respectively; zero- μ error-free traces satisfy the preservation predicate because its positive- μ antecedent never holds. Neither predicate requires a uniform solver, a correct returned answer, or polynomial bounds.
57. *column_contractive_iff_npa_psd*: slice coherence $\iff$ PSD realizability of the specified fixed completion. A biconditional with no project-local axioms (its `Print Assumptions` reports only Coq’s standard classical-reals and functional-extensionality `stdlib` axioms). Forward (*zero_marginal_npa_column_contractive_implies_psd*): column-contractive $\rightarrow$ NPA PSD. Reverse (*npa_psd_implies_column_contractive*): NPA PSD $\rightarrow$ column-contractive, proved by vector specialization plus the *quadratic_nonneg_discriminant* lemma. Zero-marginal NPA realizability is the orthogonal-observables slice of the quantum ellipsope ($\rho_{AA} = \rho_{BB} = 0$), not the full ellipsope. Classical correlations like $(1, 0, 1, 0)$ have $|S| \leq 2$ but fail column contractivity; the gate is a slice-membership test, not a truth test, and it rejects all sixteen deterministic local strategies. The classical polytope and the slice-coherent set intersect but neither contains the other.
58. *state_column_contractive_implies_tsirelson*: slice-coherent $\Rightarrow |S|^2 \leq 8$, i.e., $|S| \leq 2\sqrt{2}$. The proof factors through the repository’s fixed formal completion predicate: *state_column_contractive_implies_npa_gram* composes with *npa_psd_implies_tsirelson_bound*. The checked chain is a statement about the specified real matrix and its algebraic bound. The row-bounds lemma *tsirelson_from_row_bounds* is the algebraic Cauchy–Schwarz step inside this chain. The contrapositive ($|S| > 2\sqrt{2} \Rightarrow$ not slice-coherent) follows immediately. The converse ($|S| \leq 2\sqrt{2} \Rightarrow$ slice-coherent) is false and is not claimed.
59. *Concrete falsifications by direct arithmetic*. The PR box ($E_{00} = E_{01} = E_{10} = 1, E_{11} = -1$) fails the first column-contractivity condition: $1 - E_{00}^2 - E_{10}^2 = 1 - 1 - 1 = -1 < 0$. Any super-quantum value $|S| > 2\sqrt{2}$ falls outside the slice-coherent set by the contrapositive of *state_column_contractive_implies_tsirelson*. The Turing-classical correlation $(1, 0, 1, 0)$ also fails condition 1. These are direct consequences by `lra` once the biconditional is destructed.

60. *Elliptope completion* (Section 18.8). `elliptope_realizable`: correlator membership in the full elliptope as existential completion of the cross moments; `deterministic_strategy_elliptope`, `elliptope_convex`, `elliptope_finite_mixture`, and `lhv_mixture_elliptope` put every LHV correlator inside (with `classical_tightness_witness_elliptope` and `turing_point_elliptope` naming the two points the slice gate rejects); `elliptope_tsirelson` bounds the whole set by $S^2 \leq 8$ via `psd_cauchy_schwarz`; `pr_box_not_elliptope` excludes the PR box; `beyond_classical_elliptope` exhibits $S = 12/5 > 2$ inside, making `classical` $\subset$ `elliptope` strict. Print Assumptions on all of them: the standard classical-reals and functional-extensionality `stdlib` families, zero project-local axioms. `elliptope_check_full_sound` (in `ElliptopeGate.v`) adds the decidable two-branch $\mathbb{Z}$ -arithmetic gate (fraction-free Sylvester on completion buckets, rational LDL^T certificate reaching singular/boundary completions): a passing check entails membership, with computed acceptances (`gate_accepts_all_ones`, `gate_accepts_turing_point`, `gate_accepts_beyond_classical`, and on-boundary `gate_accepts_pythagorean_boundary`) and the universal refusal `elliptope_full_gate_never_accepts_pr_box`. Correlator level only; the runtime opcode is unchanged. This one-way soundness theorem does not establish completeness for the full set of quantum behaviors.
61. *mu_hierarchy_theorem*: for every $k \geq 1$, there exists a level- k certified trace with cost exactly k , and every trace that ends certified having executed a `CERTIFY` at level k (charge $S(\delta) \geq k$) pays total cost $\geq k$. Unbounded strict μ -cost levels: no finite budget caps every certification. Not a problem-class separation (Section 25).
62. *mu_hierarchy_no_upper_bound*: no fixed μ -budget suffices for all levels.
63. *Structural-invariant Qed group*: the 10 opcodes `PNEW`, `PSPLIT`, `PMERGE`, `MORPH`, `MORPH_ID`, `MORPH_DELETE`, `MORPH_ASSERT`, `MORPH_GET`, `COMPOSE`, `MORPH_TENSOR` carry Qed bisimulation proofs under the inductive invariant `coupling_wf` ($=$ `coupling_desc_bounded` $\wedge$ `coupling_pairs_in_range` $\wedge$ `coupling_pairs_fully_populated`) and `morph_table_wf`. The preservation theorems `coupling_wf_kami_step_preserved` and `morph_table_wf_kami_step_preserved` show these invariants hold inductively at hardware reset and through every `kami_step` operation, so the precondition is discharged in practice for any state reachable by machine execution.
64. *RTL translation boundary*: the Section Variable `rtl_step_correct` is a contract for the synthesized-RTL relationship, not a theorem that the generated Verilog automatically satisfies. The Coq theorem `driven_step_wf` covers the intermediate Kami model under `WFDDrivenPrecondition`; the emitted RTL and later FPGA artifacts are checked by the cosimulation, pipeline-manifest, synthesis, place-and-route, and bitstream gates for their recorded source/tool revisions. `bsc_kami_compilation_trusted` names the BSC/compiler and printer trust boundary. No unconditional end-to-end RTL/bitstream bisimulation is claimed.
65. *nothing_at_stake_is_free_forgery*: a finality gadget whose `finalize` step carries zero stake-at-risk admits no A2 field. Nothing-at-stake is the kernel's free forgery, stated in proof-of-stake vocabulary (Section 17).
66. *slashing_finality_floor*: for any supplied slashing gadget whose certification predicate starts false and whose false-to-true transitions satisfy the stated positive-cost premise, every run from unfinalized to finalized carries total stake-at-risk ≥ 1 ; `universal_nfi`

- `_any_substrate` instantiated at the slashing gadget.
67. *gas_schedule_exactness*: a gas schedule satisfies the commitment floor and no-overcharge exactly when its charging predicate is the commitment predicate with unit pricing; the kernel machine itself inhabits the class (`thiele_vm_commit_pricing_is_exact`).
 68. *attestation_cannot_factor_through_bare_transcript*: a sound and complete attestation verifier of the μ -dependent claim cannot factor through the report's bare transcript; the replay attack is the witness pair (`replay_is_a_two_preimage_witness`).
 69. *level_k_verification_floor*: k certification events cost at least $k \mu$ in any covered proof system, with tightness witnessed; bounds certification events only, not gate counts, circuit size, prover time, or proof length.
 70. *permanent_flip_is_not_injective*: on a finite state space with a certificate that no step revokes, any step that switches the certificate on is not injective.
 71. *a2_from_merging_price_and_permanence / permanent_certification_trace_floor*: under the same finiteness and permanence, pricing every non-injective step at ≥ 1 yields A2, and the trace-level floor follows through `universal_nfi_any_substrate`.
 72. *permanent_flips_collapse_at_least*: a step that switches on k distinct uncertified states has at least k fewer distinct images than inputs.
 73. *honest_eraser_accounting_implies_a2 / commit_without_eraser_system_is_not_honest*: erasure accounting that reports every merge charges a permanent certification flip on a finite machine at least 1; the zero-cost erasure rival leaves its own merge unreported.
 74. *permanent_flips_compression_bound / permanent_flips_log_bound*: under the counting premise that each unit of cost pays for one halving, switching k states on when m are certified costs enough that $m + k \leq 2^{\text{cost}} m$, so $\log_2^\uparrow(m + k) \leq \text{cost} + \log_2^\uparrow m$; `quad_stamp_needs_two` and `quad_stamp_cost_two_is_priced` show the bound is attained.
 75. *forced_priced_iff_merges*: an instruction is charged by every merge-pricing cost assignment exactly when its step merges two states.
 76. *flip_merges_or_revokes / injective_flip_revokes*: on a finite machine, an instruction that switches a reading on either merges two states or switches the reading off somewhere else.
 77. *permanent_record_write_is_forced_priced / forced_price_without_permanent_record*: permanent-record writes are forced to be priced; the converse fails on a three-state machine.
 78. *shadow_cannot_price_exactly / window_showing_reading_prices_exactly*: if two steps share their observed before and after and only one certifies, no price computed from the window meets the floor and never overcharges; a window that shows the reading prices exactly.
 79. *vm_bare_shadow_cannot_price_exactly / vm_forget_shadow_cannot_price_exactly*: the VM has such a pair for `bare_observable` and for `forget`, from the clean start.
 80. *permanent_step_entropy_ceiling / permanent_flip_uniform_entropy_drop*: after a permanent step, a distribution on the m certified states and the k flipping ones has Shannon entropy at most $\log_2 m$; from the uniform distribution the step removes at least $\log_2((m + k)/m)$ bits.

81. *a2_from_entropy_price_and_permanence / entropy_priced_trace_floor*: a whole-unit price that covers the bits each instruction removes satisfies A2 on a finite machine with a permanent certificate, with the trace floor through `universal_nfi_any_substrate`.
82. *permanent_flip_heat_floor / permanent_flip_heat_positive*: under the named premise `landauer_heat`, that flip dissipates at least $k_B T \ln((m+k)/m)$, which is positive when $k \geq 1$ and $T > 0$.
83. *entropy_drop_as_point_sum / entropy_drop_pos_of_support_merge*: for any distribution, the entropy a deterministic step removes is a sum of nonnegative per-state terms, positive when two states of positive probability land together.
84. *permanent_flip_full_support_heat_positive / known_state_flip_forces_no_heat*: a permanent flip forces positive heat under `landauer_heat` for every distribution positive on the states in play; from a known state it removes no entropy and forces none.
85. *fin_merging_priced / fin_compression_priced*: the eight-state machine with `FCertify`, `FJump`, `FNext` at prices 1, 2, 0 is finite, permanent, and meets both pricing premises.
86. *vm_runs_finite_trace / vm_fragment_certification_paid*: read through program counter modulo four and the certification flag, `CERTIFY 0, JUMP a 2, CHECKPOINT "" 0` run that machine, μ rises by its price, and a run of them that certifies raises μ by at least one.
87. *vm_certifying_step_is_priced_merge / vm_prices_certifying_merge_leaves_others_free*: on the full VM, every step that switches the certification flag on merges two states and costs at least one, while `JUMP 0 0` merges program counters at cost zero, so the full VM prices the certifying merge and not every merge.

Bridge-tier claims (checked in Coq, but resting on named hypotheses or documented trust boundaries):

- **Hardware correspondence**: `driven_step_wf` and `driven_trace_commutes` prove intermediate model equality under `WFDDrivenPrecondition` and `WFDDrivenRun`. Section 16 states the additional finite RTL contract and distinguishes the 47 synthesized instructions from four nonsynthesized checks. Allocation, overflow, and retirement obligations remain part of assessing an executable instance.
- **OCaml extraction**: `ocaml_observable_nofi_and_monotone` proves `NoFI` and μ -monotonicity for the Coq-side extracted observable. The external OCaml binary equivalence is a parity-test trust boundary, not a kernel theorem.
- **Structural advantage**: the kernel proves exact μ costs and the arithmetic time-tax facts for the factored-search witness; executable tests check the measured VM behavior.
- **LASSERT cost formula**: `lassert_cost_from_component_floors` and `lassert_cost_is_its_formula`. A cost meeting the state-term and description-term premises is at least the formula. That is addition over premises. Merge pricing does not force the state term (`observer_narrowing_can_be_`

free), and a physical reading also needs Landauer-Unruh calibration.

24.1 Main-result contract crosswalk

Each row identifies the result’s domain, conclusion, load-bearing premises, and instance status. A symbol prefixed by “Kernel” is in its namesake file under `coq/kernel/`; “KamiHW” names files under `coq/kami_hw/`. The stable contract probes under `tests/coq_probes/` load the exact modules. Their reports combine kernel declaration checks, expanded records and definitions, and dependency-assumption reports. The source declaration and its imported definitions remain the complete mathematical contract.

Prose label and formal symbol	Domain, conclusion, premises, and status
Certification floor <code>Kernel.UniversalCertificationCost.universal_nfi_any_substrate</code>	CertificationSystem states and instruction-list traces. An initially false, finally true certification indicator requires positive total cost. The record supplies the one-step floor. VM instances are proved.
Exact commitment pricing <code>Kernel.CommitmentPredicateAdequacy.exact_commitment_pricing_characterization</code>	LocalPredicatePricedSystem steps. The lower bound and no-overcharge characterize unit pricing of the designated state-dependent event. The record and reachability premises remain explicit.
Ledger uniqueness <code>Kernel.MuInitiality.mu_is_initial_monotone</code>	VM instruction-list execution and CostFunctional. Equality requires the specified initial value and instruction-increment law. This fixes the schedule, not the unique event every model must price.
Structural independence <code>Kernel.NecessityAbstract.mu_ledger_mutual_independence</code>	Specified projections of VMState. The theorem gives the displayed non-determination witnesses. It is neither statistical independence nor recovery of every field from a graph-free projection.
Verifier obstruction <code>VerifierExhaustiveness.V_does_not_factor_through_classical</code>	Verifier functions on the stated evidence domain. Soundness, completeness, and the supplied collision prevent factorization through the named bare projection. Authenticating external evidence requires its own contract.
Structural entitlement <code>Kernel.HonestNoFI_TheoremsWithoutAssumptions.structural_entitlement_representation</code>	Finite feasible lists and a supplied representative reduction. Initial/final certification, strict narrowing, distinguishability, covering and payment obligations are retained. Packaged witnesses prove this conditional statement.
Abstract diagonal <code>Kernel.StructuralUndecidability.structural_shortcut_undecidable</code>	Substrate and WithShortcutPredicate. No correct decider has a representable flip when the record supplies its fixed point and distinct yes/no witnesses. The conclusion is relative to that class.
Nat family <code>Kernel.NatSubstrateInstance.nat_self_undecidable</code>	For every candidate d , d fails on <code>nat_admits d</code> in the instance constructed from d . The family is proved. A fixed effective undecidable problem is not supplied by changing this quantifier order.

Prose label and formal symbol	Domain, conclusion, premises, and status
Encoded VM diagonal <code>Kernel.VMSubstrateEncoded.v</code> <code>m_structural_shortcut_undecidable_encoded</code>	Bounded <code>vm_run</code> outcomes. The code round trip is proved; representability and bounded recurrence remain premises. An applicable unbounded interpreter is not an instance of this statement without a bridge.
Bounded external decider <code>Kernel.VMBoundedDecidability.v</code> <code>m_bounded_shortcut_decide_correct</code>	VM instruction lists. A total Boolean computation decides equality with the bounded reference outcome. Proved by decidable equality of finite VM records.
Representability boundary <code>Kernel.VMBoundedDecidability.v</code> <code>m_bounded_decider_flip_not_representable</code>	Any transformer class satisfying bounded recurrence excludes the correct external decider's flip. Proved; this does not construct an unbounded interpreter.
Input-field step law <code>Kernel.VMEncodedInputAccess.v</code> <code>m_apply_logic_acc_commutes</code>	Every VM opcode commutes with replacement of the logic accumulator. Proved for all states and replacement values.
Input-field run law <code>Kernel.VMEncodedInputAccess.ru</code> <code>n_vm_logic_acc_commutates</code>	Every finite-fuel run preserves that replacement relation. Proved for every program and fuel.
Encoded interpreter obstruction <code>Kernel.VMEncodedInputAccess.no_logic_acc_encoded_interpreter</code>	No fixed interpreter using the current encoding reproduces all unbounded final-certification observations through an output observation invariant under replacement of the retained input accumulator. Other encodings and architectures are outside this conclusion.
Unbounded halting <code>Kernel.VMUnboundedExec.v</code> <code>m_halts_at_deterministic</code>	Existential fuel plus a halted state. Halting outcomes are unique. The relation and saturation foundation are proved. The separate B3 interpreter and B4 Rice results below use an explicit unbounded guest fragment.
Self-interpreter (B3) <code>Kernel.VMSelfRun.self_interpreter_correct</code>	One fixed 122-instruction host interprets twelve arithmetic/control instructions over four unbounded guest registers. Executable data encoding, positive step simulation and both directions of raw result correctness are checked. Separate contracts cover malformed words and live divergence. Guest structural fields remain ambient and unchanged.
Rice reduction (B4) <code>Kernel.VMSelfRiceUndec.self_rice</code>	Over the B3 guest model, extensional predicates separating the divergent program from a well-formed program are undecidable. Guest-program deciders and the named halting-on-zero/returning-zero predicates are covered. This closes the alternative-construction contract, without an internal recursion theorem or discharge of bounded recurrence.

Prose label and formal symbol	Domain, conclusion, premises, and status
Reachable CSR collision <code>Kernel.StructuralAxisOrthogonality.structural_shortcut_not_function_of_classical</code>	Reachable VM states and four-field forget. No decoder recovers the CSR predicate from that observation. This state predicate differs from bounded program-outcome equality.
Shadow-local execution <code>Kernel.TuringClassicalEmbedding.D2_classical_shadow_preserved</code>	The source predicate restricts the instruction fragment and its six-field observation. Graph preservation alone is insufficient when hidden graph or CSR data can be read into visible fields.
Elliptope gate soundness <code>Kernel.ElliptopeGate.elliptope_check_full_sound</code>	Witness counters and the supplied completion/certificate data. Passing the actual gate implies the defined elliptope membership. Rejection and completeness are outside this one-way conclusion.
Charge levels <code>Kernel.MuHierarchyTheorem.level_k_certification_cost_floor</code>	Executed traces from the specified initial state containing an instruction certified at level k. The ledger meets the charge floor. This differs from k certification events and from a complexity-class separation.
One-step model correspondence <code>KamiHW.GraphReconstructionBridge.driven_step_wf</code>	Gallina KamiSnapshot and VMState through the stated abstraction. Exact equality requires WFDriivenPrecondition. Physical RTL retirement, range, and toolchain obligations are separate.
Executed-run correspondence <code>KamiHW.GraphReconstructionBridge.driven_trace_commutes</code>	PC-indexed fuelled runs. WFDriivenRun checks the instruction premise at each visited snapshot. This proves the intermediate trace equality, with no claim about unbounded finite-hardware capacity.
Local fixed-bit CHSH <code>Kernel.CHSH.KernelCHSH.local_strategy_chsh_between_neg2_2</code>	Four Boolean local response bits imply the integer bound between -2 and 2. The population mixture argument in the text additionally uses a setting-independent shared-variable distribution. Finite empirical samples have a different domain.
Fixed-completion PSD <code>Kernel.QuantumPartitionPSD.column_contractive_if_fnpa_psd</code>	Four real correlators and the specified fixed matrix. PSD is equivalent to the three column-contractivity inequalities. This is an algebraic biconditional, with the real-library assumptions shown by the probe.
Integer gate <code>Kernel.QuantumPartitionPSD.chsh_assert_no_trap_implies_fnpa_psd</code>	A CHSH assertion step that advances without error yields the fixed-matrix predicate. The actual integer check supplies the premise; A2 alone does not select or imply its polynomial guard.
Relational category laws <code>Kernel.CategoryLaws.relational_compose_assoc</code>	Coupling relations modulo membership equivalence. Composition is associative; diagonal identities require the source theorem's region bounds. Allocation IDs, labels, and list equality are separate runtime observations.
Unbounded charge levels <code>Kernel.MuHierarchyTheorem.mu_hierarchy_theorem</code>	Positive k has a single CERTIFY(k-1) witness with exact cost k and the stated lower bound. This proves unbounded instruction-charge levels, with the source initialization and execution premises.

Prose label and formal symbol	Domain, conclusion, premises, and status
Restricted geometry Kernel. DiscreteGaussBonnet.di screte_gauss_bonnet	Graphs satisfying the expanded well-formed triangulation predicate, including $B=3$ chi and the incidence identities. The defined angle sum is 5π chi. This restricted identity does not derive its premises from arbitrary topology.
Permanent certificate Kernel. PermanentCertification .a2_from_merging_price _and_permanence	Any state type with a duplicate-free enumeration, any step, any certification reading. Permanence and a price of at least one on every non-injective step yield A2. The merging price is a named premise standing for Landauer's principle. The full VM is not an instance by design: its ledger is unbounded, and it prices the certifying merge while leaving other merges free. Each premise has a counterexample in the file.
Cost of a squeeze Kernel.Pe rmanentRecordPricing.p ermanent_flips_log_bou nd	Same finite, permanent setting, with the counting premise that cost c allows at most a 2^c -fold squeeze. Switching k states on beside m certified ones costs at least $\log_2^\uparrow(m+k) - \log_2^\uparrow m$. The four-state stamp attains it. Units are halvings, not joules.
Which flips are forced Kerne l.PermanentRecordPrici ng.forced_priced_iff_m erges, flip_merges_or_r evokes	Any finite machine and yes/no reading, with decidable equality on instructions for the iff. Forced price is exactly merging; a flipping instruction merges or revokes. Permanent-record writes are forced; forced does not imply permanent, by a three-state counterexample.
No exact shadow price Kerne l.ShadowPricing.shadow _cannot_price_exactly	Any machine, reading, and window with two steps that share their observed before and after while only one certifies. No window-computed price is exact. The VM has such a pair for <code>bare_observable</code> and <code>forget</code> . A window that shows the reading prices exactly.
Bits removed by a permanent flip Kernel.PermanentCer tificationEntropy.perm anent_flip_uniform_ent ropy_drop	Same finite, permanent setting, with Shannon entropy over Coq's real numbers. At least $\log_2((m+k)/m)$ bits from the uniform distribution on the states in play; a positive drop for every distribution positive on them; zero from a known state. Heat follows only through the named premise <code>landauer_heat</code> . Uses Coq's standard real-number axioms.
A finite VM instance Kernel. FiniteCertMachine.vm_r uns_finite_trace	The eight-state machine and the VM fragment CERTIFY 0, JUMP a 2, CHECKPOINT "" 0, read through program counter modulo four and the flag. The JUMP price is a program's choice the VM allows.
What the full VM prices Kern el.FiniteCertMachine.v m_prices_certifying_me rge_leaves_others_free	The full VM and its schedule. Every step that switches the certification flag on merges two states and costs at least one; JUMP 0 0 merges at cost zero; so the merging-price premise fails for the full VM, by design, while A2 holds.

Part IV

What is still open

25 Scope of the claim: what is established, what is not

This is where I draw the line around what I have proved. The named projections lose distinctions that matter to the richer machine. A2 gives an accounting law, and the cost and initiality results have the contracts stated earlier. The questions below ask how much further that argument goes. They stay open until there is an argument that closes them.

Physical interpretation of μ . The ledger is unique once the starting value and the schedule are chosen. That doesn't make it entropy, and it doesn't make it energy. A physical bridge needs a model of which states carry information, premises that can all hold at once, and a real argument tying the formal charge to dissipation. The full-ISA F1 premise pair fails the very first test: its universal collapse premise contradicts free jumps. The permanent-certificate theorem (Section 9.1) passes the first two. Its premises hold together, and finite memory accounts for retained history. Its entropy version (Section 9.5) states the price in bits, and turns bits into joules only through Landauer's principle as a named premise. Tying those units to measured heat is the part no proof here can do.

Class separation. The witnesses show specific windows losing specific structural fields. They don't stop a conventional simulator from keeping those fields in its encoding. No separation of standard complexity classes follows from these collisions, or from the charge levels climbing without bound.

Program predicates and state observations. The state-level CSR collision and the program-level diagonal live in different domains. Linking them would take a theorem of its own. The nat family and the bounded VM theorem keep their explicit quantifiers and representability premises.

Feasible-set cost bounds. The representation theorem is handed its tree and its representative reduction, along with the covering and payment conditions, the uncertified start, and the certified end. It doesn't dig the tree out of an arbitrary deterministic run, and it isn't a Shannon bound on every trace.

Full quantum behavior. The fixed-matrix biconditional is about one specified completion, with zero marginals and zero within-party moments. The ellipsope lets the within-party moments move, at the level of correlators. Full behaviors with marginals, and actual operator realizations, need more identifications than that. A sound runtime check doesn't make it complete for those bigger sets.

Those are the edges of the individual results. Every structural, conditional, and con-

sistency result keeps its hypotheses attached. What I'm proposing reaches further than certification, but no theorem here puts an unavoidable physical price on every observable change.

Trace initiality and foundational interpretation. The universal property is about evaluating instruction lists (Section 3.7). Ledger uniqueness assumes a schedule. Exact pricing assumes an event, and a reference count that says what overcharging means. Those theorems make the construction precise. They don't show that its chosen category, or its chosen event, is physically necessary.

The broader research question. Certification is one worked example. A family of observations might turn up other useful structural laws, but each would need its events, its evidence, and its cost rules justified on their own. The separation and oracle results are about what specific maps keep. They don't show infinitely many independent dimensions, zero mutual information, or a cost for changing coordinates.

26 The pointer-observable conjecture

Why put an event on the meter? The substitution contest fixes the event and compares its chargers. I want to go after the choice itself. My candidate is record proliferation: a distinction matters across an ecosystem when separate observers keep enough evidence to recover it. That gives me a criterion to test, and something specific to be wrong about.

Here is the criterion, stated as a conjecture and nothing more.

Conjecture (the pointer-observable criterion). Among the events a step rule could meter, certification is singled out by redundant record proliferation: a certification event is the kind of event whose record gets copied, checked, and stored across independent substrates that do not otherwise share state, while rival meterable events leave no such redundant trail. The event a substrate discipline is forced to price is the one whose records proliferate.

The name is borrowed deliberately. In the decoherence literature, a pointer observable is the one the environment keeps redundant records of: many independent fragments of the world each carry enough information to reconstruct the pointer value, and that redundancy, not any intrinsic label, is what makes those observables the objective, classical-looking ones. I am conjecturing that certification stands to computation the way pointer observables stand to quantum states: not privileged by fiat, privileged by being the thing everything else keeps copies of. Whether that analogy is load-bearing or just load-shaped is exactly what the conjecture is for; I am naming it, not claiming it.

Section 17 gives the idea five concrete settings: proof-of-stake finality, gas metering, trusted-execution attestation, certificate transparency, and proof-carrying verification.

These disciplines supply the motivation. I supply the formal models, including what each observer sees and which events compete for the meter. That choice is part of the argument. The proofs cannot make it for me. A public certificate may be checkable without every observer keeping a copy; a paid instruction may have no independently stored record. The criterion has to face those cases.

The permanent-certificate theorem (Section 9.1) gives the question a second handle. That theorem never uses the word certification for anything special. It prices any yes/no reading that, once on, stays on, on a machine with finite memory. So it points at a narrower candidate: the events to price are the ones that write a record no later step can take back. Certification is one of them. I tested that candidate in Section 9.3. On finite hardware, the events that must be priced are the merges, every permanent-record write is a merge, and the only free flips are ones the same step can revoke. The candidate as an exact answer is false: a merging step can also revoke. Whether the records that proliferate are the permanent ones, and whether proliferation picks out anything the merge criterion doesn't, is what the two handles have to settle between them.

How to attack it, because a conjecture without a target painted on it is just a mood. Exhibit a deployed, independently evolved metering discipline that prices a non-certification event with the same conviction these five price certification, an event whose records do *not* proliferate, and the criterion is dead. Or show the criterion fails to discriminate: that multiplication counts or head-reversals proliferate redundantly in some substrate ecosystem by the same standard I just applied, so the criterion picks out more than one event and decides nothing. Or formalize the criterion and show certification fails it. Any of the three would teach me more than another instantiation would.

`PointerObservable.v` defines the objects: an `Ecosystem` of observers each holding a boolean fragment of the state; `records`, one observer's fragment deciding an event; `redundantly_proliferating`, every observer recording it; and `unique_pointer_among`, proliferating while no named rival event does, with uniqueness deliberately relative to the rival family because absolute uniqueness is false for free (an event's complement proliferates whenever it does). The record has no predicate saying the observers are independent; that stays part of the modeling claim. The conjecture itself is stated as a schema (`pointer_criterion_holds`): over any class of ecosystems with designated metered events and rivals, every metered event is the unique pointer among its rivals.

`PointerObservableReductions.v` supplies five minimal observer models and proves `five_labeled_models_have_selected_pointer`. The choice of observable fragments, metered events, and rival families is explicit. All five are one construction with the fields renamed. These are five formal instances, not five independent empirical validations of the criterion.

The five proofs establish the criterion inside those five models. To use them as an argument about deployed systems, I owe the correspondence: why those fragments describe the observers, why those rivals are the relevant rivals, and why their records should govern the price. Five proofs about chosen models do not settle those choices. That is where the conjecture has to earn its reach.

Counterexamples are part of the argument. `PointerObservableCounterexamples.v` runs an adversarial search against the strong reading of the criterion, with its protocol written down before its results: deniable authentication, symmetric authentication, and object capabilities. That each design resists forgery and is independently deployed is argued in prose. The Coq half fixes an observer map for each and checks that the event fails to proliferate in it. It does not prove cryptographic security or establish deployment facts. On the argued cases, a rule claiming that forgery resistance alone forces every observer to retain evidence fails.

Being publicly checkable and actually getting copied everywhere are different things. A way to verify something existing doesn't mean every observer keeps a record of it. The abstract verifier examples give three ways to add evidence, each with its own guarantee. They don't show those are the only ways, or that only one of them can involve metering.

So the real question is whether a well-argued choice of observers and rival events picks out metering disciplines worth having. `V_does_not_factor_through_classical` proves that a sound and complete verifier for the distinguishing claim can't get by on the colliding bare observation alone. It doesn't pick one physical way to supply the missing evidence.

The conjecture asks which events deserve a cost law. The accounting theorems tell me what follows once that law is specified. I want both parts of the argument on the table, with neither borrowing a proof from the other.

27 Conclusion

The Thiele Machine is an abstract model of computation, and this book has been an argument made with it. The argument is that the usual account of a computation, what it reads and writes, is a shadow, and that the axis it leaves out belongs in the step. That axis holds what a computation established and what establishing it cost.

Here is where each step of the argument ended up. The shadow loses something real: proved, for every window I named, with witness pairs you can print. Establishing is not free: proved, for any system that follows A2, whatever its states and instructions. And on finite hardware with certificates that last, A2 itself follows from Landauer's principle: proved, with that principle as the named premise, in counted states and in bits. A finite piece of the VM is such a machine, and the VM runs it at exactly that price. The full VM prices the certifying merge and leaves other merges free, by

design. Leaning on a fact takes a paid history: defined, with the cost floor proved for shortcuts that hand over their receipts. The axis belongs in the step: proposed, and still mine to earn. Two parts of it are proved. An account that prices certification exactly can't compute its price from the window, because the VM has a certifying step and a non-certifying one that look the same through it; the exact price needs the reading in the step. And for a finite machine whose account prices merges, that account already charges every step that writes a permanent record, at least the logarithm of the squeeze. Which events every account has to price: the merges, on finite hardware, with every permanent-record write among them and revocation as the only escape. Proved, with Landauer's principle as the premise. The guess that the priced events are exactly the permanent records is false. What else belongs on the meter is open, with the pointer-observable conjecture as my candidate.

Two results stand out as single pieces, one in the build and one in the model.

In the build, the sharpest thing I found is this: an integer-arithmetic check on the CHSH outcome counters forces the zero-marginal NPA moment matrix to be positive semidefinite, and the argument from the one to the other never once reaches for physics. No Hilbert space, no operators, no observables anywhere in the chain. That is the claim, and it is algebra you can follow line by line. Where I checked it is `chsh_lassert_no_trap_implies_npa_psd`, and the check is where I confirmed I hadn't smuggled physics in without noticing: `Print Assumptions` on it leans on exactly three standard real-number axioms from Coq's own library: Dedekind-reals decidability (`sig_forall_dec`, `sig_not_dec`) and functional extensionality, and nothing of mine. The substrate-level result below leans on even less.

In the model, at the substrate level, the diagonal rules out a correct decider whenever its flip can be expressed inside the substrate and the instance supplies the fixed point. The nat construction proves a family result: every candidate d gets its own instance, and d fails on that instance's predicate P_d . The VM version keeps its bounded recurrence and representability premises. In L, a Turing-complete lambda calculus, the recursion theorem is proved from the reduction rules, and the same diagonal is Rice's theorem (Section 10.4). Separately, the reachable-state collision shows the CSR predicate is lost through `forget`. These are different claims. Sharing a vocabulary, or a clean assumption report, doesn't make them one.

All of this supports looking for a structural account of computation. The A2 theorem covers any system that follows its pricing rule. The irrecoverability theorem is about chosen windows. Neither needs the concrete VM to be the whole subject, and neither says every computation has to adopt this law. Which reading is the foundational one is still part of the argument.

Each of these results carries an explicit falsification condition, and they are gathered in one place (Section 23): a zero- μ trace from uncertified to certified, a column-contractive correlator with a non-PSD NPA matrix, a cost-rule-respecting `CertificationSys`

tem that certifies at total cost zero, a `nat` function that decides `nat_admits` for its own substrate. Any one of them collapses the matching theorem. You can hunt for any of them with a pencil, and the mechanised ones would stop the checker dead on top of that. I hunted. I came up empty.

I do not want this to sound like I am declaring I am right. I want to push myself, and after investigating everything, maybe the field too.

I built this. The definitions are here, the proofs are here, and the machine runs. If I have claimed more than a theorem gives me, that sentence needs fixing. If a proof fails, the proof needs fixing. I want to know where. I also want the larger question to survive the inspection: what gets left out when a computation is described only by the answer it returns? Certification gave me one place to get a grip on that question. I intend to keep going.

A Things I had to learn first

I had been around computers my whole life and paid attention to people like John Carmack on how graphics actually get made, but I came in with none of the formal fields this work turned out to need. By the time the proofs compiled I had needed to learn enough of half a dozen of them that I had to put down somewhere what those fields had words for. This appendix is that place. None of it is the argument. It is what I needed to be able to make it.

Everything below is what I understood after I went and read about it. If I have anything wrong, tell me. I am not claiming to teach you these subjects. I am putting down what I had to take from each of them to get this work to compile. If you already know one of these subsections, skip it.

A.1 Bits, bytes, and what a CPU sees

A bit is one digit that is either zero or one, the smallest fact a machine can hold. Everything else is bits stacked up. Eight of them stacked make a byte, which lands you any number from 0 to 255, and the first thing that genuinely surprised me is that the same byte is also a letter: a convention called ASCII (American Standard Code for Information Interchange) simply agrees to call 65 the letter A, 48 the digit 0, 32 a space. Nothing in the byte knows which it is. The meaning is a convention laid on top of the data, and which convention you read it through is your choice, not the byte's. Hold onto that, because it is the whole argument of this document shrunk down to one byte: the data sits there inert, and everything that matters is in the structure you read it through. I learned the byte part in February.

A word is a chunk of bits the CPU treats as one unit. The kernel math uses 64-bit words (each word can hold any number from 0 to $2^{64} - 1$, about 18 quintillion). The synthesised hardware uses 32-bit words (each word holds 0 to $2^{32} - 1$, about 4 billion). A word is just a sequence of bits with a fixed width. The substrate does not care which

width.

A register is a single named storage slot inside the CPU that holds one word. The 51-opcode VM has 16 of them, named r0 through r15. Registers are the fastest place to read or write a value. Memory is a longer list of word-sized slots, accessed by index (the “address”). The 51-opcode VM has 128 words of memory. That is the entire address space. The substrate has no address space.

The program counter (PC) is one specific register that tracks where in the program the VM is. After every instruction the PC moves forward by one unless the instruction explicitly says jump to a different address.

A stack pointer is a register that points to the top of a region of memory the program uses for nested function calls. CALL pushes a return address there; RET pops it. In the 51-opcode VM, r15 is the stack pointer by convention. The convention is mine. The substrate is not consulted.

A port is a named channel the 51-opcode VM uses to communicate with the outside world. READ_PORT brings a value in, WRITE_PORT sends a value out. The kernel does not model the outside world. It just charges μ for the communication and accepts the value the channel encoding declares. The outside world is welcome to comply. The substrate has no ports.

Modular arithmetic means counting wraps around at a maximum. If the maximum is 16, then 17 wraps to 1 and 19 wraps to 3. “Modulo 16” or “mod 16” means “wrap at 16.” The 51-opcode VM wraps register indices modulo 16 and memory indices modulo 128. If you write r17, the kernel treats that as r1.

Bitwise operations work on the bits of a value rather than treating the value as a number. Bitwise AND of two bits is 1 only when both bits are 1. Bitwise OR is 1 when either is 1. Bitwise XOR (exclusive or, sometimes written $\oplus$) is 1 exactly when the two bits differ. The bitwise version of these does the operation in parallel for every bit position. Bitwise-AND of two 64-bit words gives a 64-bit word. Bit i of the result is the AND of bit i of each input. The bits do not consult each other. The substrate does not see bits.

A shift moves the bits of a value left or right. Left shift by 3 (sometimes written $x \ll 3$) moves every bit three positions to the left, which is the same as multiplying by $2^3 = 8$. Right shift divides by a power of 2. Same idea, opposite direction.

Popcount is the count of 1-bits in a value. Popcount of the byte 11010100 is 4. Sometimes called the Hamming weight; the two names refer to the same thing. Pick one.

Hexadecimal is base 16. Each “digit” is one of 0 through 9 or A through F, where A means 10 and F means 15. Hex is convenient for working with bytes because every two hex digits make one byte. Numbers prefixed with 0x are hex: 0xFF is 255, 0xCAFEFACE is the constant unlock key in the Logic Engine accumulator (the Logic

Engine is the machine's on-chip formula checker; Section 5 builds it). It had to be something.

A trap, in the hardware sense, means the machine routes execution to a fixed address instead of advancing normally. It is the hardware equivalent of "this went wrong, go to the error handler." The 51-opcode VM traps LASSERT length mismatches to address 0xF00 (LASSERT is the logic-assertion opcode; Section 12 builds it).

An ISA (instruction set architecture) is the list of all instructions the CPU recognises, plus their formats and costs. Each instruction is identified by an opcode (a numeric tag the hardware decodes to choose which behaviour to run). The kernel has 51 opcodes. The substrate has none. The substrate has a step rule.

A.2 The notation the formal parts use

I don't think in these symbols. The formal statements in this book were written in standard notation by the models and checked in Coq, and I had to learn to read them the same way you might be learning now. Here is what each one means, in plain words, so you never have to take a symbol on faith.

$\mathbb{N}$ is the natural numbers: 0, 1, 2, 3, and so on. $\mathbb{Z}$ is the integers, which extend the natural numbers with negatives: $\dots, -2, -1, 0, 1, 2, \dots$. $\mathbb{R}$ is the real numbers, all the way along the number line including fractions and irrationals like π and $\sqrt{2}$.

$\in$ means "is an element of." $x \in S$ reads " x is in S ."

$\subseteq$ means "is a subset of" (every element of the left side is also in the right). $\subsetneq$ means strict subset: a subset but not the whole thing.

$\cup$ is set union, the set containing everything in either side. $\setminus$ is set difference: $A \setminus B$ is everything in A that is not in B .

$\times$ between two sets is the Cartesian product: $A \times B$ is the set of all (a, b) pairs with $a \in A$ and $b \in B$. Functions from a set A to a set B are written $A \rightarrow B$.

For a finite set S , $|S|$ means the number of elements in S . For a real number x , $|x|$ means the absolute value of x (its distance from zero).

$\sum_{i \in T} f(i)$ is the sum of $f(i)$ taken over every i in T . The big sigma is the summation symbol.

Δx means "the change in x ," usually the difference between a final and an initial value of x .

$\Rightarrow$ is logical implication. $P \Rightarrow Q$ means: if P is true, then Q is true. $\iff$ is the biconditional, also called "if and only if." $P \iff Q$ means both $P \Rightarrow Q$ and $Q \Rightarrow P$.

$\forall$ is "for every." $\exists$ is "there exists." $\forall x. P(x)$ reads "for every x , $P(x)$ is true." $\exists x. P(x)$ reads "there is some x that makes $P(x)$ true."

$\langle X \rangle$ around a random variable X means the expected value, the long-run average value of X . I picked up this notation reading physics papers about Bell tests.

$\circ$ between two functions is composition: $(f \circ g)(x) = f(g(x))$, first apply g , then apply f . The same symbol appears in category theory for arrow composition.

$\oplus$ in this document is bitwise XOR. $\otimes$ in this document has exactly one meaning: the MORPH_TENSOR operator of Section 12, concatenation of two coupling lists into one. Physics writes the same symbol for the tensor product of quantum operators; that sense never appears on these pages, because Section 18 does its quantum work in plain integer arithmetic.

A^T is the transpose of a matrix A (rows and columns swapped). I_n is the $n \times n$ identity matrix, which has 1 on the diagonal and 0 everywhere else.

A.3 Coq, in just enough detail

Coq is a proof assistant. It is a programming language. The type system is so expressive that you can write proofs as programs and have the compiler check whether they are valid. Section 21 has the longer build, including why I trust it. Here is the vocabulary that runs through the rest of the document.

A Theorem in Coq is a named statement followed by a proof. Lemma is the same construct, conventionally used for intermediate results. Definition introduces a named value (a number, a function, a record, anything). Inductive defines a type with a finite list of named constructors, the only ways to build a value of that type. Coq's natural numbers, for example, are an inductive type with two constructors: zero, and "successor of another natural number." Two constructors, infinitely many naturals.

A tactic is a step inside a Coq proof. The tactics that appear in this document include: induction (prove the base case, then prove that if the statement holds at step n it holds at step $n + 1$), destruct (split a hypothesis into its cases), exact (provide the literal proof term), reflexivity (prove $x = x$), simpl (simplify the goal by computing), lra (linear real arithmetic; closes goals that are linear inequalities over the reals), nra (nonlinear real arithmetic; handles polynomial inequalities), and psatz (uses Positivstellensatz certificates; it shows a polynomial is non-negative by writing it as a sum of squares). I learned each of them by needing it.

Qed is the keyword that ends a proof. It asks Coq to check the entire chain. If Coq accepts the proof, the theorem is proved.

Admitted is the keyword for "I am claiming this without proof." This is what I am not doing anywhere in the active proof tree. Every Admitted in the kernel would be a bug. Section 22 explains what zero Admitted means. And what it does not.

A Hypothesis (a Coq keyword inside a Section) is a named premise: a statement taken as given without proof. Hypotheses surface as $\forall$ -quantified premises in the conclusion.

Hypothesis only appears at named bridge points (the physical-units calibration in Section 19 is the load-bearing example). An Axiom is the global version: a top-level claim accepted without proof. The kernel tier has zero project-local Axiom declarations. The policy is bookkeeping discipline, not faith.

Print Assumptions is the Coq command that lists the global axioms a given theorem ultimately depends on. Running it on the kernel tier shows only standard Coq stdlib axioms (functional extensionality, classical logic for the reals). No project-local axiom appears anywhere in those dependency trees. One thing it does not show: a premise written into the theorem’s own statement. Those stay in the type, and “closed under the global context” does not make them go away. Whether they hold for the case you care about is something you check by reading the statement.

A total function in Coq always returns a value. Coq requires every function definition to be total. `vm_apply` is total. Every state-and-instruction pair has a defined result. Including the error case.

“Parametric in K ” means the proof works for any value of K . Register, memory, and table sizes are explicit constants in the implementation. Changing them requires rebuilding the affected definitions and checking their representation obligations. A proof about the abstract state alone does not establish that every size choice fits a physical target.

A.4 Category theory: the part I needed beyond Section 13

Section 13 builds Category, Object, Morphism, Composition, Identity, and Associativity from scratch. That build is enough for the categorical-separation argument inside this machine. Here is what I had to look up beyond that.

A functor is a structure-preserving map from one category to another. Given two categories $\mathcal{C}$ and $\mathcal{D}$, a functor F from $\mathcal{C}$ to $\mathcal{D}$ sends each object of $\mathcal{C}$ to an object of $\mathcal{D}$. It sends each arrow $f : A \rightarrow B$ in $\mathcal{C}$ to an arrow $F(f) : F(A) \rightarrow F(B)$ in $\mathcal{D}$. It respects composition ($F(g \circ f) = F(g) \circ F(f)$) and identity ($F(\text{id}_A) = \text{id}_{F(A)}$). Think of a functor as a translation rule between two categorical worlds that keeps the laws intact.

An initial object in a category has exactly one morphism into each object. Here `thiele_trace_fold_initial` proves that there is exactly one way to evaluate plain lists of instructions into a target once you pick its starting point. That is a statement about instruction lists. It doesn’t make the VM’s state space initial under certification-preserving state maps. Those maps might not exist at all. When somebody supplies them, their uniqueness on reachable states is a separate theorem.

An embedding is a functor that throws nothing away. That is the whole point of putting the word here: the classical projection of a VM state loses things on the way out, and an embedding is the move that does not. Faithful is the precise version of

“loses nothing on arrows”: if two arrows between the same pair of objects in $\mathcal{C}$ land on the same arrow under F , they were already the same arrow, so F never quietly fuses two distinct things into one. That is injectivity on each hom-set, checked one pair of objects at a time. Conservative is the version for invertibility: if $F(f)$ has an inverse over in $\mathcal{D}$, then f already had one back in $\mathcal{C}$, so F cannot manufacture an isomorphism that was not there. (An isomorphism is just an arrow with an inverse.) These are different properties, and neither implies the other. The inclusion of the one-object category of the monoid $(\mathbb{N}, +)$ into the one-object category of $(\mathbb{Z}, +)$ is faithful (distinct natural numbers stay distinct integers) but not conservative: 1 has no inverse in $\mathbb{N}$, yet its image in $\mathbb{Z}$ does. Conversely, the functor from the one-object category of the group $\mathbb{Z}$ to the trivial one-object group is conservative (every arrow in the trivial group is already invertible, so nothing needs reflecting) but not faithful: it fuses every integer onto the same single arrow. Both properties stop a map from inventing structure it shouldn’t have, but they guard against different inventions, and neither stands in for the other. An embedding needs faithfulness specifically.

A monoidal product is a way of combining two objects, or two arrows, into one. In the category of sets, the Cartesian product $A \times B$ is one. A full monoidal product also comes with a unit and with laws that make the combining behave; that whole bundle is what mathematicians mean by a monoidal structure. MORPH_TENSOR is tensor-shaped, and I want to be exact about how far that goes. Take two morphisms whose source and target regions don’t overlap. Pack them into a single morphism whose coupling is the concatenation. If the regions do overlap, the operation refuses. It also refuses unless the combined regions already exist as modules. Coq checks that local behavior. It does not build the full monoidal category with all its coherence maps. The symbol $\otimes$ below names this operation.

Yoneda’s lemma is where the early categorical engine clicked for me, the night I stopped borrowing other people’s runtimes. Nothing in the formal development uses it. It doesn’t show up in any theorem’s dependencies. The checked results rest on the category, morphism, and composition structures defined in the repository.

A.5 Bell-test vocabulary you will meet in Section 18

Section 18 builds the CHSH game from scratch. A few foundational terms thread through that section that I will not rebuild each time.

A correlation between two outcomes is the average value of their product. If Alice’s outcome a is ± 1 and Bob’s outcome b is ± 1 , the correlation is $\langle ab \rangle$, which lies between -1 and $+1$. A correlation of $+1$ means they always match. -1 means they always disagree. 0 means there is no average correlation. A correlator is one specific correlation in a labelled setting: for example, E_{00} is the correlation when Alice’s setting is 0 and Bob’s setting is 0 .

The marginal of a joint distribution on (a, b) is the distribution on a alone, with b averaged out. Zero-marginal means Alice’s marginal gives $\langle a \rangle = 0$ and Bob’s gives

$\langle b \rangle = 0$. Each outcome is $+1$ or -1 equally often, on average. The Section 18 derivation works in this symmetric slice.

A local hidden variable model is a classical-physics-style theory where Alice’s and Bob’s outcomes are determined by a shared random variable λ they agreed on before the game started, plus their own measurement setting. “Local” means no communication during the game. “Hidden” means λ need not be directly observable. “Variable” because λ is random. Bell’s theorem says no local hidden variable model can produce certain correlations that quantum mechanics can.

A no-signaling correlation is one where Alice’s marginal does not depend on Bob’s setting, and Bob’s marginal does not depend on Alice’s setting. You cannot learn Bob’s setting by looking only at Alice’s outputs. Not even in principle. Most correlations in this book (classical, quantum, and the PR-box below) are no-signaling.

Quantum-realizable, in this document, means “compatible with some quantum-mechanical state-and-measurement description.” The NPA framework (Navascués-Pironio-Acín) gives a hierarchy of tests for it, each one positive-semidefiniteness of a moment matrix. Every level is necessary. No single level is sufficient in general. Section 18 builds the first level and part of the next. The theorems here prove PSD of those matrices. They don’t prove quantum realizability.

The PR-box (Popescu-Rohrlich box) is a hypothetical correlation that wins the CHSH game with probability 1, achieving $|S| = 4$. It is no-signaling, but stronger than anything quantum mechanics allows. Nature does not provide it. It is the standard counterexample to the false claim “any no-signaling correlation is quantum-realizable.”

The Cauchy-Schwarz inequality is the basic inner-product inequality $|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$. It reads: the inner product of two vectors is bounded by the product of their lengths. It is one of the workhorse algebraic moves in the Section 18 derivation.

The Tsirelson bound is the value $2\sqrt{2} \approx 2.828$, the largest CHSH score $|S|$ that any quantum-mechanical strategy can achieve. The classical bound is 2. The PR-box bound is 4. Tsirelson sits strictly between classical and PR-box. Nature picked one of these three.

The elliptope, named in Section 24 and Section 25, is a set of correlators: the ones that can be filled out into a positive-semidefinite matrix. In the usual vector or quantum reading it means something physical. The Coq development here proves the real-matrix statement and what follows from it. It doesn’t prove a full physical realization theorem. The proof in Section 18 covers one slice of the set, the one with zero marginals and a fixed completion.

A.6 Hardware and synthesis vocabulary

Section 16 builds the scaffolding stack (Coq kernel, OCaml runner, Kami RTL) carefully. Here is the synthesis-tooling vocabulary that Section 16’s third subsection uses without

expanding each tool.

A hardware description language (HDL) is a programming language for describing circuits. Bluespec SystemVerilog and Verilog are two HDLs. Bluespec is higher-level: you describe rules and the compiler figures out the gate-level details. Verilog is lower-level: you describe gates and wires more directly. Verilog RTL specifically means “register-transfer level” Verilog, which describes how data moves between registers each clock cycle. It is what synthesis tools consume. They do not eat Bluespec.

Synthesis is the process of compiling a hardware description down to actual logic gates (the building blocks of chips). yosys is an open-source synthesis tool. After synthesis, place-and-route assigns the gates to physical positions on a chip and connects them. nextpnr-xilinx is the place-and-route tool for the Xilinx FPGAs this design targets. The last stretch is producing a bitstream, the binary file the FPGA reads to configure itself, and prjxray does it in two moves: fasm2frames packs the routed design into configuration frames, then xc7frames2bit writes the frames out as the bitstream. Four steps, four tools, one chip.

An FPGA (field-programmable gate array) is a chip you can program after manufacturing. The gates are physical. The connections between them are configurable. An ASIC (application-specific integrated circuit) is the non-programmable version: gates fixed at manufacturing time for one specific design.

A LUT (look-up table) is the basic logic primitive in an FPGA: a small memory that maps a fixed-length list of inputs to an output, programmed at configuration time to implement any small boolean function. A flip-flop is a one-bit memory cell that holds its value across clock cycles. A carry chain is specialised wiring for fast addition. Distributed RAM is RAM built from LUTs. Block RAM is dedicated RAM blocks on the chip. DSP slices are specialised hardware for digital-signal-processing arithmetic. The synthesised K325T design uses none of them. yosys runs with `-nodsp`, so the multiplier in the multi-cycle witness checker (Section 16) is built out of LUTs. Put on DSP slices instead, the same multiplier couldn’t get through the open-source place-and-route in time. It computes the same thing either way. The choice is about what the toolchain can place and route, not about correctness.

A JSON netlist is the gate-level circuit serialised in a JSON text file (JSON is a standard text format for structured data). A VCD waveform is a Value Change Dump, a record of every signal’s value at every time step in a simulation. The name describes the file. iverilog is an open-source Verilog simulator. It runs the design and produces a VCD output you can inspect.

A bisimulation is a precise notion of “two systems behave the same.” You pick a relation between the states of one system and the states of the other. Related states have to agree on whatever you chose to observe, and whenever one of them takes a step, the other has a matching step that lands in related states again. That is looser than

“same output state, field by field.” Field-by-field equality is one particular, stronger contract. Section 16 says exactly which observations and transitions the hardware relation covers. A parity test is the empirical cousin: run two implementations on the same battery of inputs and check they agree, a correspondence earned by testing rather than by proof. The kernel-to-OCaml boundary is closed that way, and the document says so where it happens.

A.7 Other terms I use without rebuilding

Provenance, in software, means the recorded chain of where an object came from. A morphism’s provenance is the specific sequence of MORPH and COMPOSE instructions that built it. You can replay the chain.

Simulation, in computer science, means one system imitating another’s input-output behaviour. A Turing machine simulates a Thiele Machine when it produces the same outputs on the same inputs, by writing the whole Thiele state out across its tape and grinding through it. That is real, and it matters, and it is exactly the thing people reach for when they want to wave the whole question away: if a tape can stand in for the state, what was the difference? The difference is that the tape carries a picture of the state, not the state, the way a photograph of a fire keeps you exactly as warm as any other photograph. The cost ledger and the A2 rule live in the step relation, not in the bytes, and an encoding copies the bytes. Simulating the substrate is not being the substrate. A Turing machine can be one, though: build its transition table so the rule is its own step rule, and it is an instance, not a simulation. What no machine can do is make a simulated rule binding just by writing it down on tape.

Projection, in mathematics, is a function that drops information on purpose. The classical projection of a VM state is the one that does the dropping I keep pointing at: it throws away μ , certification, and the morphism graph, and keeps registers, memory, and the program counter. That is the exact list. The three it keeps are the three a Turing machine already has, which is the whole reason the projection lands you back in ordinary computation. The three it drops are the three that were doing the new work. Nothing about the projection is wrong; it is a perfectly good function. It just costs you everything that made the state more than a tape.

Quotient, in mathematics, is the operation of declaring two things the same when they agree on what you decided to look at, and then working with the lumped-together classes instead of the originals. The classical projection is exactly that kind of move: any two VM states that match on registers, memory, and the program counter get swept into the same class, even when their μ , their certification, and their morphism graph were completely different. So the projection is not just dropping fields one at a time; it is gluing together states that the dropped fields kept apart. That is the precise sense in which it loses information, and it is comforting that the math already had the word sitting there waiting.

A kernel of a map is context-dependent. For a group or module homomorphism it

is the inverse image of the identity element; for the quotient induced by a projection π , the relevant equivalence relation is $s \sim t$ exactly when $\pi(s) = \pi(t)$. The classical projection in this document uses that second quotient reading. “Kernel” wears three different hats in this document and I would rather call that out than let it ambush you: this mathematical usage; the Coq proof core, meaning the kernel-tier theorems in Section 24, which is scaffolding I lean on, not the substrate itself; and the operating-system kernel, which I imply at most by analogy and never actually use. Whenever the word shows up, the surrounding sentence tells you which one is meant.

Embedding is a function that preserves structure and is injective. Faithful embedding additionally preserves all the structure-relevant distinctions.

An injection is a function where no two inputs map to the same output. A surjection is a function whose image covers every possible output. Their negations are noninjective and nonsurjective.

Decidable, in computer science, means there is an algorithm that always halts with a yes-or-no answer. Keep the question and the algorithm apart. The predicate is the mathematical claim $P(x)$. The decision procedure is the algorithm, and it only decides P once there is a proof that its answer matches. The integer-only Z-arithmetic check in CHSH_LASSERT always halts with a Boolean. Its soundness theorem says a pass means the stated real-matrix predicate holds.

Polynomial complexity: an algorithm is polynomial-time if there is some power k such that the algorithm’s running time on inputs of size n is at most a constant times n^k . Polynomial-space is the analogous concept for memory used. The class P is decision problems solvable in polynomial time. The class NP is decision problems where, given a candidate solution, a polynomial-time algorithm can check whether the solution is correct. The famous P-vs-NP question asks whether finding is fundamentally harder than checking. Nobody has proved either direction. The predicates `error_free_preservation_budget` and `positive_input_certification_budget` only constrain traces chosen separately for each input. They do not express these complexity classes.

Determinant: a number computed from a square matrix that says whether the matrix is singular (zero determinant means the matrix collapses some direction to zero). For a 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the determinant is $ad - bc$. For larger matrices, you can compute it by “expansion along a row”: pick a row, for each entry take the determinant of the submatrix you get by deleting that entry’s row and column (with alternating signs), multiply by the entry, and sum. The recursion bottoms out at the 2×2 case above. Section 18 walks the explicit calculations for the 3×3 cases the argument needs.

Schur complement: a way of turning a question about a block matrix into a question

about a smaller matrix. If $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ is a block matrix where A is invertible, the Schur complement of A is $D - CA^{-1}B$. The fact the argument uses: for a symmetric real block matrix whose pivot A is positive definite, the whole matrix is positive-semidefinite exactly when the Schur complement is. Invertible alone is not enough. The pivot has to be positive definite. The Section 18 build uses this for the 4×4 block of the specified moment matrix.

A principal submatrix is the square subblock you get by picking a subset of indices and keeping only the rows and columns at those indices. A standard result is that a matrix is positive-semidefinite if and only if every principal submatrix is positive-semidefinite. Every one. Section 18 relies on this.

A.8 The axioms (A1 through A6)

The Thiele Machine has a small numbered list of substrate-level axioms. They appear by name (“A2,” “A3,” ...) throughout the document and the formalization. The list below is what each name means, where it lives in the kernel, and what its conclusion is. The axioms are substrate-level; the kernel is scaffolding-level; the list bridges the two.

The kernel keeps two parallel axiom-numbering systems because two parallel substrate abstractions are formalised. The `AbstractCertMachine` record carries A1 (the cert indicator exists, by record existence) and A3 (only cert-setters flip it, the `acm_preserve` field). A2 (cert starts unset) and A4 (cert-setters cost ≥ 1) live alongside the record, not in it: A2 is the `acm_cert M s0 = false` hypothesis to `abstract_nf1`, A4 is the standalone lemma `cert_addr_setter_cost_pos`. The consolidated `CertificationSystem` record folds the cost-on-certification semantic content into a single field kept by the name A2 (the load-bearing piece). The `QuantitativeCertificationSystem` extension adds A3–A6, which lift the cost-floor result from ≥ 1 to $\geq K$ for an arbitrary threshold K . The two numbering systems agree on the cost-on-certification semantic content; they disagree on which axiom carries that content (A4 in the `AbstractCertMachine` list, A2 in the consolidated `CertificationSystem` list). When the rest of this document says “A2,” it means the cost-on-certification rule, in the consolidated `CertificationSystem` sense.

A1 (state includes a certification indicator).

An `AbstractCertMachine` record bundles a state type S , a step function `acm_step`, and a boolean cert indicator `acm_cert` : $S \rightarrow \text{bool}$. A1 is the structural requirement that this indicator exists at all. In Coq, A1 is enforced by the record type itself. There is no way to hand Coq an `AbstractCertMachine` value with `acm_cert` left blank; it reads the field as missing and declines to build the record, the way a clerk declines a form with an empty box. A1 is discharged at type-check time.

A2 (cost-on-certification).

In the consolidated `CertificationSystem` record, Coq field `cs_cert_costs`. Statement: for every state s and instruction i , if `cs_cert s = false` and `cs_cert (cs_step s i) = true`, then `cs_cost i` ≥ 1 . In words: any single step that flips the certification predicate from no to yes must have positive cost. This is the load-bearing axiom for No Free Insight. `universal_nfi_any_substrate` concludes from A2 alone that every trace from uncertified to certified has total cost ≥ 1 . Note: in the `AbstractCertMachine` numbering, the same content appears as A4 (cert-setters cost ≥ 1). The `AbstractCertMachine` A2 is the separate precondition that the cert indicator starts unset, which the consolidated system handles as a theorem hypothesis instead of a numbered axiom.

A3 (cost bounds witness growth).

Coq field `qcs_cost_bounds_witness` on the record `QuantitativeCertificationSystem`. Statement: for every state s and instruction i , `qcs_witness s + cs_cost i` $\geq$ `qcs_witness (cs_step s i)`. In words: each instruction's cost is an upper bound on how much the system's accumulated evidence (the "witness") can grow in that step. The cost is the speed-limit on evidence accumulation. (In the `AbstractCertMachine` numbering, A3 is the field `acm_preserve`: non-cert-setter instructions preserve the cert indicator. The two A3s are distinct properties on distinct record types; the monograph's A3 is the quantitative one above.)

A4 (witness nondecreasing).

Coq field `qcsf_witness_nondecreasing` on the record `QuantitativeCertificationSystem_full`. Statement: for every s and i , `qcs_witness s` $\leq$ `qcs_witness (cs_step s i)`. A3 and A4 point in opposite directions: A3 caps how fast the witness can grow ($w' \leq w + \text{cost}$), A4 stops it from shrinking ($w \leq w'$). Neither implies the other. $(w, \text{cost}, w') = (1, 0, 0)$ satisfies A3 with non-negative cost yet violates A4, and $(0, 0, 1)$ does the reverse, so A4 is its own field, not a consequence of A3. It is also idle: the $\geq K$ floor telescopes through A3, A5, and A6 alone and never touches A4. The field is kept so downstream consumers that want witness monotonicity in hand have it without re-deriving it; the consolidated `QuantitativeCertificationSystem` record drops A4 entirely, carrying only A3 and A5.

A5 (certification requires witness).

Coq field `qcs_cert_threshold_witness` on `QuantitativeCertificationSystem`. Statement: `cs_cert s = true` $\rightarrow$ `qcs_witness s` $\geq$ `qcs_threshold`. In words: certification cannot fire until the system has accumulated at least `qcs_threshold`-worth of evidence. This is what lets the quantitative theorem push the cost floor from ≥ 1 to $\geq K$, where K is the threshold. Pay more, certify more.

A6 (witness initial zero).

The hypothesis that the run starts with zero witness, stated as a precondition on the quantitative theorem rather than a record field. Without A6 the telescoping bound becomes $K - \text{initial_witness}$ instead of K . Zero is a clean place to start.

A1 through A6, as the consolidated records state them, are independent of any specific instruction set; the `AbstractCertMachine` variant is the one exception, typing its step over the VM's own instruction type. They are conditions on Coq record types; supplying instances of those record types is the only way to invoke the corresponding theorems. The 51-opcode VM supplies them by construction: the kernel builds two concrete `CertificationSystem` values for the VM (one for the `csr_cert_addr` channel, one for the `vm_certified` channel), each discharging A2 from the existing `no_free_certification` kernel theorems. A separate substrate-level abstraction (the `Substrate` typeclass) carries a related but distinct field `Substrate.mu_monotone` stating that the cost ledger never decreases on a step. This is a different, more general property than A2. A2 says the cert transition costs ≥ 1 . Monotonicity only says the ledger never goes down. Neither is stated as implying the other. Both hold on the 51-opcode VM, where every instruction adds non-negative cost and the cert transition adds at least 1. Two locks on the same door, by different keys.

That is the vocabulary. The rest of the document uses these words without rebuilding them.